



# Cambridge IGCSE™

## PHYSICS

0625/12

Paper 1 Multiple Choice (Core)

October/November 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall =  $10 \text{ m/s}^2$ ).

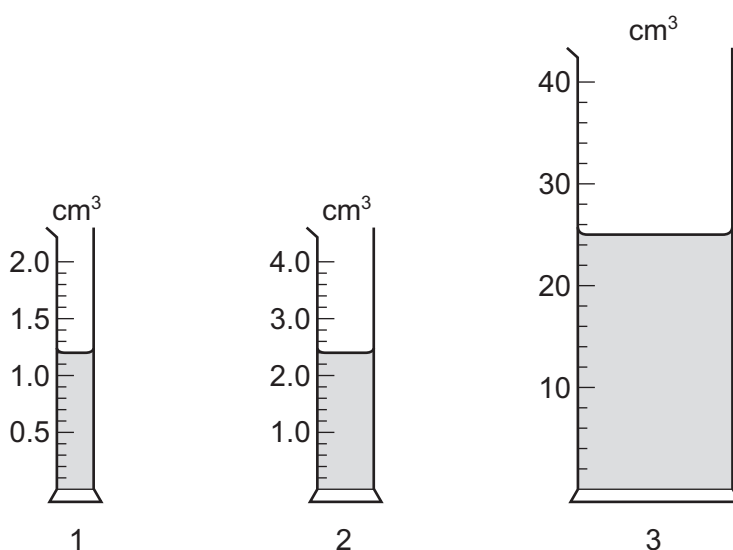
## INFORMATION

- The total mark for this paper is 40.
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- Any rough working should be done on this question paper.

This document has **16** pages. Blank pages are indicated.



- 1 A student measures the volumes of three liquids using three different measuring cylinders.



The table shows the volumes recorded by the student.

| measuring cylinder | volume / cm <sup>3</sup> |
|--------------------|--------------------------|
| 1                  | 1.2                      |
| 2                  | 2.2                      |
| 3                  | 25                       |

Which readings are correctly recorded?

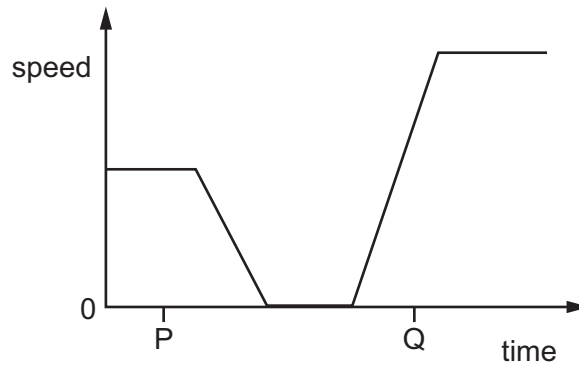
- A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 1 only

- 2 A car driver measures the time taken to complete four separate journeys.

In which journey does the driver have the greatest average speed?

|          | distance / km | time / h |
|----------|---------------|----------|
| <b>A</b> | 60            | 2        |
| <b>B</b> | 60            | 3        |
| <b>C</b> | 120           | 3        |
| <b>D</b> | 120           | 4        |

- 3 The graph shows how the speed of an object varies with time.



Which row describes the motion of the object at times P and Q?

|          | P                          | Q            |
|----------|----------------------------|--------------|
| <b>A</b> | at rest                    | accelerating |
| <b>B</b> | at rest                    | decelerating |
| <b>C</b> | moving with constant speed | accelerating |
| <b>D</b> | moving with constant speed | decelerating |

- 4 A stone has a weight of 4.1 N.

What is the mass of the stone?

- A** 0.41 kg      **B** 2.4 kg      **C** 4.1 kg      **D** 41 kg

- 5 Which quantity is weight an example of?

- A** acceleration  
**B** force  
**C** mass  
**D** pressure

- 6 A rectangular metal block is 20 cm long.

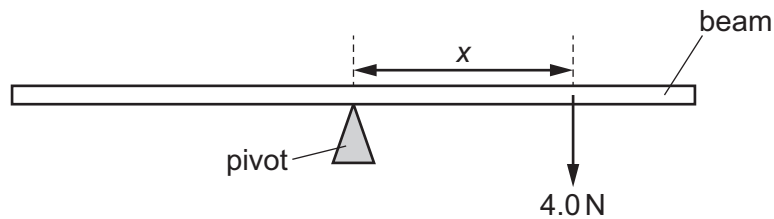
The cross-sectional area of the block is 25 cm<sup>2</sup>.

The mass of the block is 4000 g.

What is the density of the metal?

- A** 0.13 g/cm<sup>3</sup>      **B** 0.32 g/cm<sup>3</sup>      **C** 8.0 g/cm<sup>3</sup>      **D** 2000 g/cm<sup>3</sup>

- 7 A force of 4.0 N acts on a beam as shown.

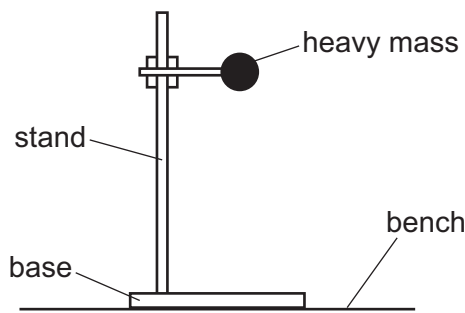


The line of action of the force is a distance  $x$  from a pivot. The moment of this force about this pivot is 8.0 N cm.

What is distance  $x$ ?

- A** 0.50 cm      **B** 2.0 cm      **C** 12 cm      **D** 32 cm

- 8 The diagram shows a stand. The stand holds a heavy mass above the bench.



Which two changes would definitely make the stand more stable?

- A** Lower the mass and make the base narrower.  
**B** Lower the mass and make the base wider.  
**C** Raise the mass and make the base narrower.  
**D** Raise the mass and make the base wider.

- 9 Two wind farms supply electrical energy to consumers in different ways.

Wind farm 1 supplies energy directly to consumers and for every 1000 J of energy extracted from the wind, 360 J is transferred to consumers as electrical energy.

Wind farm 2 stores electrical energy in batteries and then supplies energy from the batteries to the consumer. For every 1000 J of energy extracted from the wind, 270 J is transferred to consumers as electrical energy.

Which statement about the two wind farms is correct?

- A** Wind farm 1 is less reliable and less efficient than wind farm 2.  
**B** Wind farm 1 is less reliable but more efficient than wind farm 2.  
**C** Wind farm 1 is more reliable but less efficient than wind farm 2.  
**D** Wind farm 1 is more reliable and more efficient than wind farm 2.

10 Work is done.

Which physical quantity is transferred?

- A distance
- B energy
- C force
- D temperature

11 To calculate the power produced by a force, the size of the force must be known.

What else needs to be known to calculate the power?

|   | the distance that the force moves the object | the time for which the force acts on the object |
|---|----------------------------------------------|-------------------------------------------------|
| A | ✓                                            | ✓                                               |
| B | ✓                                            | x                                               |
| C | x                                            | ✓                                               |
| D | x                                            | x                                               |

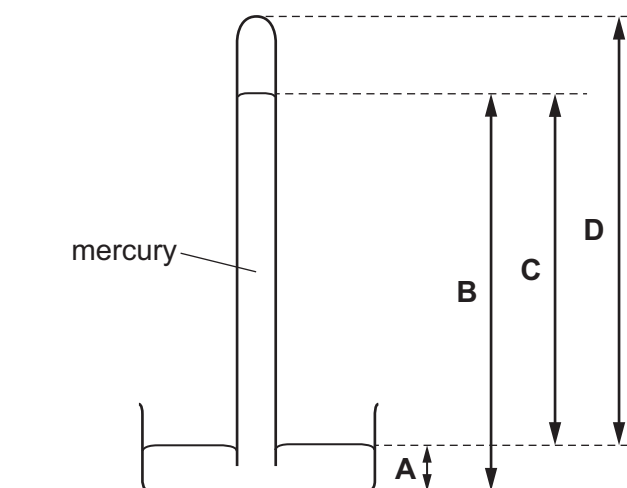
key

✓ = needed

x = not needed

12 The diagram shows a mercury barometer.

Which height is used as a measurement of atmospheric pressure?



- 13** A man who weighs 540 N stands with both feet on the ground. The area of contact between one foot and the ground is  $0.015\text{ m}^2$ .

What is the pressure exerted on the floor by the man?

- A**  $1800\text{ N/m}^2$
  - B**  $3600\text{ N/m}^2$
  - C**  $18\,000\text{ N/m}^2$
  - D**  $36\,000\text{ N/m}^2$
- 14** On a warm day, a driver checks the air pressure in a car tyre. At night, the temperature drops and the air pressure in the tyre decreases. There are no air leaks in the tyre.

Why does the pressure decrease?

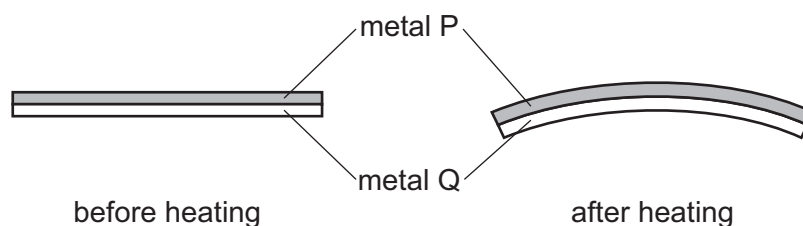
- A** The air molecules in the tyre move more slowly.
  - B** The air molecules in the tyre stop moving.
  - C** The volume of the air in the tyre decreases.
  - D** The volume of the air in the tyre increases.
- 15** Wet clothes are put out on a line to allow the water in the clothes to evaporate.

Which type of weather would cause the water to evaporate most quickly?

- A** a cold day with no wind
- B** a cold day with wind
- C** a hot day with no wind
- D** a hot day with wind

- 16** A bimetallic strip is used to control the temperature of electrical appliances. It is made of two different metals fixed together.

The diagram shows the shape of the bimetallic strip before and after heating.



Which statement is correct?

- A** Metal P contracts more than metal Q on heating.
  - B** Metal Q contracts more than metal P on heating.
  - C** Metal P expands more than metal Q on heating.
  - D** Metal Q expands more than metal P on heating.
- 17** Samples of four different substances are heated in turn by the same heater, for the same time duration and in the same conditions. The increase in temperature for each sample is recorded in the table.

Which sample has the highest thermal capacity?

|          | sample    | temperature rise / °C |
|----------|-----------|-----------------------|
| <b>A</b> | aluminium | 6.0                   |
| <b>B</b> | copper    | 8.0                   |
| <b>C</b> | mercury   | 10.0                  |
| <b>D</b> | water     | 5.0                   |

- 18** Thermal energy is supplied at the same rate to four bars made from different materials.



After several minutes, there is a temperature difference between X and Y for each bar.

Which statement identifies the bar with the greatest temperature difference between points X and Y, and gives the correct reason?

- A** copper because it is a good conductor of thermal energy
- B** lead because it is the densest metal
- C** plastic because it is a poor conductor of thermal energy
- D** wood because it is a good conductor of thermal energy

- 19** A scientist measures the air temperature at different heights from the floor in a cave. The results are recorded in the table.

| height / m | temperature / °C |
|------------|------------------|
| 0          | 10               |
| 10         | 11               |
| 20         | 13               |
| 30         | 14               |
| 40         | 16               |

Why does altering the height affect the temperature of the air?

- A** The molecules in warm air have less energy than the molecules in cool air.
  - B** The molecules in cool air are further apart than the molecules in warm air.
  - C** Warm air is less dense than cool air.
  - D** Cool air rises above warm air.
- 20** Which row correctly describes light waves?

|          | wave type    | direction of vibrations                   |
|----------|--------------|-------------------------------------------|
| <b>A</b> | longitudinal | parallel to direction of wave travel      |
| <b>B</b> | longitudinal | perpendicular to direction of wave travel |
| <b>C</b> | transverse   | parallel to direction of wave travel      |
| <b>D</b> | transverse   | perpendicular to direction of wave travel |

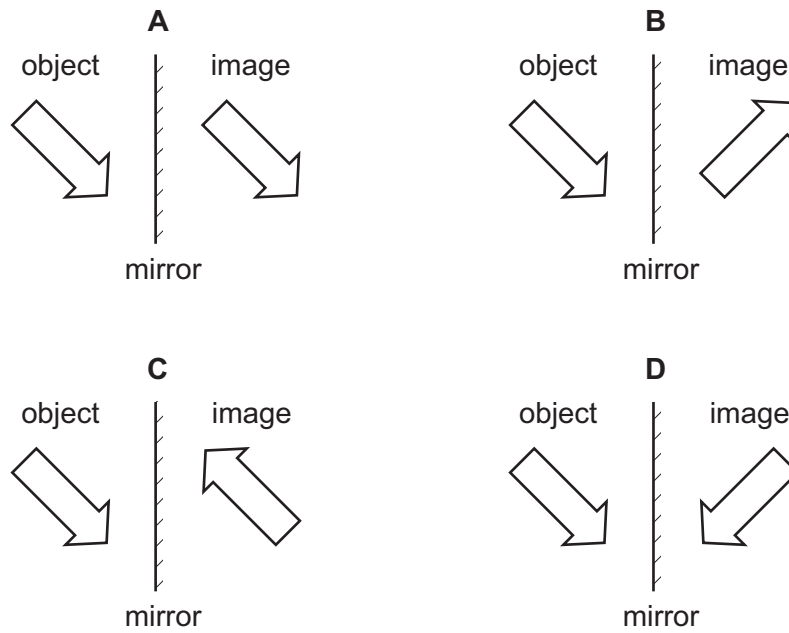
- 21** Waves travel across the surface of water.

What is meant by the amplitude of the wave?

- A** the maximum distance of a water particle from its mean position
- B** how far the wave travels every second
- C** the number of waves passing a point every second
- D** the distance between the top of consecutive waves

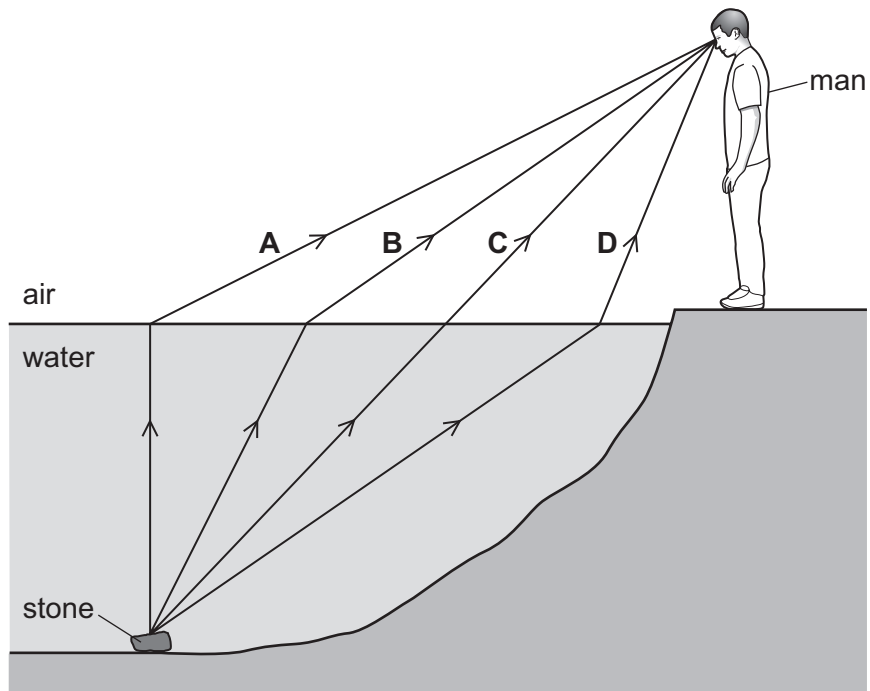


22 Which diagram shows the image correctly formed by reflection?



23 A man sees a stone at the bottom of a pool of water.

Which path could be taken by light from the stone to the man?



- 24** White light can be split into different colours by passing it through a prism.

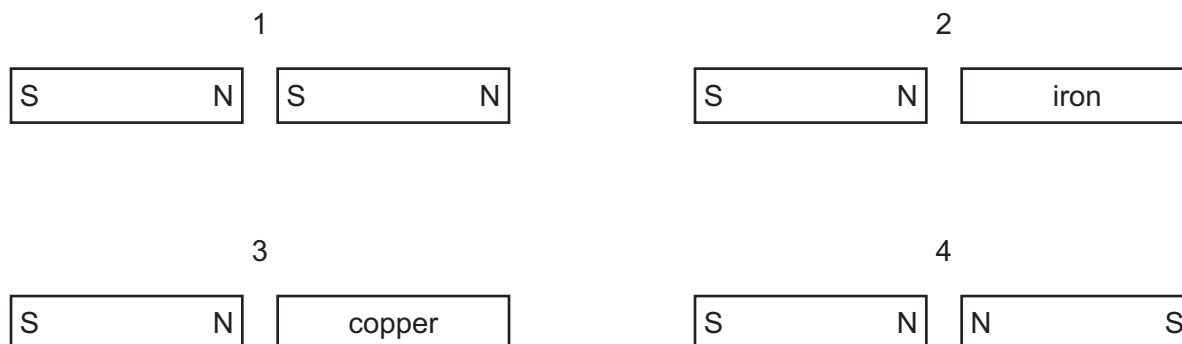
What is the name of this process?

- A** diffraction
  - B** dispersion
  - C** reflection
  - D** total internal reflection
- 25** Which type of radiation does a remote controller use to send its instructions to a TV set?
- A**  $\gamma$ -rays
  - B** infrared radiation
  - C** ultraviolet radiation
  - D** X-rays
- 26** A police car with its siren sounding is stationary in heavy traffic. A pedestrian notices that, although the loudness of the sound produced does not change, the pitch varies.

Which row describes the amplitude and the frequency of the sound?

|          | amplitude | frequency |
|----------|-----------|-----------|
| <b>A</b> | constant  | constant  |
| <b>B</b> | constant  | varying   |
| <b>C</b> | varying   | constant  |
| <b>D</b> | varying   | varying   |

- 27** A student sets up four experiments using bar magnets and other metal objects. The N and S poles of the bar magnets are labelled N and S.

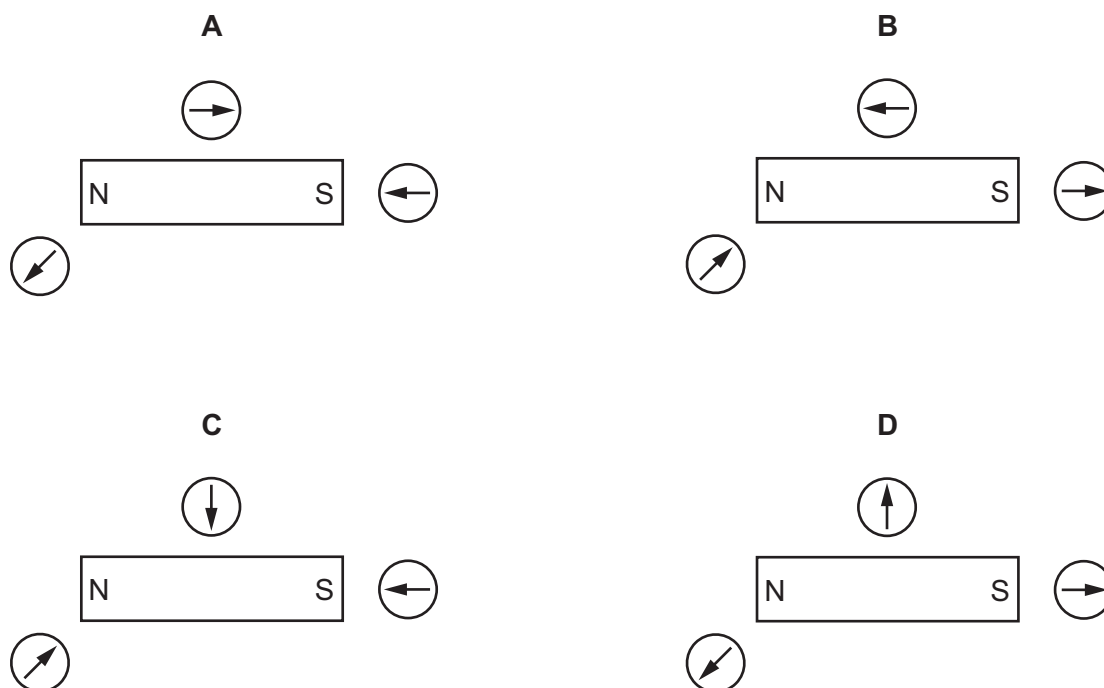


Which pairs attract each other?

- A** 1 and 2
- B** 1 and 3
- C** 2 and 4
- D** 3 and 4

- 28** A student uses three small plotting compasses to investigate the magnetic field around a bar magnet.

Which diagram shows the directions in which the compass needles point?

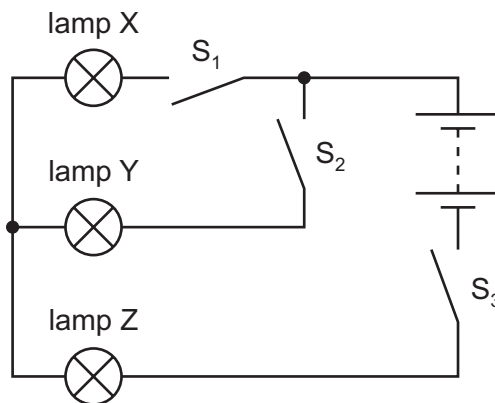


- 29** A polythene rod is charged negatively by rubbing it with a cloth.

Which statement explains why the rod has become charged?

- A** The rod has gained electrons.
  - B** The rod has gained protons.
  - C** The rod has lost electrons.
  - D** The rod has lost protons.
- 30** There is a current in a variable resistor when a potential difference (p.d.) is applied across it.
- In which situation is the current increased?
- A** Decrease the p.d. and keep the resistance the same.
  - B** Decrease the p.d. and increase the resistance.
  - C** Keep the p.d. the same and decrease the resistance.
  - D** Keep the p.d. the same and increase the resistance.

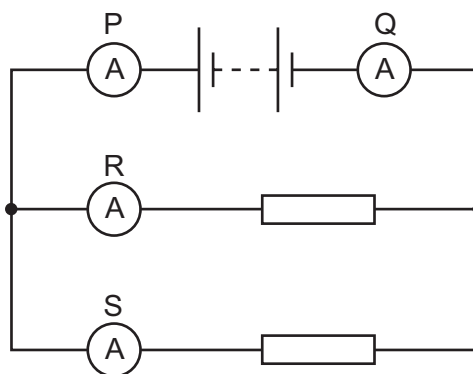
- 31 The circuit shown includes a battery, three lamps X, Y and Z and three switches  $S_1$ ,  $S_2$  and  $S_3$ .



Lamp X is off but lamps Y and Z are lit.

Which switches are closed?

- A**  $S_1$  and  $S_2$  only  
**B**  $S_1$  and  $S_3$  only  
**C**  $S_2$  and  $S_3$  only  
**D**  $S_1$ ,  $S_2$  and  $S_3$
- 32 A student uses four ammeters P, Q, R and S to measure the current in different parts of the circuit shown.



Which two ammeters read the largest current?

- A** P and Q      **B** P and R      **C** R and Q      **D** R and S
- 33 An electrical appliance is powered from a mains supply.

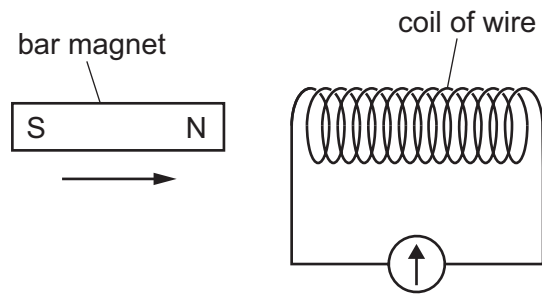
The appliance normally uses a current of 3 A, but the current briefly rises to 4 A at the instant the appliance is switched on. The cable to the appliance is designed for currents up to 6 A.

A fuse is used to protect the circuit.

What should be the rating of the fuse?

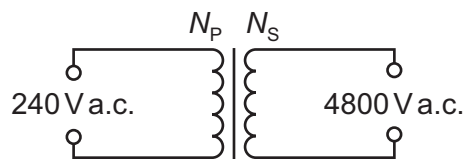
- A** 1 A      **B** 3 A      **C** 5 A      **D** 13 A

- 34** A student investigates electromagnetic induction. She moves the N pole of a magnet quickly towards a coil of wire. There is a reading on the galvanometer.



What can she do to get a greater reading on the galvanometer?

- A** Hold the bar magnet stationary inside the coil.
  - B** Move the bar magnet slowly away from the coil.
  - C** Use a coil of wire with fewer turns on it.
  - D** Use a stronger bar magnet.
- 35** A transformer is needed to convert a supply of 240 V a.c. into 4800 V a.c..

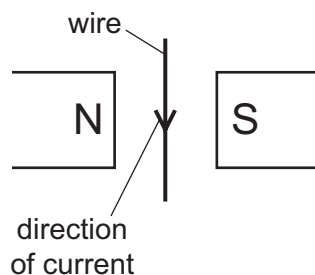


Which pair of coils would be suitable for this transformer?

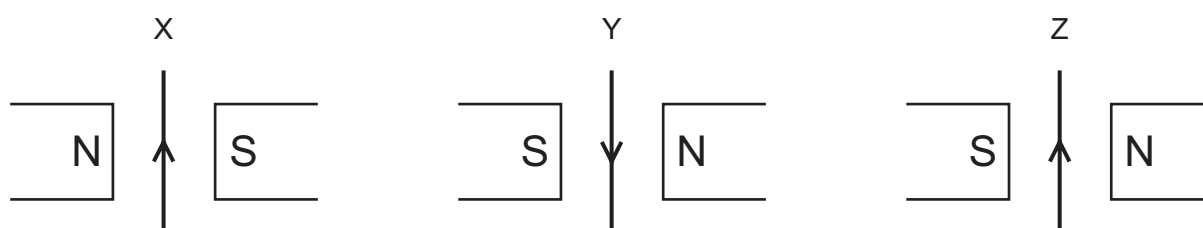
|          | number of turns<br>on primary coil $N_P$ | number of turns<br>on secondary coil $N_S$ |
|----------|------------------------------------------|--------------------------------------------|
| <b>A</b> | 50                                       | 1 000                                      |
| <b>B</b> | 240                                      | 48 000                                     |
| <b>C</b> | 480                                      | 24                                         |
| <b>D</b> | 2000                                     | 100                                        |

- 36 The diagram shows a wire carrying a current in the direction shown. The wire is between the poles of a magnet.

A force is produced on the wire out of the page.



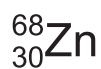
The wire and magnet are now put into different arrangements X, Y and Z. The arrow shows the direction of the current in each case.



In which arrangements is the force on the wire out of the page?

- A X only      B X and Y      C X and Z      D Z only
- 37 Which statement is correct for the nucleus of **any** atom?
- A The nucleus contains electrons, neutrons and protons.
- B The nucleus contains the same number of protons as neutrons.
- C The nucleus has a total charge of zero.
- D The nucleus is very small compared with the size of the atom.

- 38 The symbol represents a nucleus of zinc.



Which row gives the numbers of protons and neutrons in this nucleus?

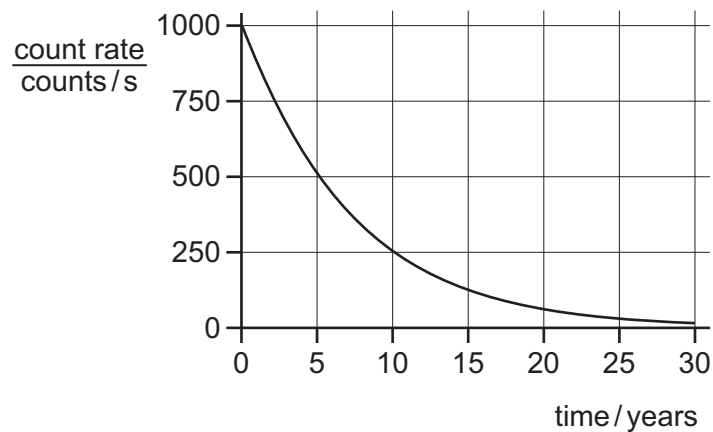
|   | number of protons | number of neutrons |
|---|-------------------|--------------------|
| A | 30                | 38                 |
| B | 30                | 68                 |
| C | 38                | 30                 |
| D | 38                | 68                 |

39 Three types of radiation are  $\alpha$ -radiation,  $\beta$ -radiation and  $\gamma$ -radiation.

Which statement is correct?

- A  $\alpha$ -radiation is less ionising than  $\beta$ -radiation.
- B  $\alpha$ -radiation is less ionising than  $\gamma$ -radiation.
- C  $\gamma$ -radiation produces no ionisation.
- D  $\beta$ -radiation is more ionising than  $\gamma$ -radiation.

40 The graph shows the radioactive decay curve of a substance.



What is the half-life of this substance?

- A 0.5 years
- B 5 years
- C 15 years
- D 30 years

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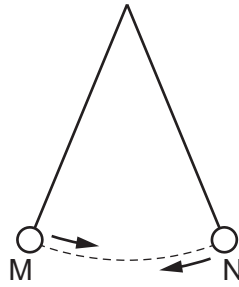
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- 1 The diagram shows a pendulum. The pendulum bob swings repeatedly between points M and N.



A student starts a stop-watch when the bob reaches point M.

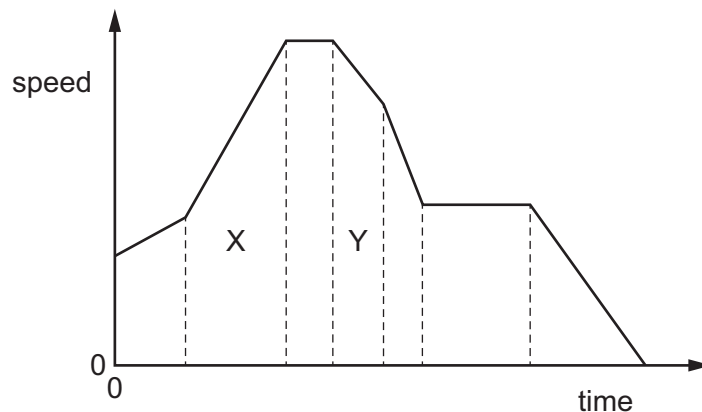
He counts each time the bob changes direction and stops the watch on the tenth change in direction.

The watch shows a time of 12.0 seconds.

What is the period of the pendulum?

- A** 0.60 s      **B** 1.2 s      **C** 2.4 s      **D** 12.0 s

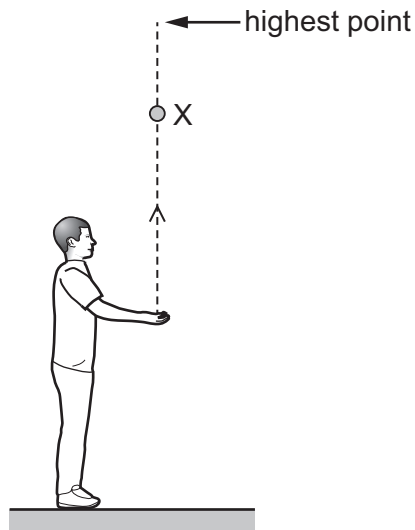
- 2 The speed–time graph represents a journey.



How does the graph show that the distance travelled in section X of the journey is greater than the distance travelled in section Y?

- A** The area below section X of the graph is greater than the area below section Y.  
**B** The gradient of section X of the graph is greater than the gradient of section Y.  
**C** The speed at the end of section X of the journey is greater than the speed at the end of section Y.  
**D** The time for section X of the journey is greater than the time for section Y.

- 3 A boy throws a ball vertically upwards with a speed  $v$ .



Which row describes the speed and the acceleration of the ball at point X on the way upwards?

|          | speed      | acceleration |
|----------|------------|--------------|
| <b>A</b> | decreasing | upwards      |
| <b>B</b> | decreasing | downwards    |
| <b>C</b> | increasing | upwards      |
| <b>D</b> | increasing | downwards    |

- 4 A student compares the weights of different objects.

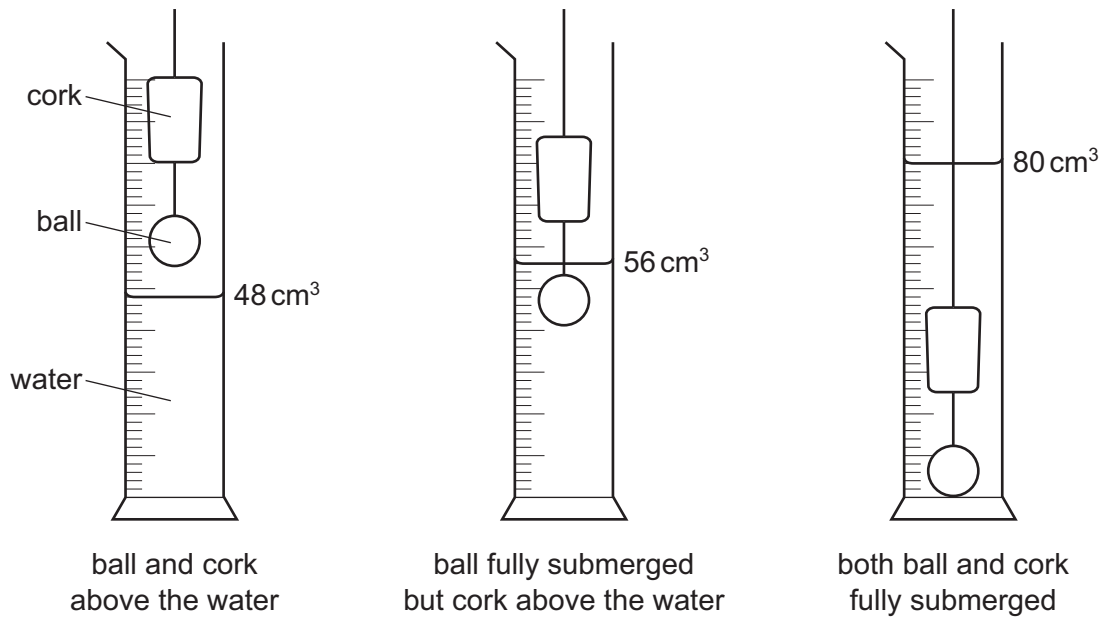
Which apparatus does he use?

- A** balance
- B** measuring cylinder
- C** stop-watch
- D** thermometer

- 5 Which quantity is weight an example of?

- A** acceleration
- B** force
- C** mass
- D** pressure

- 6 A metal ball is attached to a cork and is lowered into a measuring cylinder, pulling the cork into the water, as shown.



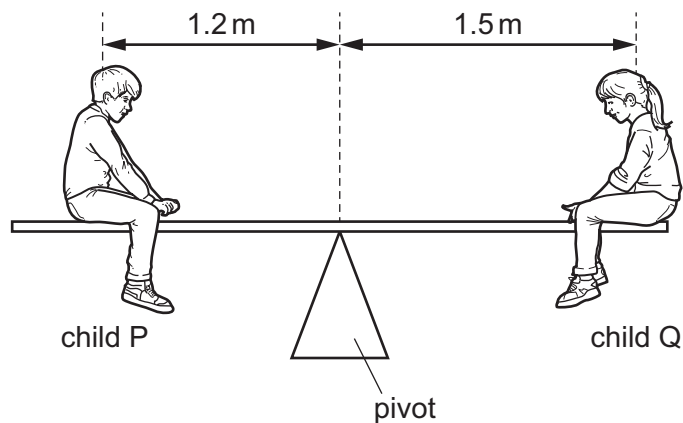
The mass of the cork is 4.8 g.

What is the density of the cork?

- A**  $0.15 \text{ g/cm}^3$     **B**  $0.20 \text{ g/cm}^3$     **C**  $0.60 \text{ g/cm}^3$     **D**  $5.0 \text{ g/cm}^3$

- 7 A uniform plank rests on a pivot at its centre.

Two children P and Q sit on the plank in the positions shown.



The mass of child P is 25 kg.

The plank is balanced.

What is the mass of child Q?

- A** 20 kg    **B** 25 kg    **C** 31 kg    **D** 45 kg

- 8 An object is in equilibrium on the Earth.

Which statement is correct?

- A All the forces acting on the object are in the same direction.
- B All the forces acting on the object have the same value.
- C The object is weightless.
- D The resultant force acting on the object is zero.

- 9 A man jumps from a stationary balloon. After falling several hundred metres, he opens his parachute.

At which position is his kinetic energy greatest?

- A just after he jumps from the balloon
- B just before he opens his parachute
- C just after his parachute opens
- D just before he lands

- 10 The arrows show an outline of the processes in an oil-fired power station.

oil is burnt → turns turbine → turns generator → output transformer

What are the processes for the transfer of energy between the turbine and the generator and between the generator and the output transformer?

|   | turbine to generator       | generator to output transformer |
|---|----------------------------|---------------------------------|
| A | electrical working         | electrical working              |
| B | mechanical working         | electrical working              |
| C | mechanical working         | transfer of thermal energy      |
| D | transfer of thermal energy | mechanical working              |

- 11 To calculate the power produced by a force, the size of the force must be known.

What else needs to be known to calculate the power?

|   | the distance that the force moves the object | the time for which the force acts on the object |
|---|----------------------------------------------|-------------------------------------------------|
| A | ✓                                            | ✓                                               |
| B | ✓                                            | ✗                                               |
| C | ✗                                            | ✓                                               |
| D | ✗                                            | ✗                                               |

key

✓ = needed

✗ = not needed

- 12** A rectangular marble block has dimensions 1 m by 1 m by 5 m and weighs 125 000 N.

The marble block is stored with the long side resting on the ground, as in diagram 1.

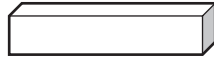


diagram 1

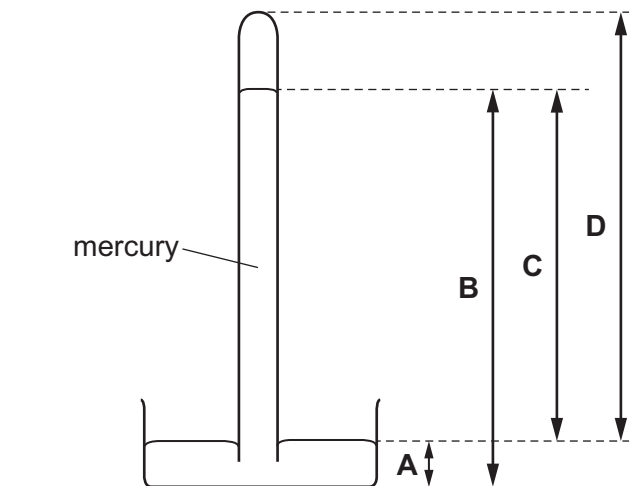


diagram 2

What is the change in the pressure on the ground due to the block when the block is stored as in diagram 2 rather than diagram 1?

- A** a decrease of 25 000 Pa
  - B** an increase of 100 000 Pa
  - C** an increase of 125 000 Pa
  - D** no change
- 13** The diagram shows a mercury barometer.

Which height is used as a measurement of atmospheric pressure?

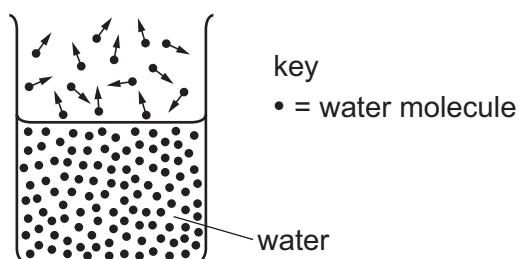


- 14** On a warm day, a driver checks the air pressure in a car tyre. At night, the temperature drops and the air pressure in the tyre decreases. There are no air leaks in the tyre.

Why does the pressure decrease?

- A** The air molecules in the tyre move more slowly.
- B** The air molecules in the tyre stop moving.
- C** The volume of the air in the tyre decreases.
- D** The volume of the air in the tyre increases.

- 15** The diagram shows the more energetic water molecules escaping from the surface of liquid water.



What is this process called?

- A** Brownian motion
  - B** condensation
  - C** evaporation
  - D** conduction
- 16** When a bridge is built, a gap is left between each concrete slab.

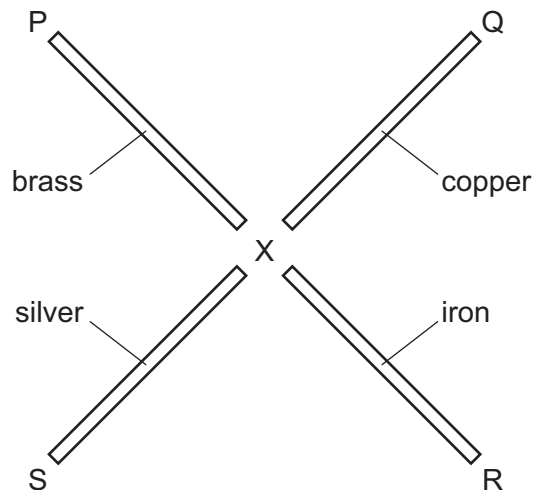
Why are these gaps left?

- A** Concrete expands on warm days.
  - B** Concrete contracts on warm days.
  - C** The gaps expand on warm days.
  - D** The gaps contract on cold days.
- 17** The table shows the melting and boiling points of four different substances.

Which substance becomes a liquid when placed in a beaker of boiling water?

|          | melting point / °C | boiling point / °C |
|----------|--------------------|--------------------|
| <b>A</b> | 10                 | 90                 |
| <b>B</b> | 40                 | 150                |
| <b>C</b> | 105                | 230                |
| <b>D</b> | 110                | 320                |

**18** The diagram shows four rods. Each rod is made of a different metal.



Wax is used to attach small metal balls at the rod ends P, Q, R and S.

Each rod is the same size. They are heated uniformly by a Bunsen burner at point X.

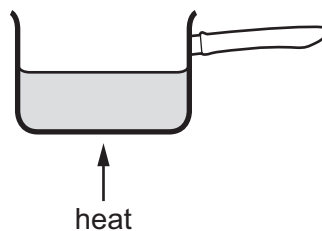
As the rods warm up, the wax melts and the balls fall off.

Why does the ball on the silver rod fall first?

- A** Silver is the best conductor of heat.
- B** Silver is the worst conductor of heat.
- C** Silver is the best radiator of heat.
- D** Silver is the worst radiator of heat.

**19** The diagram shows a pan of water being heated.

After a short time, all the water in the pan begins to boil.



What is the main process by which thermal energy is transferred through the water?

- A** conduction
- B** convection
- C** evaporation
- D** radiation

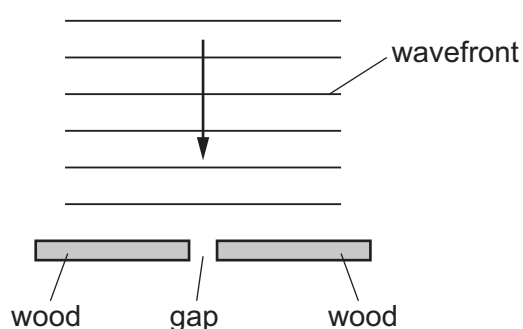


20 Which row correctly describes light waves?

|          | wave type    | direction of vibrations                   |
|----------|--------------|-------------------------------------------|
| <b>A</b> | longitudinal | parallel to direction of wave travel      |
| <b>B</b> | longitudinal | perpendicular to direction of wave travel |
| <b>C</b> | transverse   | parallel to direction of wave travel      |
| <b>D</b> | transverse   | perpendicular to direction of wave travel |

21 The diagram shows two pieces of wood resting in shallow water of constant depth.

Straight, parallel wavefronts approach the pieces of wood as indicated.



The gap between the pieces of wood is 2.0 cm wide.

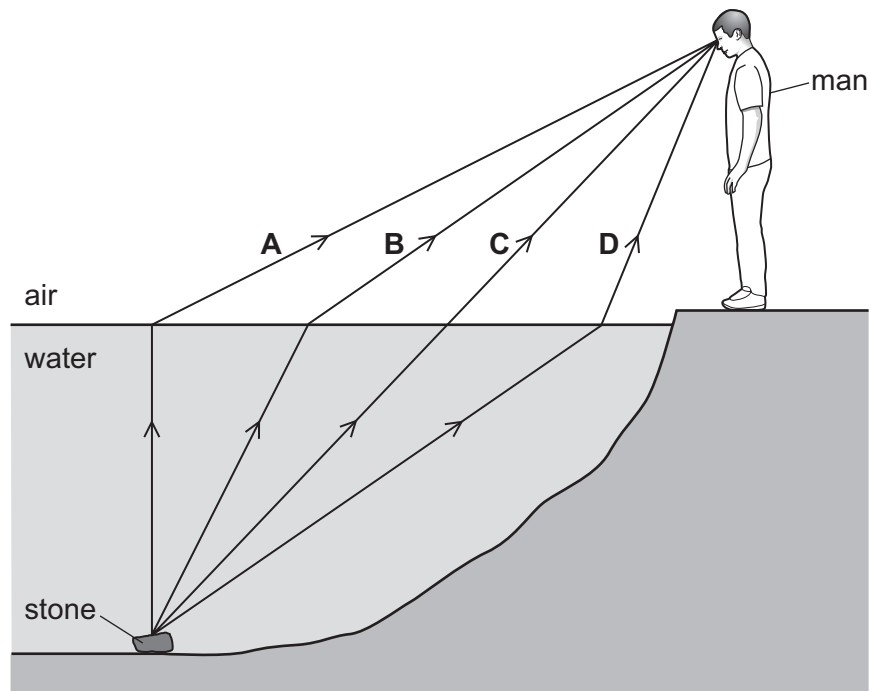
The wavefronts are 3.0 cm apart.

What is the appearance of the wavefronts after they pass through the gap?

- A** semicircular and 2.0 cm apart
- B** semicircular and 3.0 cm apart
- C** straight and 2.0 cm apart
- D** straight and 3.0 cm apart

**22** A man sees a stone at the bottom of a pool of water.

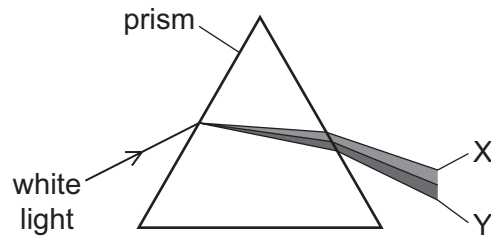
Which path could be taken by light from the stone to the man?



**23** Which statement about a thin converging lens is correct?

- A** All rays of light refracted by the lens pass through the principal focus.
- B** All rays initially parallel to the principal axis of the lens are refracted through the principal focus.
- C** The focal length of the lens is the distance between the image and the principal focus.
- D** The focal length of the lens is the distance between the object and the image.

- 24** The diagram shows a beam of white light passing through a triangular prism. A spectrum is produced.



Which row correctly shows a wave property involved in producing the colours at X and Y?

|          | wave property | X      | Y      |
|----------|---------------|--------|--------|
| <b>A</b> | diffraction   | red    | violet |
| <b>B</b> | dispersion    | red    | violet |
| <b>C</b> | reflection    | violet | red    |
| <b>D</b> | refraction    | violet | red    |

- 25** The diagram shows the electromagnetic spectrum.

|                |   |             |   |          |            |   |
|----------------|---|-------------|---|----------|------------|---|
| $\gamma$ -rays | E | ultraviolet | F | infrared | microwaves | G |
|----------------|---|-------------|---|----------|------------|---|

Which types of wave are E, F and G?

|          | E      | F             | G          |
|----------|--------|---------------|------------|
| <b>A</b> | radio  | visible light | X-rays     |
| <b>B</b> | radio  | X-rays        | ultrasound |
| <b>C</b> | X-rays | radio         | ultrasound |
| <b>D</b> | X-rays | visible light | radio      |

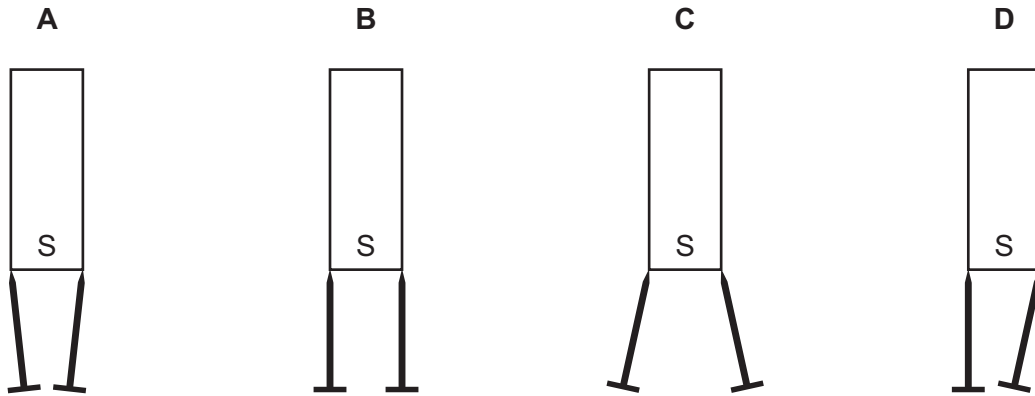
- 26** A police car with its siren sounding is stationary in heavy traffic. A pedestrian notices that, although the loudness of the sound produced does not change, the pitch varies.

Which row describes the amplitude and the frequency of the sound?

|          | amplitude | frequency |
|----------|-----------|-----------|
| <b>A</b> | constant  | constant  |
| <b>B</b> | constant  | varying   |
| <b>C</b> | varying   | constant  |
| <b>D</b> | varying   | varying   |

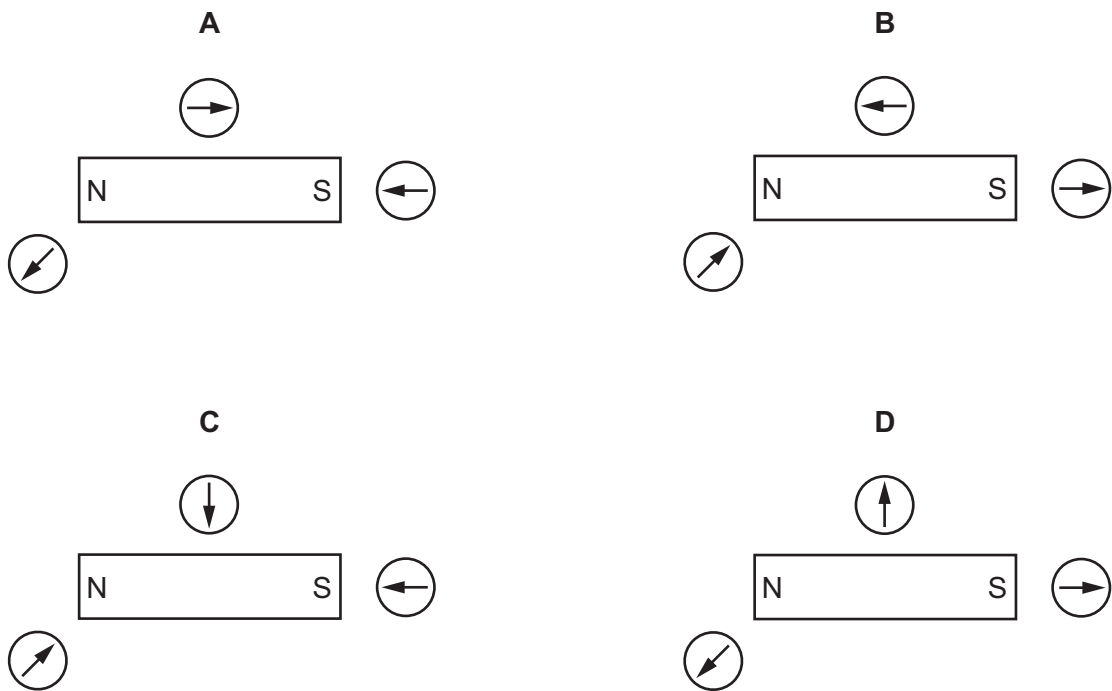
- 27** Two soft-iron pins are suspended from the S pole of a bar magnet.

Which diagram shows how the pins are deflected?



- 28** A student uses three small plotting compasses to investigate the magnetic field around a bar magnet.

Which diagram shows the directions in which the compass needles point?



- 29** What is an electric current in a metal wire?

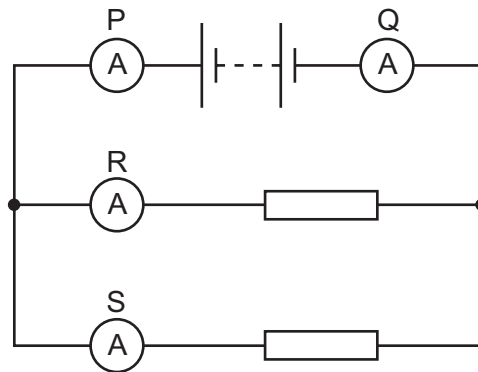
- A** a flow of electrons
- B** a flow of neutrons
- C** a flow of nucleons
- D** a flow of protons

- 30 Four wires are made of the same metal.

Which wire has the greatest resistance?

- A a 100 cm long wire with a diameter of 3.0 mm
- B a 100 cm long wire with a diameter of 6.0 mm
- C a 10 cm long wire with a diameter of 3.0 mm
- D a 10 cm long wire with a diameter of 6.0 mm

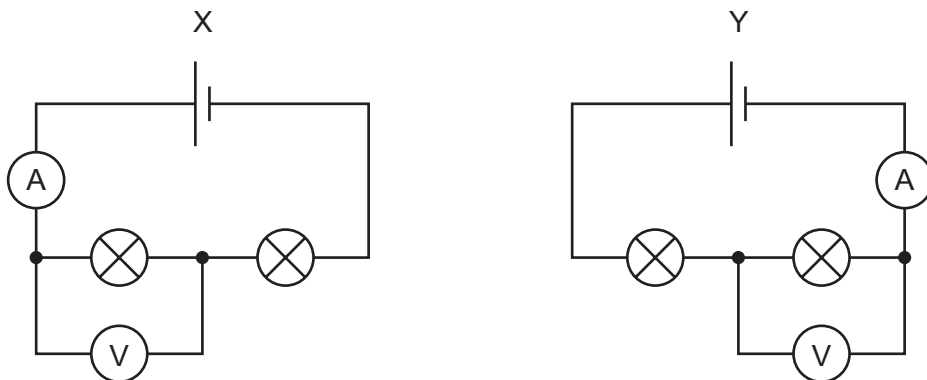
- 31 A student uses four ammeters P, Q, R and S to measure the current in different parts of the circuit shown.



Which two ammeters read the largest current?

- A P and Q
- B P and R
- C R and Q
- D R and S

- 32 A circuit X is set up with two identical lamps. Circuit Y is then set up, changing the positions of the meters.



On which meters do the readings change?

- A both the ammeter and the voltmeter
- B the ammeter only
- C the voltmeter only
- D neither the ammeter nor the voltmeter

- 33** An electrical appliance is powered from a mains supply.

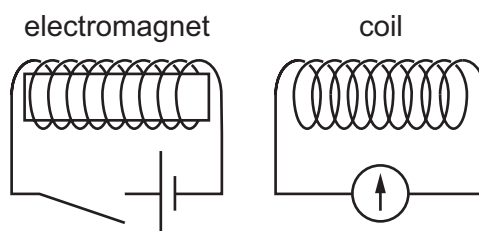
The appliance normally uses a current of 3 A, but the current briefly rises to 4 A at the instant the appliance is switched on. The cable to the appliance is designed for currents up to 6 A.

A fuse is used to protect the circuit.

What should be the rating of the fuse?

- A** 1 A                      **B** 3 A                      **C** 5 A                      **D** 13 A

- 34** An electromagnet is positioned close to a coil of wire.



The electromagnet is switched on, remains on for a short time, and is then switched off.

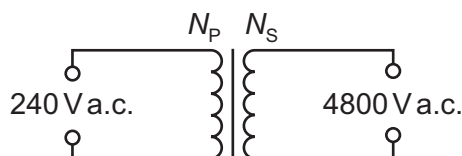
Three statements about the pointer on the galvanometer during this sequence are given.

- 1 The pointer kicks to one side as the electromagnet is switched on.
- 2 The pointer records a steady non-zero value while the electromagnet remains switched on.
- 3 The pointer kicks to the other side as the electromagnet is switched off.

Which statements are correct?

- A** 1 and 2 only    **B** 1 and 3 only    **C** 2 and 3 only    **D** 1, 2 and 3

- 35** A transformer is needed to convert a supply of 240 V a.c. into 4800 V a.c..

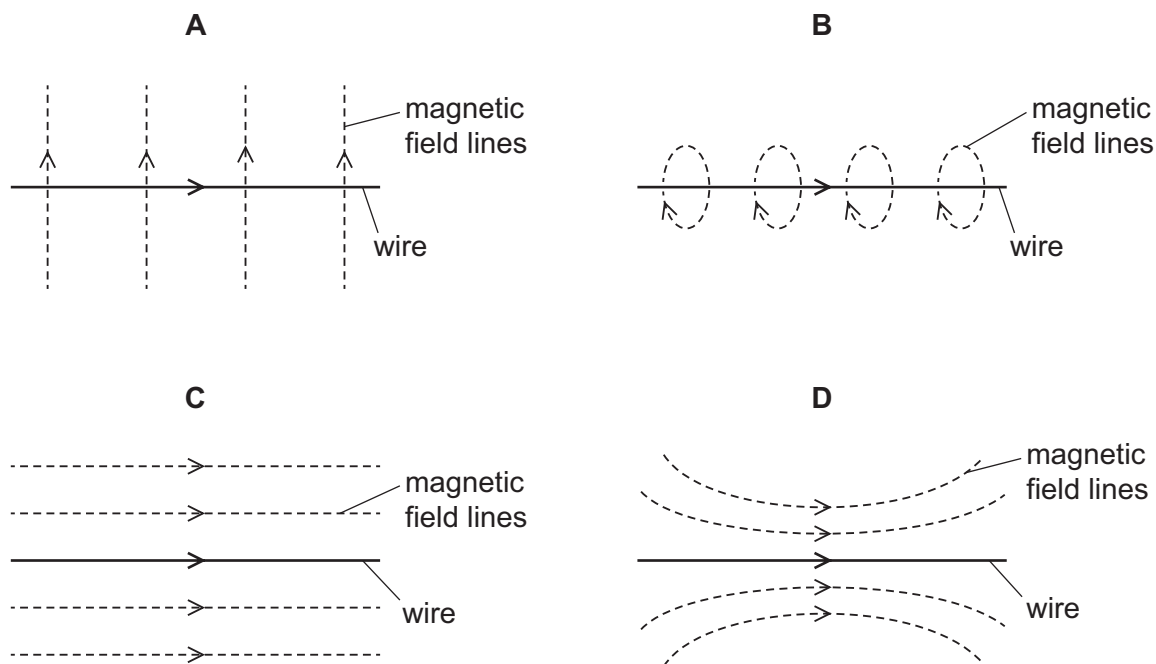


Which pair of coils would be suitable for this transformer?

|          | number of turns<br>on primary coil $N_P$ | number of turns<br>on secondary coil $N_S$ |
|----------|------------------------------------------|--------------------------------------------|
| <b>A</b> | 50                                       | 1 000                                      |
| <b>B</b> | 240                                      | 48 000                                     |
| <b>C</b> | 480                                      | 24                                         |
| <b>D</b> | 2000                                     | 100                                        |

36 The diagrams show a current-carrying wire with an arrow in the direction of the current.

Which diagram shows the magnetic field produced by the current?



37 Which statement is correct for the nucleus of **any** atom?

- A The nucleus contains electrons, neutrons and protons.
- B The nucleus contains the same number of protons as neutrons.
- C The nucleus has a total charge of zero.
- D The nucleus is very small compared with the size of the atom.

38 The nucleus of an atom X is represented by the notation shown.



How many protons and how many neutrons are in this nucleus?

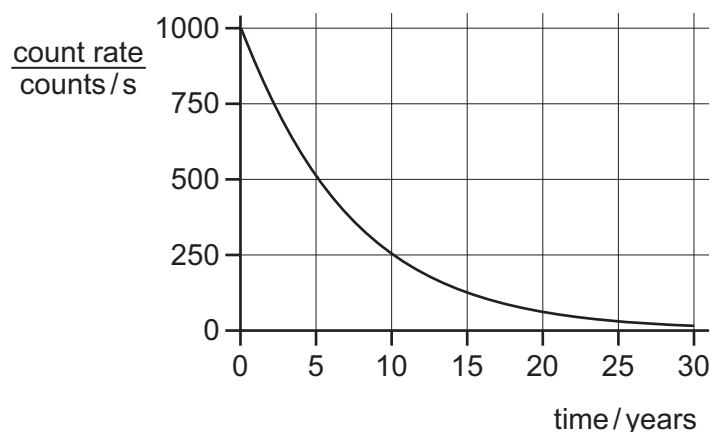
|   | number of protons | number of neutrons |
|---|-------------------|--------------------|
| A | P                 | Q                  |
| B | P                 | $Q - P$            |
| C | Q                 | P                  |
| D | Q                 | $P - Q$            |

- 39 A radiation detector records a low reading even when no radioactive source is close. This is due to background radiation.

What does **not** contribute to this background radiation?

- A rocks on Earth
- B cosmic rays from the Sun
- C satellite TV signals
- D waste from nuclear power stations

- 40 The graph shows the radioactive decay curve of a substance.



What is the half-life of this substance?

- A 0.5 years
- B 5 years
- C 15 years
- D 30 years

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# Cambridge IGCSE™

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**PHYSICS**

**0625/13**

Paper 1 Multiple Choice (Core)

**October/November 2020**

**MARK SCHEME**

Maximum Mark: 40

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| Question | Answer | Marks |
|----------|--------|-------|
| 1        | C      | 1     |
| 2        | B      | 1     |
| 3        | B      | 1     |
| 4        | A      | 1     |
| 5        | B      | 1     |
| 6        | B      | 1     |
| 7        | C      | 1     |
| 8        | A      | 1     |
| 9        | D      | 1     |
| 10       | B      | 1     |
| 11       | A      | 1     |
| 12       | C      | 1     |
| 13       | D      | 1     |
| 14       | A      | 1     |
| 15       | D      | 1     |
| 16       | D      | 1     |
| 17       | D      | 1     |
| 18       | B      | 1     |
| 19       | D      | 1     |
| 20       | D      | 1     |
| 21       | B      | 1     |
| 22       | B      | 1     |
| 23       | B      | 1     |
| 24       | A      | 1     |
| 25       | A      | 1     |
| 26       | B      | 1     |
| 27       | C      | 1     |
| 28       | A      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | C      | 1     |
| 30       | D      | 1     |
| 31       | A      | 1     |
| 32       | C      | 1     |
| 33       | C      | 1     |
| 34       | B      | 1     |
| 35       | A      | 1     |
| 36       | A      | 1     |
| 37       | D      | 1     |
| 38       | A      | 1     |
| 39       | A      | 1     |
| 40       | B      | 1     |



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**PHYSICS**

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| Question | Answer | Marks |
|----------|--------|-------|
| 1        | C      | 1     |
| 2        | C      | 1     |
| 3        | C      | 1     |
| 4        | A      | 1     |
| 5        | B      | 1     |
| 6        | C      | 1     |
| 7        | B      | 1     |
| 8        | B      | 1     |
| 9        | B      | 1     |
| 10       | B      | 1     |
| 11       | A      | 1     |
| 12       | C      | 1     |
| 13       | C      | 1     |
| 14       | A      | 1     |
| 15       | D      | 1     |
| 16       | C      | 1     |
| 17       | D      | 1     |
| 18       | C      | 1     |
| 19       | C      | 1     |
| 20       | D      | 1     |
| 21       | A      | 1     |
| 22       | D      | 1     |
| 23       | B      | 1     |
| 24       | B      | 1     |
| 25       | B      | 1     |
| 26       | B      | 1     |
| 27       | A      | 1     |
| 28       | A      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | A      | 1     |
| 30       | C      | 1     |
| 31       | C      | 1     |
| 32       | A      | 1     |
| 33       | C      | 1     |
| 34       | D      | 1     |
| 35       | A      | 1     |
| 36       | D      | 1     |
| 37       | D      | 1     |
| 38       | A      | 1     |
| 39       | D      | 1     |
| 40       | B      | 1     |



# Cambridge IGCSE™

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**PHYSICS**

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**October/November 2020**

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| Question | Answer | Marks |
|----------|--------|-------|
| 1        | C      | 1     |
| 2        | A      | 1     |
| 3        | B      | 1     |
| 4        | A      | 1     |
| 5        | B      | 1     |
| 6        | B      | 1     |
| 7        | A      | 1     |
| 8        | D      | 1     |
| 9        | B      | 1     |
| 10       | B      | 1     |
| 11       | A      | 1     |
| 12       | B      | 1     |
| 13       | C      | 1     |
| 14       | A      | 1     |
| 15       | C      | 1     |
| 16       | A      | 1     |
| 17       | B      | 1     |
| 18       | A      | 1     |
| 19       | B      | 1     |
| 20       | D      | 1     |
| 21       | B      | 1     |
| 22       | B      | 1     |
| 23       | B      | 1     |
| 24       | B      | 1     |
| 25       | D      | 1     |
| 26       | B      | 1     |
| 27       | C      | 1     |
| 28       | A      | 1     |



| Question | Answer | Marks |
|----------|--------|-------|
| 29       | A      | 1     |
| 30       | A      | 1     |
| 31       | A      | 1     |
| 32       | D      | 1     |
| 33       | C      | 1     |
| 34       | B      | 1     |
| 35       | A      | 1     |
| 36       | B      | 1     |
| 37       | D      | 1     |
| 38       | D      | 1     |
| 39       | C      | 1     |
| 40       | B      | 1     |



**Cambridge Assessment International Education**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

**0625/13**

Paper 1 Multiple Choice (Core)

**October/November 2019**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)

**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

**DO NOT WRITE IN ANY BARCODES.**

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

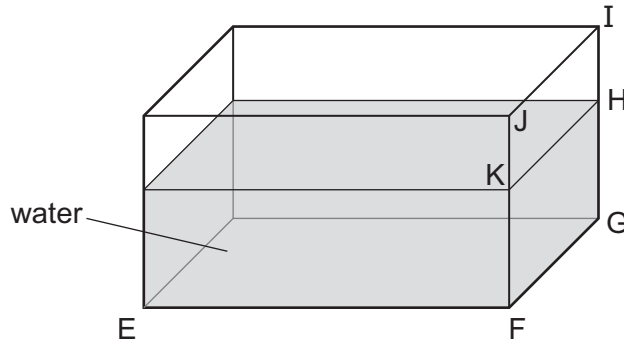
Take the weight of 1.0 kg to be 10 N (acceleration of free fall =  $10 \text{ m/s}^2$ ).

This syllabus is regulated for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **15** printed pages and **1** blank page.



- 1 A student uses a ruler to find the volume of water in a tank.  
She measures the lengths EF and FG.



What other length does she need to measure?

- A** FJ                      **B** FK                      **C** HI                      **D** IJ
- 2 An object begins to fall close to the Earth's surface. Air resistance can be ignored.  
Which statement about the object's acceleration is correct?
- A** The acceleration is constant.  
**B** The acceleration decreases as the body falls.  
**C** The acceleration increases as the body falls.  
**D** The acceleration is zero.
- 3 A car travels at an average speed of 60 km/h for 15 minutes.  
How far does the car travel in 15 minutes?
- A** 4.0 km                      **B** 15 km                      **C** 240 km                      **D** 900 km
- 4 Which equation shows the relationship between the weight  $W$  and the mass  $m$  of an object?
- A**  $W = \frac{m}{g}$   
**B**  $W = mg$   
**C**  $W = m + g$   
**D**  $W = \frac{g}{m}$

- 5 Four hollow glass spheres P, Q, R and S each have a mass of 72 g.

Their volumes are given in the table.

|   | volume / cm <sup>3</sup> |
|---|--------------------------|
| P | 55                       |
| Q | 65                       |
| R | 75                       |
| S | 85                       |

Which spheres sink in a liquid of density 0.9 g/cm<sup>3</sup>?

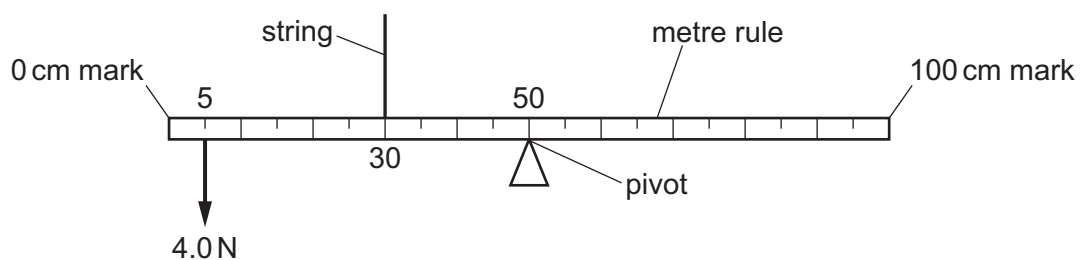
- A** P, Q and R    **B** Q, R and S    **C** R and S only    **D** S only
- 6 A student measures the length of a spring. She then attaches different weights to the spring. She measures the length of the spring for each weight.

The table shows her results.

| weight / N | length / mm |
|------------|-------------|
| 0          | 520         |
| 1.0        | 524         |
| 2.0        | 528         |
| 3.0        | 533         |
| 4.0        | 537         |
| 5.0        | 540         |

What is the extension of the spring with a weight of 3.0 N attached to it?

- A** 4 mm    **B** 5 mm    **C** 12 mm    **D** 13 mm
- 7 The diagram shows a uniform metre rule. The rule is pivoted at its mid-point. A downward force of 4.0 N acts on the rule at the 5 cm mark. The rule is held by a string at the 30 cm mark. The rule is in equilibrium.

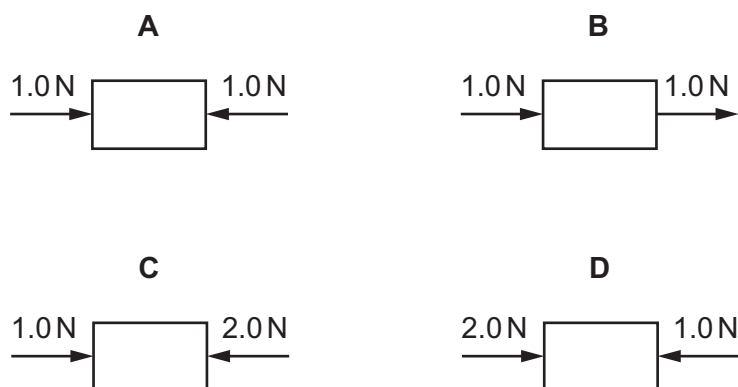


What is the upward force that the string exerts on the rule?

- A** 0.67 N    **B** 4.0 N    **C** 6.0 N    **D** 9.0 N

- 8 The diagrams represent the only two forces acting on an object.

Which object could be moving to the right at constant speed?



- 9 Brakes are used to slow down a moving car.

Into which form of energy is most of the kinetic energy converted as the car slows down?

- A chemical
- B elastic
- C thermal
- D sound

- 10 Three situations are listed.

- 1 someone blowing air into a party balloon
- 2 a crane lifting a block of concrete
- 3 a pile of books at rest on a shelf

In which situations is work being done?

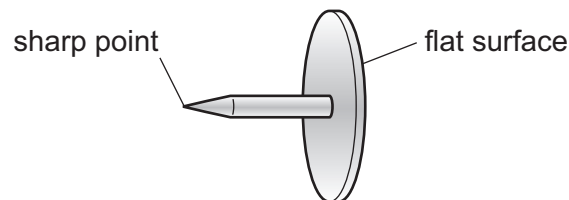
- A 1 only
- B 1 and 2 only
- C 2 and 3 only
- D 1, 2 and 3

- 11 A boy lifts a brick from the ground and places it at rest on a higher shelf. He does 30 J of work against gravity.

Which row correctly describes the final energies of the brick?

|          | gravitational potential energy gained by the brick / J | kinetic energy gained by the brick / J |
|----------|--------------------------------------------------------|----------------------------------------|
| <b>A</b> | 0                                                      | 30                                     |
| <b>B</b> | 15                                                     | 15                                     |
| <b>C</b> | 30                                                     | 0                                      |
| <b>D</b> | 27                                                     | 3                                      |

- 12 A drawing pin (thumb tack) has a sharp point at one end and a flat surface at the other end.



The pin is pushed into a wooden board.

How do the pressure and the force at the sharp point compare with the pressure and the force on the flat surface?

|          | force at the sharp point         | pressure at the sharp point      |
|----------|----------------------------------|----------------------------------|
| <b>A</b> | greater than on the flat surface | greater than on the flat surface |
| <b>B</b> | greater than on the flat surface | less than on the flat surface    |
| <b>C</b> | the same as on the flat surface  | greater than on the flat surface |
| <b>D</b> | the same as on the flat surface  | less than on the flat surface    |

- 13 Pressure is related to force and area.

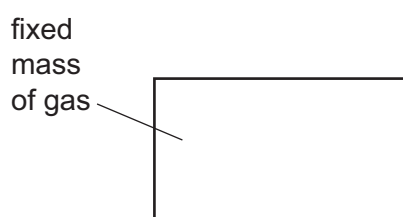
Which situation **cannot** be explained using this relationship?

- A** Using a longer spanner than normal to undo a tight nut.
- B** Hammering a nail into a piece of wood.
- C** Tractors using wide tyres in a muddy field.
- D** A sharp kitchen knife cutting vegetables more easily than a blunt one.

- 14 Which row compares the separation and the motion of the molecules of a hot gas with those of a cool liquid? (Both the gas and the liquid are at the same pressure.)

|          | separation        | motion           |
|----------|-------------------|------------------|
| <b>A</b> | greater for a gas | faster for a gas |
| <b>B</b> | greater for a gas | slower for a gas |
| <b>C</b> | smaller for a gas | faster for a gas |
| <b>D</b> | smaller for a gas | slower for a gas |

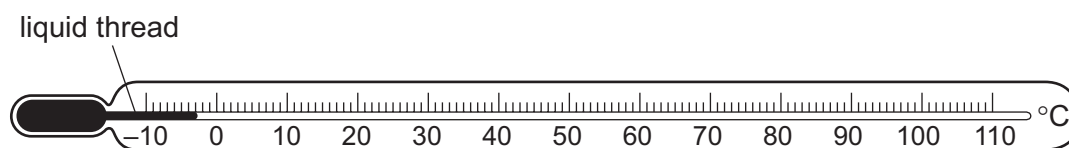
- 15 A fixed mass of gas is trapped in a container. The temperature of the gas is increased but the volume of the gas is kept constant.



How does this change affect the kinetic energy of the molecules and the pressure on the walls of the container?

|          | kinetic energy | pressure  |
|----------|----------------|-----------|
| <b>A</b> | increases      | increases |
| <b>B</b> | stays the same | increases |
| <b>C</b> | increases      | decreases |
| <b>D</b> | decreases      | increases |

- 16 The diagram shows a liquid-in-glass thermometer.



When the temperature of the thermometer rises, the changes produced cause the liquid thread to move to the right.

Why does this happen when the temperature of the thermometer rises?

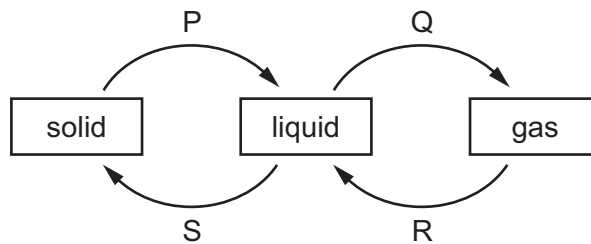
- A** Gases contract and liquids expand.
- B** Gases contract and solids expand.
- C** Liquids expand more than gases.
- D** Liquids expand more than solids.

- 17 A liquid-in-glass thermometer has a range from  $-10^{\circ}\text{C}$  to  $110^{\circ}\text{C}$ .

What are the fixed point temperatures used when calibrating this thermometer in  $^{\circ}\text{C}$ ?

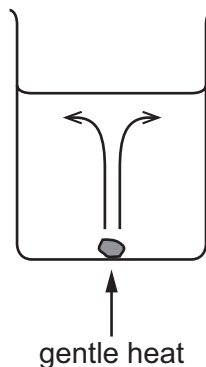
- A  $-10^{\circ}\text{C}$  and  $0^{\circ}\text{C}$
- B  $-10^{\circ}\text{C}$  and  $110^{\circ}\text{C}$
- C  $0^{\circ}\text{C}$  and  $100^{\circ}\text{C}$
- D  $0^{\circ}\text{C}$  and  $110^{\circ}\text{C}$

- 18 The diagram shows four labelled changes of state between solid, liquid and gas.



Which changes need an energy input?

- A P and Q
  - B Q and R
  - C R and S
  - D S and P
- 19 A beaker contains some cold water. A purple crystal is placed on the bottom of the beaker. The beaker is gently heated beneath the crystal. The crystal dissolves in the water. The colour spreads, as shown in the diagram.



Three students each make a statement about the experiment.

Student 1 says the purple water is less dense than the rest of the water.

Student 2 says the purple water is warmer than the rest of the water.

Student 3 says all of the water will eventually get heated, even though water is a poor conductor of thermal energy.

Which students are correct?

- A 1 and 2 only
- B 1 and 3 only
- C 2 and 3 only
- D 1, 2 and 3



20 Why are the outside walls of houses often painted white in very hot countries?

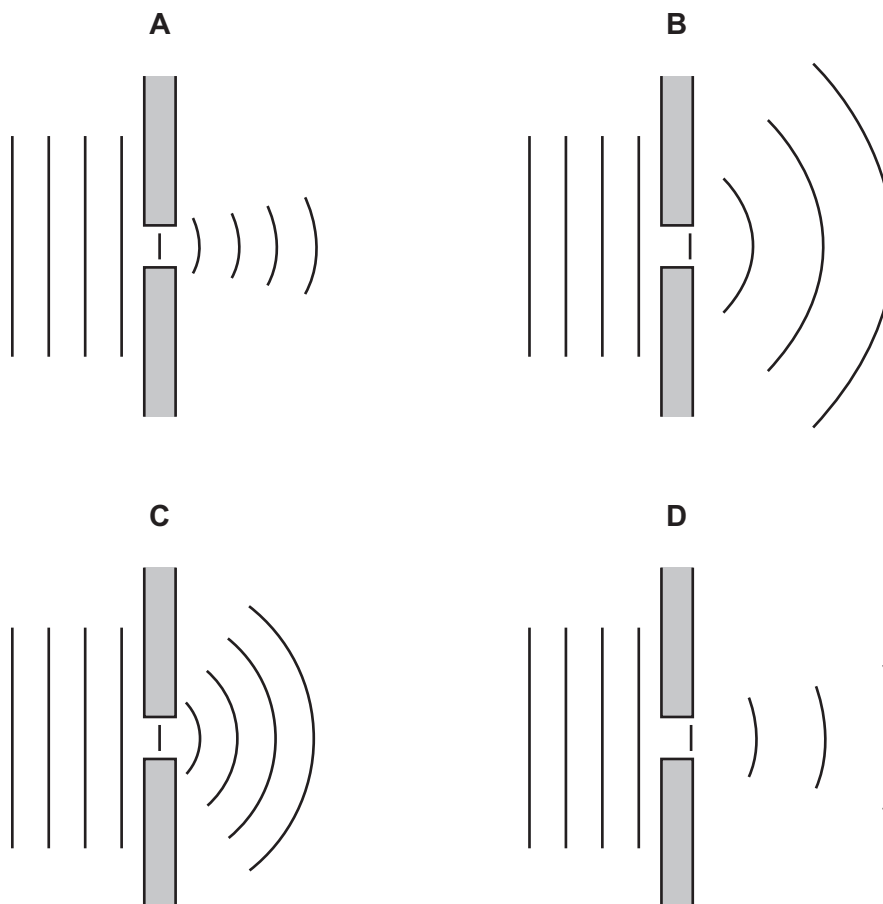
- A White surfaces are good absorbers of infrared radiation.
- B White surfaces are good emitters of infrared radiation.
- C White surfaces are poor absorbers of infrared radiation.
- D White surfaces are poor reflectors of infrared radiation.

21 Which type of wave is **not** an example of a transverse wave?

- A sound wave
- B microwave
- C infrared wave
- D radio wave

22 Plane water waves approach a narrow gap in a barrier.

Which diagram shows the diffraction pattern that would occur?

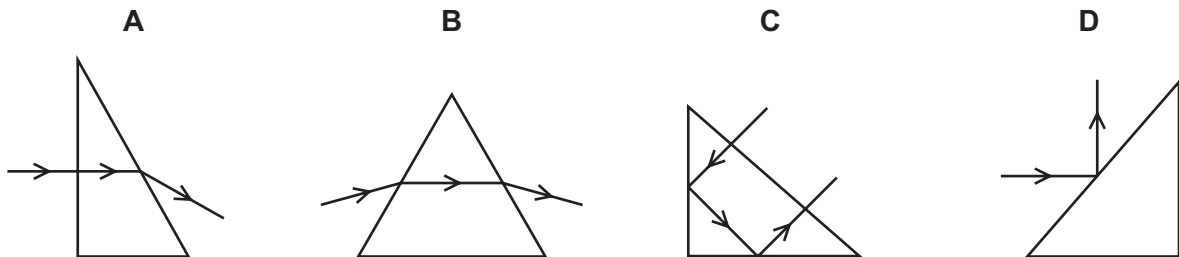


- 23** A person stands 1.0 m in front of a plane mirror. The mirror is moved away from the person at a speed of 1.0 m/s.

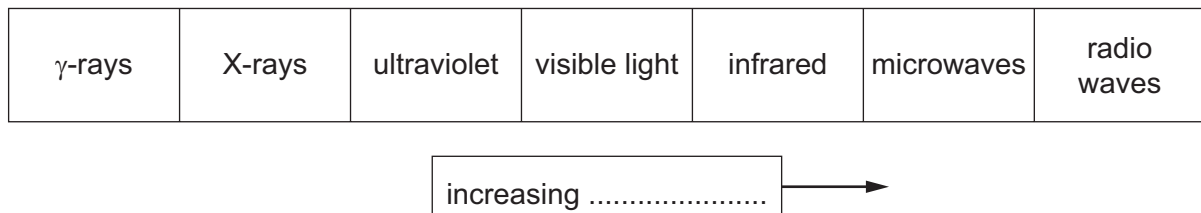
Which statement is correct?

- A** The image moves away from the person at a speed of 1.0 m/s.
- B** The image moves away from the person at a speed of 2.0 m/s.
- C** The image moves towards the person at a speed of 1.0 m/s.
- D** The image moves towards the person at a speed of 2.0 m/s.

- 24** Which diagram shows total internal reflection of light by a glass prism?



- 25** The diagram shows the electromagnetic spectrum.



A word is missing from the label below the spectrum.

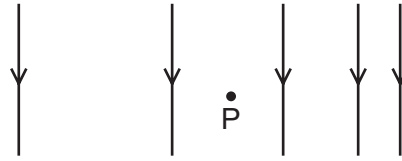
Which word is missing?

- A** amplitude
  - B** frequency
  - C** speed
  - D** wavelength
- 26** A man hears a starting pistol fire 1.5 seconds after he sees a puff of smoke from the pistol. The sound and the smoke are made at the same time. The starting pistol is 450 metres away from the man.

What is the speed of sound calculated from this observation?

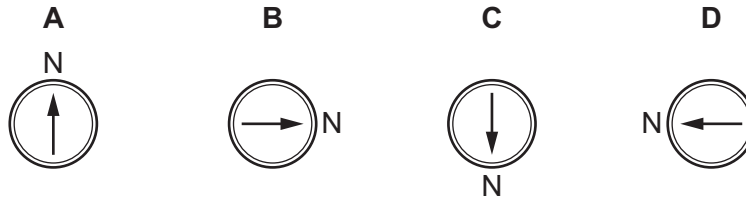
- A** 150 m/s
- B** 300 m/s
- C** 330 m/s
- D** 625 m/s

- 27 The diagram represents a magnetic field. The field increases in strength from left to right.

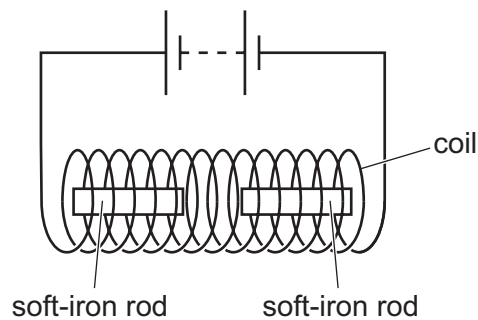


A small compass is placed at P.

Which way will the compass needle point?



- 28 Two soft-iron rods are placed end-to-end inside a coil. The coil is connected to a battery.



The connections from the battery to the coil are now reversed.

What happens to the soft-iron rods in each case?

|          | battery connections<br>as shown | battery connections<br>reversed |
|----------|---------------------------------|---------------------------------|
| <b>A</b> | rods attract                    | rods attract                    |
| <b>B</b> | rods attract                    | rods repel                      |
| <b>C</b> | rods repel                      | rods attract                    |
| <b>D</b> | rods repel                      | rods repel                      |

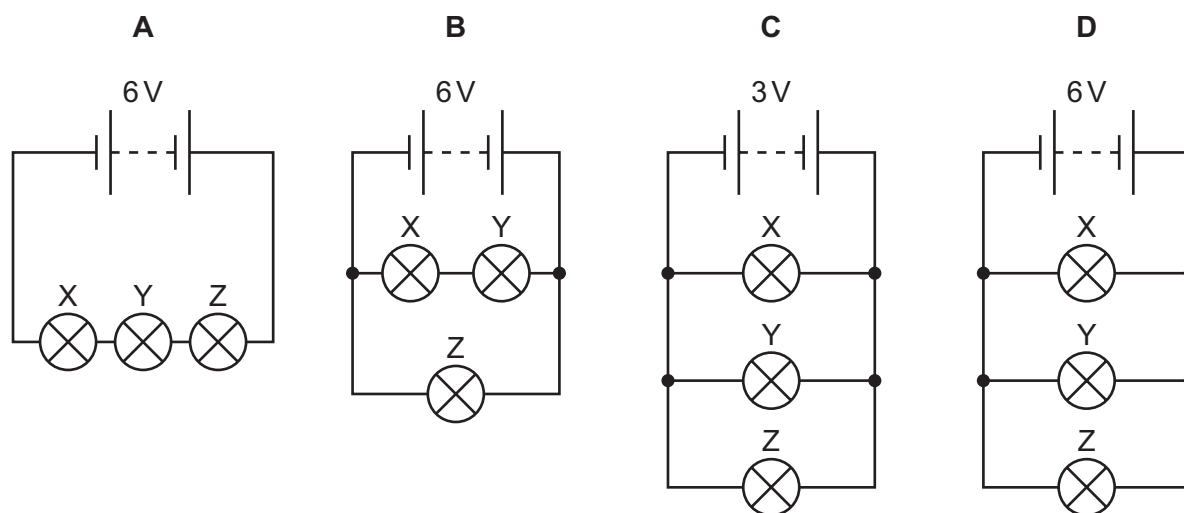
- 29** A plastic rod is rubbed with a dry cloth. The rod becomes positively charged.

Why has the rod become positively charged?

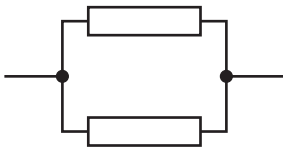
- A** It has gained electrons.
  - B** It has gained neutrons.
  - C** It has lost electrons.
  - D** It has lost neutrons.
- 30** A resistor is connected to a battery. There is a current in the resistor.
- What is the main energy change?
- A** Chemical energy is converted into thermal energy.
  - B** Chemical energy is converted into gravitational potential energy.
  - C** Nuclear energy is converted into thermal energy.
  - D** Nuclear energy is converted into gravitational potential energy.
- 31** Lamps X and Y are designed to operate at normal brightness when each are connected to a 3.0 V supply.

Lamp Z is designed to operate at normal brightness when connected to a 6.0 V supply.

In which circuit do all three lamps operate at normal brightness?



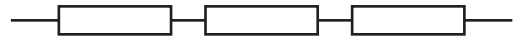
**32** Identical resistors are connected together to form arrangements X, Y and Z.



X



Y



Z

What is the correct order of the resistances of the arrangements from the largest to the smallest?

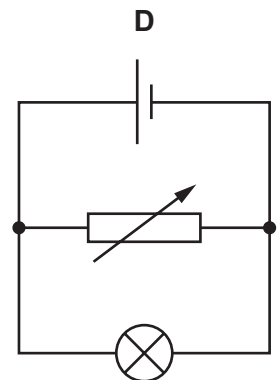
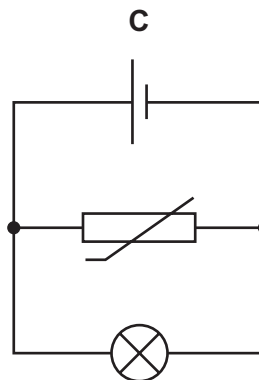
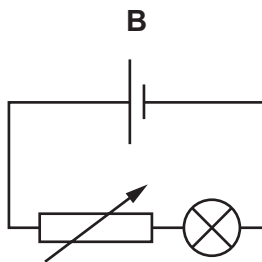
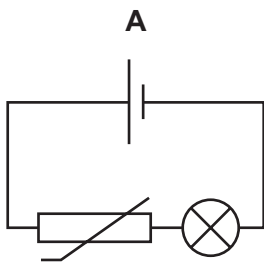
**A**  $X \rightarrow Y \rightarrow Z$

**B**  $Y \rightarrow X \rightarrow Z$

**C**  $Z \rightarrow X \rightarrow Y$

**D**  $Z \rightarrow Y \rightarrow X$

**33** Which circuit shows a variable resistor used to control the brightness of a lamp?



**34** Where must a fuse be connected in a mains electric circuit?

**A** the earth wire only

**B** the live wire only

**C** the neutral wire only

**D** the live wire and the earth wire

**35** A step-up transformer produces a 60 V a.c. output from a 12 V a.c. input.

There are 50 turns on the secondary coil.

How many turns are there on the primary coil?

**A** 5

**B** 10

**C** 50

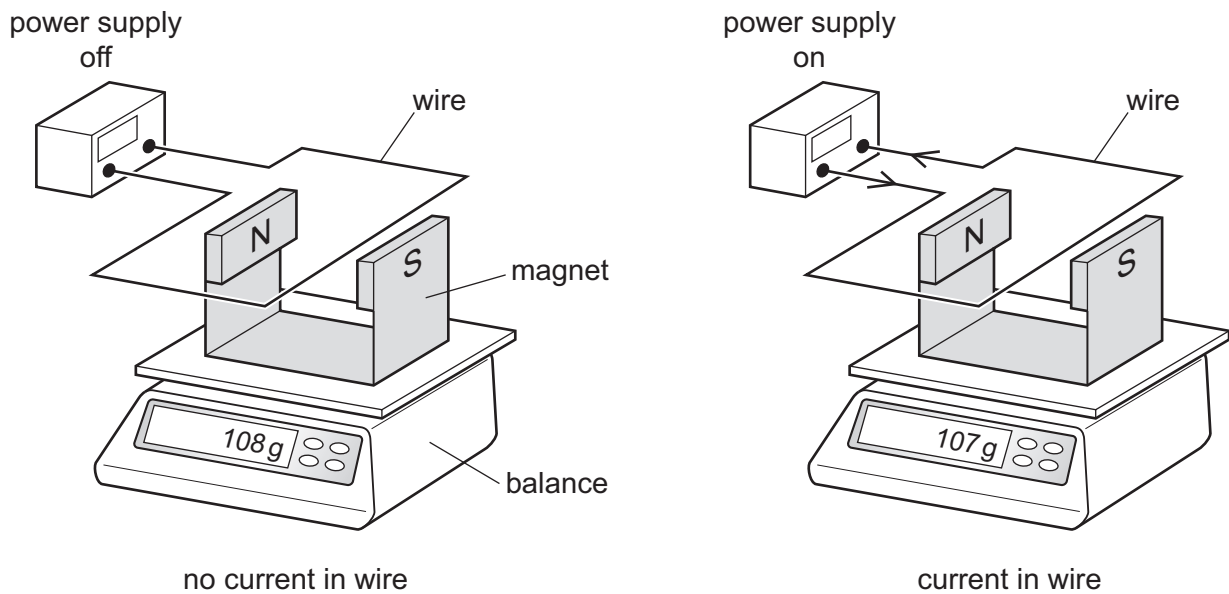
**D** 250

- 36** A student uses a balance, a magnet and a power supply to determine the force on a wire in a magnetic field.

The wire is held between the poles of the magnet.

The student switches on the power supply.

The diagrams show the readings with and without a current in the wire.

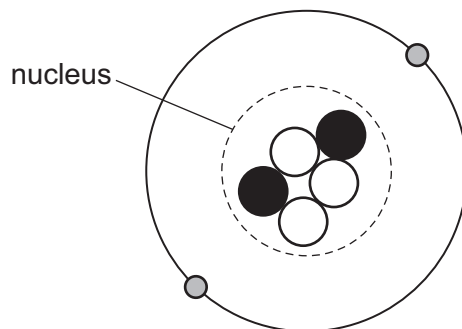


The student reverses the current in the wire. The magnitude of the current does not change.

What is the new reading on the balance?

- A** 106g      **B** 107g      **C** 108g      **D** 109g

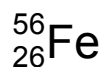
37 The diagram represents a neutral atom.



Which row identifies each type of particle in the diagram?

|          |  |  |  |
|----------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>A</b> | electron                                                                          | neutron                                                                           | proton                                                                            |
| <b>B</b> | electron                                                                          | proton                                                                            | neutron                                                                           |
| <b>C</b> | neutron                                                                           | electron                                                                          | proton                                                                            |
| <b>D</b> | proton                                                                            | electron                                                                          | neutron                                                                           |

38 An iron nuclide is represented by the symbol shown.



Which statements about a nucleus of this iron nuclide are correct?

- 1 The nucleus contains 56 neutrons.
- 2 The nucleon number is 30.
- 3 The proton number is 26.

**A** 1 and 2 only    **B** 1 and 3 only    **C** 2 and 3 only    **D** 3 only

**39** Three types of radiation that can cause ionisation are  $\alpha$ -,  $\beta$ - and  $\gamma$ -radiation.

Which row identifies the least and the most ionising of these radiations?

|          | least ionising | most ionising |
|----------|----------------|---------------|
| <b>A</b> | $\alpha$       | $\beta$       |
| <b>B</b> | $\alpha$       | $\gamma$      |
| <b>C</b> | $\gamma$       | $\beta$       |
| <b>D</b> | $\gamma$       | $\alpha$      |

**40** Why are some radioactive sources stored in boxes made from lead?

- A** Lead absorbs emissions from the radioactive sources.
- B** Lead decreases the half-life of radioactive sources.
- C** Lead increases the half-life of radioactive sources.
- D** Lead repels emissions from the radioactive sources.



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**Cambridge Assessment International Education**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

**0625/12**

Paper 1 Multiple Choice (Core)

**October/November 2019**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)

**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

**DO NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall =  $10 \text{ m/s}^2$ ).

This syllabus is regulated for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **15** printed pages and **1** blank page.

- 1 A measuring cylinder contains  $10\text{ cm}^3$  of water.

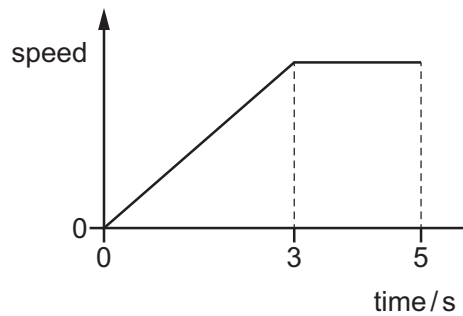
A piece of steel is lowered into the measuring cylinder until it is fully submerged. The volume reading increases to  $12\text{ cm}^3$ .

A second piece of steel is lowered into the measuring cylinder so that it is also fully submerged. The volume reading increases to  $15\text{ cm}^3$ .

Which row shows the volumes of the two pieces of steel?

|          | volume of first piece of steel / $\text{cm}^3$ | volume of second piece of steel / $\text{cm}^3$ |
|----------|------------------------------------------------|-------------------------------------------------|
| <b>A</b> | 2                                              | 3                                               |
| <b>B</b> | 2                                              | 5                                               |
| <b>C</b> | 12                                             | 3                                               |
| <b>D</b> | 12                                             | 15                                              |

- 2 The graph shows the motion of a car for a five-second period.



Which row is correct?

|          | the car is at rest at | the car is moving at a constant speed at |
|----------|-----------------------|------------------------------------------|
| <b>A</b> | 0.0 s                 | 2.0 s                                    |
| <b>B</b> | 0.0 s                 | 4.0 s                                    |
| <b>C</b> | 4.0 s                 | 0.0 s                                    |
| <b>D</b> | 4.0 s                 | 2.0 s                                    |

- 3 A car travels at an average speed of  $60\text{ km/h}$  for 15 minutes.

How far does the car travel in 15 minutes?

- A** 4.0 km      **B** 15 km      **C** 240 km      **D** 900 km

- 4 A box is placed on the ground. An upward force of 15 N is needed to lift the box at constant speed.

Which row correctly describes the box?

|          | mass of the box | weight of the box |
|----------|-----------------|-------------------|
| <b>A</b> | 1.5 kg          | 15 N              |
| <b>B</b> | 15 N            | 1.5 kg            |
| <b>C</b> | 15 N            | 150 kg            |
| <b>D</b> | 150 kg          | 15 N              |

- 5 The table gives the mass and the volume of three objects P, Q and R.

| object | mass / g | volume / cm <sup>3</sup> |
|--------|----------|--------------------------|
| P      | 23       | 36                       |
| Q      | 170      | 720                      |
| R      | 240      | 340                      |

Which objects can float in a liquid of density 0.85 g/cm<sup>3</sup>?

- A** P and Q only  
**B** P and R only  
**C** Q and R only  
**D** P, Q and R
- 6 A student measures the length of a spring. She then attaches different weights to the spring. She measures the length of the spring for each weight.

The table shows her results.

| weight / N | length / mm |
|------------|-------------|
| 0          | 520         |
| 1.0        | 524         |
| 2.0        | 528         |
| 3.0        | 533         |
| 4.0        | 537         |
| 5.0        | 540         |

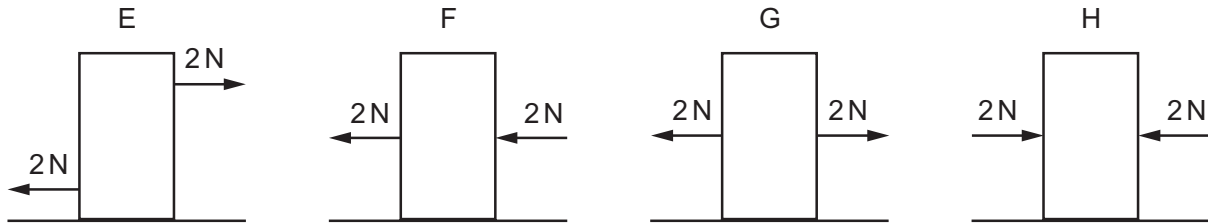
What is the extension of the spring with a weight of 3.0 N attached to it?

- A** 4 mm      **B** 5 mm      **C** 12 mm      **D** 13 mm

7 What is the unit of the moment of a force?

- A** N                      **B** N/kg                      **C** N/m                      **D** Nm

8 The diagrams show a block of wood on a frictionless surface. In each diagram, the block has two forces acting on its sides.



Which diagrams show the block in equilibrium?

- A** E, G and H only  
**B** E and F only  
**C** G and H only  
**D** E, F, G and H

9 Brakes are used to slow down a moving car.

Into which form of energy is most of the kinetic energy converted as the car slows down?

- A** chemical  
**B** elastic  
**C** thermal  
**D** sound

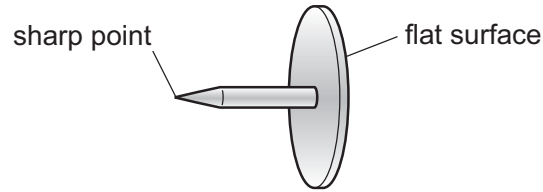
10 What is the unit of thermal energy?

- A** °C                      **B** N                      **C** J                      **D** W

11 Which source of energy is non-renewable?

- A** oil  
**B** solar  
**C** tidal  
**D** wind

- 12** A drawing pin (thumb tack) has a sharp point at one end and a flat surface at the other end.



The pin is pushed into a wooden board.

How do the pressure and the force at the sharp point compare with the pressure and the force on the flat surface?

|          | force at the sharp point         | pressure at the sharp point      |
|----------|----------------------------------|----------------------------------|
| <b>A</b> | greater than on the flat surface | greater than on the flat surface |
| <b>B</b> | greater than on the flat surface | less than on the flat surface    |
| <b>C</b> | the same as on the flat surface  | greater than on the flat surface |
| <b>D</b> | the same as on the flat surface  | less than on the flat surface    |

- 13** A metal block of weight  $W$  rests on a table. In order to calculate the pressure that the block exerts on the table, one other quantity must be known.

What is the other quantity?

- A** the area of contact between the block and the table
- B** the density of the block
- C** the mass of the block
- D** the volume of the block

- 14** Which row describes the arrangement and the motion of the molecules in a gas?

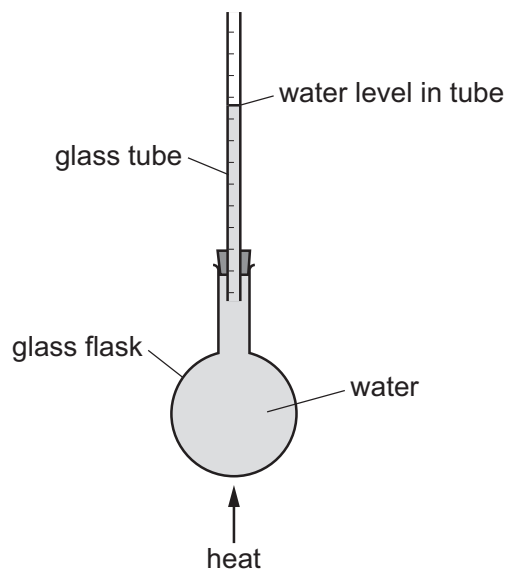
|          | arrangement    | motion       |
|----------|----------------|--------------|
| <b>A</b> | far apart      | move freely  |
| <b>B</b> | far apart      | vibrate only |
| <b>C</b> | tightly packed | move freely  |
| <b>D</b> | tightly packed | vibrate only |

- 15** A driver of a car measures the pressure of the air in the tyres. He measures the pressure again after a long journey. The pressure reading has increased.

Which row states what has happened to the speed of the air molecules and the temperature of the air in the tyres?

|          | speed of molecules | temperature |
|----------|--------------------|-------------|
| <b>A</b> | decreases          | decreases   |
| <b>B</b> | decreases          | increases   |
| <b>C</b> | increases          | decreases   |
| <b>D</b> | increases          | increases   |

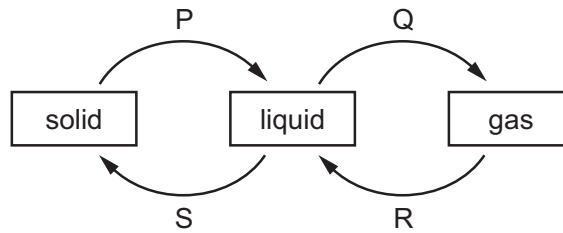
- 16** Some water in a glass flask is gently heated.



Why does the water level in the glass tube rise during heating?

- A** Only the glass flask expands.  
**B** Only the water expands.  
**C** The glass flask expands more than the water.  
**D** The water expands more than the glass flask.
- 17** What is the temperature difference between the fixed points on the °C temperature scale?
- A** 10 °C      **B** 100 °C      **C** 110 °C      **D** 120 °C

18 The diagram shows four labelled changes of state between solid, liquid and gas.



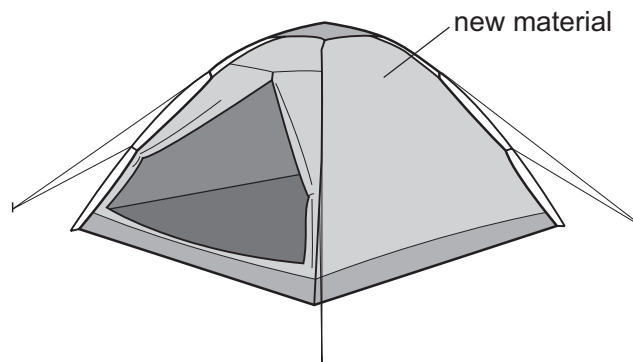
Which changes need an energy input?

- A** P and Q      **B** Q and R      **C** R and S      **D** S and P

19 Which method of thermal transfer occurs when the density of some of a liquid decreases and the liquid moves upwards?

- A** conduction  
**B** convection  
**C** evaporation  
**D** radiation

20 The diagram shows a tent made from a new material.



What type of material should the tent be made of to reflect the radiant energy from the Sun?

|          | material texture | material surface colour |
|----------|------------------|-------------------------|
| <b>A</b> | dull             | black                   |
| <b>B</b> | dull             | white                   |
| <b>C</b> | shiny            | black                   |
| <b>D</b> | shiny            | white                   |

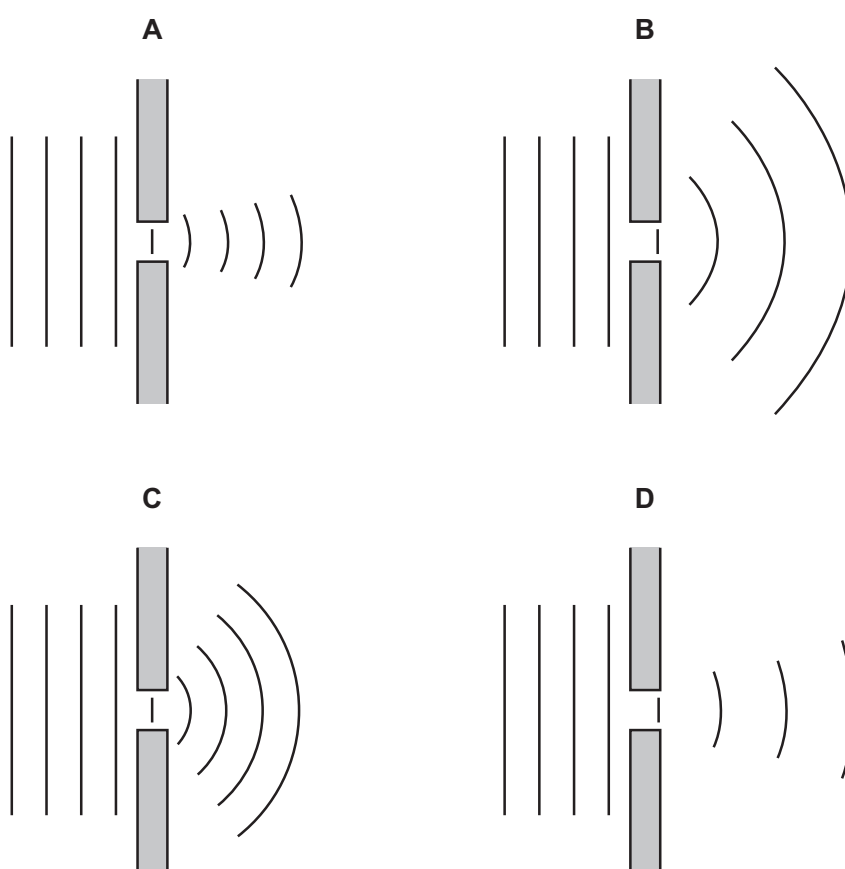


- 21** Which row correctly describes the vibrations of a transverse wave and also gives a correct example of a transverse wave?

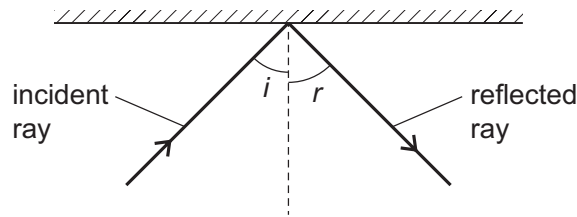
|          | description of vibration           | example of a transverse wave |
|----------|------------------------------------|------------------------------|
| <b>A</b> | right-angles to the wave direction | sound                        |
| <b>B</b> | right-angles to the wave direction | water wave                   |
| <b>C</b> | parallel to the wave direction     | sound                        |
| <b>D</b> | parallel to the wave direction     | water wave                   |

- 22** Plane water waves approach a narrow gap in a barrier.

Which diagram shows the diffraction pattern that would occur?

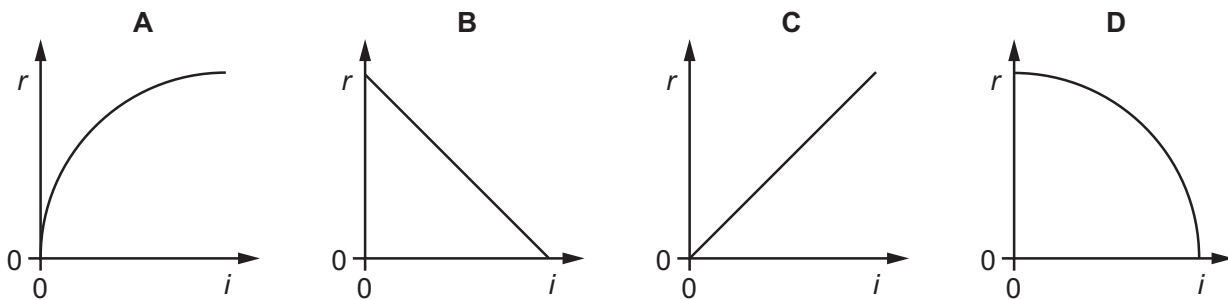


- 23** A ray of light is incident on a plane mirror. A student measures the angle of incidence  $i$  and the angle of reflection  $r$ .



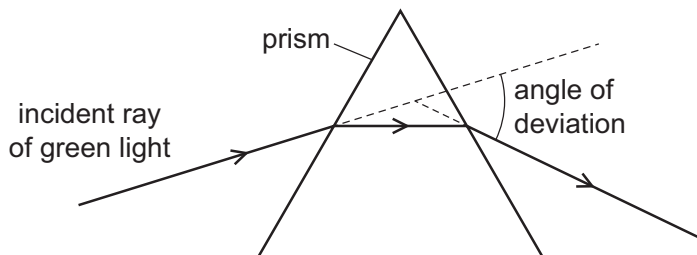
The student varies the angle of incidence and then plots a graph of  $r$  against  $i$ .

What does the graph look like?



- 24** The diagram shows the path of a ray of green light through a glass prism.

The angle of deviation is the angle between the incident ray and the ray leaving the prism.

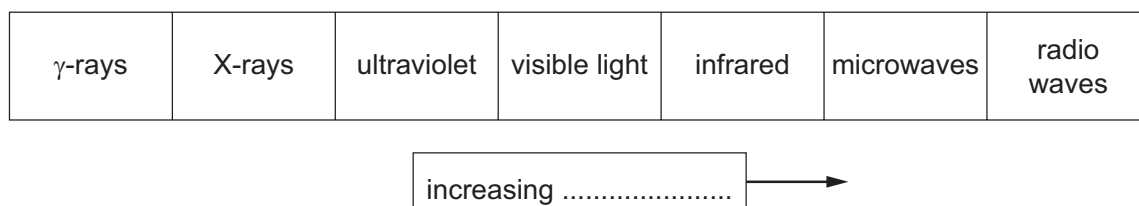


When a ray of white light is incident on the prism, it separates into the colours of the visible spectrum.

What is the name of this effect and which colour of light has the smallest angle of deviation?

|          | name of effect | colour with smallest angle of deviation |
|----------|----------------|-----------------------------------------|
| <b>A</b> | diffraction    | red                                     |
| <b>B</b> | diffraction    | violet                                  |
| <b>C</b> | dispersion     | red                                     |
| <b>D</b> | dispersion     | violet                                  |

- 25 The diagram shows the electromagnetic spectrum.



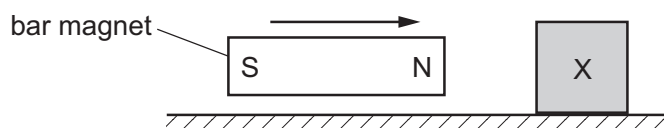
A word is missing from the label below the spectrum.

Which word is missing?

- A** amplitude
  - B** frequency
  - C** speed
  - D** wavelength
- 26 A man hears a starting pistol fire 1.5 seconds after he sees a puff of smoke from the pistol. The sound and the smoke are made at the same time. The starting pistol is 450 metres away from the man.

What is the speed of sound calculated from this observation?

- A** 150 m/s      **B** 300 m/s      **C** 330 m/s      **D** 625 m/s
- 27 A bar magnet is slowly moved towards an unmagnetised metal object X.

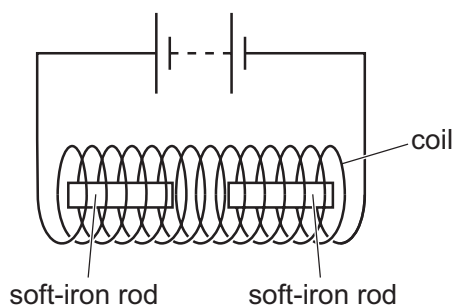


When it is a few centimetres away, the object begins to slide towards the magnet.

Why does this happen?

- A** X is a non-magnetic material and magnetism is induced in it.
- B** X is a non-magnetic material and magnetism is not induced in it.
- C** X is a magnetic material and magnetism is induced in it.
- D** X is a magnetic material and magnetism is not induced in it.

- 28** Two soft-iron rods are placed end-to-end inside a coil. The coil is connected to a battery.



The connections from the battery to the coil are now reversed.

What happens to the soft-iron rods in each case?

|          | battery connections<br>as shown | battery connections<br>reversed |
|----------|---------------------------------|---------------------------------|
| <b>A</b> | rods attract                    | rods attract                    |
| <b>B</b> | rods attract                    | rods repel                      |
| <b>C</b> | rods repel                      | rods attract                    |
| <b>D</b> | rods repel                      | rods repel                      |

- 29** A plastic rod is rubbed with a dry cloth. The rod becomes positively charged.

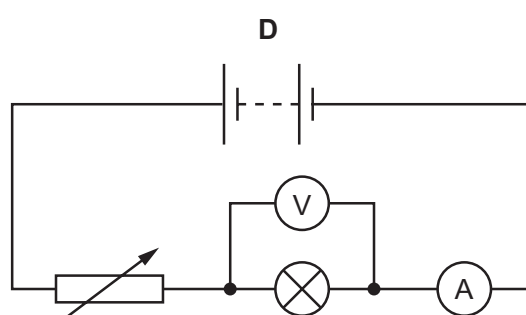
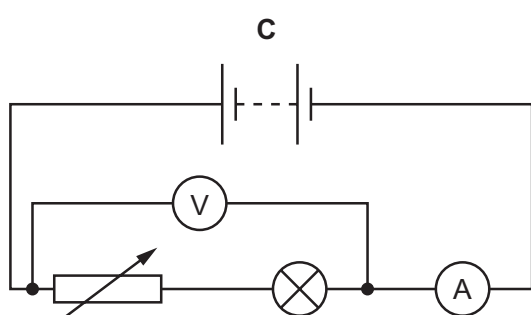
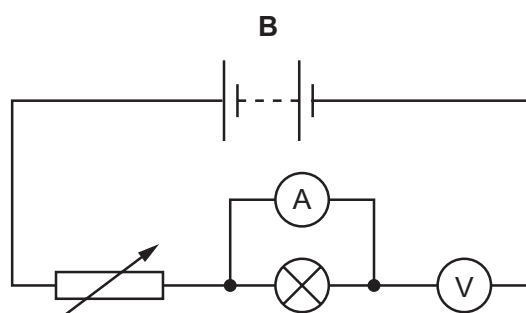
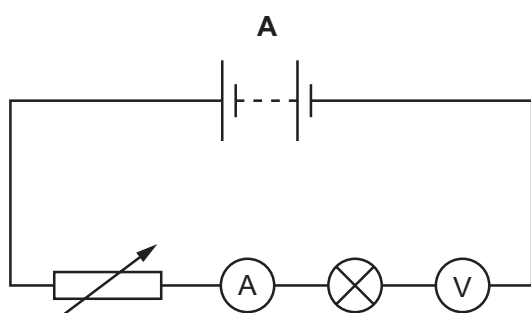
Why has the rod become positively charged?

- A** It has gained electrons.
  - B** It has gained neutrons.
  - C** It has lost electrons.
  - D** It has lost neutrons.
- 30** Which quantity has the same unit as potential difference (p.d.)?
- A** current
  - B** electromotive force (e.m.f.)
  - C** resistance
  - D** moment of a force

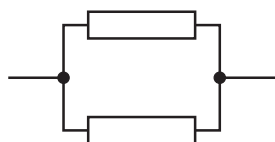
**31** A student determines the resistance of an electric lamp.

He measures the current in the lamp and the potential difference (p.d.) across it.

Which circuit did he use?



**32** Identical resistors are connected together to form arrangements X, Y and Z.



X



Y



Z

What is the correct order of the resistances of the arrangements from the largest to the smallest?

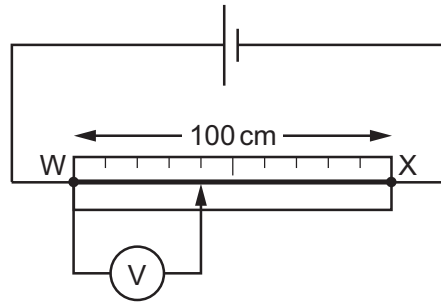
**A**  $X \rightarrow Y \rightarrow Z$

**B**  $Y \rightarrow X \rightarrow Z$

**C**  $Z \rightarrow X \rightarrow Y$

**D**  $Z \rightarrow Y \rightarrow X$

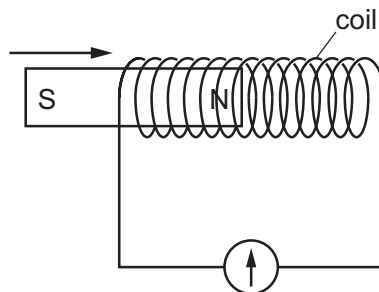
- 33 The circuit shows a wire WX connected to a cell.



The potential difference (p.d.) between W and X is 1.5 V.

What is the reading on the voltmeter?

- A 0.4 V                      B 0.6 V                      C 0.9 V                      D 4.0 V
- 34 Where must a fuse be connected in a mains electric circuit?
- A the earth wire only
- B the live wire only
- C the neutral wire only
- D the live wire and the earth wire
- 35 The N-pole of a magnet is moved into a coil of wire connected to a galvanometer.



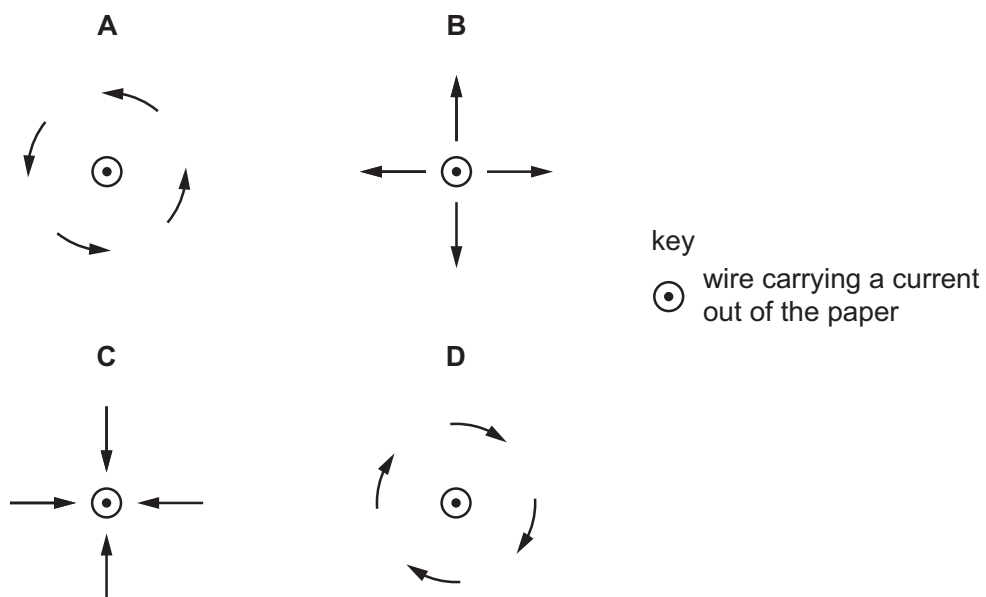
The needle of the galvanometer moves.

Which situation **must** give a smaller galvanometer reading?

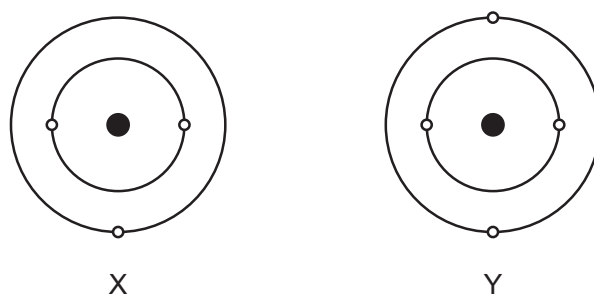
- A Use a coil with fewer turns and a stronger magnet.
- B Use a coil with fewer turns and a weaker magnet.
- C Use a coil with more turns and a stronger magnet.
- D Use a coil with more turns and a weaker magnet.

**36** The diagram shows a conductor carrying current in a direction out of the plane of the page.

Which set of arrows represents the direction of the magnetic field due to this current?



**37** The diagrams show the simple atomic structure for two neutral atoms X and Y of different elements.



Which row is correct?

|          | atom with more electrons | atom with a more positively charged nucleus |
|----------|--------------------------|---------------------------------------------|
| <b>A</b> | X                        | X                                           |
| <b>B</b> | X                        | Y                                           |
| <b>C</b> | Y                        | X                                           |
| <b>D</b> | Y                        | Y                                           |

38 A nuclide of chlorine has the symbol shown.



What is the nucleon number of this nuclide of chlorine?

- A 17                      B 18                      C 35                      D 52

39 Which type of radiation can be stopped by a sheet of paper?

- A  $\alpha$ -particles  
B  $\beta$ -particles  
C  $\gamma$ -rays  
D X-rays

40 Why are some radioactive sources stored in boxes made from lead?

- A Lead absorbs emissions from the radioactive sources.  
B Lead decreases the half-life of radioactive sources.  
C Lead increases the half-life of radioactive sources.  
D Lead repels emissions from the radioactive sources.



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**Cambridge Assessment International Education**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

**0625/11**

Paper 1 Multiple Choice (Core)

**October/November 2019**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)



**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

**DO NOT WRITE IN ANY BARCODES.**

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall =  $10 \text{ m/s}^2$ ).

This syllabus is regulated for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

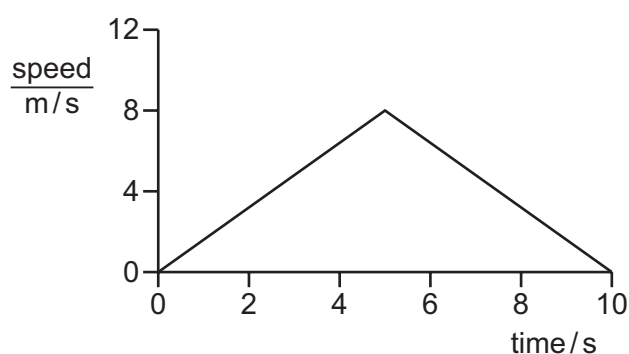
This document consists of **16** printed pages.

- 1 A student measures the volume of a small irregularly-shaped stone.

Which apparatus must be used?

- A a measuring cylinder containing water and a ruler only
- B a measuring cylinder containing water only
- C an empty measuring cylinder and a ruler only
- D a ruler only

- 2 The graph shows how the speed of an object changes with time.



How far does the object travel in 10 seconds?

- A 8 m
  - B 10 m
  - C 40 m
  - D 80 m
- 3 A car travels at an average speed of 60 km/h for 15 minutes.
- How far does the car travel in 15 minutes?
- A 4.0 km
  - B 15 km
  - C 240 km
  - D 900 km
- 4 Which quantity is a force due to a gravitational field?
- A density
  - B mass
  - C weight
  - D volume
- 5 The density of air is  $1.2 \text{ kg/m}^3$ .
- A room has dimensions  $5.0 \text{ m} \times 4.0 \text{ m} \times 3.0 \text{ m}$ .
- What is the mass of the air in the room?
- A 0.02 kg
  - B 0.10 kg
  - C 50 kg
  - D 72 kg

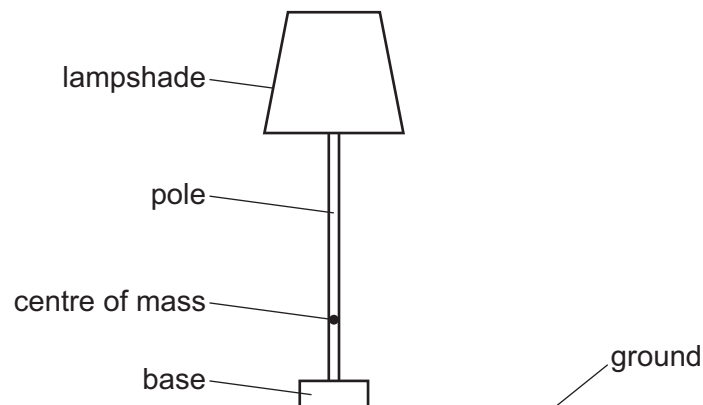
- 6 A student measures the length of a spring. She then attaches different weights to the spring. She measures the length of the spring for each weight.

The table shows her results.

| weight / N | length / mm |
|------------|-------------|
| 0          | 520         |
| 1.0        | 524         |
| 2.0        | 528         |
| 3.0        | 533         |
| 4.0        | 537         |
| 5.0        | 540         |

What is the extension of the spring with a weight of 3.0 N attached to it?

- A** 4 mm      **B** 5 mm      **C** 12 mm      **D** 13 mm
- 7 Which statement gives a complete description of any object that is in equilibrium?
- A** There are no forces acting.  
**B** There is no resultant force.  
**C** There is no resultant force and no resultant turning effect.  
**D** There is no resultant turning effect.
- 8 The diagram shows a lamp.



Changing which feature increases the stability of the lamp?

- A** a larger lampshade  
**B** a longer pole  
**C** a heavier base  
**D** a higher centre of mass

- 9 Brakes are used to slow down a moving car.

Into which form of energy is most of the kinetic energy converted as the car slows down?

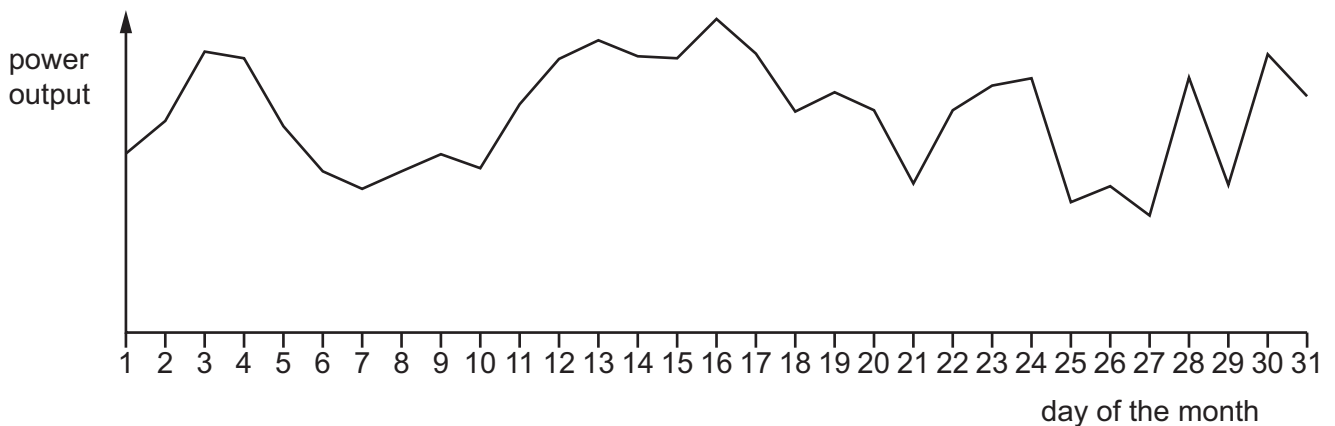
- A chemical
- B elastic
- C thermal
- D sound

- 10 Which does **not** transfer useful energy?

- A emitting a sound wave
- B measuring a temperature
- C passing an electric current
- D pushing a box along the floor

- 11 The power output from solar panels is recorded every day for a month.

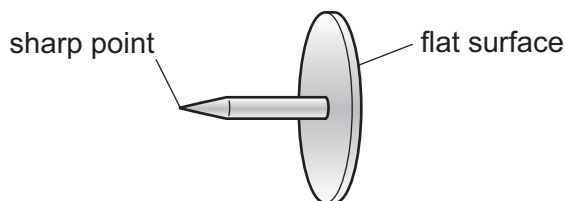
The graph shows the output recorded.



Which conclusion can be drawn from this graph?

- A The power output from the solar panels changes from day to day.
- B The power output from the solar panels is cheap to produce.
- C Solar panels create no pollution.
- D Solar energy is renewable.

- 12 A drawing pin (thumb tack) has a sharp point at one end and a flat surface at the other end.



The pin is pushed into a wooden board.

How do the pressure and the force at the sharp point compare with the pressure and the force on the flat surface?

|          | force at the sharp point         | pressure at the sharp point      |
|----------|----------------------------------|----------------------------------|
| <b>A</b> | greater than on the flat surface | greater than on the flat surface |
| <b>B</b> | greater than on the flat surface | less than on the flat surface    |
| <b>C</b> | the same as on the flat surface  | greater than on the flat surface |
| <b>D</b> | the same as on the flat surface  | less than on the flat surface    |

- 13 The table shows four forces. Each force acts on a different surface.

Which row shows the **least** pressure?

|          | size of the force / N | area of the surface / m <sup>2</sup> |
|----------|-----------------------|--------------------------------------|
| <b>A</b> | 0.30                  | 0.040                                |
| <b>B</b> | 10                    | 2.0                                  |
| <b>C</b> | 60                    | 15                                   |
| <b>D</b> | 1200                  | 40                                   |

- 14 A test-tube contains 1.0 cm<sup>3</sup> of liquid water at 100 °C. The liquid water boils to form 1600 cm<sup>3</sup> of steam.

What is the reason for the large increase in volume?

- A** Steam molecules are bigger than water molecules.
- B** The average distance between the molecules is much greater in the steam.
- C** The molecules do not move until the water turns into a gas.
- D** There are more steam molecules than there were water molecules.

15 Which statement about the evaporation of a liquid is correct?

- A The least energetic molecules escape from the surface and the temperature of the liquid decreases.
- B The least energetic molecules escape from the surface and the temperature of the liquid increases.
- C The most energetic molecules escape from the surface and the temperature of the liquid decreases.
- D The most energetic molecules escape from the surface and the temperature of the liquid increases.

16 Which effect is caused by thermal expansion?

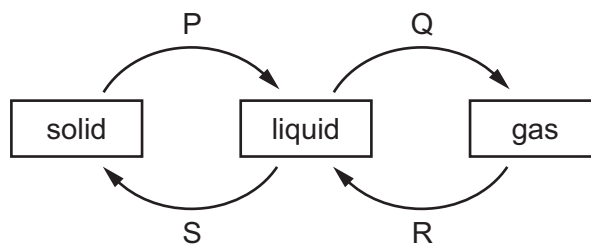
- A a metal surface heating up in direct sunlight
- B ice-cream melting on a hot day
- C a railway track buckling on a hot day
- D ice forming on a pond on a cold day

17 A liquid-in-glass thermometer uses a change in a property of a liquid to measure temperature.

Which property is used?

- A mass
- B thermal capacity
- C volume
- D weight

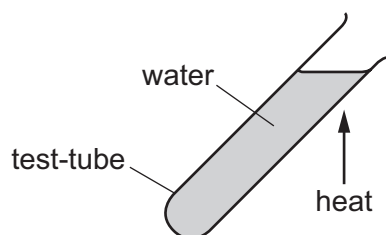
18 The diagram shows four labelled changes of state between solid, liquid and gas.



Which changes need an energy input?

- A P and Q
- B Q and R
- C R and S
- D S and P

- 19** A glass test-tube containing water is heated at the top. The water at the top boils, but the water at the bottom remains cold.



Which row explains why the water at the bottom of the test-tube remains cold?

|          | glass                  | water                  |
|----------|------------------------|------------------------|
| <b>A</b> | good thermal conductor | good thermal conductor |
| <b>B</b> | good thermal conductor | poor thermal conductor |
| <b>C</b> | poor thermal conductor | good thermal conductor |
| <b>D</b> | poor thermal conductor | poor thermal conductor |

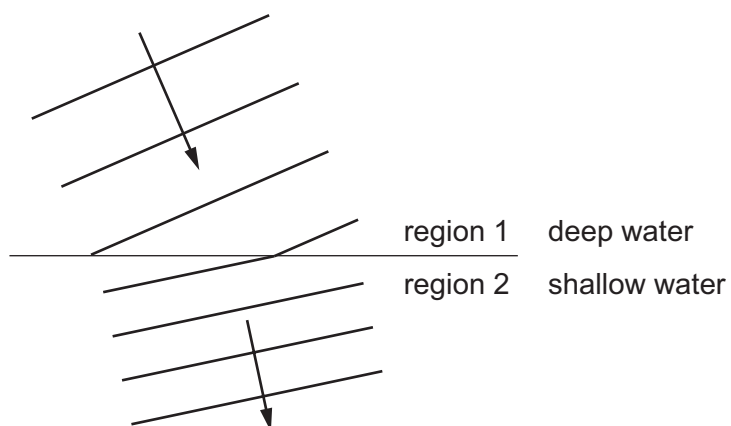
- 20** In countries where it is usually hot, houses are often painted white.

What is the reason for this?

- A** White surfaces are good reflectors of radiant energy.
- B** White surfaces are good transmitters of radiant energy.
- C** White surfaces are good absorbers of radiant energy.
- D** White surfaces are good emitters of radiant energy.



21 The diagram shows wavefronts on the surface of water, viewed from above.

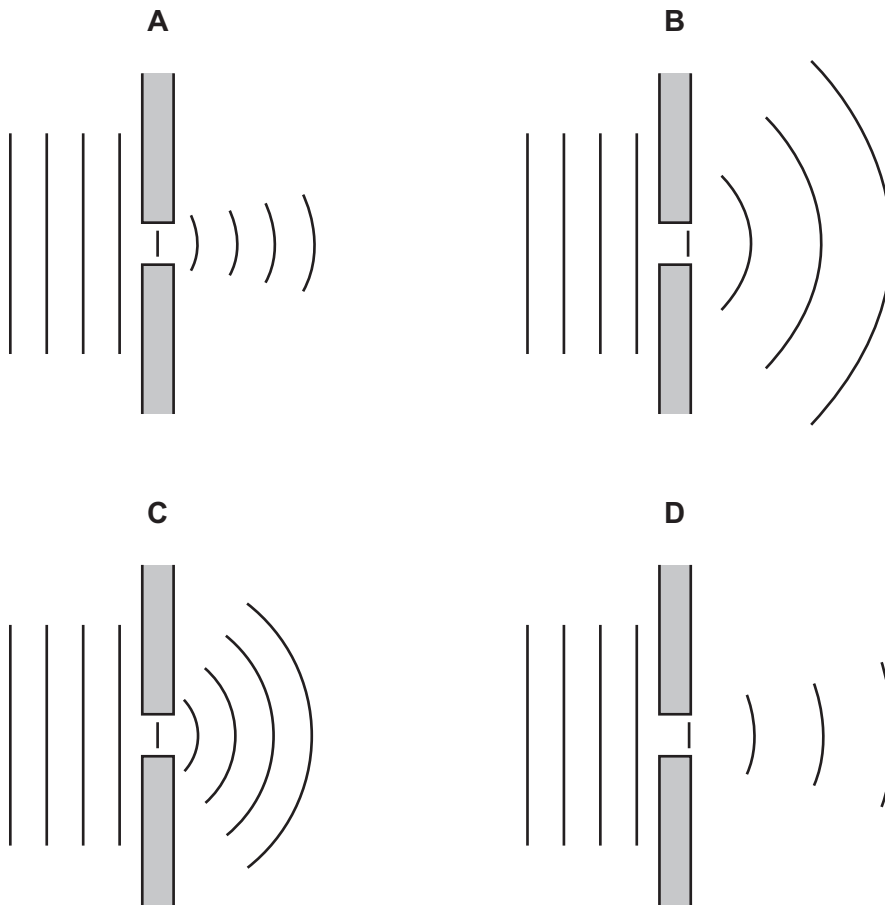


Which row is correct for the wavefronts moving from region 1 to region 2?

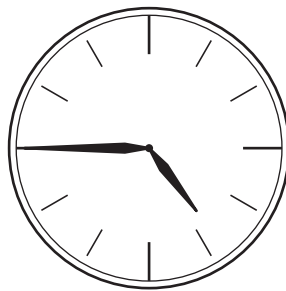
|          | name of process | the speeds of the wavefronts in regions 1 and 2 are |
|----------|-----------------|-----------------------------------------------------|
| <b>A</b> | diffraction     | different                                           |
| <b>B</b> | diffraction     | the same                                            |
| <b>C</b> | refraction      | different                                           |
| <b>D</b> | refraction      | the same                                            |

**22** Plane water waves approach a narrow gap in a barrier.

Which diagram shows the diffraction pattern that would occur?



**23** The diagram shows the image of a clock in a plane mirror.



Which is the actual time?

**A** 04:15

**B** 04:45

**C** 07:15

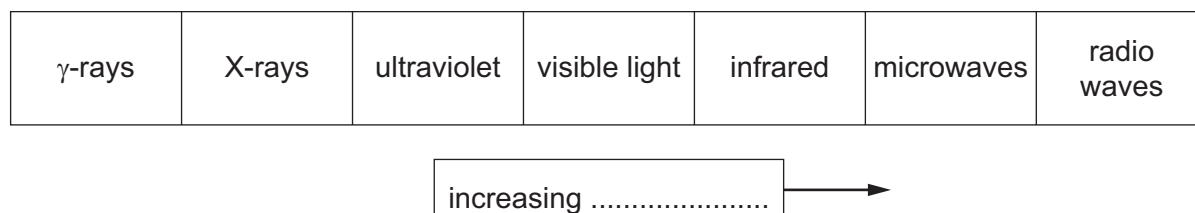
**D** 07:45

- 24** Total internal reflection may occur when light reaches an air-glass boundary.

Under which conditions is light totally internally reflected?

|          | medium in which light travels towards the boundary | angle of incidence              |
|----------|----------------------------------------------------|---------------------------------|
| <b>A</b> | air                                                | greater than the critical angle |
| <b>B</b> | air                                                | less than the critical angle    |
| <b>C</b> | glass                                              | greater than the critical angle |
| <b>D</b> | glass                                              | less than the critical angle    |

- 25** The diagram shows the electromagnetic spectrum.



A word is missing from the label below the spectrum.

Which word is missing?

- A** amplitude
- B** frequency
- C** speed
- D** wavelength
- 26** A man hears a starting pistol fire 1.5 seconds after he sees a puff of smoke from the pistol. The sound and the smoke are made at the same time. The starting pistol is 450 metres away from the man.

What is the speed of sound calculated from this observation?

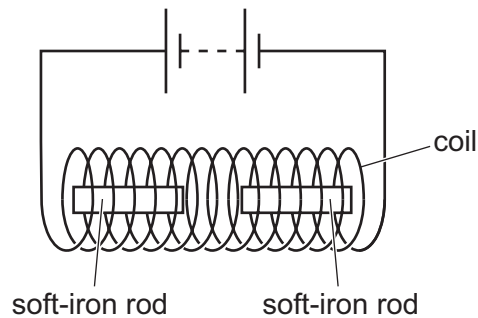
- A** 150 m/s      **B** 300 m/s      **C** 330 m/s      **D** 625 m/s

- 27** A student has a bar magnet. He brings the magnet close to an object. The magnet and the object repel each other.

What must the object be?

- A** another permanent magnet
- B** any magnetic material
- C** a block of wood
- D** a piece of copper

- 28** Two soft-iron rods are placed end-to-end inside a coil. The coil is connected to a battery.



The connections from the battery to the coil are now reversed.

What happens to the soft-iron rods in each case?

|          | battery connections<br>as shown | battery connections<br>reversed |
|----------|---------------------------------|---------------------------------|
| <b>A</b> | rods attract                    | rods attract                    |
| <b>B</b> | rods attract                    | rods repel                      |
| <b>C</b> | rods repel                      | rods attract                    |
| <b>D</b> | rods repel                      | rods repel                      |

- 29** A plastic rod is rubbed with a dry cloth. The rod becomes positively charged.

Why has the rod become positively charged?

- A** It has gained electrons.
- B** It has gained neutrons.
- C** It has lost electrons.
- D** It has lost neutrons.

- 30** A teacher asks her class “What quantity can be recorded in volts?”.

Student 1 says “The potential difference across a resistor”.

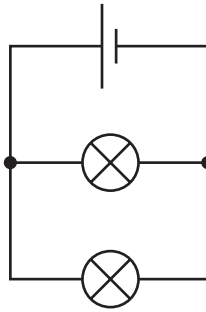
Student 2 says “The rating of a fuse”.

Student 3 says “The electromotive force of a battery”.

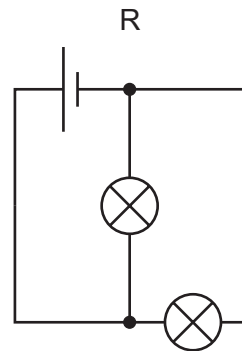
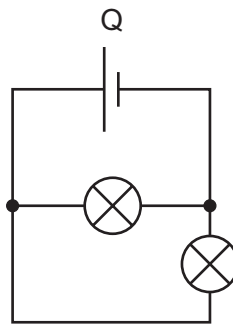
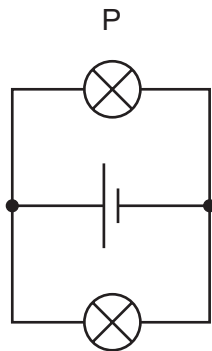
Which students are correct?

- A** 1 only                      **B** 1 and 2                      **C** 1 and 3                      **D** 2 and 3

- 31 The circuit shows two lamps connected to a d.c. supply.



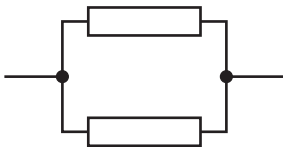
The same lamps and power supply are arranged in different ways, as shown.



In which circuits will the lamps be the same brightness as in the original circuit?

- A** Q only      **B** P and Q only      **C** P and R only      **D** P, Q and R

- 32 Identical resistors are connected together to form arrangements X, Y and Z.



X



Y

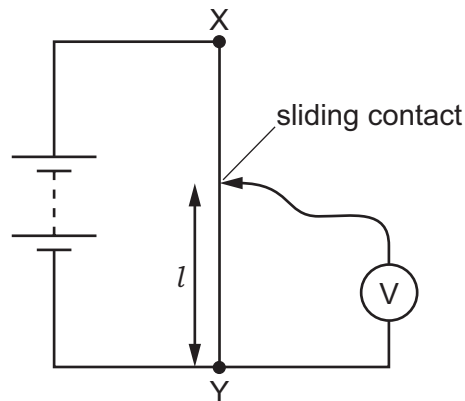


Z

What is the correct order of the resistances of the arrangements from the largest to the smallest?

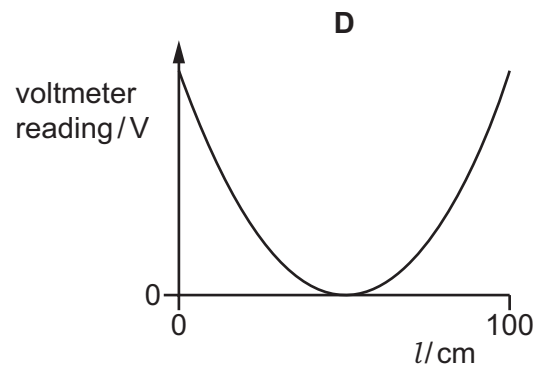
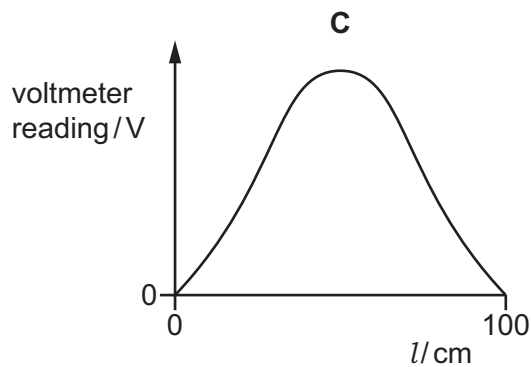
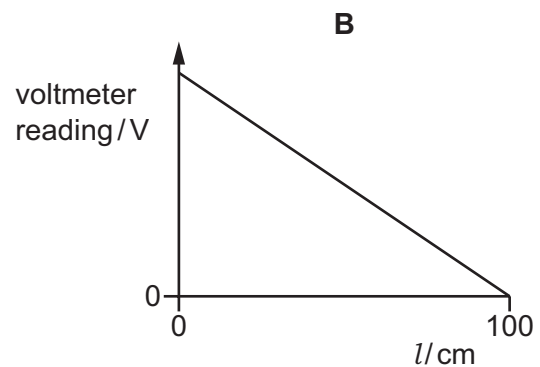
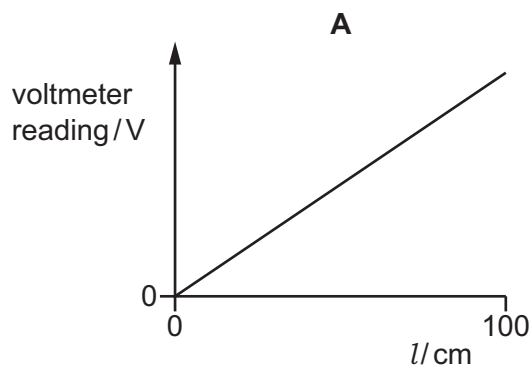
- A**  $X \rightarrow Y \rightarrow Z$   
**B**  $Y \rightarrow X \rightarrow Z$   
**C**  $Z \rightarrow X \rightarrow Y$   
**D**  $Z \rightarrow Y \rightarrow X$

33 A student uses 100 cm of resistance wire XY in a circuit to make a potential divider.



He changes the length of wire  $l$  by moving the sliding contact along the resistance wire.

Which graph shows how the voltmeter reading changes as the length of wire  $l$  is increased from zero to 100 cm?



34 Where must a fuse be connected in a mains electric circuit?

- A** the earth wire only
- B** the live wire only
- C** the neutral wire only
- D** the live wire and the earth wire

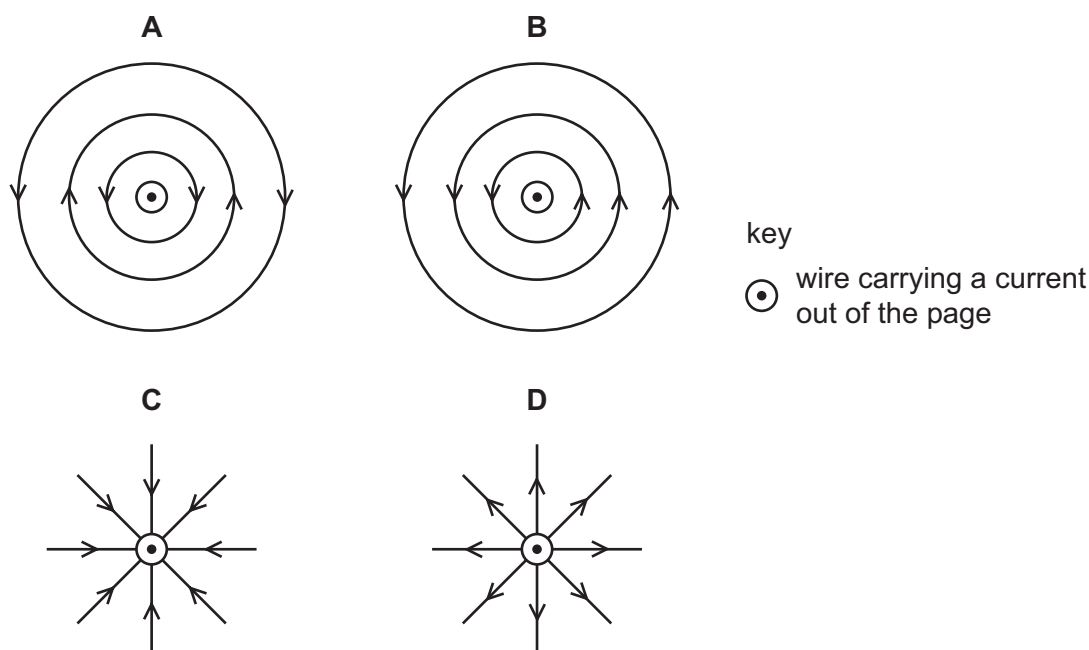
- 35** A student investigates the output voltage induced across a coil of wire by a bar magnet.

When will the induced voltage have the greatest value?

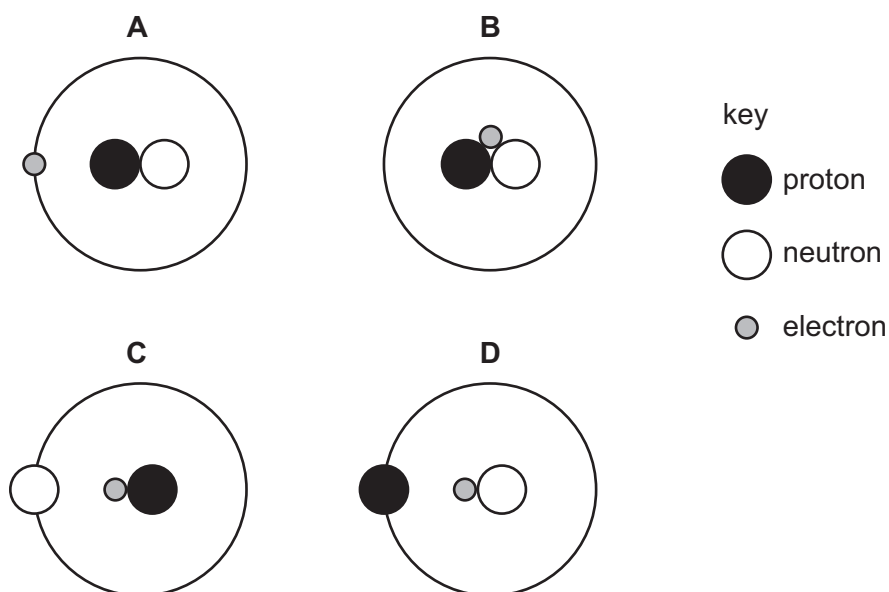
- A** The student slowly moves the bar magnet into the coil of wire.
- B** The student leaves the bar magnet stationary in the coil of wire.
- C** The student quickly removes the bar magnet from the coil of wire.
- D** The student places the bar magnet at rest outside the coil of wire.

- 36** There is a current in a wire. The direction of the current is out of the page.

Which diagram shows the magnetic field pattern produced?



- 37** Which diagram shows a possible structure of a neutral atom?





- 38 A nuclide of cobalt contains 27 protons and 32 neutrons.

Which symbol represents this nuclide?

- A  ${}^{27}_{59}\text{Co}$       B  ${}^{32}_{27}\text{Co}$       C  ${}^{32}_{59}\text{Co}$       D  ${}^{59}_{27}\text{Co}$

- 39 An isotope of radon is radioactive. It decays by emitting an  $\alpha$ -particle.

What happens to the nucleus of a radon atom during the emission of the  $\alpha$ -particle?

- A It becomes the nucleus of a different isotope of radon with fewer neutrons.  
B It becomes the nucleus of a different isotope of radon with more neutrons.  
C It becomes the nucleus of an element with a higher proton number.  
D It becomes the nucleus of an element with a lower proton number.
- 40 Why are some radioactive sources stored in boxes made from lead?
- A Lead absorbs emissions from the radioactive sources.  
B Lead decreases the half-life of radioactive sources.  
C Lead increases the half-life of radioactive sources.  
D Lead repels emissions from the radioactive sources.

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**PHYSICS**

**0625/13**

Paper 1 Multiple Choice (Core)

**October/November 2019**

MARK SCHEME

Maximum Mark: 40

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**Cambridge Assessment**  
International Education

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | B      | 1     |
| 2        | A      | 1     |
| 3        | B      | 1     |
| 4        | B      | 1     |
| 5        | A      | 1     |
| 6        | D      | 1     |
| 7        | D      | 1     |
| 8        | A      | 1     |
| 9        | C      | 1     |
| 10       | B      | 1     |
| 11       | C      | 1     |
| 12       | C      | 1     |
| 13       | A      | 1     |
| 14       | A      | 1     |
| 15       | A      | 1     |
| 16       | D      | 1     |
| 17       | C      | 1     |
| 18       | A      | 1     |
| 19       | D      | 1     |
| 20       | C      | 1     |
| 21       | A      | 1     |
| 22       | C      | 1     |
| 23       | B      | 1     |
| 24       | C      | 1     |
| 25       | D      | 1     |
| 26       | B      | 1     |
| 27       | C      | 1     |
| 28       | A      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | C      | 1     |
| 30       | A      | 1     |
| 31       | B      | 1     |
| 32       | D      | 1     |
| 33       | B      | 1     |
| 34       | B      | 1     |
| 35       | B      | 1     |
| 36       | D      | 1     |
| 37       | A      | 1     |
| 38       | D      | 1     |
| 39       | D      | 1     |
| 40       | A      | 1     |



**Cambridge Assessment International Education**  
Cambridge International General Certificate of Secondary Education

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**PHYSICS**

**0625/12**

Paper 1 Multiple Choice (Core)

**October/November 2019**

MARK SCHEME

Maximum Mark: 40

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| Question | Answer | Mark |
|----------|--------|------|
| 1        | A      | 1    |
| 2        | B      | 1    |
| 3        | B      | 1    |
| 4        | A      | 1    |
| 5        | D      | 1    |
| 6        | D      | 1    |
| 7        | D      | 1    |
| 8        | C      | 1    |
| 9        | C      | 1    |
| 10       | C      | 1    |
| 11       | A      | 1    |
| 12       | C      | 1    |
| 13       | A      | 1    |
| 14       | A      | 1    |
| 15       | D      | 1    |
| 16       | D      | 1    |
| 17       | B      | 1    |
| 18       | A      | 1    |
| 19       | B      | 1    |
| 20       | D      | 1    |
| 21       | B      | 1    |
| 22       | C      | 1    |
| 23       | C      | 1    |
| 24       | C      | 1    |
| 25       | D      | 1    |
| 26       | B      | 1    |
| 27       | C      | 1    |
| 28       | A      | 1    |

| Question | Answer | Mark |
|----------|--------|------|
| 29       | C      | 1    |
| 30       | B      | 1    |
| 31       | D      | 1    |
| 32       | D      | 1    |
| 33       | B      | 1    |
| 34       | B      | 1    |
| 35       | B      | 1    |
| 36       | A      | 1    |
| 37       | D      | 1    |
| 38       | C      | 1    |
| 39       | A      | 1    |
| 40       | A      | 1    |



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**PHYSICS**

**0625/11**

Paper 1 Multiple Choice (Core)

**October/November 2019**

MARK SCHEME

Maximum Mark: 40

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| Question | Answer | Marks |
|----------|--------|-------|
| 1        | B      | 1     |
| 2        | C      | 1     |
| 3        | B      | 1     |
| 4        | C      | 1     |
| 5        | D      | 1     |
| 6        | D      | 1     |
| 7        | C      | 1     |
| 8        | C      | 1     |
| 9        | C      | 1     |
| 10       | B      | 1     |
| 11       | A      | 1     |
| 12       | C      | 1     |
| 13       | C      | 1     |
| 14       | B      | 1     |
| 15       | C      | 1     |
| 16       | C      | 1     |
| 17       | C      | 1     |
| 18       | A      | 1     |
| 19       | D      | 1     |
| 20       | A      | 1     |
| 21       | C      | 1     |
| 22       | C      | 1     |
| 23       | C      | 1     |
| 24       | C      | 1     |
| 25       | D      | 1     |
| 26       | B      | 1     |
| 27       | A      | 1     |
| 28       | A      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | C      | 1     |
| 30       | C      | 1     |
| 31       | D      | 1     |
| 32       | D      | 1     |
| 33       | A      | 1     |
| 34       | B      | 1     |
| 35       | C      | 1     |
| 36       | B      | 1     |
| 37       | A      | 1     |
| 38       | D      | 1     |
| 39       | D      | 1     |
| 40       | A      | 1     |



**Cambridge International Examinations**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

Paper 1 Multiple Choice (Core)

**0625/13**

**October/November 2018**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)



**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

**DO NOT WRITE IN ANY BARCODES.**

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall =  $10 \text{ m/s}^2$ ).

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This document consists of **16** printed pages.



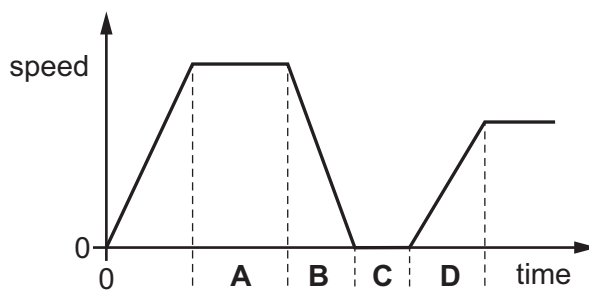
- 1 A pendulum is set in motion and timed. The time measured for 20 complete swings is 30 s.  
What is the time for one complete swing of the pendulum?

**A** 0.67 s                      **B** 0.75 s                      **C** 1.5 s                      **D** 3.0 s

- 2 A girl goes for a ride on her bicycle.

The diagram shows how her speed changes with time for part of her journey.

In which labelled section is she moving with constant speed?



- 3 A boy runs 400 m at an average speed of 4.0 m/s.

He runs the first 200 m in 40 s.

How long does he take to run the second 200 m?

**A** 60 s                      **B** 66.7 s                      **C** 80 s                      **D** 140 s

- 4 Which quantity is measured in newtons?

**A** energy  
**B** mass  
**C** weight  
**D** work

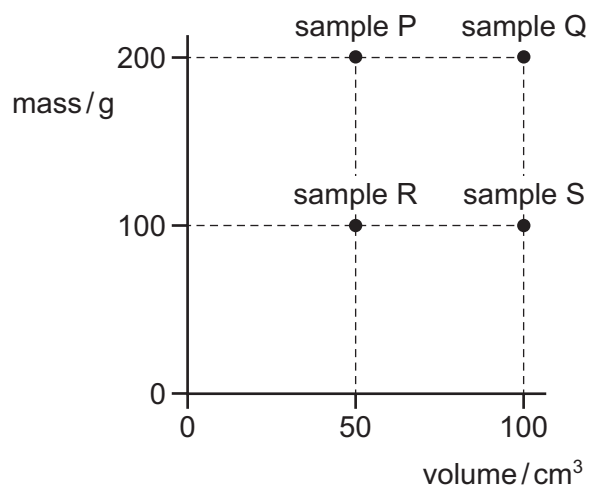
- 5 The acceleration of free fall on Mars is  $3.7 \text{ m/s}^2$ .

What is the weight of a 100 g mass on Mars?

**A** 0.037 N                      **B** 0.37 N                      **C** 37 N                      **D** 370 N

- 6 A student carries out experiments to find the mass and the volume of four samples of rock.

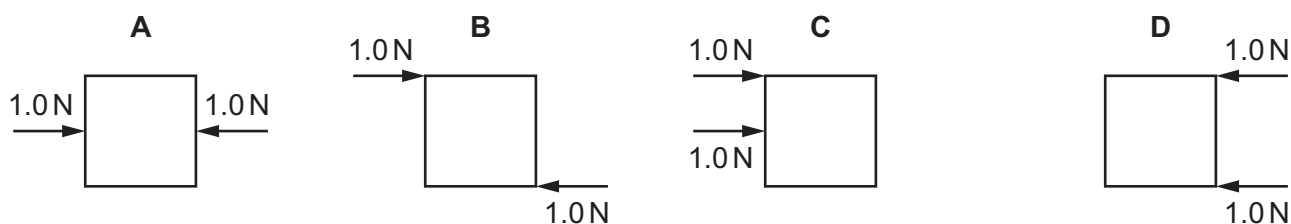
The graph shows the results.



Which pair are samples of the same type of rock?

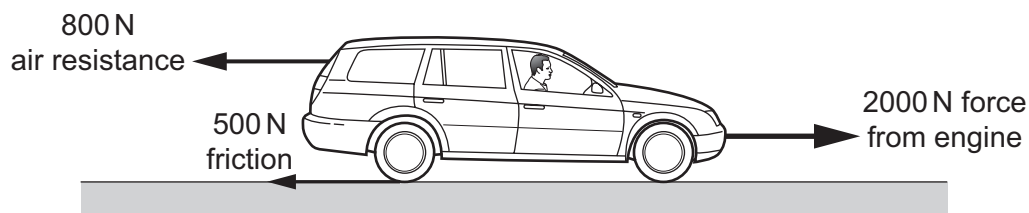
- A** P and Q      **B** P and S      **C** R and Q      **D** Q and S
- 7 Four objects each have two forces acting on them.

Which object is in equilibrium?



- 8 A car moves along a level road.

The diagram shows all of the horizontal forces acting on the car.

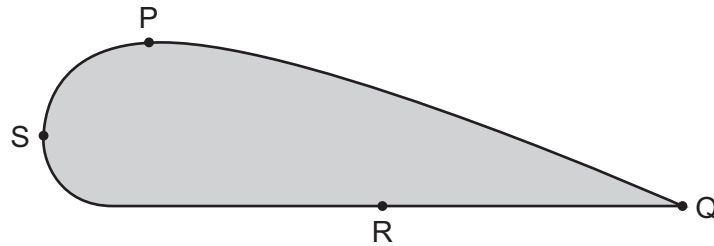


Which statement is correct?

- A** The car is slowing down.  
**B** The car is speeding up.  
**C** The car is moving at a constant speed.  
**D** The car is moving backwards.

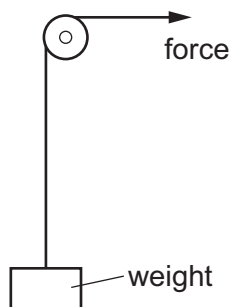
- 9 The diagram shows an aluminium sheet of constant thickness.

Four significant points on the aluminium sheet are labelled P, Q, R and S.



How can the centre of mass be found?

- A by finding where a line from P to R crosses a line from S to Q
  - B by suspending the sheet and a plumb-line from Q and finding where the plumb-line crosses a line from P to R
  - C by suspending the sheet and a plumb-line from P and then from S and finding where the plumb-lines cross
  - D by suspending the sheet and a plumb-line from P and finding where the plumb-line crosses a line from S to Q
- 10 Which device is designed to convert chemical energy into kinetic energy?
- A an a.c. generator
  - B a battery-powered torch
  - C a car engine
  - D a wind-up mechanical clock
- 11 A weight hangs from a rope.



In which case does the force do the most work on the weight?

- A The force raises the weight 1.0 m in 15 s.
- B The force raises the weight 2.0 m in 45 s.
- C The weight is held at the same level for 10 s.
- D The weight is held at the same level for 60 s.

- 12 Diagram 1 shows a conical flask containing water.

Diagram 2 shows a beaker with the same base area as the flask and containing water of the same depth  $d$ .

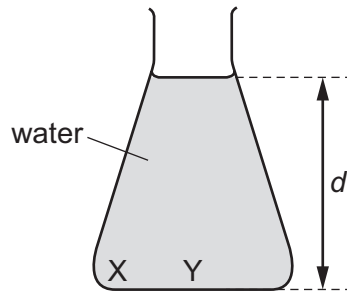


diagram 1

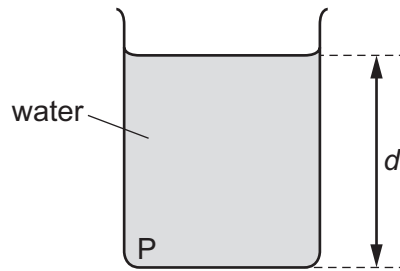
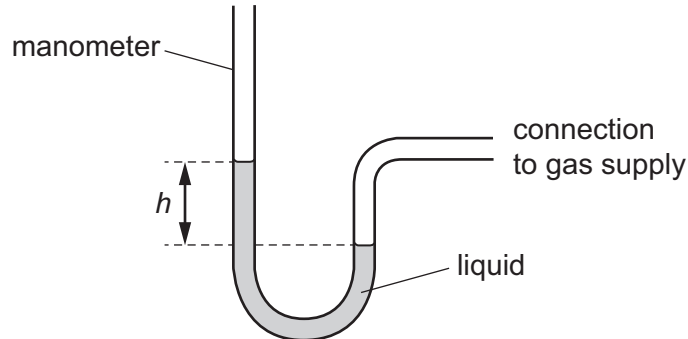


diagram 2

Which statement is correct?

- A The pressure at P is equal to the pressure at X.
  - B The pressure at P is greater than the pressure at Y.
  - C The pressure at P is greater than the pressure at X.
  - D The pressure at Y is greater than the pressure at X.
- 13 A manometer is used to measure the pressure of a gas supply.



Which change gives a greater value of height  $h$ ?

- A using a less dense liquid
- B using a more dense liquid
- C using a narrower tube
- D using a wider tube

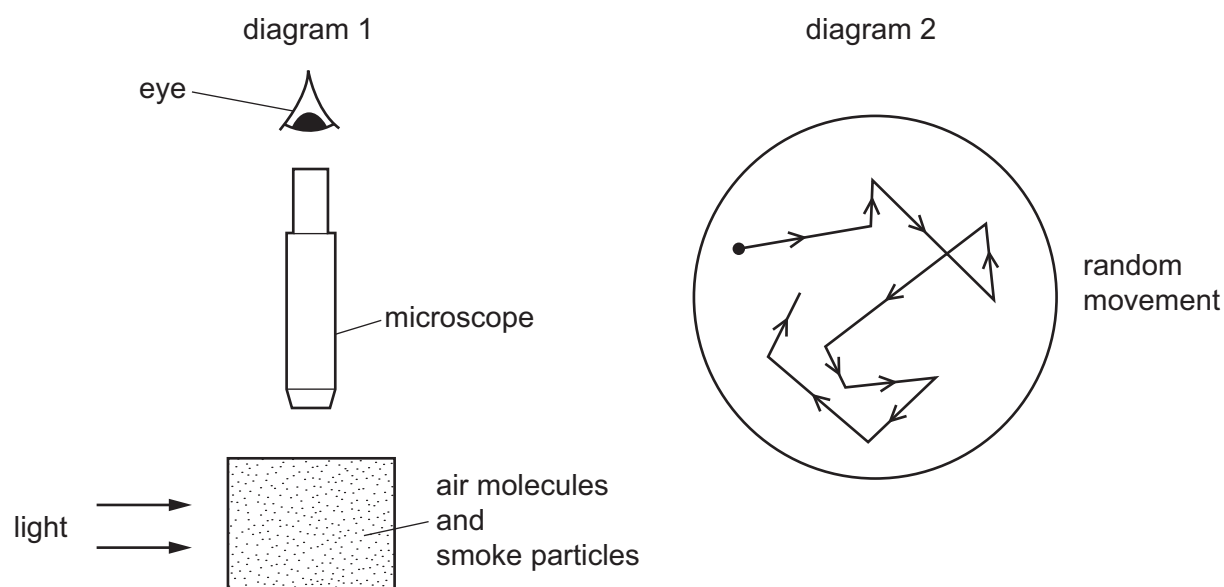
- 14** Liquid in a dish evaporates. This causes the temperature of the remaining liquid to change.

Which row is correct?

|          | change in temperature | reason for temperature change                       |
|----------|-----------------------|-----------------------------------------------------|
| <b>A</b> | increase              | the most energetic molecules remain in the liquid   |
| <b>B</b> | increase              | the most energetic molecules escape from the liquid |
| <b>C</b> | decrease              | the most energetic molecules remain in the liquid   |
| <b>D</b> | decrease              | the most energetic molecules escape from the liquid |

- 15** Diagram 1 shows apparatus being used to observe smoke particles.

Diagram 2 shows how a smoke particle moves randomly.



Why do the smoke particles move randomly?

- A** They are hit by air molecules.
- B** They are less dense than air.
- C** They are moved by convection currents.
- D** They gain energy from the light.



16 Which row shows the relative order of thermal expansion of solids, liquids and gases?

|          | most expansion | → | least expansion |
|----------|----------------|---|-----------------|
| <b>A</b> | solids         |   | liquids gases   |
| <b>B</b> | solids         |   | gases liquids   |
| <b>C</b> | gases          |   | solids liquids  |
| <b>D</b> | gases          |   | liquids solids  |

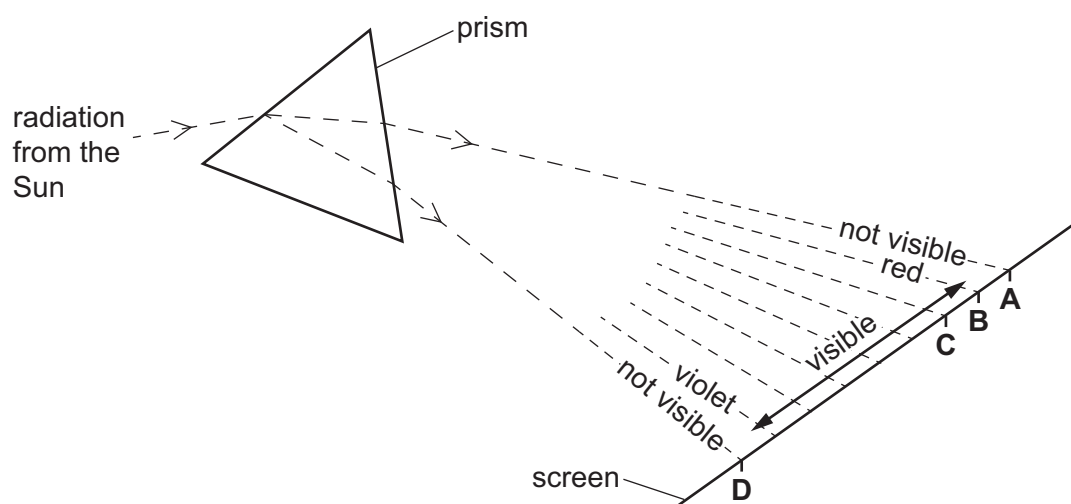
17 Steam at 100 °C condenses to form water at 100 °C.

What happens to the average separation and what happens to the average speed of the water molecules?

|          | separation | average speed    |
|----------|------------|------------------|
| <b>A</b> | decreases  | decreases        |
| <b>B</b> | decreases  | remains the same |
| <b>C</b> | increases  | decreases        |
| <b>D</b> | increases  | remains the same |

18 Some of the Sun's radiation passes through a prism. The diagram shows the spectrum of the radiation.

Which point on the screen does the infra-red radiation reach?



**19** One end of a shiny metal rod is heated and the other end quickly gets hot.

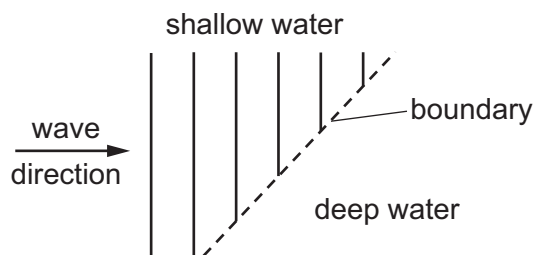
Which statement describes why the other end quickly gets hot?

- A** Metals are good thermal conductors.
- B** Metals are poor thermal conductors.
- C** Shiny surfaces are good emitters of infra-red radiation.
- D** Shiny surfaces are poor emitters of infra-red radiation.

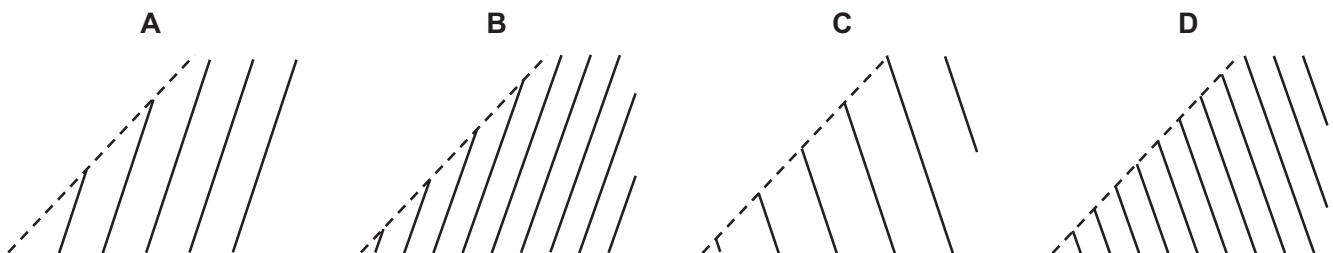
**20** Which statement about waves is correct?

- A** Waves do not transfer either energy or matter.
- B** Waves transfer both energy and matter.
- C** Waves transfer energy without transferring matter.
- D** Waves transfer matter without transferring energy.

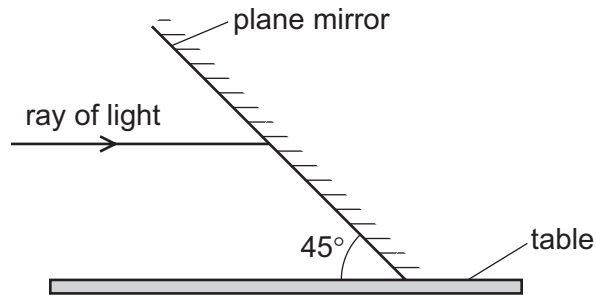
**21** Plane water waves travel from a shallow region into a deeper region. They travel more quickly in the deeper water.



Which diagram shows the wave pattern in the deeper water?

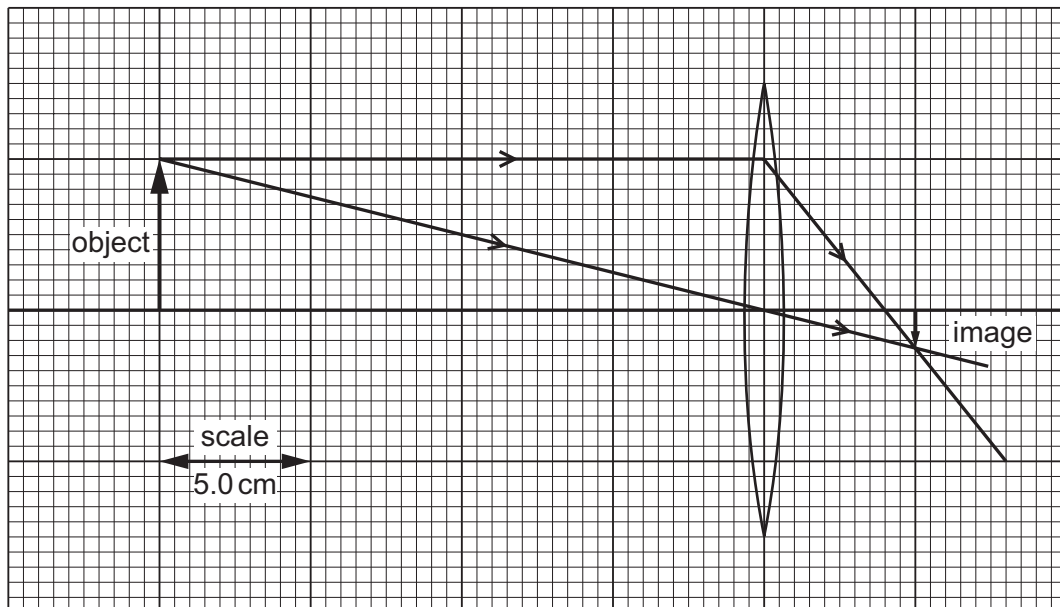


- 22** A plane mirror is fixed so that it is at an angle of  $45^\circ$  to a table. A ray of light, initially parallel to the table, is incident on the mirror.



Which angle does the reflected ray make with the table?

- A**  $0^\circ$                       **B**  $22.5^\circ$                       **C**  $45^\circ$                       **D**  $90^\circ$
- 23** An object is placed 20 cm in front of a thin converging lens. The scale drawing shows how the lens forms a real, inverted image.



Which row gives the focal length of the lens and the distance of the image from the lens?

|          | focal length<br>of lens / cm | distance of image<br>from lens / cm |
|----------|------------------------------|-------------------------------------|
| <b>A</b> | 4.0                          | 5.0                                 |
| <b>B</b> | 5.0                          | 4.0                                 |
| <b>C</b> | 8.0                          | 10.0                                |
| <b>D</b> | 10.0                         | 8.0                                 |

- 24** Different parts of the electromagnetic spectrum are used for different purposes. Below are four statements about parts of the spectrum.

statement 1: Infra-red waves are used in television remote controllers.

statement 2: Radio waves are used to transmit television pictures from satellites to Earth.

statement 3: Ultraviolet waves are used for intruder alarms.

statement 4: X-rays are used for security checks.

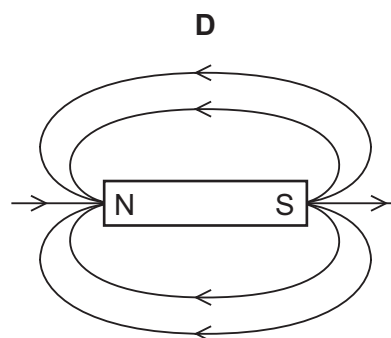
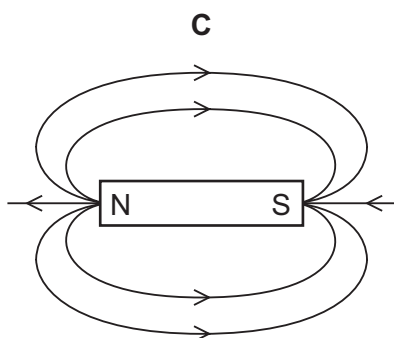
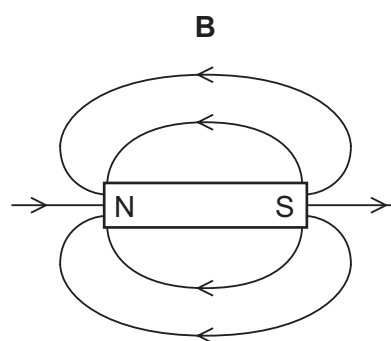
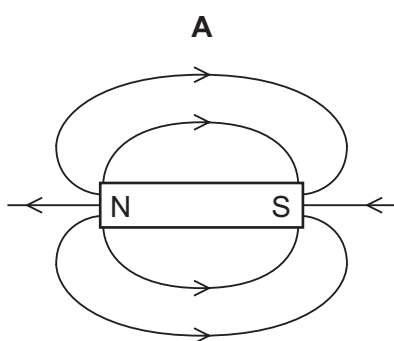
Which statements are correct?

- A** 1 and 2      **B** 1 and 4      **C** 2 and 3      **D** 3 and 4

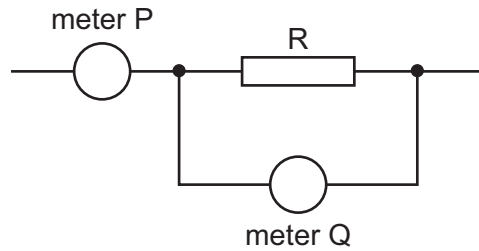
- 25** Which row gives the nature of sound waves and the name of the effect that causes an echo of a sound?

|          | nature of sound waves | effect causing an echo |
|----------|-----------------------|------------------------|
| <b>A</b> | longitudinal          | reflection             |
| <b>B</b> | longitudinal          | refraction             |
| <b>C</b> | transverse            | reflection             |
| <b>D</b> | transverse            | refraction             |

- 26** Which diagram shows the pattern and the direction of the magnetic field lines around a bar magnet?

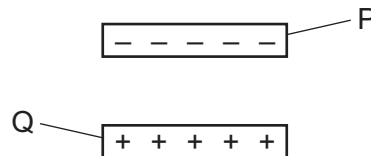


- 27 The diagram shows part of a circuit which contains an ammeter, a voltmeter and a resistor R. The ammeter and the voltmeter are in the correct positions to determine the resistance of R.



Which expression gives the value of R?

- A** reading of meter P  $\div$  reading of meter Q  
**B** (reading of meter P)<sup>2</sup>  $\times$  reading of meter Q  
**C** reading of meter Q  $\times$  reading of meter P  
**D** reading of meter Q  $\div$  reading of meter P
- 28 A negatively charged plastic rod P is placed above a positively charged plastic rod Q.



What are the directions of the electrostatic forces on rod P and on rod Q?

|          | electrostatic force<br>on rod P | electrostatic force<br>on rod Q |
|----------|---------------------------------|---------------------------------|
| <b>A</b> | downwards                       | downwards                       |
| <b>B</b> | downwards                       | upwards                         |
| <b>C</b> | upwards                         | downwards                       |
| <b>D</b> | upwards                         | upwards                         |

- 29 Which row shows the meaning of the quantity e.m.f. and the unit in which it is measured?

|          | meaning               | unit |
|----------|-----------------------|------|
| <b>A</b> | electromagnetic force | N    |
| <b>B</b> | electromagnetic force | V    |
| <b>C</b> | electromotive force   | N    |
| <b>D</b> | electromotive force   | V    |

- 30** Diagram 1 shows a voltmeter that is not connected to anything. It shows a small reading, called a zero error, because it is not correctly adjusted.

Diagram 2 shows the same meter used to measure the p.d. across a resistor in a circuit.

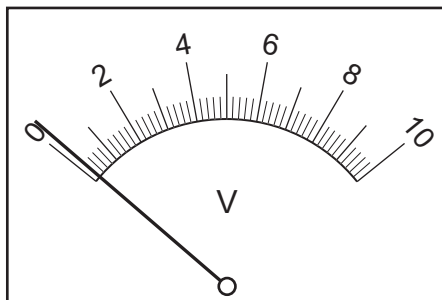


diagram 1

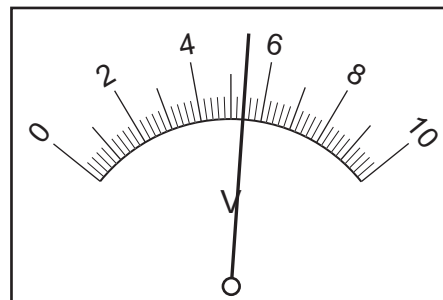
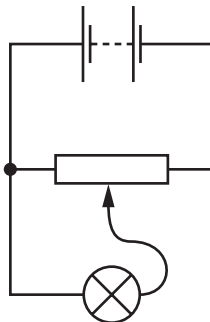


diagram 2

What is the p.d. across the resistor?

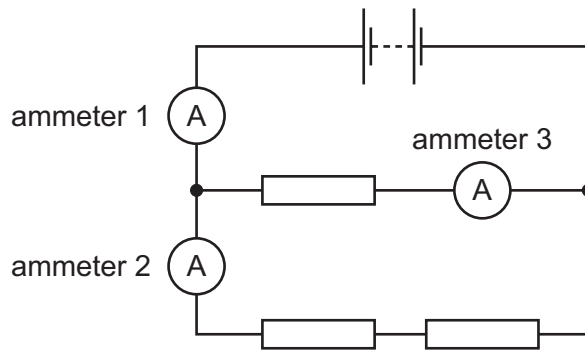
- A** 4.6 V      **B** 4.7 V      **C** 5.2 V      **D** 5.4 V
- 31** The diagram shows a potential divider used to control the brightness of a lamp.



What happens as the slider on the potential divider is moved to the right?

- A** The potential difference across the lamp decreases and the lamp gets brighter.  
**B** The potential difference across the lamp decreases and the lamp gets dimmer.  
**C** The potential difference across the lamp increases and the lamp gets brighter.  
**D** The potential difference across the lamp increases and the lamp gets dimmer.

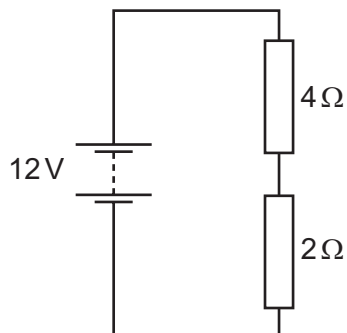
32 The diagram shows three identical resistors connected to a battery.



Which row gives the possible readings on the three ammeters?

|          | ammeter 1 / A | ammeter 2 / A | ammeter 3 / A |
|----------|---------------|---------------|---------------|
| <b>A</b> | 1.5           | 3.0           | 4.5           |
| <b>B</b> | 3.0           | 1.5           | 4.5           |
| <b>C</b> | 3.0           | 4.5           | 1.5           |
| <b>D</b> | 4.5           | 1.5           | 3.0           |

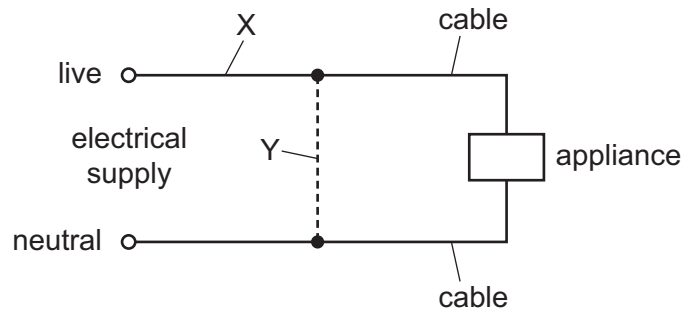
33 A  $4\ \Omega$  resistor and a  $2\ \Omega$  resistor are connected to a 12 V battery in a circuit.



What is the current in the  $2\ \Omega$  resistor?

- A** 0.5 A      **B** 2.0 A      **C** 3.0 A      **D** 6.0 A

- 34** Either a fuse or a circuit-breaker can be used to protect electrical cables from large currents that could cause overheating.



When a fuse is used, where should it be connected, and when a circuit-breaker is used, where should it be connected?

|          | position of fuse | position of circuit-breaker |
|----------|------------------|-----------------------------|
| <b>A</b> | X                | X                           |
| <b>B</b> | X                | Y                           |
| <b>C</b> | Y                | X                           |
| <b>D</b> | Y                | Y                           |

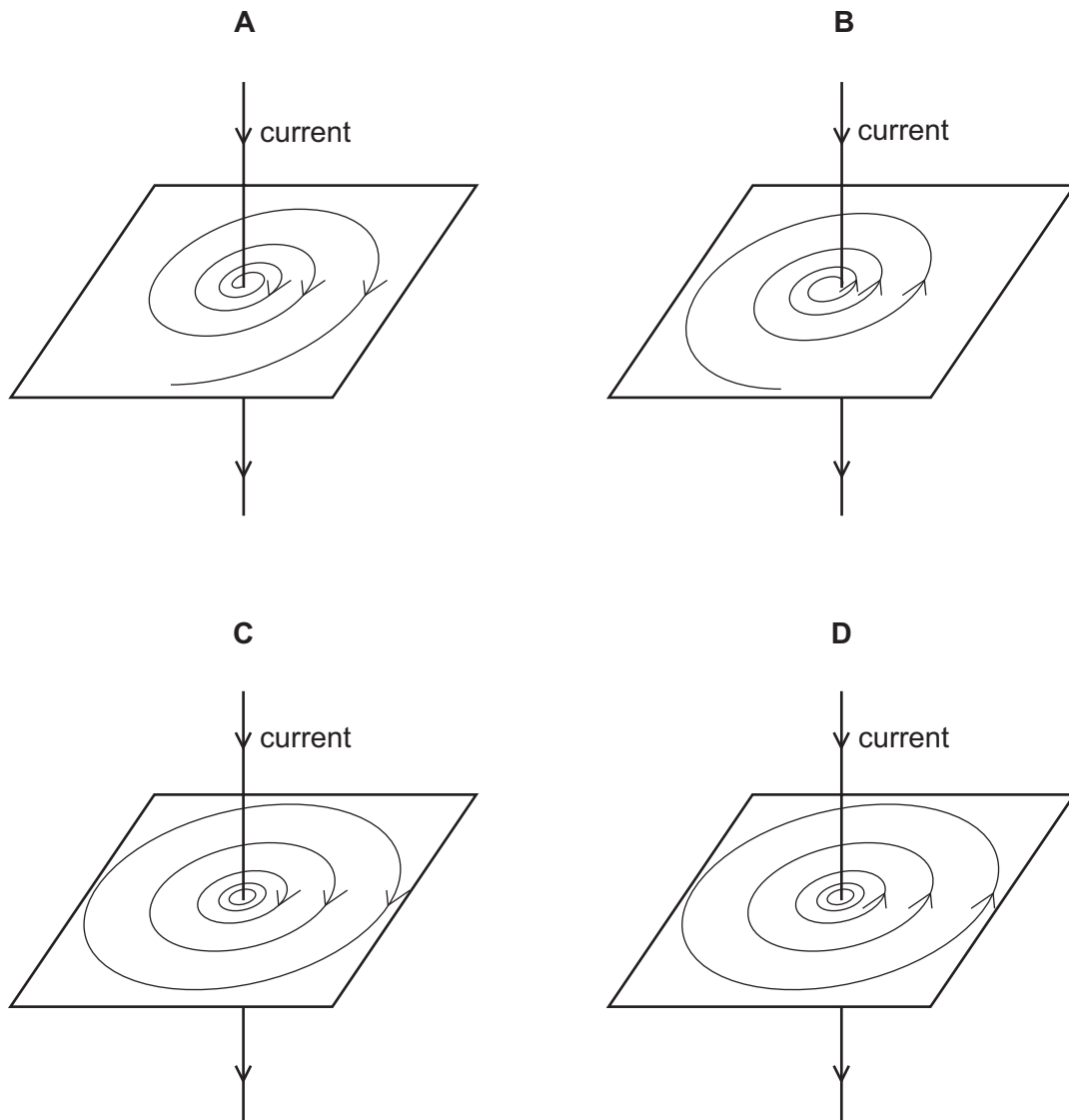
- 35** There is an electric current in a wire. The wire is placed in a magnetic field. A force acts on the wire due to the current.

Which statement is correct?

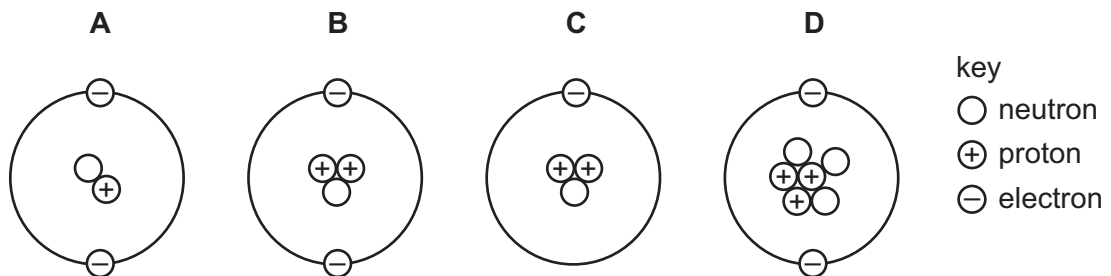
- A** The magnetic field must be produced by a permanent magnet and not by an electromagnet.
- B** The wire must be made from a magnetic material.
- C** When both the current and the magnetic field are reversed, the direction of the force is unchanged.
- D** When the current is reversed, but not the magnetic field, there will be no force on the wire.



- 36 Which diagram shows the pattern and direction of the magnetic field due to a current in a straight wire?



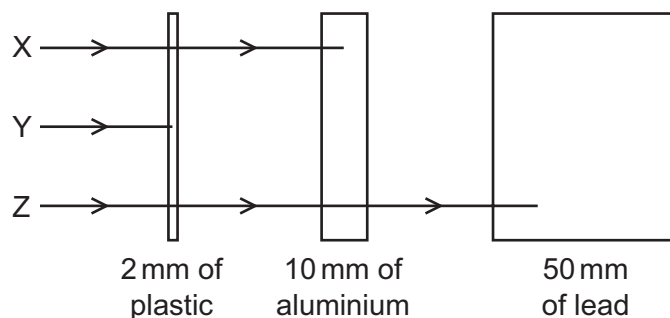
- 37 Which diagram represents the structure of a neutral atom?



38 What is a carbon nucleus,  $^{12}_6\text{C}$ , composed of?

- A 6 neutrons and 6 electrons
- B 6 neutrons, 6 protons and 6 electrons
- C 6 protons and 6 electrons
- D 6 protons and 6 neutrons

39 The diagram shows the paths of three different types of radiation X, Y and Z.



Which row correctly identifies X, Y and Z?

|   | X                   | Y                   | Z                   |
|---|---------------------|---------------------|---------------------|
| A | $\alpha$ -particles | $\beta$ -particles  | $\gamma$ -rays      |
| B | $\beta$ -particles  | $\alpha$ -particles | $\gamma$ -rays      |
| C | $\beta$ -particles  | $\gamma$ -rays      | $\alpha$ -particles |
| D | $\gamma$ -rays      | $\alpha$ -particles | $\beta$ -particles  |

40 The count rate due to a sample of a radioactive isotope is measured for 80 minutes.

| time<br>/ minutes | count rate<br>counts/second |
|-------------------|-----------------------------|
| 0                 | 480                         |
| 20                | 380                         |
| 40                | 300                         |
| 60                | 240                         |
| 80                | 190                         |

What is the half-life of the isotope?

- A 20 minutes
- B 40 minutes
- C 60 minutes
- D 80 minutes



**Cambridge International Examinations**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

Paper 1 Multiple Choice (Core)

**0625/12**

**October/November 2018**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)



**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

**DO NOT WRITE IN ANY BARCODES.**

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall =  $10 \text{ m/s}^2$ ).

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **16** printed pages.



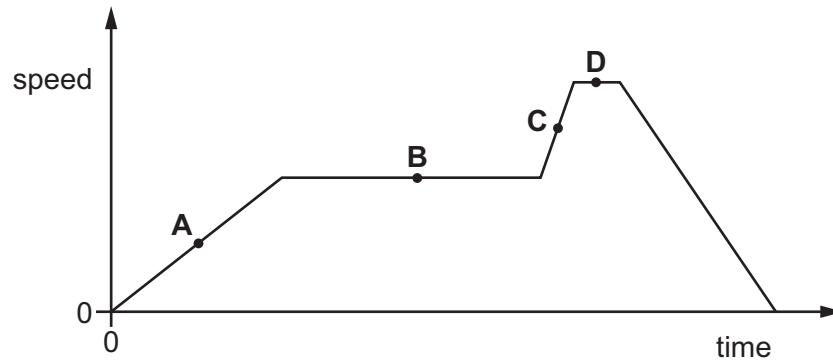
- 1 A pendulum is set in motion and timed. The time measured for 20 complete swings is 30 s.

What is the time for one complete swing of the pendulum?

- A 0.67 s      B 0.75 s      C 1.5 s      D 3.0 s

- 2 The graph shows how the speed of an object varies with time.

At which point on the graph is the greatest distance travelled per second?



- 3 An athlete runs at a speed of 8 m/s for 10 s, and then at a speed of 6 m/s for 12 s.

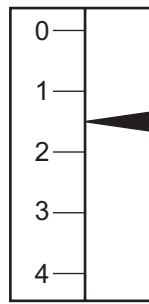
Which calculation gives the average speed of the athlete in m/s?

- A  $\frac{8+6}{2}$   
 B  $\frac{(8 \times 10) + (6 \times 12)}{22}$   
 C  $\frac{(8 \div 10) + (6 \div 12)}{22}$   
 D  $\frac{(10 \div 8) + (12 \div 6)}{22}$

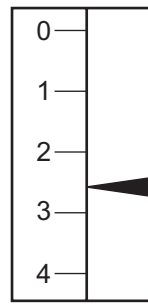
- 4 Which quantity is measured in newtons?

- A energy  
 B mass  
 C weight  
 D work

- 5 An object is suspended from a spring balance on the Earth. The same object is suspended from the same spring balance on another planet.



reading on  
Earth

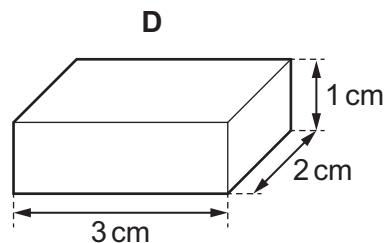
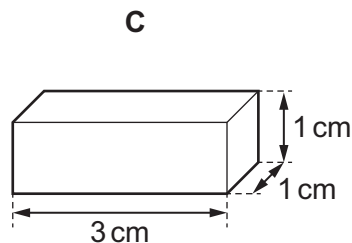
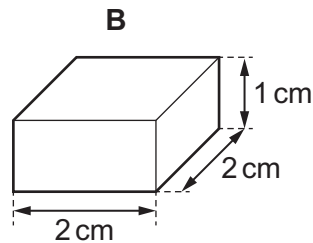
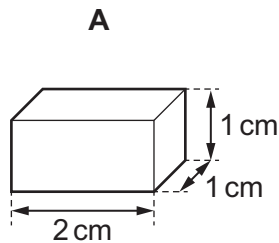


reading on  
another planet

Which statement explains the difference between the two readings?

- A Both the mass and the weight of the object are greater on the other planet.
  - B The mass of the object is greater on the other planet than on Earth, but the weight is unchanged.
  - C The spring stretches more easily when on the other planet.
  - D The weight of the object is greater on the other planet than on Earth, but the mass is unchanged.
- 6 The diagram shows four blocks of different metals. Each block has a mass of 12g.

Which metal has the largest density?

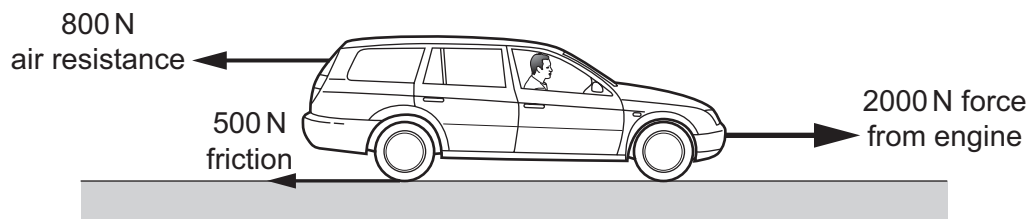


7 Which is an example of a force?

- A density
- B mass
- C volume
- D weight

8 A car moves along a level road.

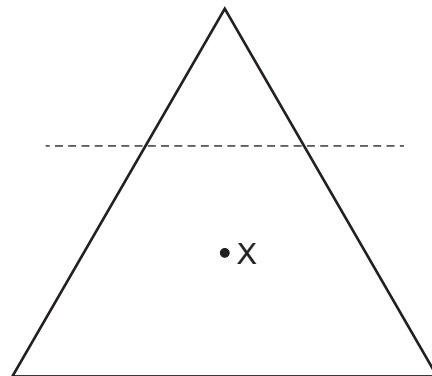
The diagram shows all of the horizontal forces acting on the car.



Which statement is correct?

- A The car is slowing down.
- B The car is speeding up.
- C The car is moving at a constant speed.
- D The car is moving backwards.

9 Point X is the centre of mass of a lamina in the shape of a triangle with sides of equal length. The top of the triangle is cut off along the dotted line shown.

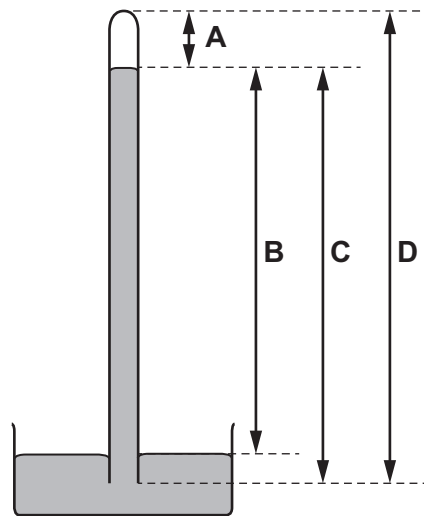


What happens to the centre of mass, X?

- A moves towards the bottom of the page
- B moves to the left
- C moves to the right
- D moves towards the top of the page

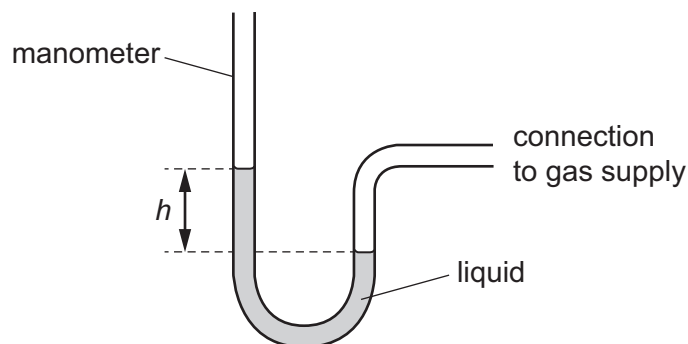
- 10 Which device is designed to convert chemical energy into kinetic energy?
- A an a.c. generator
  - B a battery-powered torch
  - C a car engine
  - D a wind-up mechanical clock
- 11 Which two physical quantities must be used to calculate the power developed by a student running up a flight of steps?
- A force exerted and the vertical height of the steps only
  - B force exerted and the time taken only
  - C work done and the vertical height of the steps only
  - D work done and the time taken only
- 12 A simple mercury barometer is used to measure the atmospheric pressure.

Which height is measured to find the atmospheric pressure?



- 13 The particles of a gas, in a container of fixed volume, are given more energy.
- Which effect does this have on the gas?
- A Both the pressure and temperature of the gas increase.
  - B Only the temperature of the gas increases.
  - C Neither the pressure nor temperature of the gas increases.
  - D Only the pressure of the gas increases.

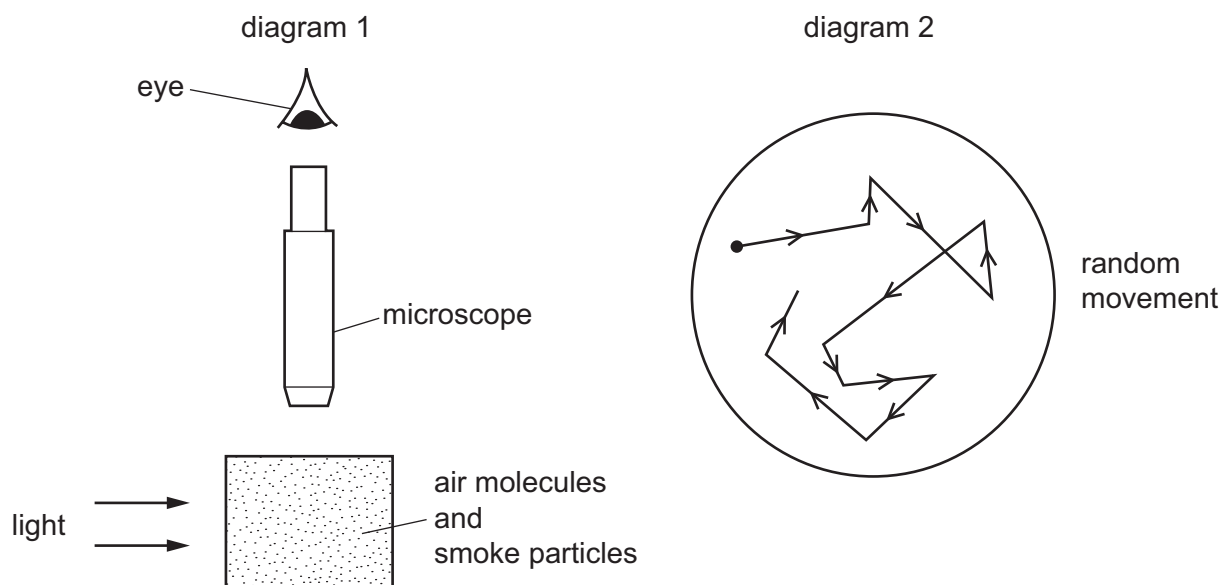
- 14 A manometer is used to measure the pressure of a gas supply.



Which change gives a greater value of height  $h$ ?

- A using a less dense liquid
  - B using a more dense liquid
  - C using a narrower tube
  - D using a wider tube
- 15 Diagram 1 shows apparatus being used to observe smoke particles.

Diagram 2 shows how a smoke particle moves randomly.



Why do the smoke particles move randomly?

- A They are hit by air molecules.
- B They are less dense than air.
- C They are moved by convection currents.
- D They gain energy from the light.



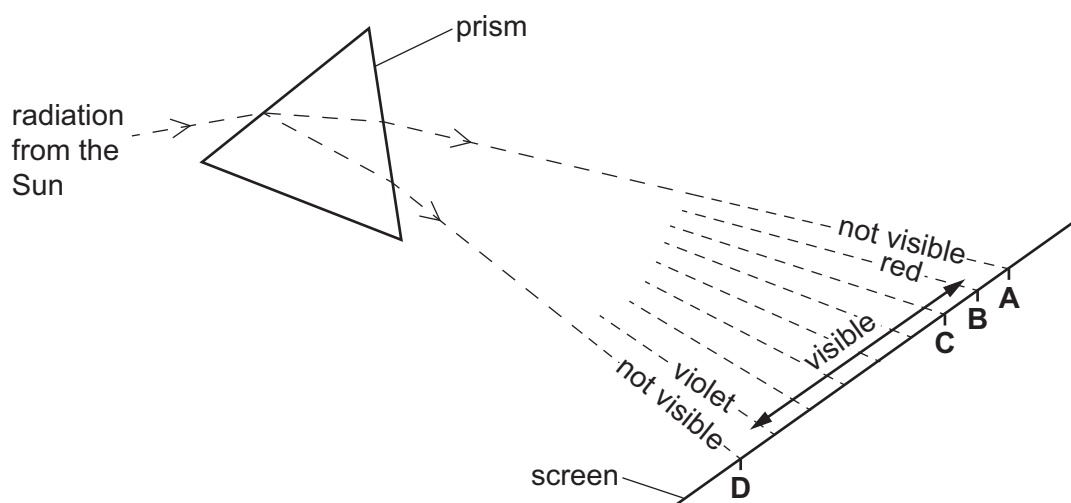
- 16** The metal lid on a glass jar is difficult to unscrew.

The jar is placed in a warm oven until the jar and the lid reach the same temperature. The lid is now easily unscrewed.

Which property accounts for this?

- A** thermal capacity of the jar
  - B** thermal capacity of the lid
  - C** thermal conduction
  - D** thermal expansion
- 17** Two cold metal containers X and Y are both filled with the same mass of hot water at the same temperature.
- Both containers are well insulated.
- The thermal capacity of container X is greater than that of container Y.
- How do the final temperatures of the water in X and in Y compare?
- A** The temperature in X is higher than in Y.
  - B** The temperature in X is lower than in Y.
  - C** The temperatures in X and in Y are equal but lower than the initial temperature of the water.
  - D** The temperatures in X and in Y are the same as the initial temperature of the water.
- 18** Some of the Sun's radiation passes through a prism. The diagram shows the spectrum of the radiation.

Which point on the screen does the infra-red radiation reach?



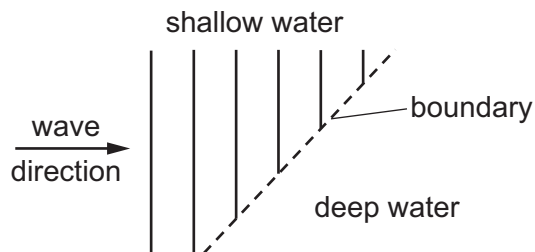
19 Which statement about convection is **not** correct?

- A It enables water in a pan on a cooker to get evenly heated.
- B It happens in liquids and gases.
- C It means that heat rises.
- D It occurs because the density of a fluid decreases when it is heated.

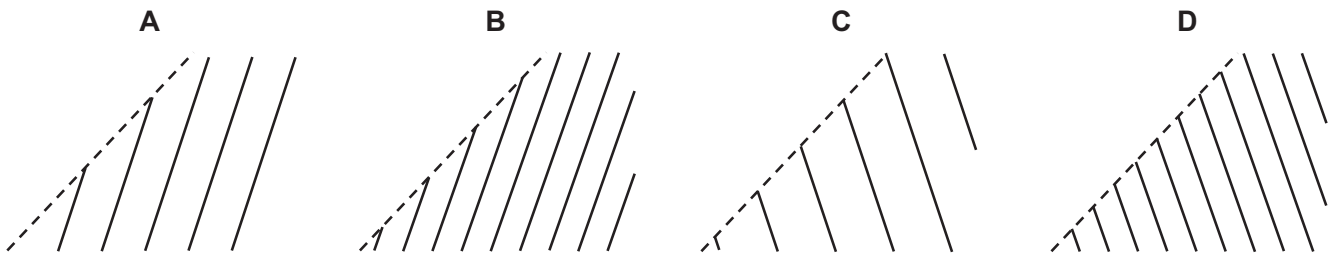
20 What is meant by the frequency of a water wave?

- A the distance between consecutive crests of the wave
- B the distance moved by the wave per unit time
- C the maximum vertical displacement of the wave
- D the number of crests passing a point per unit time

21 Plane water waves travel from a shallow region into a deeper region. They travel more quickly in the deeper water.

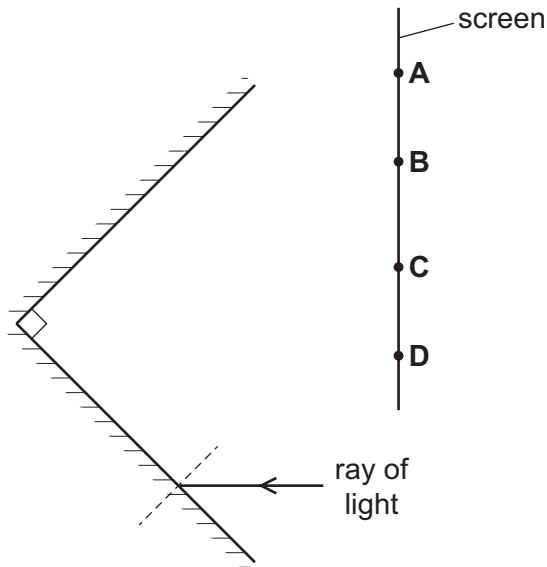


Which diagram shows the wave pattern in the deeper water?

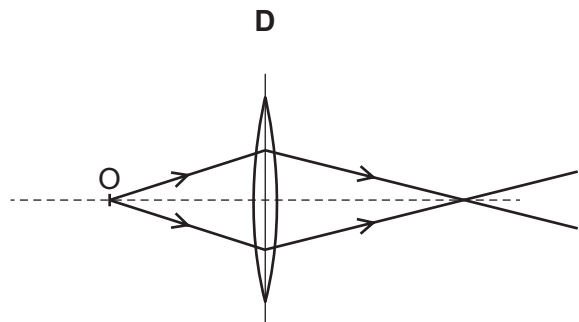
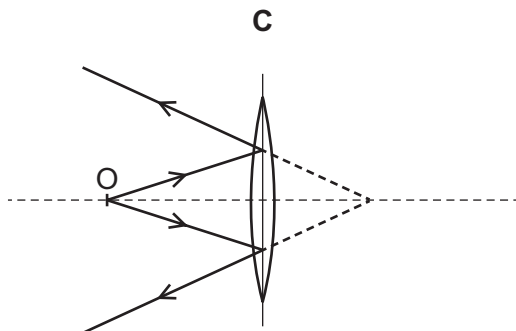
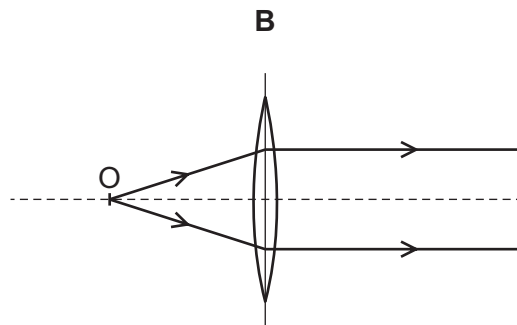
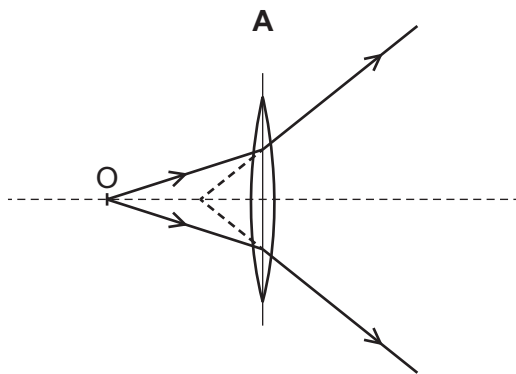


- 22** The diagram shows two plane mirrors at  $90^\circ$  to each other. A ray of light is incident on one of the mirrors. The ray reflects off both mirrors before reaching a screen.

At which labelled point does the ray reach the screen?



- 23** Which diagram shows the formation of a real image of an object O placed in front of a converging lens?



- 24** Different parts of the electromagnetic spectrum are used for different purposes. Below are four statements about parts of the spectrum.

statement 1: Infra-red waves are used in television remote controllers.

statement 2: Radio waves are used to transmit television pictures from satellites to Earth.

statement 3: Ultraviolet waves are used for intruder alarms.

statement 4: X-rays are used for security checks.

Which statements are correct?

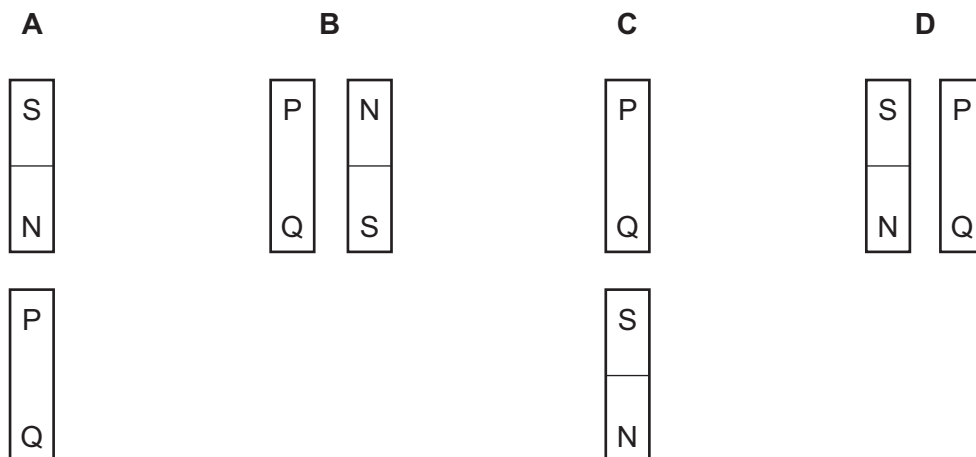
- A** 1 and 2      **B** 1 and 4      **C** 2 and 3      **D** 3 and 4

- 25** Which statement about ultrasound is correct?

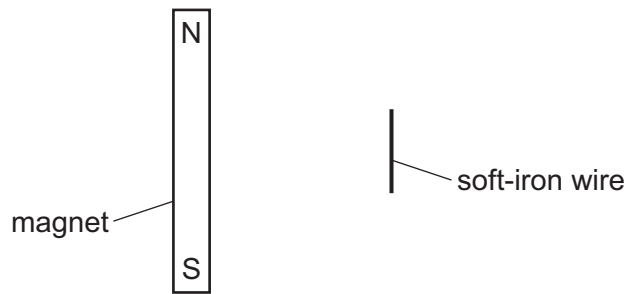
- A** Ultrasound must have greater amplitude than audible sound.  
**B** Ultrasound must have greater frequency than audible sound.  
**C** Ultrasound must have lower amplitude than audible sound.  
**D** Ultrasound must have lower frequency than audible sound.

- 26** An iron bar PQ is placed close to a bar magnet.

In which diagram does end P become an induced north pole?



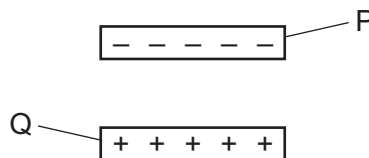
- 27 The diagram shows a bar magnet at rest on a smooth horizontal surface. A length of soft-iron wire is held parallel to the magnet.



The wire is released.

What happens?

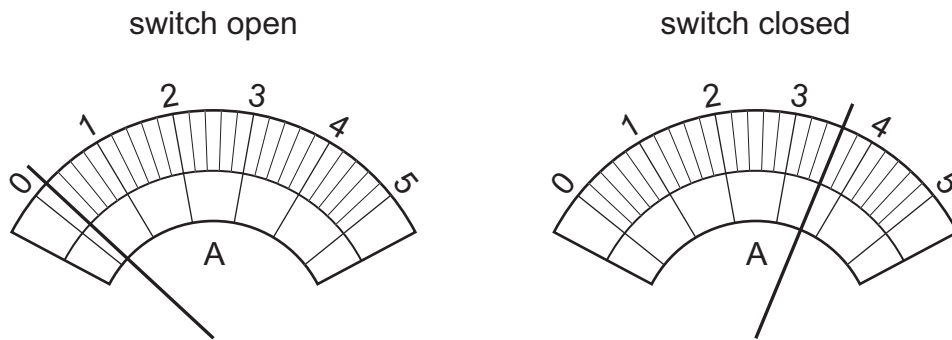
- A The wire moves away from the magnet.
  - B The wire moves towards the magnet.
  - C The wire's centre stays in its present position and the wire rotates through  $90^\circ$  in a clockwise direction.
  - D The wire's centre stays in its present position and the wire rotates through  $90^\circ$  in an anticlockwise direction.
- 28 A negatively charged plastic rod P is placed above a positively charged plastic rod Q.



What are the directions of the electrostatic forces on rod P and on rod Q?

|          | electrostatic force<br>on rod P | electrostatic force<br>on rod Q |
|----------|---------------------------------|---------------------------------|
| <b>A</b> | downwards                       | downwards                       |
| <b>B</b> | downwards                       | upwards                         |
| <b>C</b> | upwards                         | downwards                       |
| <b>D</b> | upwards                         | upwards                         |

- 29 The diagrams show the readings on an ammeter in a series circuit before and after the switch in the circuit has been closed.



What is the current in the circuit when the switch is closed?

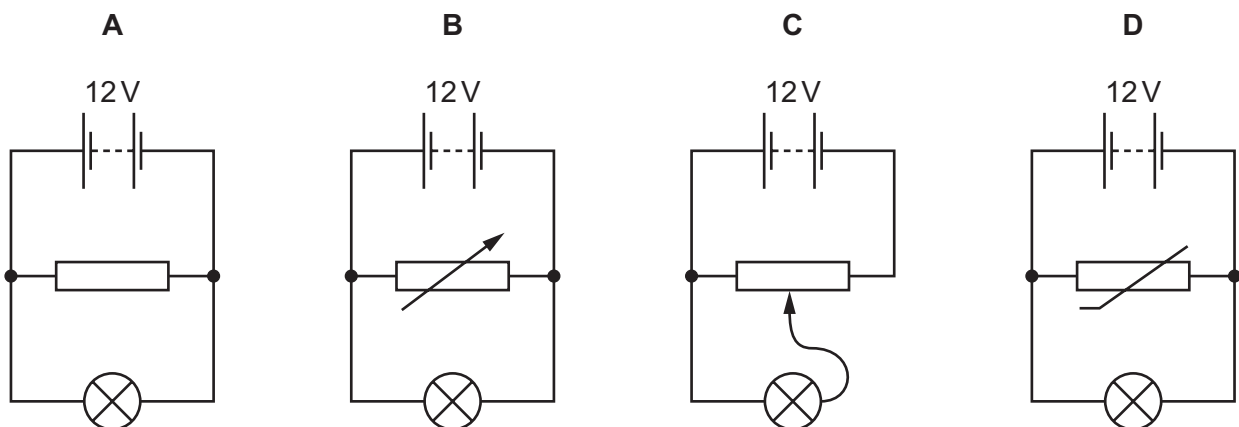
- A** 3.2 A                      **B** 3.4 A                      **C** 3.6 A                      **D** 3.8 A
- 30 A student investigates the resistance of a lamp.

Which row states how the meters must be connected?

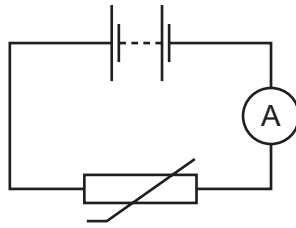
|          | ammeter                   | voltmeter                 |
|----------|---------------------------|---------------------------|
| <b>A</b> | in parallel with the lamp | in parallel with the lamp |
| <b>B</b> | in parallel with the lamp | in series with the lamp   |
| <b>C</b> | in series with the lamp   | in parallel with the lamp |
| <b>D</b> | in series with the lamp   | in series with the lamp   |

- 31 A lamp is connected into a circuit so that the potential difference across it can be varied from 0 to 12 V.

Which circuit is suitable?



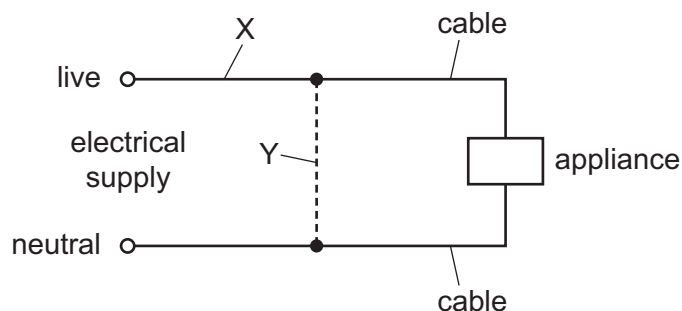
32 The diagram shows a circuit.



Which statement describes and explains how the circuit behaves?

- A** The ammeter reading decreases when the light intensity decreases.
  - B** The ammeter reading decreases when the temperature decreases.
  - C** The ammeter reading increases when the light intensity decreases.
  - D** The ammeter reading increases when the temperature decreases.
- 33 Which component uses a small direct current (d.c.) in one circuit to switch on a much larger current in a second circuit?
- A** potential divider
  - B** relay
  - C** transformer
  - D** variable resistor

- 34 Either a fuse or a circuit-breaker can be used to protect electrical cables from large currents that could cause overheating.

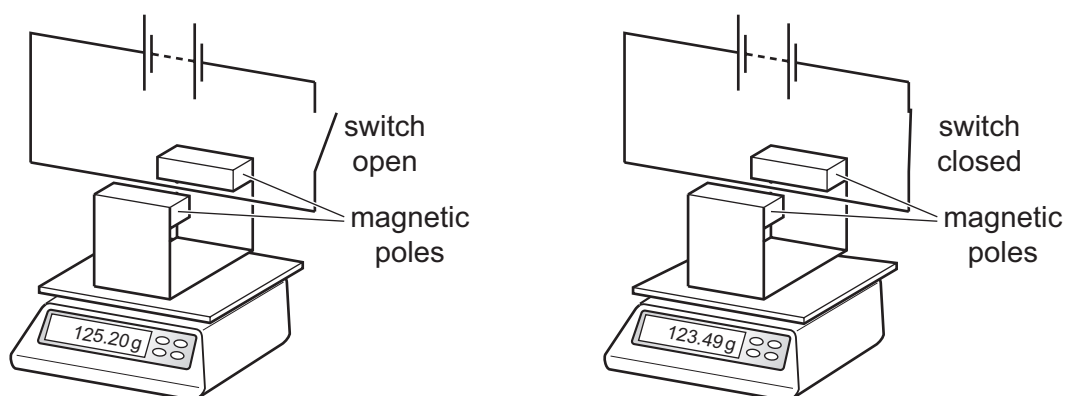


When a fuse is used, where should it be connected, and when a circuit-breaker is used, where should it be connected?

|          | position of fuse | position of circuit-breaker |
|----------|------------------|-----------------------------|
| <b>A</b> | X                | X                           |
| <b>B</b> | X                | Y                           |
| <b>C</b> | Y                | X                           |
| <b>D</b> | Y                | Y                           |

- 35 The diagram shows a horizontal wire between the two magnetic poles of a U-shaped magnet. The U-shaped magnet is on a balance.

When the switch closed, the reading on the balance decreases.



The experiment is carried out three more times with the following changes made.

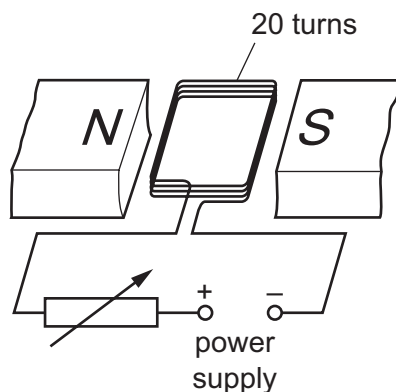
- 1 Only the current is reversed.
- 2 Only the magnetic field is reversed.
- 3 Both the current and the magnetic field are reversed at the same time.

Which changes cause an increase in the reading on the balance?

- A** 1 only      **B** 2 only      **C** 3 only      **D** 1 and 2 only



- 36 The diagram shows a coil of wire between the poles of a magnet.

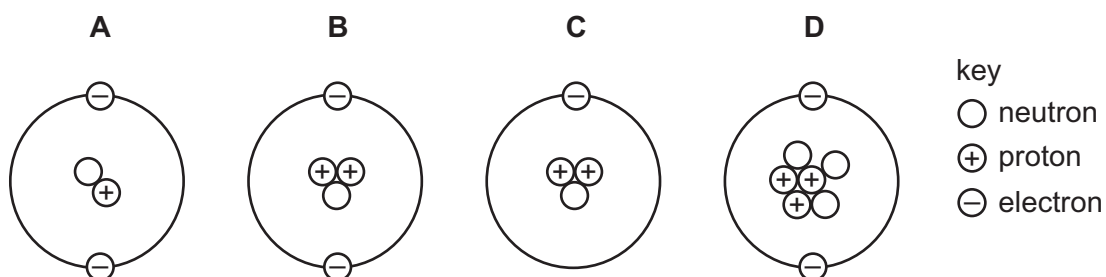


The coil consists of 20 turns of insulated wire.

The coil is connected to a variable resistor and a power supply.

How can the turning effect on the coil be increased?

- A** by moving the poles of the magnet closer to the coil
  - B** by reducing the number of turns on the coil while keeping the current constant
  - C** by increasing the resistance of the variable resistor
  - D** by reversing the terminals of the power supply
- 37 Which diagram represents the structure of a neutral atom?

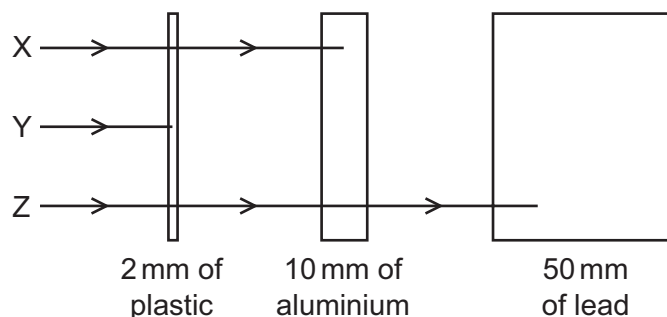


- 38 The nuclide notation for the isotope lithium-7 is  ${}^7_3\text{Li}$ .

How many neutrons are there in an atom of lithium-7?

- A** 3
- B** 4
- C** 7
- D** 10

39 The diagram shows the paths of three different types of radiation X, Y and Z.



Which row correctly identifies X, Y and Z?

|          | X                   | Y                   | Z                   |
|----------|---------------------|---------------------|---------------------|
| <b>A</b> | $\alpha$ -particles | $\beta$ -particles  | $\gamma$ -rays      |
| <b>B</b> | $\beta$ -particles  | $\alpha$ -particles | $\gamma$ -rays      |
| <b>C</b> | $\beta$ -particles  | $\gamma$ -rays      | $\alpha$ -particles |
| <b>D</b> | $\gamma$ -rays      | $\alpha$ -particles | $\beta$ -particles  |

40 Samples of four different radioactive isotopes all emit radiation at the same rate.

As time passes, the rates of emission decrease for all the samples.

Which sample of isotope has the greatest decrease in rate?

|          | isotope       | half-life    |
|----------|---------------|--------------|
| <b>A</b> | americium-241 | 470 years    |
| <b>B</b> | iodine-128    | 25 minutes   |
| <b>C</b> | thoron-220    | 54.5 seconds |
| <b>D</b> | uranium-237   | 6.75 days    |

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**Cambridge International Examinations**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

**0625/11**

Paper 1 Multiple Choice (Core)

**October/November 2018**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)



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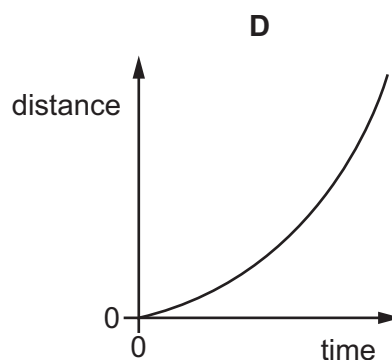
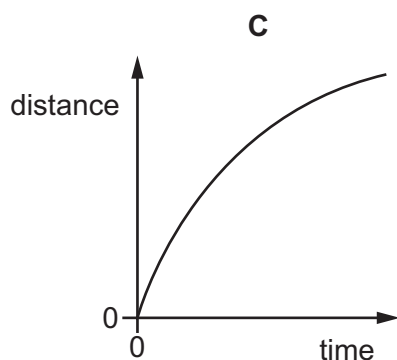
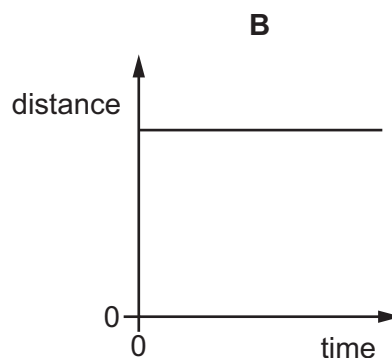
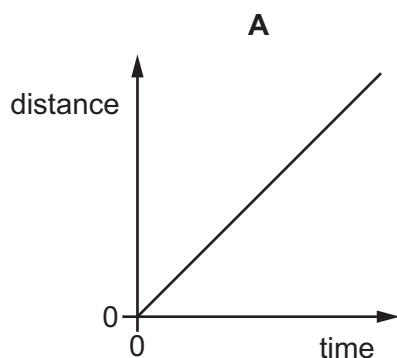
- 1 A pendulum is set in motion and timed. The time measured for 20 complete swings is 30 s.

What is the time for one complete swing of the pendulum?

- A** 0.67 s      **B** 0.75 s      **C** 1.5 s      **D** 3.0 s

- 2 The diagrams are distance-time graphs for four bodies.

Which body is moving with an increasing speed?



- 3 A runner runs 300 m at an average speed of 3.0 m/s. She then runs another 300 m at an average speed of 6.0 m/s.

What is her average speed for the total distance of 600 m?

- A** 2.0 m/s      **B** 4.0 m/s      **C** 4.5 m/s      **D** 8.0 m/s

- 4 Which quantity is measured in newtons?

- A** energy  
**B** mass  
**C** weight  
**D** work

- 5 Four identical steel blocks are placed on a balance. The reading on the balance is 220 N.

What is the mass of one steel block?

- A 5.5 kg      B 22 kg      C 55 kg      D 88 kg

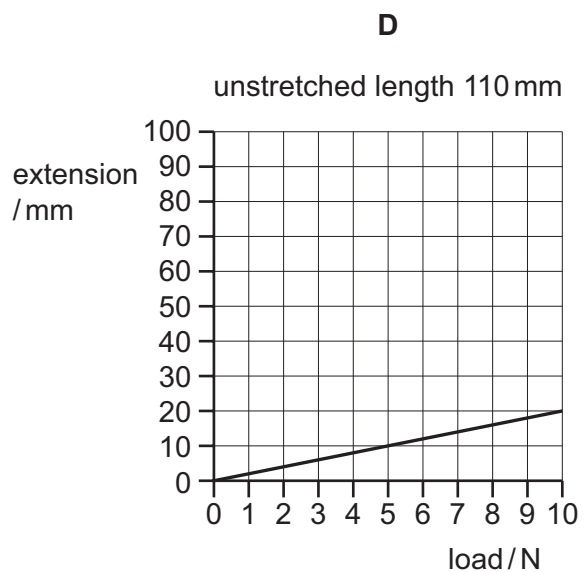
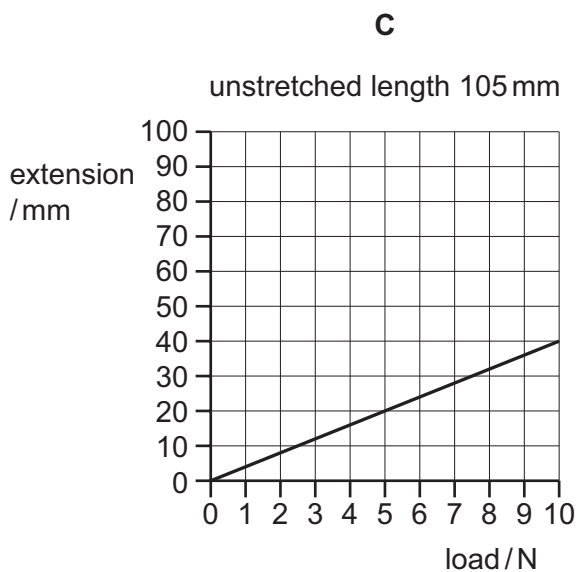
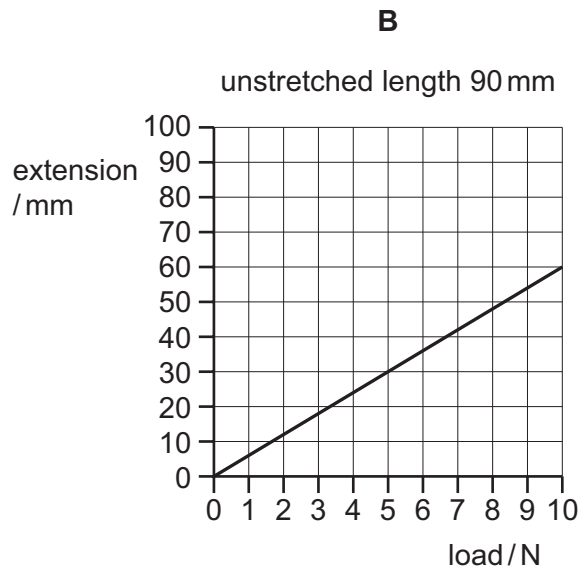
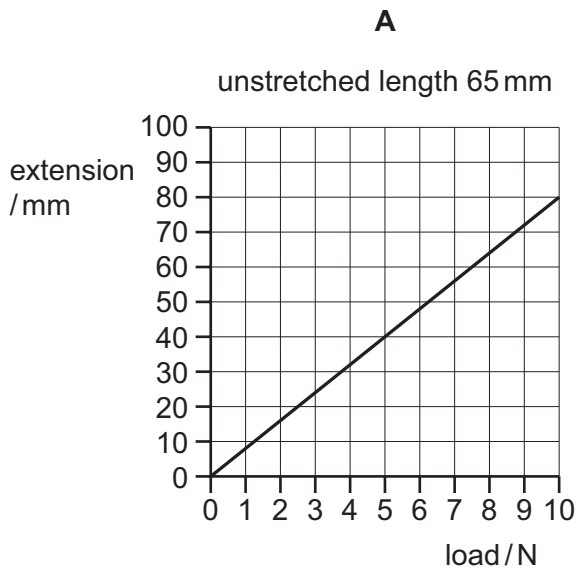
- 6 A liquid has a volume of  $0.040 \text{ m}^3$  and a mass of 30 000 g.

What is the density of the liquid?

- A  $0.075 \text{ kg/m}^3$     B  $7.5 \text{ kg/m}^3$     C  $750 \text{ kg/m}^3$     D  $7500 \text{ kg/m}^3$

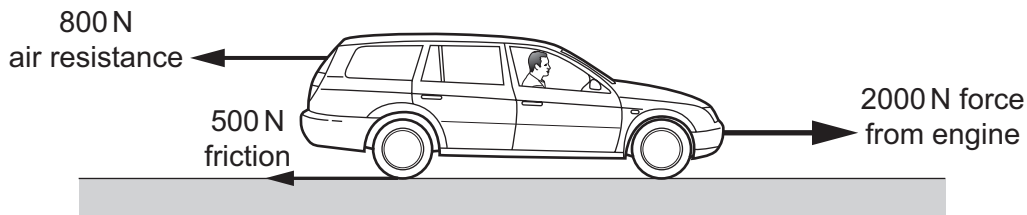
- 7 The unstretched lengths and extension-load graphs are shown for each of four different springs.

Which spring is the longest when a load of 5.0 N is hung from each spring?



- 8 A car moves along a level road.

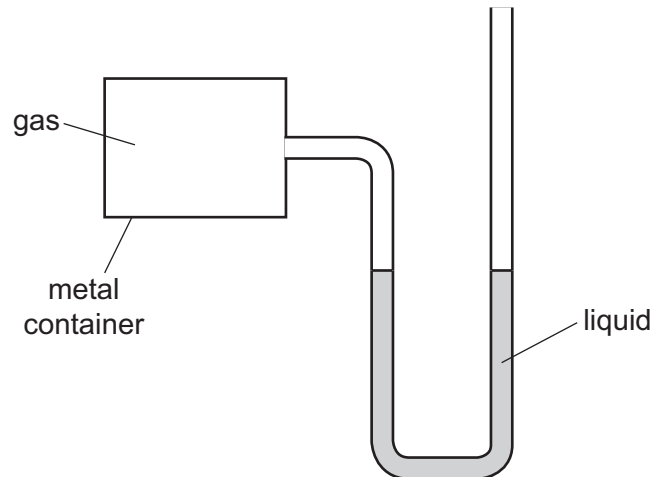
The diagram shows all of the horizontal forces acting on the car.



Which statement is correct?

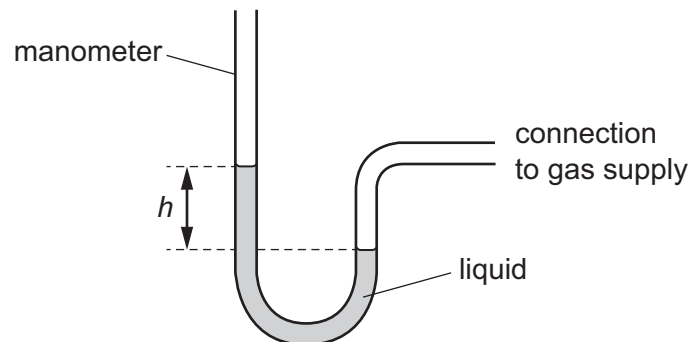
- A The car is slowing down.
  - B The car is speeding up.
  - C The car is moving at a constant speed.
  - D The car is moving backwards.
- 9 Which force and distance produce the smallest moment about a pivot?
- A a 6.0 N force at a perpendicular distance of 4.0 m from the pivot
  - B a 7.0 N force at a perpendicular distance of 6.0 m from the pivot
  - C a 10 N force at a perpendicular distance of 4.0 m from the pivot
  - D a 12 N force at a perpendicular distance of 3.0 m from the pivot
- 10 Which device is designed to convert chemical energy into kinetic energy?
- A an a.c. generator
  - B a battery-powered torch
  - C a car engine
  - D a wind-up mechanical clock
- 11 A man climbs a ladder.
- Which two quantities can be used to calculate the useful power of the man?
- A the weight of the man and the time taken only
  - B the weight of the man and the vertical distance moved only
  - C the work done by the man and the time taken only
  - D the work done by the man and the vertical distance moved only

- 12 A manometer is used to measure the pressure of gas inside a metal container.



What is the pressure inside the container?

- A equal to the atmospheric pressure
  - B greater than the atmospheric pressure
  - C less than the atmospheric pressure but greater than zero
  - D zero
- 13 A manometer is used to measure the pressure of a gas supply.



Which change gives a greater value of height  $h$ ?

- A using a less dense liquid
- B using a more dense liquid
- C using a narrower tube
- D using a wider tube

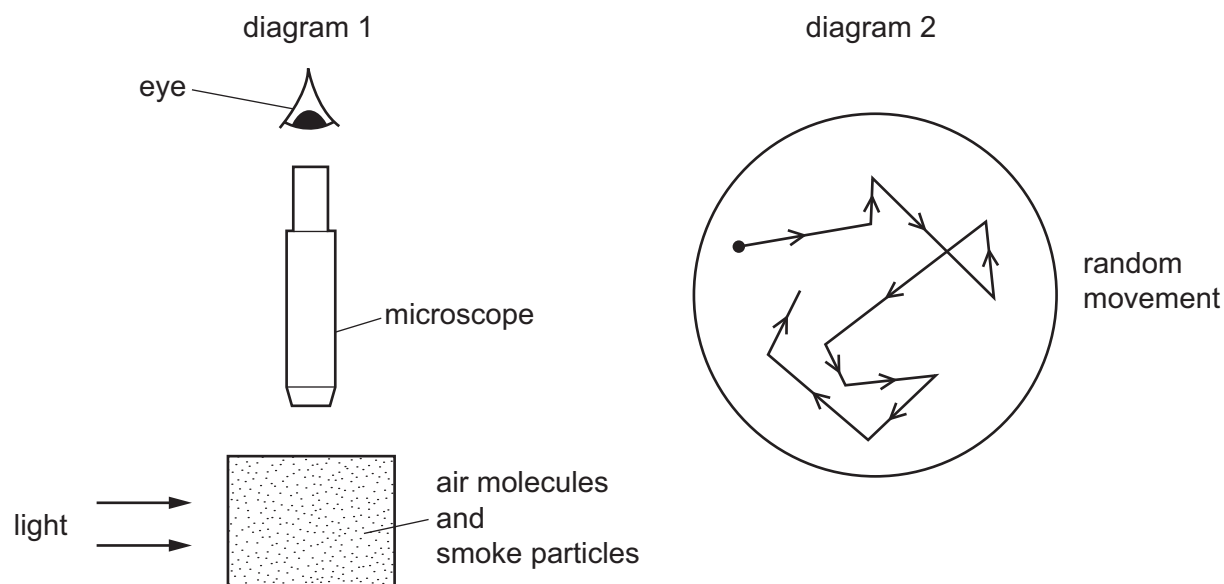
- 14** During evaporation of a liquid, the more energetic molecules escape. The temperature of the remaining liquid changes.

Which row identifies where these molecules escape from and describes the temperature change?

|          | molecules escape from the | temperature of remaining liquid |
|----------|---------------------------|---------------------------------|
| <b>A</b> | body of the liquid        | decreases                       |
| <b>B</b> | body of the liquid        | increases                       |
| <b>C</b> | surface of the liquid     | decreases                       |
| <b>D</b> | surface of the liquid     | increases                       |

- 15** Diagram 1 shows apparatus being used to observe smoke particles.

Diagram 2 shows how a smoke particle moves randomly.



Why do the smoke particles move randomly?

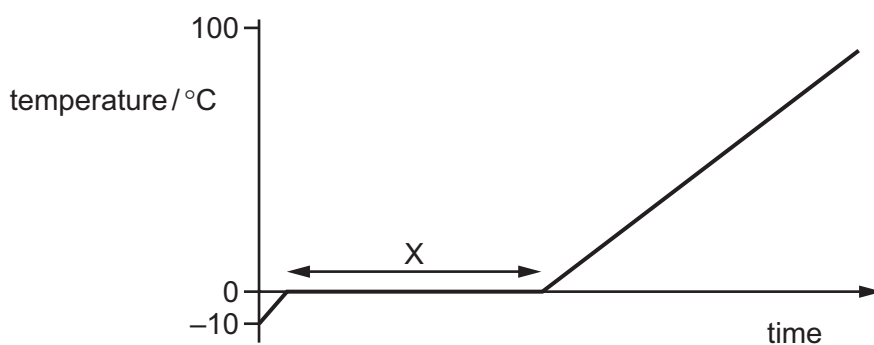
- A** They are hit by air molecules.
- B** They are less dense than air.
- C** They are moved by convection currents.
- D** They gain energy from the light.



- 16** The distance between two electricity pylons is 60 m. An engineer fits a cable of length 62 m between the pylons.

Why does the engineer choose a cable that is longer than the distance between the two pylons?

- A** to allow for contraction of the cable in cold weather
  - B** to create a slope in the cable for electrons to flow down
  - C** to keep the current low and the voltage high
  - D** to reduce magnetic fields around the cable
- 17** Some ice is slowly heated and its temperature is measured. A graph is plotted of temperature against time.

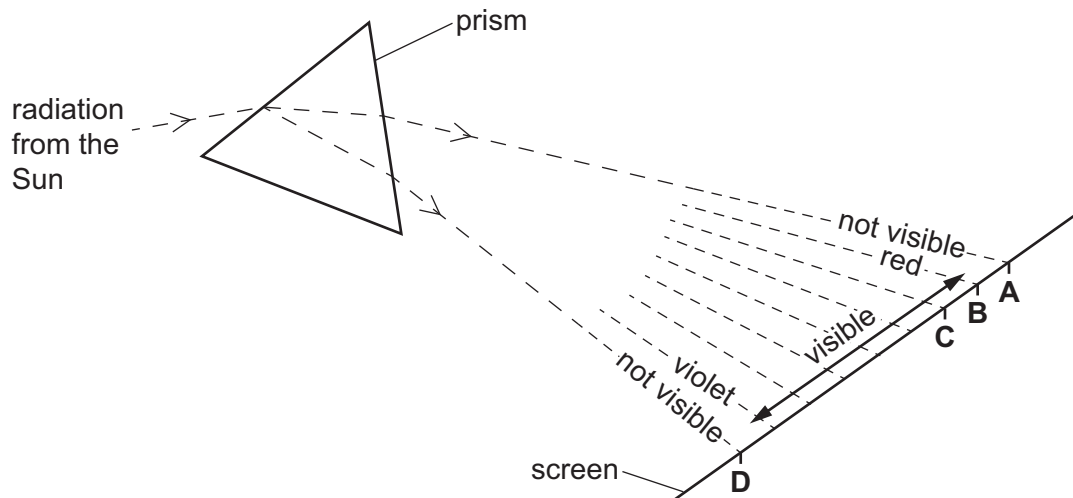


Which row describes what happens to the thermal energy and to the temperature in section X?

|          | thermal energy    | temperature of ice |
|----------|-------------------|--------------------|
| <b>A</b> | gained by ice     | rises              |
| <b>B</b> | gained by ice     | stays the same     |
| <b>C</b> | not gained by ice | rises              |
| <b>D</b> | not gained by ice | stays the same     |

- 18** Some of the Sun's radiation passes through a prism. The diagram shows the spectrum of the radiation.

Which point on the screen does the infra-red radiation reach?



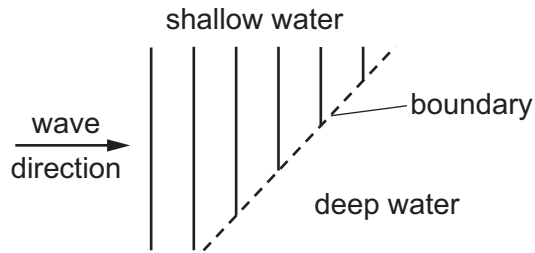
- 19** On a cold day, a shiny metal rod feels colder to the touch than a black plastic rod.

Which statement explains this observation?

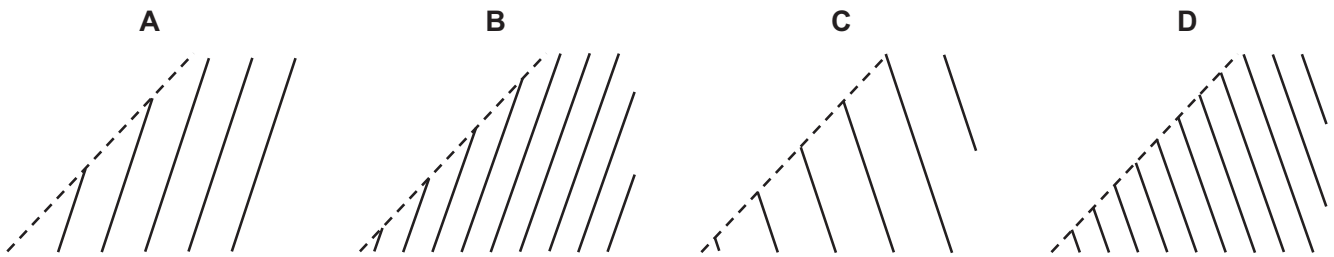
- A** The metal rod is a better absorber of infra-red radiation than the plastic rod.
  - B** The metal rod is a better thermal conductor than the plastic rod.
  - C** The metal rod is a worse absorber of infra-red radiation than the plastic rod.
  - D** The metal rod is a worse thermal conductor than the plastic rod.
- 20** A boat moves up and down repeatedly as a water wave passes it.
- Which name is given to the number of up-and-down movements of the boat per unit time?

- A** amplitude
- B** frequency
- C** speed
- D** wavelength

- 21** Plane water waves travel from a shallow region into a deeper region. They travel more quickly in the deeper water.

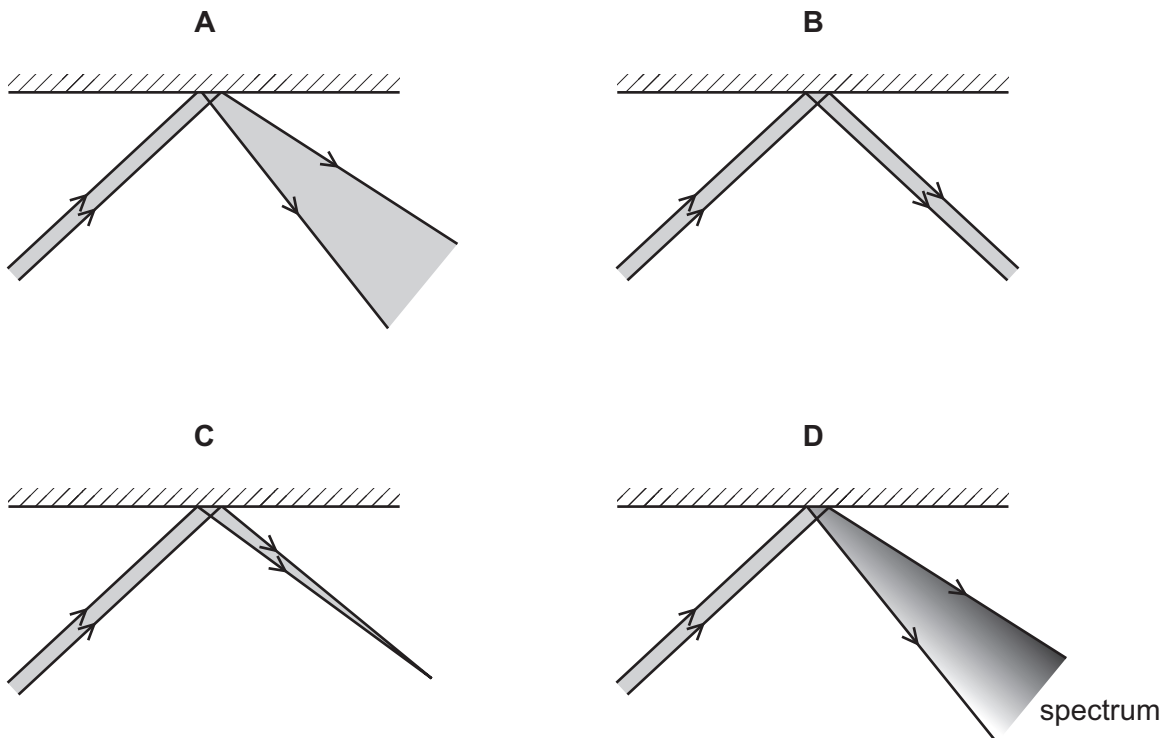


Which diagram shows the wave pattern in the deeper water?

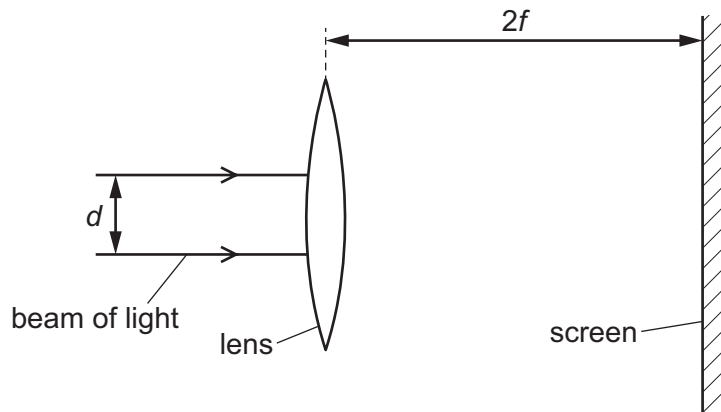


- 22** A parallel beam of light is incident on a plane mirror.

Which diagram shows how the beam is reflected by the mirror?



- 23** The diagram shows a parallel, cylindrical light beam of diameter  $d$  incident on a thin converging lens. A screen is placed a distance equal to two focal lengths  $2f$  from the lens.



Which diagram shows the size of the spot of light seen on the screen?

**A**



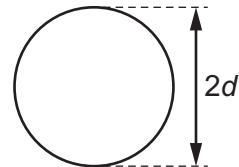
**B**



**C**



**D**



- 24** Different parts of the electromagnetic spectrum are used for different purposes. Below are four statements about parts of the spectrum.

statement 1: Infra-red waves are used in television remote controllers.

statement 2: Radio waves are used to transmit television pictures from satellites to Earth.

statement 3: Ultraviolet waves are used for intruder alarms.

statement 4: X-rays are used for security checks.

Which statements are correct?

**A** 1 and 2

**B** 1 and 4

**C** 2 and 3

**D** 3 and 4

- 25** A woman hears the first note produced by a clarinet.

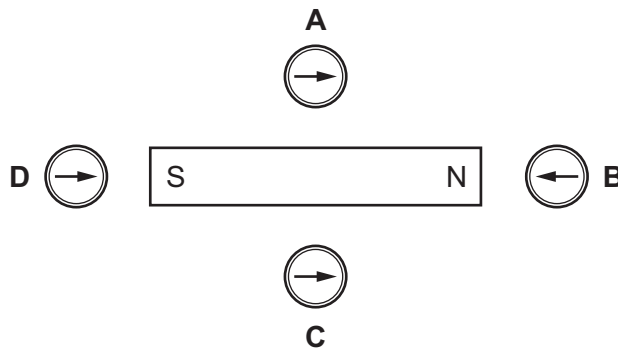
She then hears a second note that has a higher pitch and is quieter.

Which row compares the frequency and the amplitude of the two notes?

|          | first note       | second note       |
|----------|------------------|-------------------|
| <b>A</b> | higher frequency | larger amplitude  |
| <b>B</b> | higher frequency | smaller amplitude |
| <b>C</b> | lower frequency  | larger amplitude  |
| <b>D</b> | lower frequency  | smaller amplitude |

- 26** The diagram shows a bar magnet and four plotting compasses.

Which compass shows the direction of the magnetic field due to the magnet?

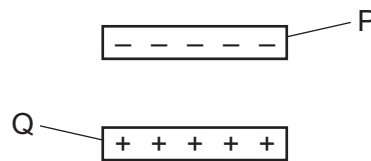


- 27** Iron is used for the core of a transformer and steel is used to make a bar magnet.

Which statement explains these uses of iron and of steel?

- A** Iron is a magnetic material and steel is a non-magnetic material.
- B** Iron is a permanent magnetic material and steel is a temporary magnetic material.
- C** Iron is a temporary magnetic material and steel is a permanent magnetic material.
- D** Iron is a non-magnetic material and steel is a magnetic material.

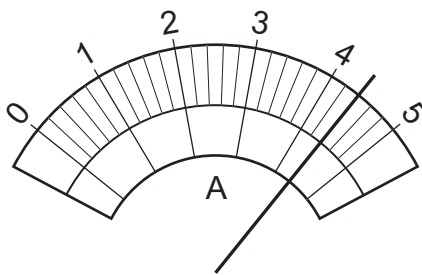
- 28 A negatively charged plastic rod P is placed above a positively charged plastic rod Q.



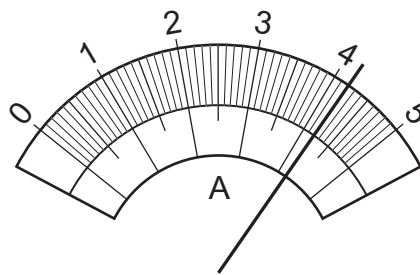
What are the directions of the electrostatic forces on rod P and on rod Q?

|          | electrostatic force on rod P | electrostatic force on rod Q |
|----------|------------------------------|------------------------------|
| <b>A</b> | downwards                    | downwards                    |
| <b>B</b> | downwards                    | upwards                      |
| <b>C</b> | upwards                      | downwards                    |
| <b>D</b> | upwards                      | upwards                      |

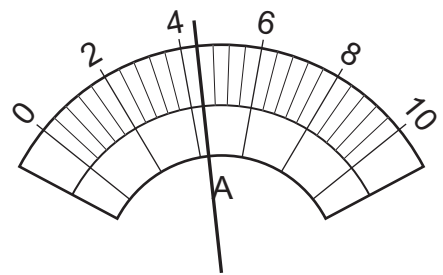
- 29 The diagrams show the scales of three ammeters.



ammeter 1



ammeter 2

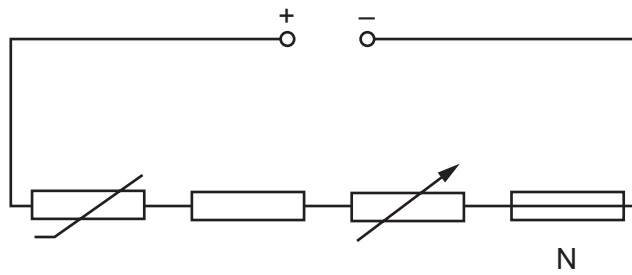


ammeter 3

Which ammeters show the same current reading?

- A** all three ammeters
  - B** ammeter 1 and ammeter 2 only
  - C** ammeter 1 and ammeter 3 only
  - D** ammeter 2 and ammeter 3 only
- 30 The resistance of a component in a circuit is found using an ammeter and a voltmeter.
- How are the ammeter and the voltmeter connected?
- A** the voltmeter and ammeter in parallel with the component
  - B** the voltmeter and ammeter in series with the component
  - C** the voltmeter in parallel with the component and the ammeter in series with the component
  - D** the voltmeter in series with the component and the ammeter in parallel with the component

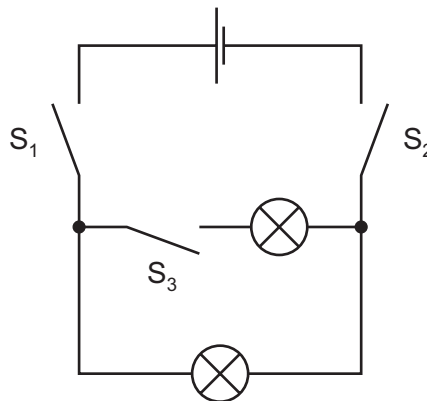
31 The diagram shows a circuit with a power supply and four components.



What is component N?

- A fixed resistor
- B fuse
- C thermistor
- D variable resistor

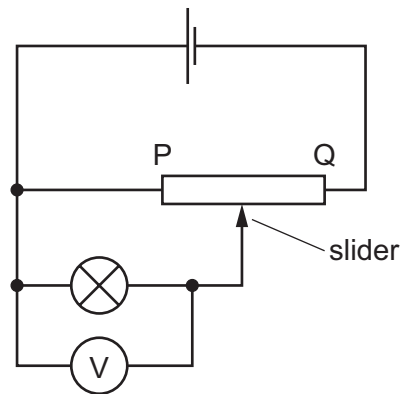
32 Two lamps are connected in parallel.



Which switches must be closed so that both lamps light?

- A  $S_1$  and  $S_2$  only
- B  $S_1$  and  $S_3$  only
- C  $S_2$  and  $S_3$  only
- D  $S_1$ ,  $S_2$  and  $S_3$

- 33 The circuit diagram shows a variable potential divider.

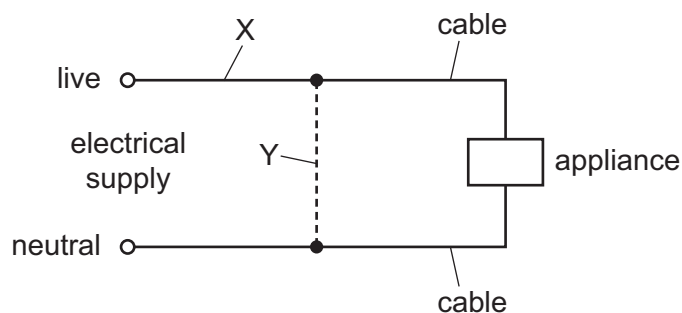


The slider is moved from P towards Q.

What happens to the reading on the voltmeter and to the brightness of the lamp?

|          | reading on voltmeter | brightness of lamp |
|----------|----------------------|--------------------|
| <b>A</b> | decreases            | decreases          |
| <b>B</b> | decreases            | increases          |
| <b>C</b> | increases            | decreases          |
| <b>D</b> | increases            | increases          |

- 34 Either a fuse or a circuit-breaker can be used to protect electrical cables from large currents that could cause overheating.

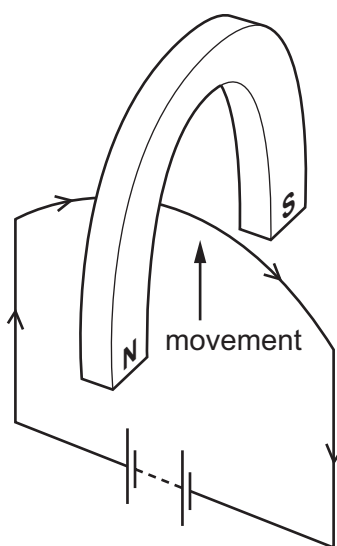


When a fuse is used, where should it be connected, and when a circuit-breaker is used, where should it be connected?

|          | position of fuse | position of circuit-breaker |
|----------|------------------|-----------------------------|
| <b>A</b> | X                | X                           |
| <b>B</b> | X                | Y                           |
| <b>C</b> | Y                | X                           |
| <b>D</b> | Y                | Y                           |



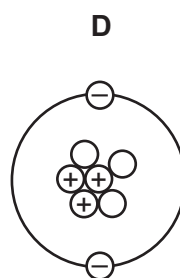
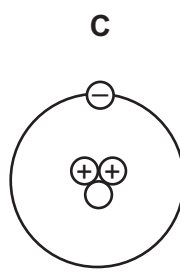
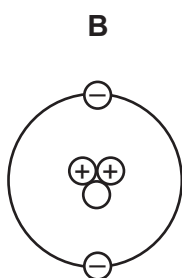
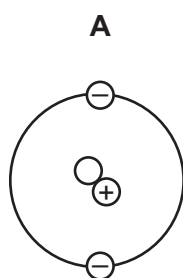
- 35 Which electrical device uses the turning effect produced by a current-carrying coil in a magnetic field?
- A a.c. generator  
B d.c. motor  
C relay  
D transformer
- 36 A wire is placed in a strong magnetic field. When a current is passed through the wire it moves upwards, as shown.



The current is reversed.

In which direction does the wire move?

- A downwards  
B towards the north pole  
C towards the south pole  
D upwards
- 37 Which diagram represents the structure of a neutral atom?



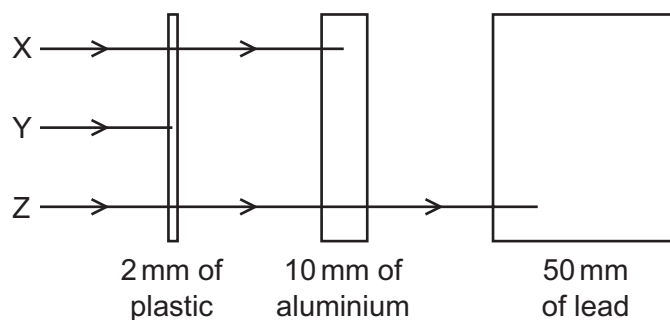
key  
○ neutron  
⊕ proton  
⊖ electron

- 38 Each nucleus of nuclide X contains 10 protons and a total of 22 nucleons.

Which notation is correct for this nuclide?

- A  ${}_{10}^{22}\text{X}$       B  ${}_{32}^{22}\text{X}$       C  ${}_{10}^{12}\text{X}$       D  ${}_{22}^{10}\text{X}$

- 39 The diagram shows the paths of three different types of radiation X, Y and Z.



Which row correctly identifies X, Y and Z?

|   | X                   | Y                   | Z                   |
|---|---------------------|---------------------|---------------------|
| A | $\alpha$ -particles | $\beta$ -particles  | $\gamma$ -rays      |
| B | $\beta$ -particles  | $\alpha$ -particles | $\gamma$ -rays      |
| C | $\beta$ -particles  | $\gamma$ -rays      | $\alpha$ -particles |
| D | $\gamma$ -rays      | $\alpha$ -particles | $\beta$ -particles  |

- 40 A sample of a radioactive isotope emits 9600  $\alpha$ -particles per second.

After 40 hours the rate of emission has fallen to 600  $\alpha$ -particles per second.

What is the half-life of this isotope?

- A 4.0 hours      B 8.0 hours      C 10 hours      D 20 hours

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| 3        | A      | 1     |
| 4        | C      | 1     |
| 5        | B      | 1     |
| 6        | C      | 1     |
| 7        | A      | 1     |
| 8        | B      | 1     |
| 9        | C      | 1     |
| 10       | C      | 1     |
| 11       | B      | 1     |
| 12       | A      | 1     |
| 13       | A      | 1     |
| 14       | D      | 1     |
| 15       | A      | 1     |
| 16       | D      | 1     |
| 17       | B      | 1     |
| 18       | A      | 1     |
| 19       | A      | 1     |
| 20       | C      | 1     |
| 21       | C      | 1     |
| 22       | D      | 1     |
| 23       | A      | 1     |
| 24       | B      | 1     |
| 25       | A      | 1     |
| 26       | A      | 1     |
| 27       | D      | 1     |
| 28       | B      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | D      | 1     |
| 30       | C      | 1     |
| 31       | C      | 1     |
| 32       | D      | 1     |
| 33       | B      | 1     |
| 34       | A      | 1     |
| 35       | C      | 1     |
| 36       | C      | 1     |
| 37       | B      | 1     |
| 38       | D      | 1     |
| 39       | B      | 1     |
| 40       | C      | 1     |



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| 2        | D      | 1     |
| 3        | B      | 1     |
| 4        | C      | 1     |
| 5        | D      | 1     |
| 6        | A      | 1     |
| 7        | D      | 1     |
| 8        | B      | 1     |
| 9        | A      | 1     |
| 10       | C      | 1     |
| 11       | D      | 1     |
| 12       | B      | 1     |
| 13       | A      | 1     |
| 14       | A      | 1     |
| 15       | A      | 1     |
| 16       | D      | 1     |
| 17       | B      | 1     |
| 18       | A      | 1     |
| 19       | C      | 1     |
| 20       | D      | 1     |
| 21       | C      | 1     |
| 22       | B      | 1     |
| 23       | D      | 1     |
| 24       | B      | 1     |
| 25       | B      | 1     |
| 26       | D      | 1     |
| 27       | B      | 1     |
| 28       | B      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | B      | 1     |
| 30       | C      | 1     |
| 31       | C      | 1     |
| 32       | B      | 1     |
| 33       | B      | 1     |
| 34       | A      | 1     |
| 35       | D      | 1     |
| 36       | A      | 1     |
| 37       | B      | 1     |
| 38       | B      | 1     |
| 39       | B      | 1     |
| 40       | C      | 1     |





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| 2        | D      | 1     |
| 3        | B      | 1     |
| 4        | C      | 1     |
| 5        | A      | 1     |
| 6        | C      | 1     |
| 7        | C      | 1     |
| 8        | B      | 1     |
| 9        | A      | 1     |
| 10       | C      | 1     |
| 11       | C      | 1     |
| 12       | A      | 1     |
| 13       | A      | 1     |
| 14       | C      | 1     |
| 15       | A      | 1     |
| 16       | A      | 1     |
| 17       | B      | 1     |
| 18       | A      | 1     |
| 19       | B      | 1     |
| 20       | B      | 1     |
| 21       | C      | 1     |
| 22       | B      | 1     |
| 23       | C      | 1     |
| 24       | B      | 1     |
| 25       | D      | 1     |
| 26       | D      | 1     |
| 27       | C      | 1     |
| 28       | B      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | C      | 1     |
| 30       | C      | 1     |
| 31       | B      | 1     |
| 32       | D      | 1     |
| 33       | D      | 1     |
| 34       | A      | 1     |
| 35       | B      | 1     |
| 36       | A      | 1     |
| 37       | B      | 1     |
| 38       | A      | 1     |
| 39       | B      | 1     |
| 40       | C      | 1     |



**Cambridge International Examinations**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

**0625/13**

Paper 1 Multiple Choice (Core)

**October/November 2017**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)



**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

**DO NOT WRITE IN ANY BARCODES.**

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall =  $10 \text{ m/s}^2$ ).

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

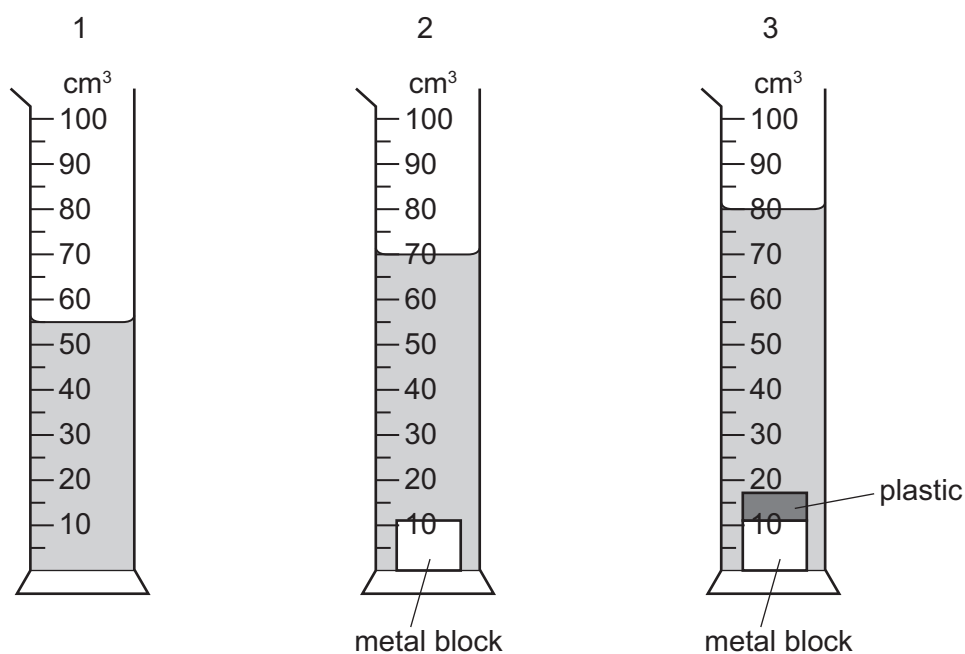
This document consists of **19** printed pages and **1** blank page.



- 1 A measuring cylinder contains some water. A small metal block is slowly lowered into the water and is then removed.

Finally a piece of plastic is attached to the metal block and the block is again slowly lowered into the water.

The diagrams show the measuring cylinder at each stage of this process.

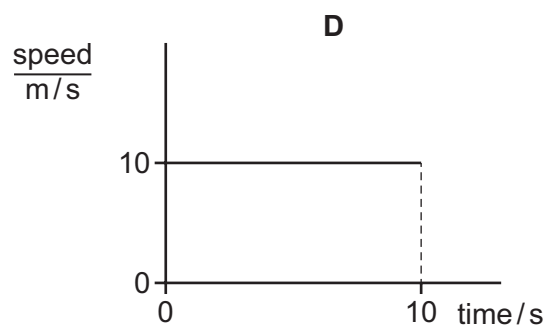
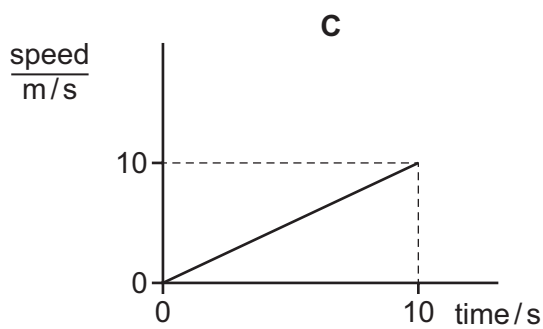
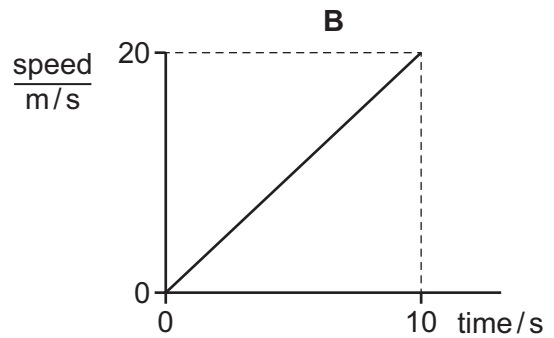
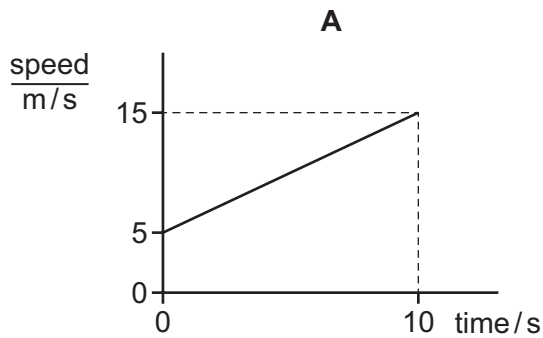


What is the volume of the piece of plastic?

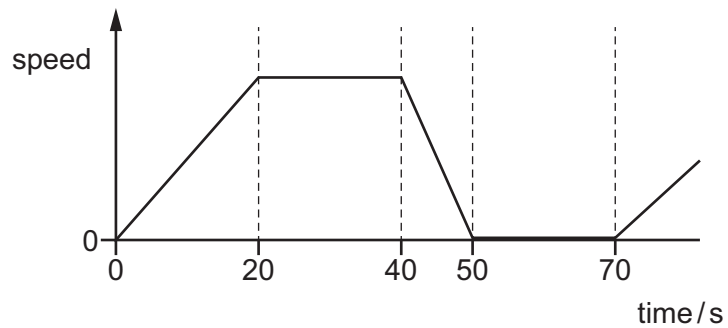
- A** 10  $\text{cm}^3$       **B** 25  $\text{cm}^3$       **C** 70  $\text{cm}^3$       **D** 80  $\text{cm}^3$

- 2 A car accelerates from rest and travels a distance of 100 m in 10 seconds.

Which speed-time graph represents the motion of this car?



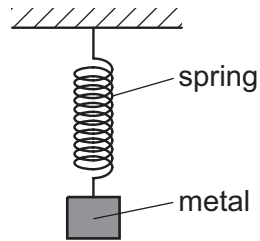
- 3 The diagram is the speed-time graph for a bicycle journey.



When is the bicycle moving at a constant speed?

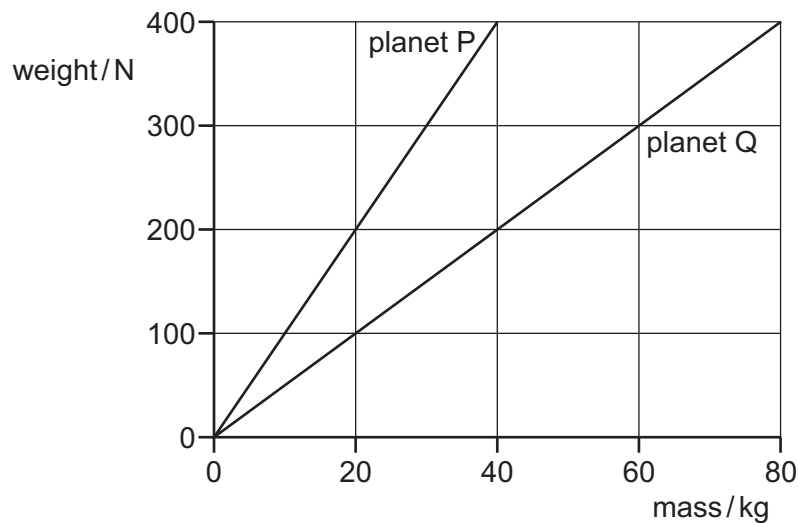
- A** between 0 and 20 s
- B** between 20 s and 40 s
- C** between 40 s and 50 s
- D** between 50 s and 70 s

- 4 A spring is stretched by hanging a piece of metal from it.



Which name is given to the force that stretches the spring?

- A friction
  - B mass
  - C pressure
  - D weight
- 5 The graph shows how weight varies with mass on planet P and on planet Q.



An object weighs 400 N on planet P. The object is taken to planet Q.

Which row is correct?

|          | mass of object<br>on planet Q / kg | weight of object<br>on planet Q / N |
|----------|------------------------------------|-------------------------------------|
| <b>A</b> | 40                                 | 200                                 |
| <b>B</b> | 40                                 | 400                                 |
| <b>C</b> | 80                                 | 200                                 |
| <b>D</b> | 80                                 | 400                                 |

6 What is needed to determine the density of a regularly shaped block?

- A a balance and a beaker
- B a balance and a ruler
- C a measuring cylinder and a beaker
- D a measuring cylinder and a ruler

7 A cart experiences a forward force of 500 N.

The cart also experiences a backward force of 200 N.

There are no other forward or backward forces on the cart.

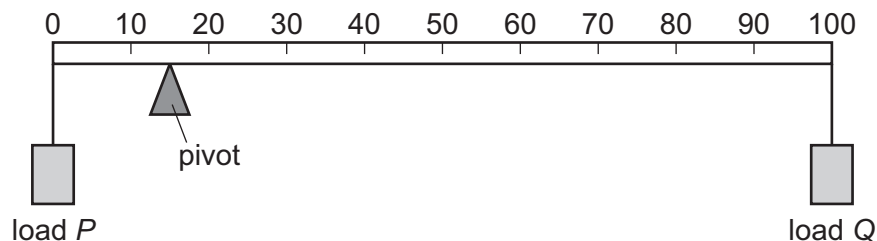
What is the resultant force on the cart?

- A 300 N backwards
- B 300 N forwards
- C 700 N backwards
- D 700 N forwards

8 The diagram shows a beam with a pivot, a load  $P$  at one end and a load  $Q$  at the other end.

The pivot can be moved left or right and the loads can be increased or decreased.

The weight of the beam can be ignored.

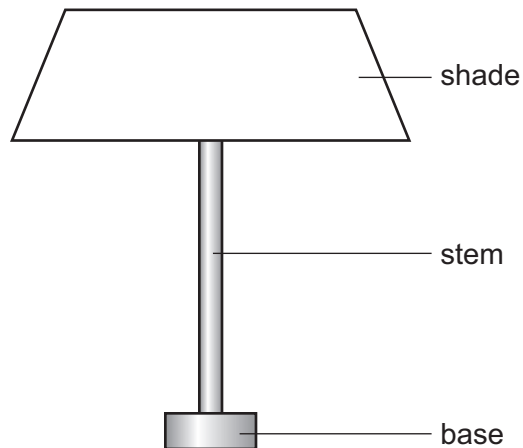


Which combination of pivot position and loads causes the beam to be in equilibrium?

|   | pivot position<br>/ cm | load $P$<br>/ N | load $Q$<br>/ N |
|---|------------------------|-----------------|-----------------|
| A | 30                     | 6.0             | 4.0             |
| B | 40                     | 6.0             | 4.0             |
| C | 50                     | 5.0             | 9.0             |
| D | 60                     | 3.0             | 9.0             |



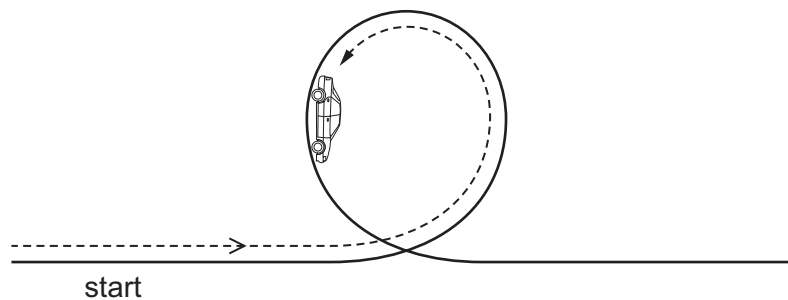
- 9 The lamp in the diagram is not very stable and falls over easily.



Which row shows changes that would definitely make the lamp more stable?

|          | base     | centre of gravity |
|----------|----------|-------------------|
| <b>A</b> | narrower | higher            |
| <b>B</b> | narrower | lower             |
| <b>C</b> | wider    | higher            |
| <b>D</b> | wider    | lower             |

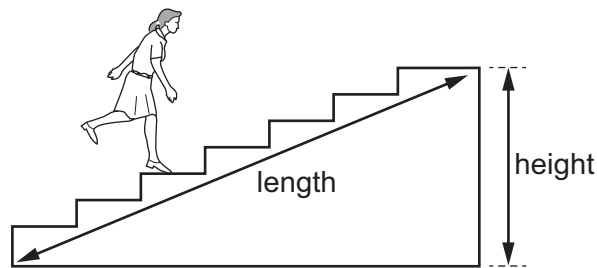
- 10 A toy car without a motor is moving at high speed along a track. The toy car follows a vertical loop as shown.



Which row describes the changes that are taking place in the kinetic energy and in the gravitational potential energy of the car in the position shown?

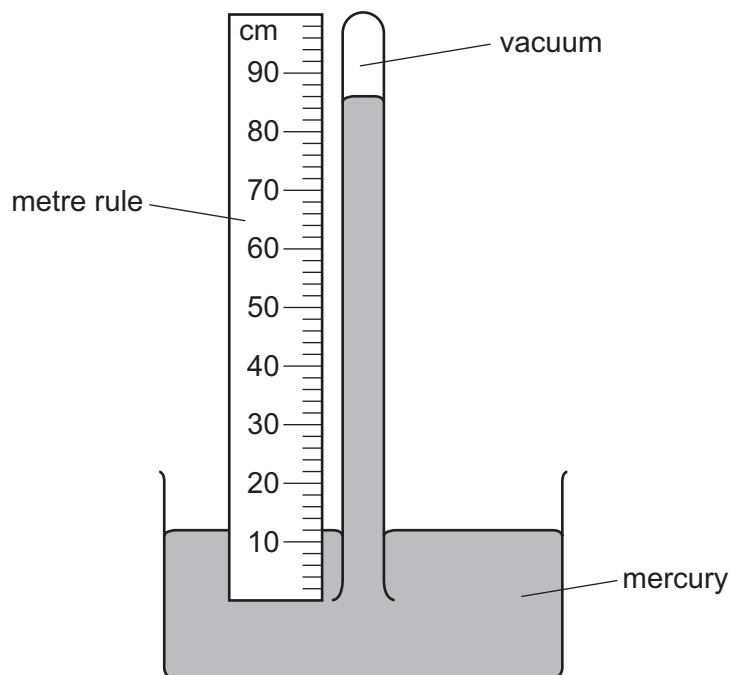
|          | kinetic energy | gravitational potential energy |
|----------|----------------|--------------------------------|
| <b>A</b> | decreasing     | decreasing                     |
| <b>B</b> | decreasing     | increasing                     |
| <b>C</b> | increasing     | decreasing                     |
| <b>D</b> | increasing     | increasing                     |

- 11 A student runs up a flight of stairs.



Which information is **not** needed to calculate the rate at which the student is doing work against gravity?

- A the height of the flight of stairs
  - B the length of the flight of stairs
  - C the time taken to run up the stairs
  - D the weight of the student
- 12 The diagram shows a simple mercury barometer.



Which length is used to find the value of atmospheric pressure?

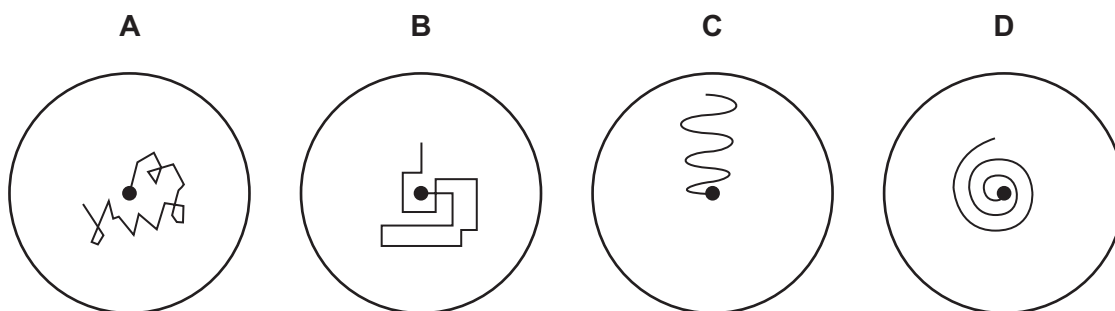
- A 12 cm
  - B 74 cm
  - C 86 cm
  - D 100 cm
- 13 A polystyrene cube of mass 5.0 kg is placed on a horizontal surface. The pressure due to the cube is  $89 \text{ N/m}^2$ .

What is the length of the sides of the cube?

- A 0.56 m
- B 0.75 m
- C 1.3 m
- D 1.8 m

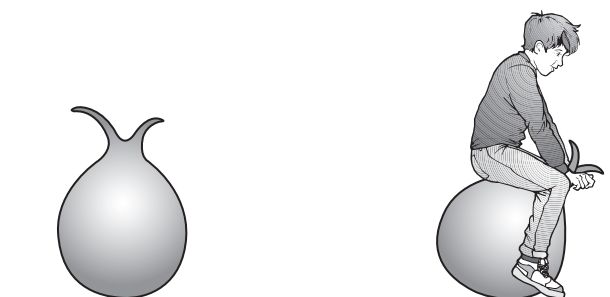
14 A pollen grain in a beaker of still water is viewed through a microscope.

Which diagram shows the most likely movement of the pollen grain?



15 The diagram shows an air-filled rubber toy. A child sits on the toy and its volume decreases.

The temperature of the air in the toy does not change.



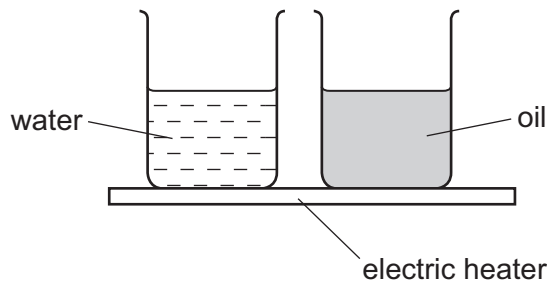
How does the air pressure in the toy change and why?

|          | pressure  | reason                                          |
|----------|-----------|-------------------------------------------------|
| <b>A</b> | decreases | air molecules move more slowly                  |
| <b>B</b> | decreases | air molecules strike the rubber less frequently |
| <b>C</b> | increases | air molecules move more quickly                 |
| <b>D</b> | increases | air molecules strike the rubber more frequently |

16 Which property **cannot** be used for the measurement of temperature?

- A** half-life of a radioactive isotope
- B** length of a solid metal bar
- C** pressure of a gas
- D** volume of a liquid

- 17 The diagram shows an electric heater being used to heat a beaker of water and an identical beaker containing oil. Both are heated for one minute.



The temperature of the water and the temperature of the oil increase steadily. The increase in temperature of the oil is much greater than that of the water.

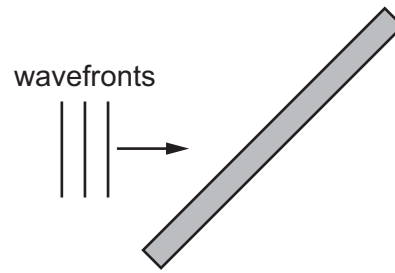
Why is this?

- A Oil has a higher boiling point than water.
  - B Oil has a lower boiling point than water.
  - C The oil has a larger thermal capacity than the water.
  - D The oil has a smaller thermal capacity than the water.
- 18 Why is the heating coil of a domestic immersion heater placed at the bottom of the tank?
- A Cold water is less dense than hot water and therefore sinks.
  - B Cold water is more dense than hot water and therefore rises.
  - C Hot water is less dense than cold water and therefore rises.
  - D Hot water is more dense than cold water and therefore sinks.
- 19 A plane wave in a shallow tank of water of uniform depth is incident normally on the small gap in a barrier.

What happens after the wave passes through the gap?

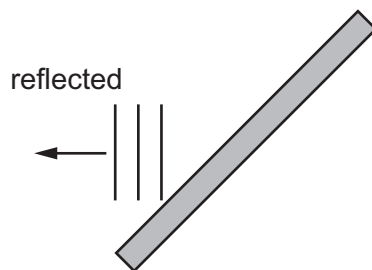
- A The frequency increases.
- B The speed decreases.
- C The wavelength decreases.
- D The wave spreads out.

20 The diagram represents plane wavefronts of a water wave about to strike a solid barrier.

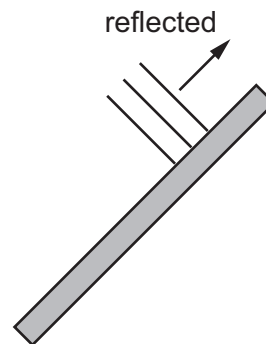


Which diagram shows the position of the wavefronts after reflection at the barrier?

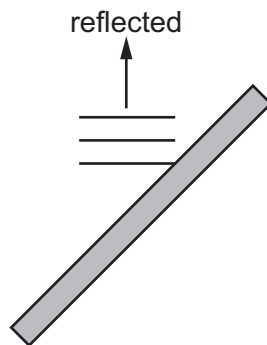
**A**



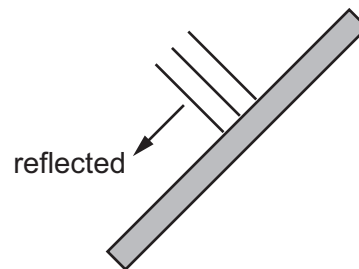
**B**



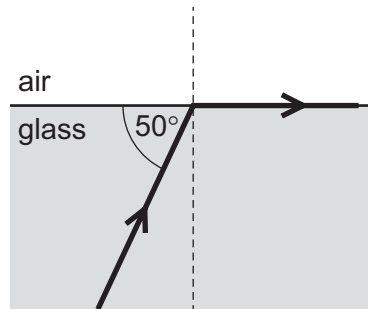
**C**



**D**

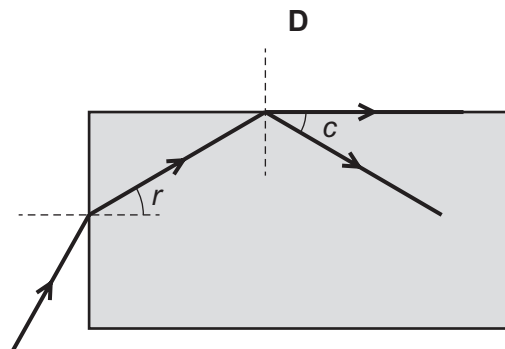
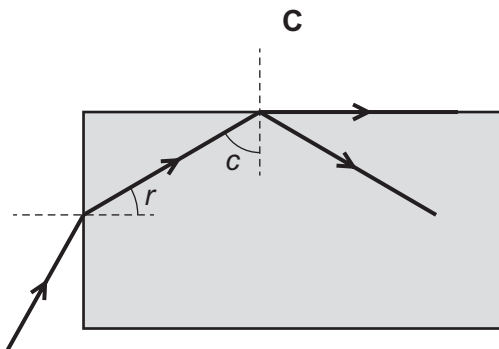
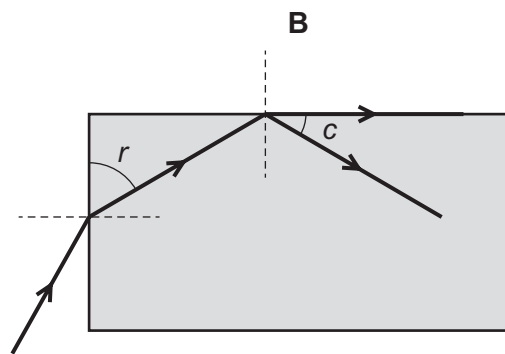
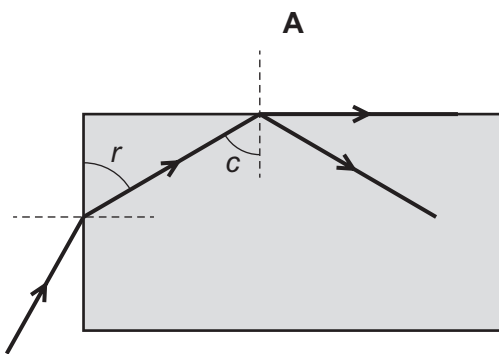


- 21 The diagram shows a ray of light in glass. The ray reaches a boundary with air. One weak ray of light is missing from the diagram.



Which statement is correct?

- A At the boundary, the speed of the light will become less.
  - B The critical angle for light at this boundary is  $50^\circ$ .
  - C The diagram shows an example of diffraction of light.
  - D The missing ray is a weak reflected ray.
- 22 Light enters a transparent block.
- The light is refracted into the block and then strikes the top edge of the block at the critical angle.
- Which diagram correctly shows the angle of refraction  $r$  and the critical angle  $c$ ?



- 23** Which statement about radio waves is correct?
- A** They are used in television remote controllers.
  - B** They can be detected by the human eye.
  - C** They travel as longitudinal waves.
  - D** They have the same speed in a vacuum as ultraviolet waves.
- 24** Which types of thermal energy transfer are due to electromagnetic waves?
- A** conduction and convection
  - B** convection and radiation
  - C** convection only
  - D** radiation only
- 25** The speed of sound in air is 330 m/s.

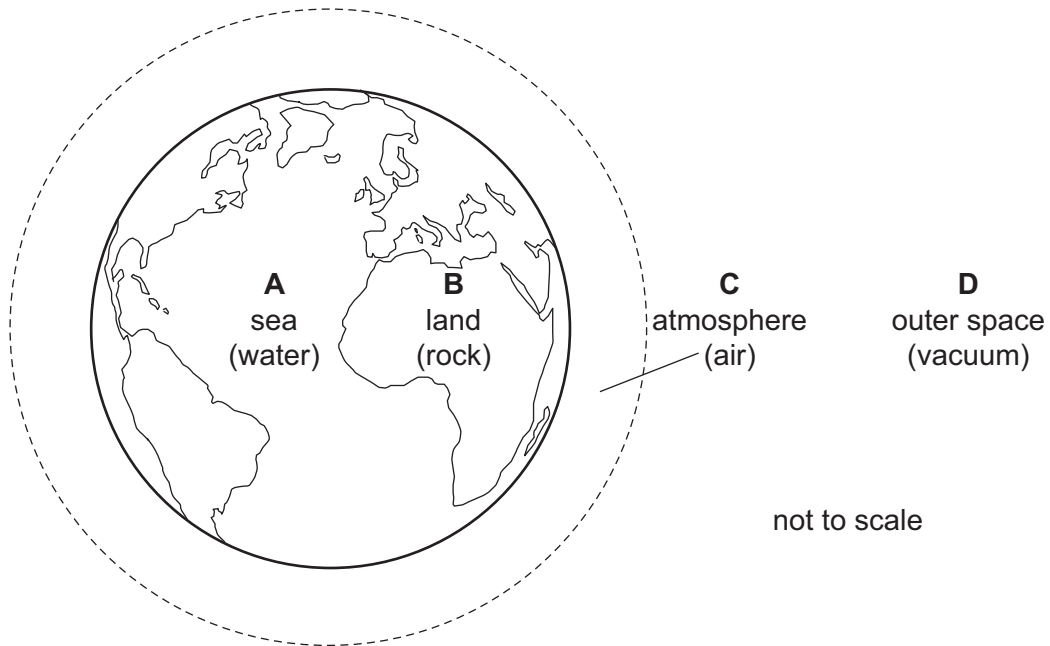
The speed of ultraviolet waves in air is 300 000 000 m/s.

Which row gives a possible frequency and speed of an ultrasound wave in air?

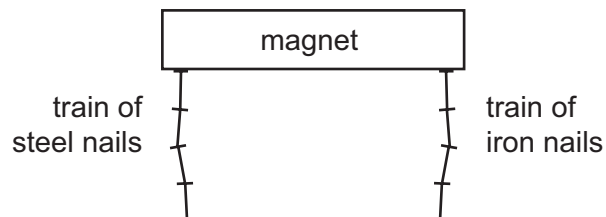
|          | frequency / Hz | <u>speed</u><br>m/s |
|----------|----------------|---------------------|
| <b>A</b> | 4000           | 330                 |
| <b>B</b> | 4000           | 300 000 000         |
| <b>C</b> | 40 000         | 330                 |
| <b>D</b> | 40 000         | 300 000 000         |

26 The diagram shows the Earth and its surroundings.

Through which labelled region can sound **not** be transmitted?



27 A train of steel nails and a train of iron nails hang from a strong magnet.



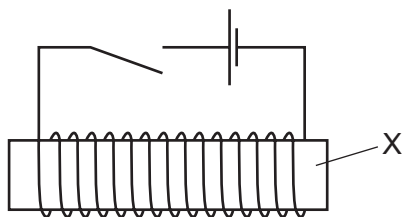
The trains are then carefully removed from the magnet.

What happens to the trains?

- A** Both trains fall apart.
- B** Both trains stay together.
- C** Only the train of iron nails falls apart.
- D** Only the train of steel nails falls apart.



28 The diagram shows an electromagnet.



Which material is part X made from?

- A** copper
- B** iron
- C** steel
- D** tin

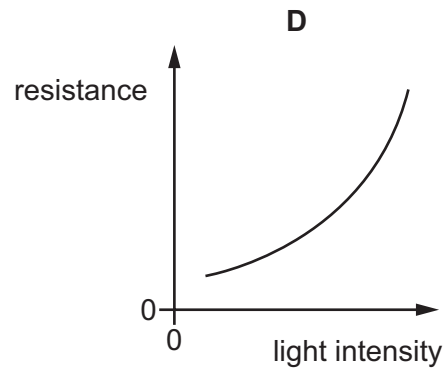
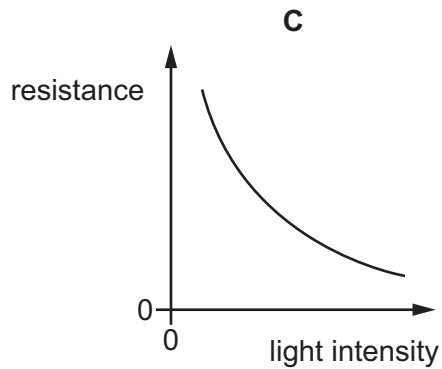
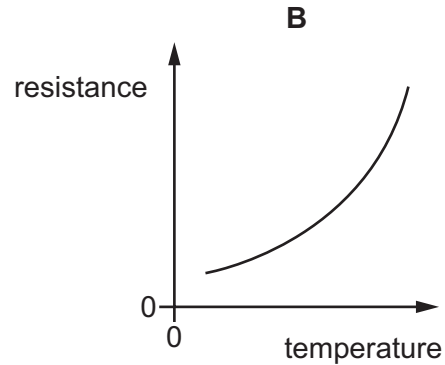
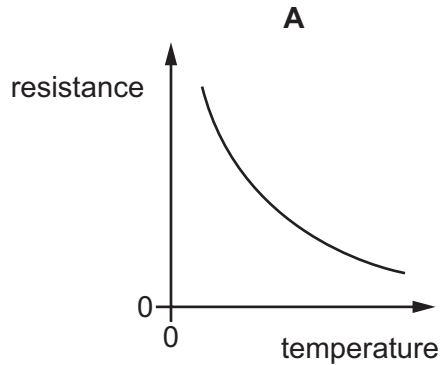
29 In which unit is the electromotive force (e.m.f.) of a battery measured?

- |            |            |            |            |
|------------|------------|------------|------------|
| <b>A</b> A | <b>B</b> J | <b>C</b> N | <b>D</b> V |
|------------|------------|------------|------------|

- 30 The diagram shows a circuit symbol for a component that can be used as an input transducer.



Which graph shows how the resistance of the component varies with its intended input variable?

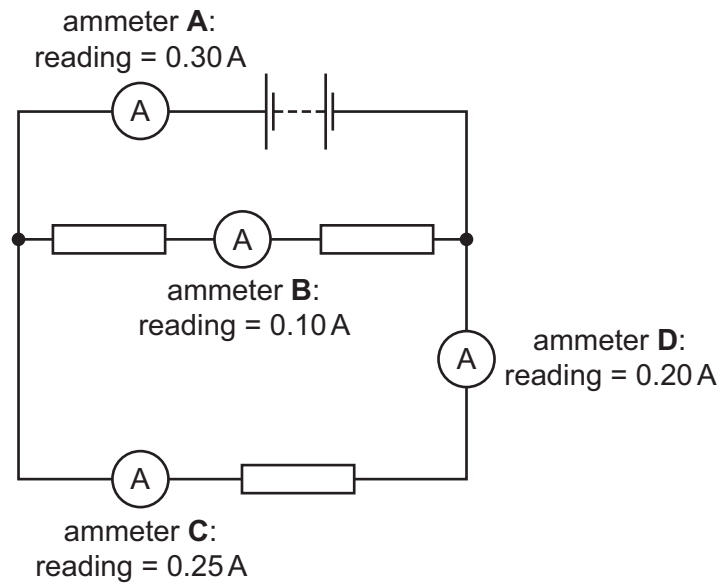


- 31 Which statement is **not** correct for two identical lamps connected in parallel?

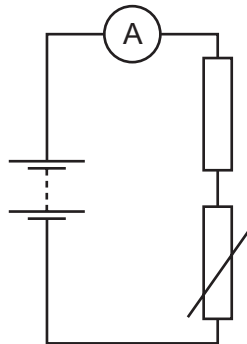
- A** The potential difference across each lamp is half the supply voltage.
- B** The remaining lamp still operates when one lamp is removed.
- C** They can be switched on and off separately.
- D** They remain bright if another lamp is connected in parallel.

- 32 Only one of the ammeters in the circuit is showing an incorrect reading.

Which ammeter is showing the **incorrect** reading?



- 33 The diagram shows a circuit with a fixed resistor connected in series with a thermistor and an ammeter.

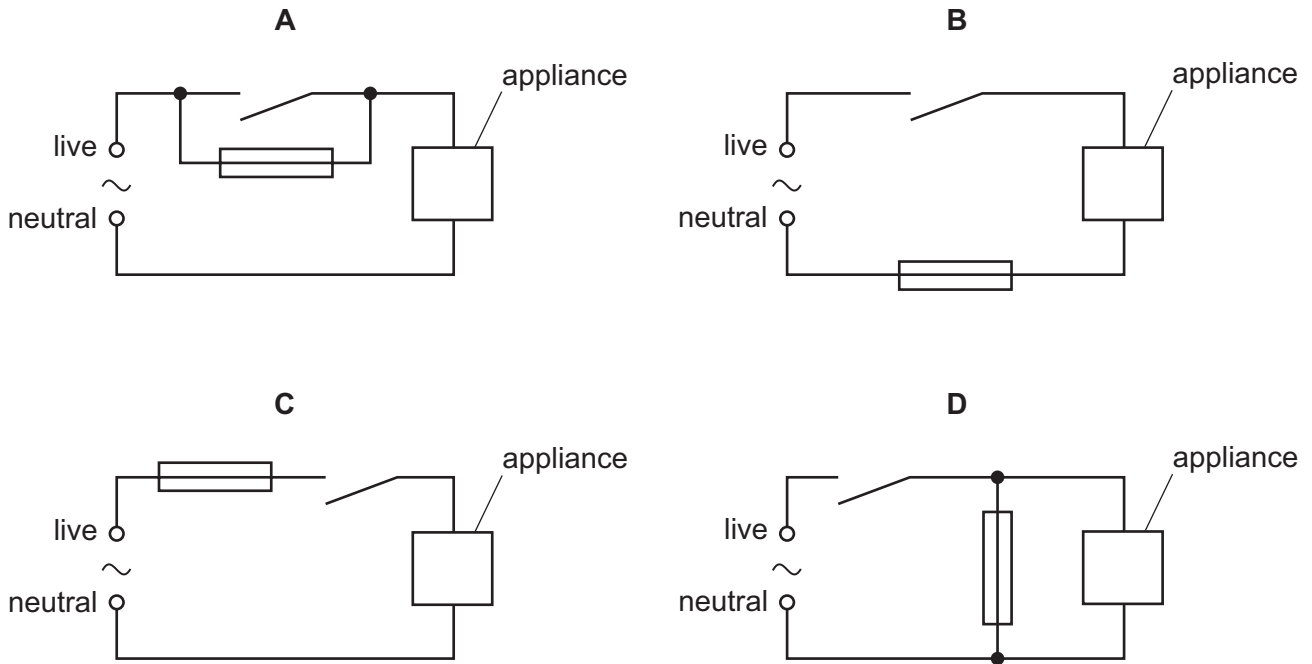


Which row shows how temperature change affects the resistance of the thermistor and the current in the circuit?

|          | temperature | resistance of thermistor | current in circuit |
|----------|-------------|--------------------------|--------------------|
| <b>A</b> | decreases   | decreases                | increases          |
| <b>B</b> | decreases   | increases                | decreases          |
| <b>C</b> | increases   | decreases                | decreases          |
| <b>D</b> | increases   | increases                | increases          |

- 34** An appliance is connected to a mains supply. Its circuit also contains a switch and a fuse.

Which circuit shows the fuse in the correct position?



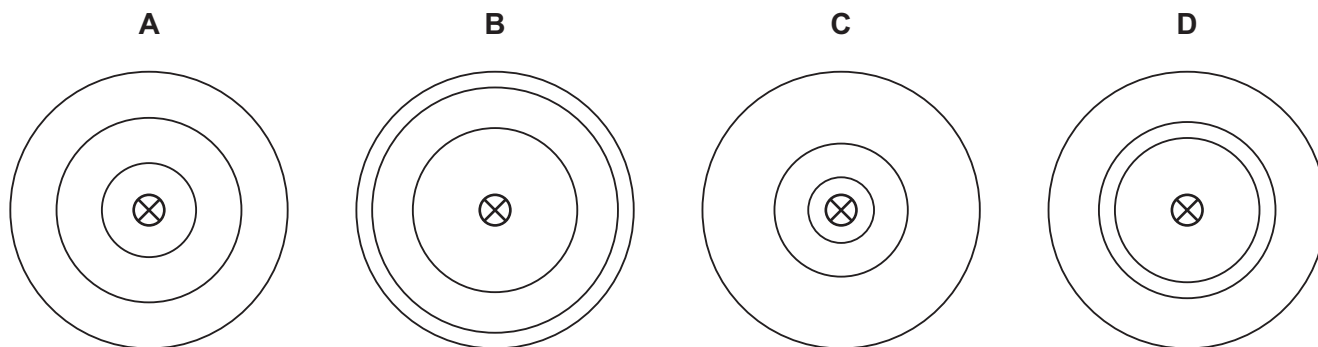
- 35** An electrical device requires a voltage of 5.0 V to operate normally. The mains supply voltage is 250 V.

Which row shows a transformer that provides output voltage that enables the device to operate normally when connected to the mains supply?

|          | number of turns on the primary coil | number of turns on the secondary coil |
|----------|-------------------------------------|---------------------------------------|
| <b>A</b> | 500                                 | 2500                                  |
| <b>B</b> | 500                                 | 25 000                                |
| <b>C</b> | 2500                                | 500                                   |
| <b>D</b> | 25 000                              | 500                                   |

36 A straight current-carrying wire has a magnetic field around it.

Which diagram best shows the magnetic field pattern around the wire?

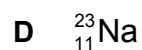
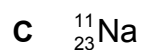
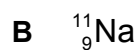
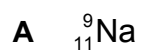


37 Which row states the charge on an electron and states whether electrons are located inside the nucleus of an atom?

|          | charge   | located inside the nucleus |
|----------|----------|----------------------------|
| <b>A</b> | negative | no                         |
| <b>B</b> | negative | yes                        |
| <b>C</b> | positive | no                         |
| <b>D</b> | positive | yes                        |

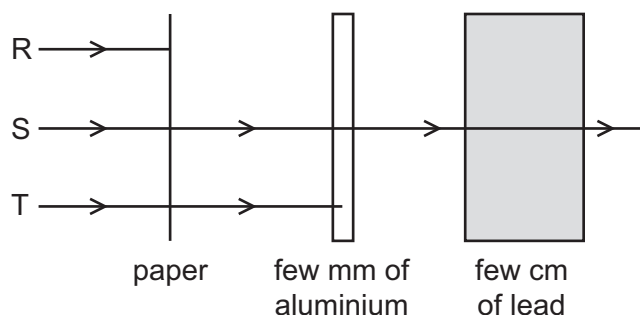
38 The element sodium, symbol Na, has a proton number of 11.

Which is a possible symbol for a sodium nuclide?



- 39** A radioactive source emits three types of radiation R, S and T.

The diagram shows an experiment set up to study the penetrating properties of R, S and T.



Which types of radiation are R, S and T?

|          | R                   | S                   | T                   |
|----------|---------------------|---------------------|---------------------|
| <b>A</b> | $\alpha$ -particles | $\beta$ -particles  | $\gamma$ -rays      |
| <b>B</b> | $\alpha$ -particles | $\gamma$ -rays      | $\beta$ -particles  |
| <b>C</b> | $\beta$ -particles  | $\alpha$ -particles | $\gamma$ -rays      |
| <b>D</b> | $\gamma$ -rays      | $\beta$ -particles  | $\alpha$ -particles |

- 40** Radioactive iodine-131 emits  $\beta$ -particles and has a half-life of 8 days. It decays to produce xenon-131.

Which statement about this decay is correct?

- A** After 8 days no more  $\beta$ -particles are emitted.
- B** After 8 days the number of xenon-131 atoms has halved.
- C** After 16 days the iodine-131 has decayed completely.
- D** After 16 days the number of iodine-131 atoms has reduced to one quarter.

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**Cambridge International Examinations**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

Paper 1 Multiple Choice (Core)

**0625/12**

**October/November 2017**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)



**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

**DO NOT WRITE IN ANY BARCODES.**

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall =  $10 \text{ m/s}^2$ ).

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

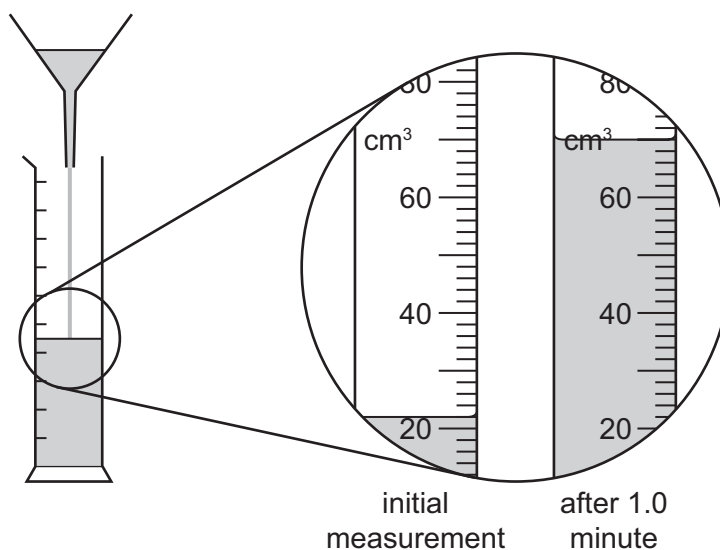
This document consists of **19** printed pages and **1** blank page.





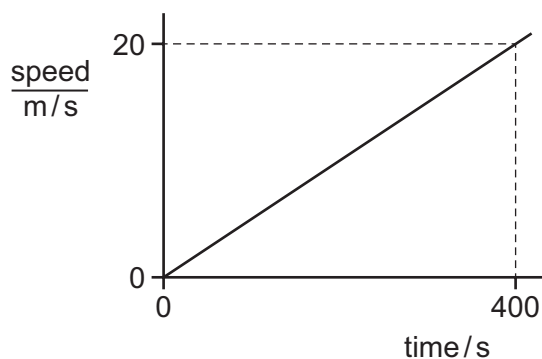
- 1 A student investigates the rate of flow of oil through a funnel.

The diagrams show the experiment and the volume of oil in the measuring cylinder at the start of the experiment, and one minute later.



What is the rate of flow of oil through the funnel during the one minute?

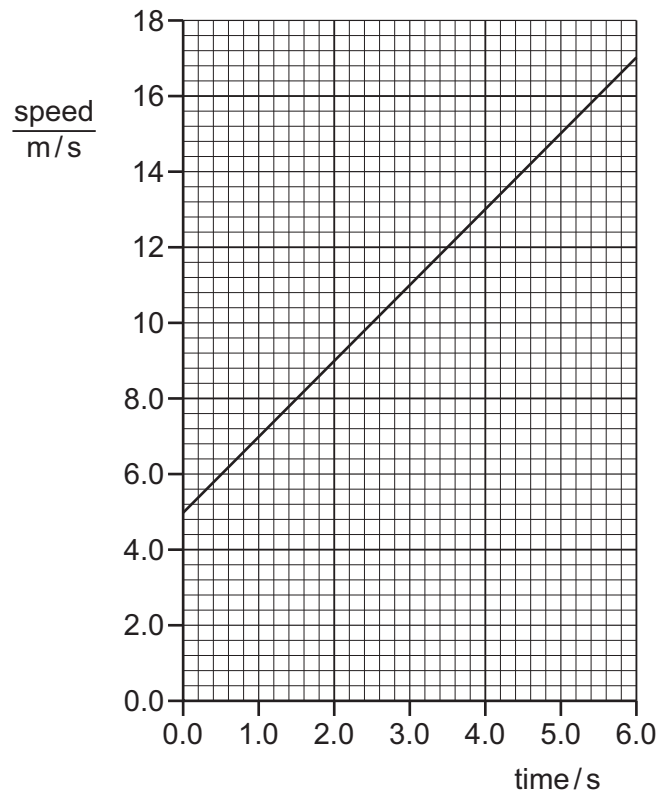
- A**  $0.73 \text{ cm}^3/\text{s}$     **B**  $0.80 \text{ cm}^3/\text{s}$     **C**  $44 \text{ cm}^3/\text{s}$     **D**  $48 \text{ cm}^3/\text{s}$
- 2 The graph represents the motion of a vehicle.



What is the distance travelled by the vehicle in 400 s?

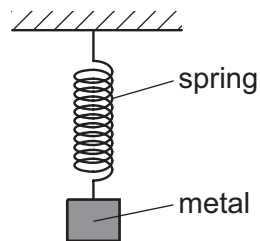
- A** 20 m    **B** 400 m    **C** 4000 m    **D** 8000 m

- 3 The speed-time graph represents the motion of an object.



What is the average speed of the object?

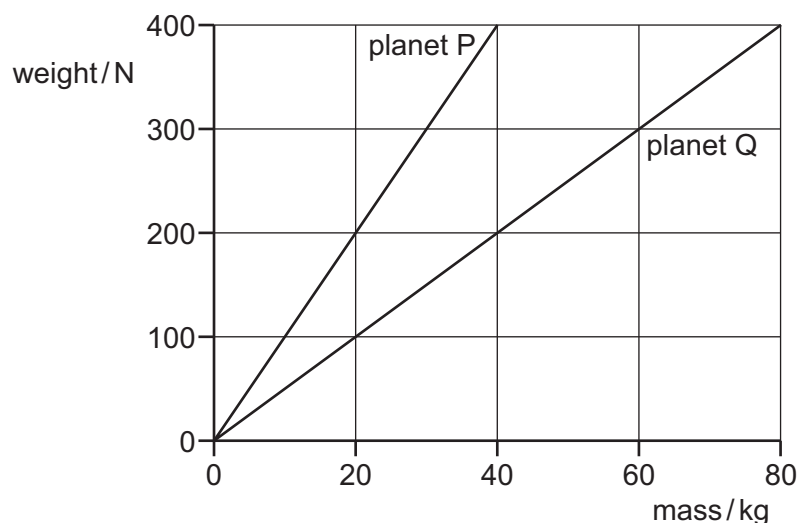
- A** 2.0 m/s      **B** 6.0 m/s      **C** 8.5 m/s      **D** 11 m/s
- 4 A spring is stretched by hanging a piece of metal from it.



Which name is given to the force that stretches the spring?

- A** friction  
**B** mass  
**C** pressure  
**D** weight

- 5 The graph shows how weight varies with mass on planet P and on planet Q.



An object weighs 400 N on planet P. The object is taken to planet Q.

Which row is correct?

|          | mass of object<br>on planet Q / kg | weight of object<br>on planet Q / N |
|----------|------------------------------------|-------------------------------------|
| <b>A</b> | 40                                 | 200                                 |
| <b>B</b> | 40                                 | 400                                 |
| <b>C</b> | 80                                 | 200                                 |
| <b>D</b> | 80                                 | 400                                 |

- 6 What is needed to determine the density of a regularly shaped block?

- A** a balance and a beaker
- B** a balance and a ruler
- C** a measuring cylinder and a beaker
- D** a measuring cylinder and a ruler

- 7 The diagram shows an aircraft flying in a straight horizontal line at constant speed.

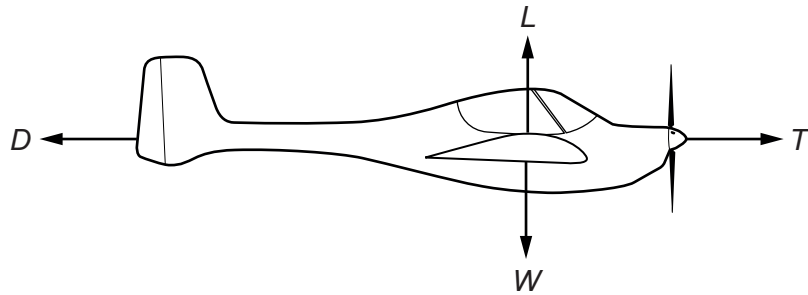
$W$  is the weight of the aircraft.

$L$  is the lift (upward force) due to air flow over the wings.

$T$  is the thrust force due to the engine.

$D$  is the air resistance (drag).

The diagram shows the direction of these forces.

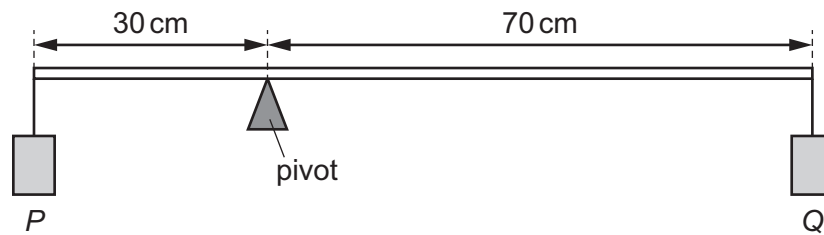


What are the relationships between the forces?

- A  $L = W$  and  $T = D$
- B  $L = W$  and  $T$  is greater than  $D$
- C  $L$  is greater than  $W$  and  $T = D$
- D  $L$  is greater than  $W$  and  $T$  is greater than  $D$

- 8 A pivoted beam supports a load  $P$  at one end and a load  $Q$  at the other end.

The weight of the beam can be ignored.



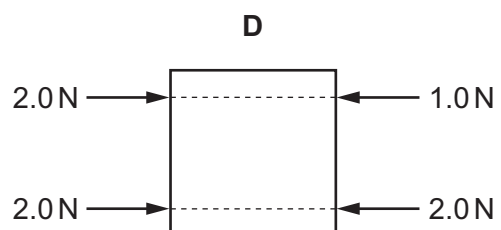
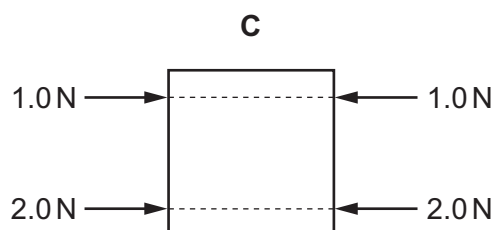
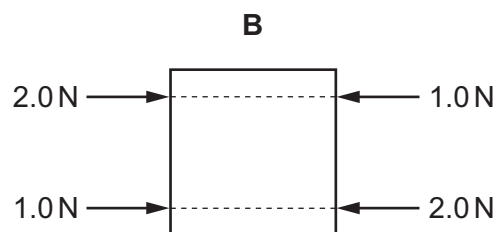
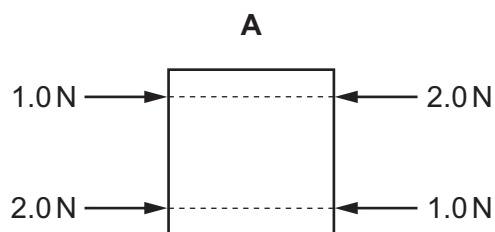
The beam is balanced.

Which row gives possible values for  $P$  and for  $Q$ ?

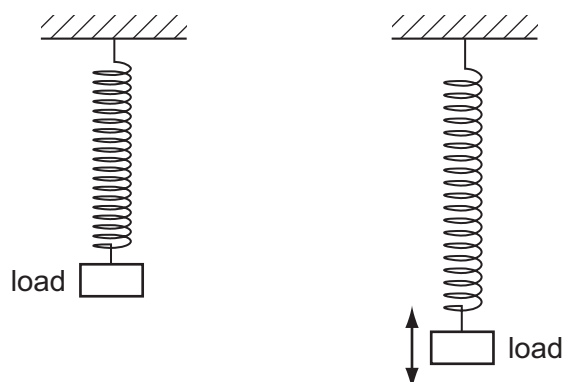
|          | $P/\text{N}$ | $Q/\text{N}$ |
|----------|--------------|--------------|
| <b>A</b> | 21           | 7            |
| <b>B</b> | 21           | 9            |
| <b>C</b> | 30           | 10           |
| <b>D</b> | 30           | 70           |

- 9 Each diagram shows a metal plate with four parallel forces acting on it. These are the only forces acting on the plates.

In which diagram is the plate in equilibrium?



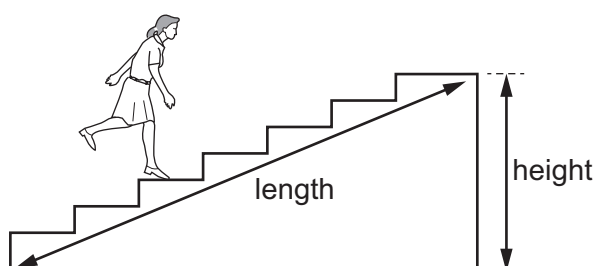
- 10 A load is attached to the end of a spring. A student pulls the load down from its rest position and releases it. It oscillates vertically and eventually comes to rest.



Which row gives the type of energy transferred to the apparatus, and the type of energy to which this has then been transferred when the load comes to rest?

|          | type of energy transferred to the apparatus | type of energy to which this has then been transferred when the load comes to rest |
|----------|---------------------------------------------|------------------------------------------------------------------------------------|
| <b>A</b> | elastic energy in the spring                | chemical energy in the spring and in the air                                       |
| <b>B</b> | elastic energy in the spring                | thermal energy in the spring and in the air                                        |
| <b>C</b> | gravitational potential energy in the load  | chemical energy in the spring and in the air                                       |
| <b>D</b> | gravitational potential energy in the load  | thermal energy in the spring and in the air                                        |

- 11 A student runs up a flight of stairs.



Which information is **not** needed to calculate the rate at which the student is doing work against gravity?

- A** the height of the flight of stairs
- B** the length of the flight of stairs
- C** the time taken to run up the stairs
- D** the weight of the student

- 12** A book has a mass of 400 g.

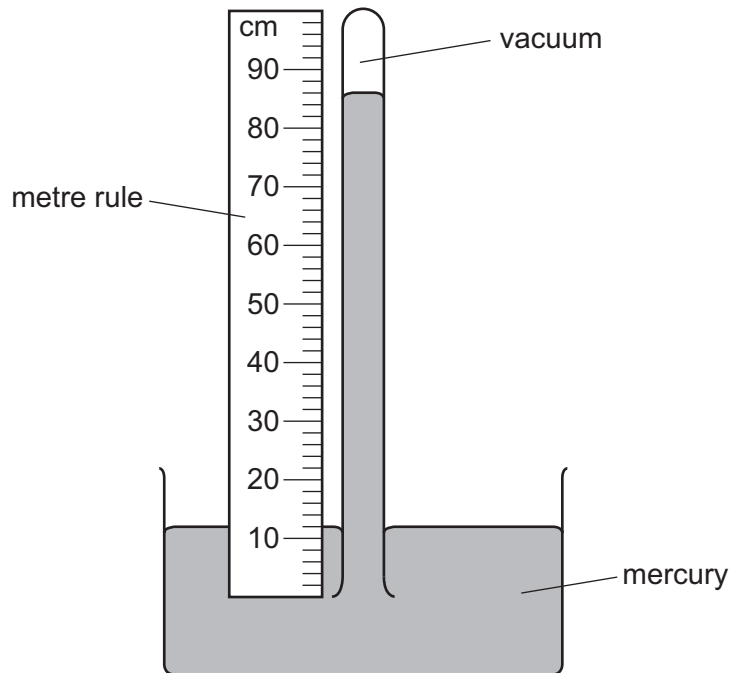
One of its surfaces measures  $0.10\text{ m} \times 0.20\text{ m}$ . This surface is in contact with a table.

The gravitational field strength  $g$  is  $10\text{ N/kg}$ .

What is the pressure exerted on the table due to the book?

- A**  $0.08\text{ N/m}^2$       **B**  $8.0\text{ N/m}^2$       **C**  $20\text{ N/m}^2$       **D**  $200\text{ N/m}^2$

- 13** The diagram shows a simple mercury barometer.

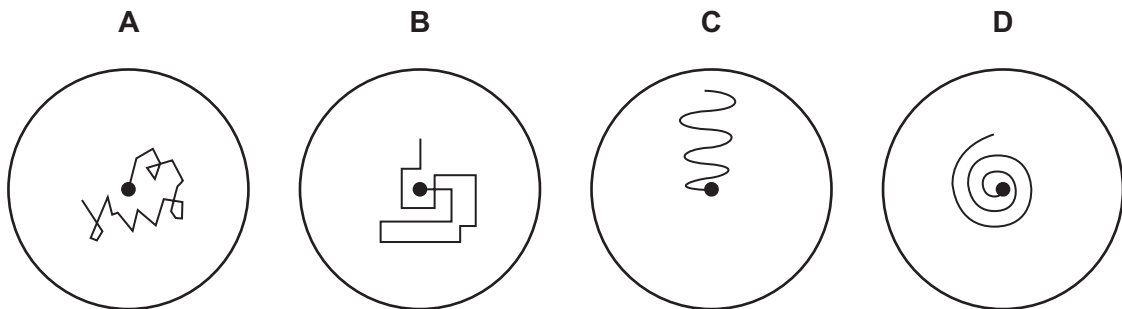


Which length is used to find the value of atmospheric pressure?

- A** 12 cm      **B** 74 cm      **C** 86 cm      **D** 100 cm

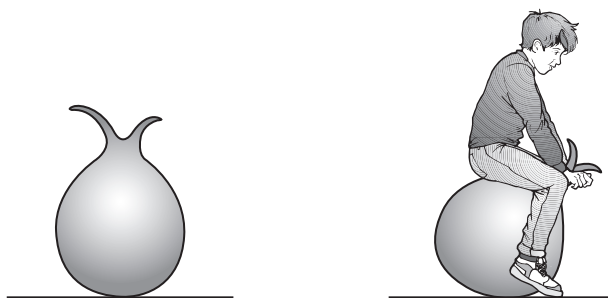
- 14** A pollen grain in a beaker of still water is viewed through a microscope.

Which diagram shows the most likely movement of the pollen grain?



- 15** The diagram shows an air-filled rubber toy. A child sits on the toy and its volume decreases.

The temperature of the air in the toy does not change.



How does the air pressure in the toy change and why?

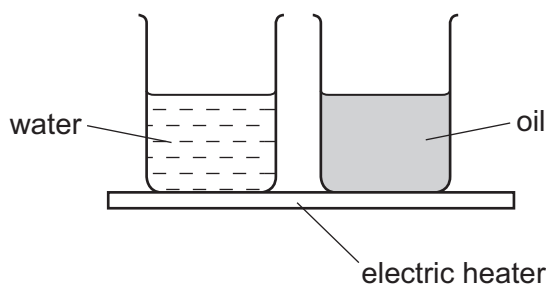
|          | pressure  | reason                                          |
|----------|-----------|-------------------------------------------------|
| <b>A</b> | decreases | air molecules move more slowly                  |
| <b>B</b> | decreases | air molecules strike the rubber less frequently |
| <b>C</b> | increases | air molecules move more quickly                 |
| <b>D</b> | increases | air molecules strike the rubber more frequently |

- 16** Which row identifies the fixed points on the Celsius scale?

|          | lower fixed point           | upper fixed point           |
|----------|-----------------------------|-----------------------------|
| <b>A</b> | boiling point of mercury    | melting point of pure ice   |
| <b>B</b> | boiling point of pure water | melting point of pure ice   |
| <b>C</b> | melting point of mercury    | boiling point of pure water |
| <b>D</b> | melting point of pure ice   | boiling point of pure water |



- 17 The diagram shows an electric heater being used to heat a beaker of water and an identical beaker containing oil. Both are heated for one minute.

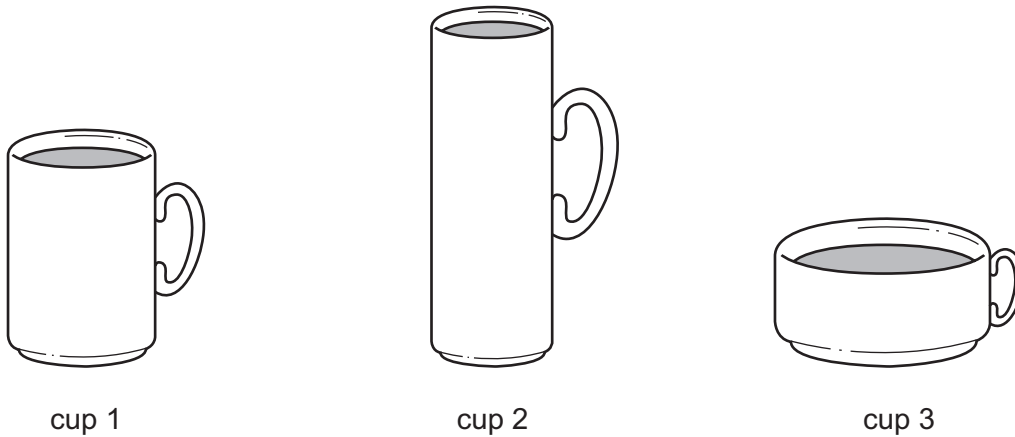


The temperature of the water and the temperature of the oil increase steadily. The increase in temperature of the oil is much greater than that of the water.

Why is this?

- A Oil has a higher boiling point than water.
  - B Oil has a lower boiling point than water.
  - C The oil has a larger thermal capacity than the water.
  - D The oil has a smaller thermal capacity than the water.
- 18 Which statement about the direction of a change of state is correct?
- A Evaporation is the reverse process to boiling.
  - B Evaporation is the reverse process to condensation.
  - C Evaporation is the reverse process to melting.
  - D Evaporation is the reverse process to solidification.

- 19 Three cups are made from the same insulating material.



The cups are each filled with the same volume of hot tea.

In which cup does the tea cool most quickly and in which cup does it cool most slowly?

|          | most quickly | most slowly |
|----------|--------------|-------------|
| <b>A</b> | 1            | 3           |
| <b>B</b> | 2            | 1           |
| <b>C</b> | 3            | 2           |
| <b>D</b> | 3            | 1           |

- 20 Why does a balloon filled with hot air rise?

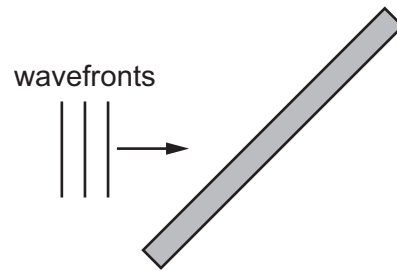
- A** Cold air is less dense than hot air.
- B** Cold air is more dense than hot air.
- C** Heat rises.
- D** The density of the balloon is greater than the density of the surrounding gas.

- 21 A wave moves along the surface of water.

What is the wavelength of the wave?

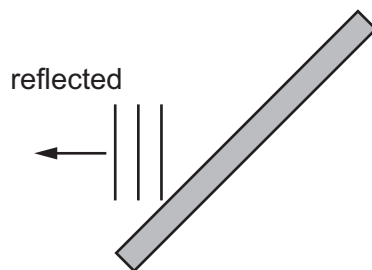
- A** the distance between one crest and the next crest
- B** the distance that a crest moves along the surface in one second
- C** the distance that a particle of water moves up and down
- D** the number of waves that pass a fixed point in one second

22 The diagram represents plane wavefronts of a water wave about to strike a solid barrier.

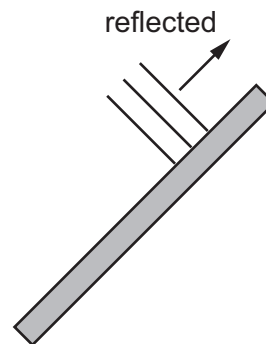


Which diagram shows the position of the wavefronts after reflection at the barrier?

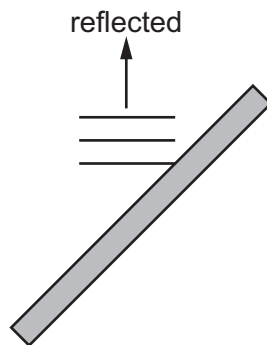
**A**



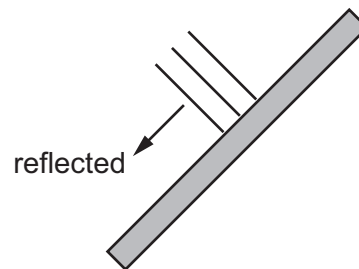
**B**



**C**

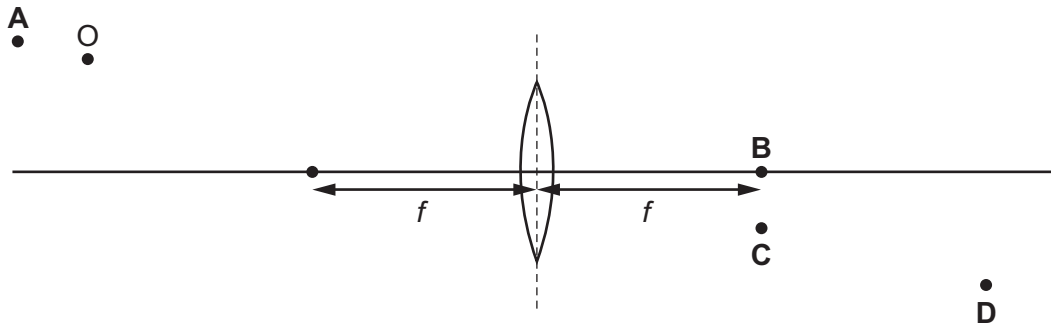


**D**

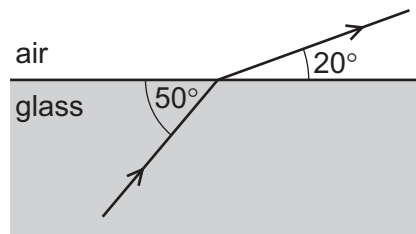


- 23 The diagram shows an object O in front of a thin converging lens of focal length  $f$ .

At which point will the lens form a sharp image of the object?



- 24 The diagram shows light travelling from glass to air.



What is the angle of refraction for this ray of light?

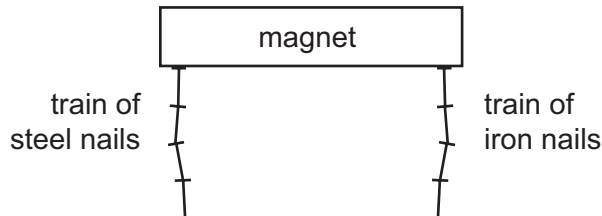
- A  $20^\circ$                       B  $40^\circ$                       C  $50^\circ$                       D  $70^\circ$
- 25 Which piece of equipment is designed to produce a type of electromagnetic wave?
- A electric fire  
B electric generator  
C electric motor  
D electromagnet
- 26 What is ultrasound?
- A sound waves that are so loud that they damage human hearing  
B sound waves that are too high-pitched for humans to hear  
C sound waves that are too low-pitched for humans to hear  
D sound waves that are too quiet for humans to hear

- 27 A student finds that it takes sound 0.33 seconds to travel 100 metres.

From this information, what is the speed of sound?

- A 30 m/s      B 60 m/s      C 300 m/s      D 600 m/s

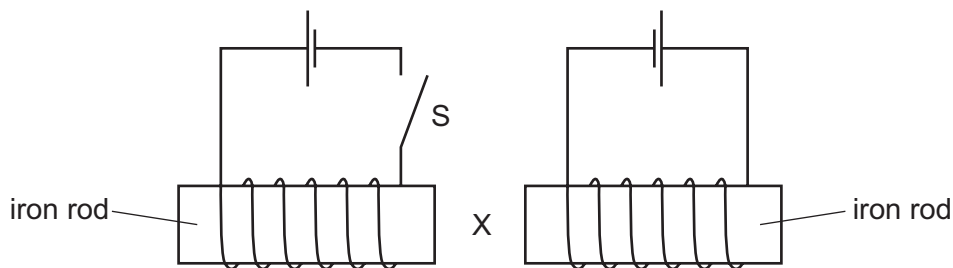
- 28 A train of steel nails and a train of iron nails hang from a strong magnet.



The trains are then carefully removed from the magnet.

What happens to the trains?

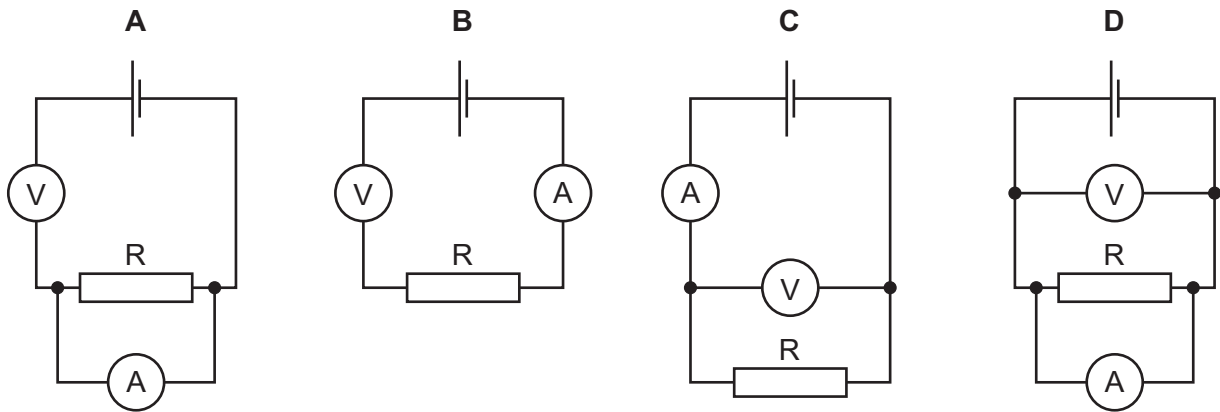
- A Both trains fall apart.  
 B Both trains stay together.  
 C Only the train of iron nails falls apart.  
 D Only the train of steel nails falls apart.
- 29 Two circuits are set up as shown. The iron rods are placed close together and are able to move.



What happens to the size of the gap at X when switch S is closed?

- A It decreases.  
 B It decreases then increases.  
 C It increases.  
 D It does not change.

30 Which circuit can be used to determine the resistance of resistor R?



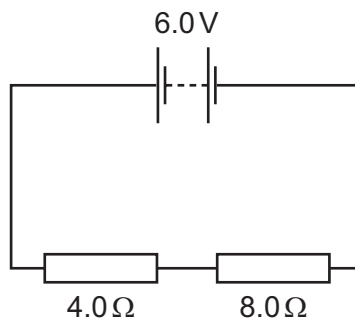
31 A wire has a resistance  $R$ .

A second wire made from the same metal has double the length and has double the diameter of the first wire.

What is the resistance of the second wire?

- A**  $\frac{R}{2}$       **B**  $R$       **C**  $4R$       **D**  $8R$

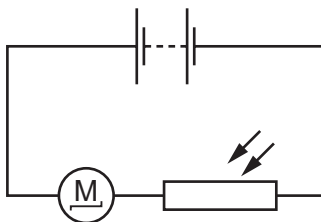
32 The circuit diagram shows a  $4.0\Omega$  resistor and an  $8.0\Omega$  resistor connected to a  $6.0\text{V}$  battery.



What is the potential difference (p.d.) across the  $4.0\Omega$  resistor?

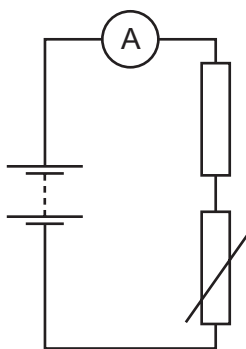
- A**  $0.5\text{V}$       **B**  $2.0\text{V}$       **C**  $4.0\text{V}$       **D**  $6.0\text{V}$

- 33 The diagram shows a motor and a light-dependent resistor (LDR) connected in a circuit.



Which change decreases the current in the motor so that it turns more slowly?

- A connecting a resistor in parallel with the LDR
  - B connecting a resistor in series with the LDR
  - C exchanging the positions of the motor and the LDR in the circuit
  - D increasing the brightness of the light falling on the LDR
- 34 The diagram shows a circuit with a fixed resistor connected in series with a thermistor and an ammeter.

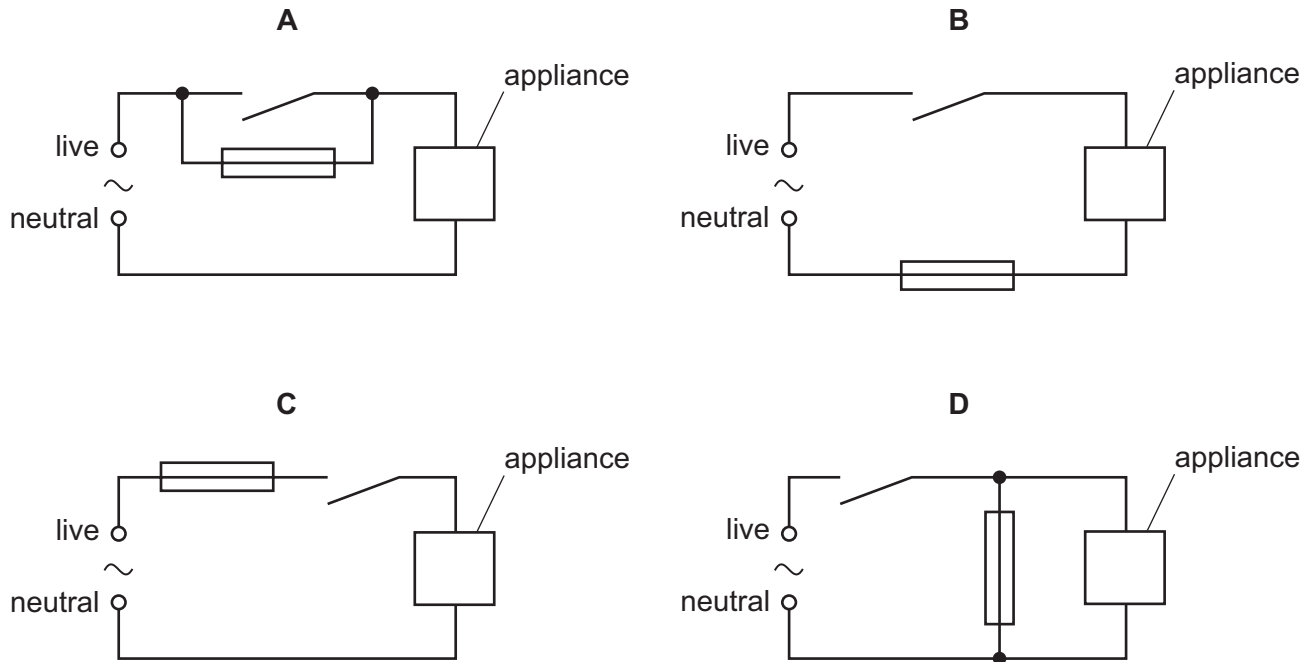


Which row shows how temperature change affects the resistance of the thermistor and the current in the circuit?

|   | temperature | resistance of thermistor | current in circuit |
|---|-------------|--------------------------|--------------------|
| A | decreases   | decreases                | increases          |
| B | decreases   | increases                | decreases          |
| C | increases   | decreases                | decreases          |
| D | increases   | increases                | increases          |

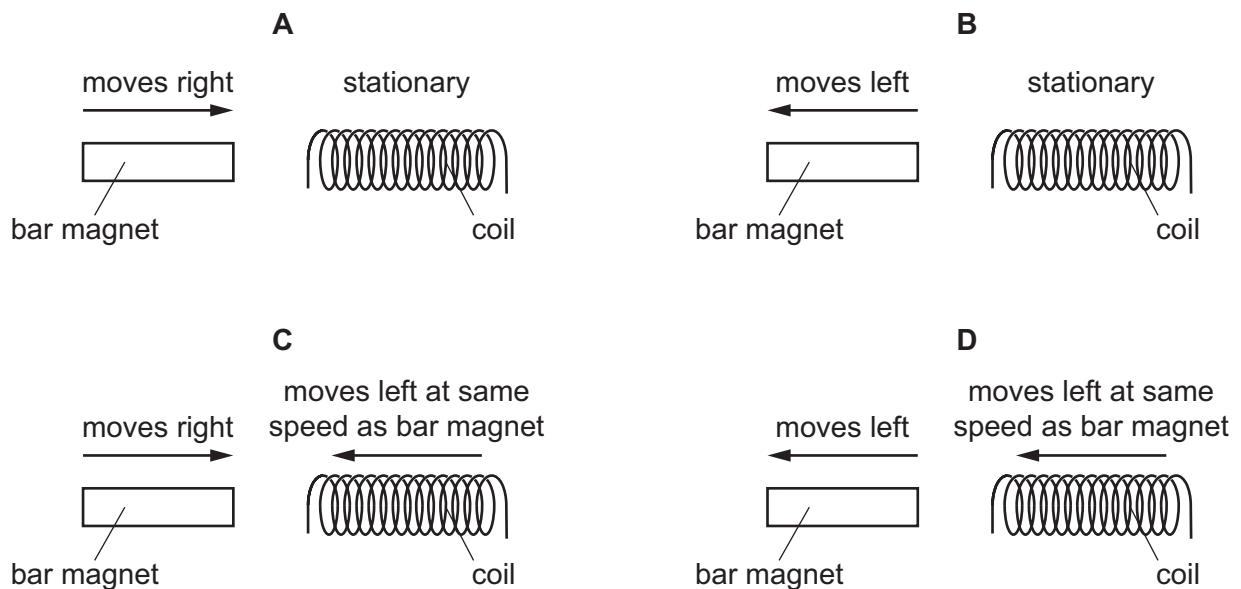
- 35** An appliance is connected to a mains supply. Its circuit also contains a switch and a fuse.

Which circuit shows the fuse in the correct position?



- 36** The diagram shows a bar magnet and a coil of wire. The bar magnet is moved at the same speed in each experiment.

In which situation is the largest electromotive force (e.m.f.) induced?





- 37** Diagram 1 shows a wire carrying an electric current. The wire is between the poles of a magnet. The current and the magnetic field produce a force on the wire which is upwards, as shown.

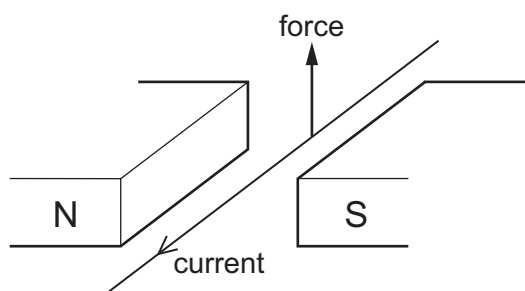


diagram 1

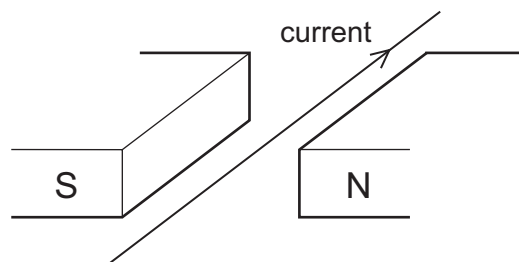


diagram 2

The magnetic poles and the current are now both reversed, as in diagram 2.

In which direction is the force on the wire in diagram 2?

- A** downwards
- B** towards the North pole of the magnet
- C** towards the South pole of the magnet
- D** upwards

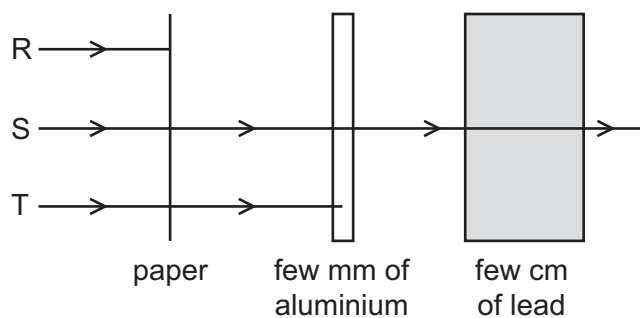
- 38** A nuclide of radon-222 is represented by  $^{222}_{86}\text{Rn}$ .

How many protons and how many neutrons are in a nucleus of this nuclide?

|          | protons | neutrons |
|----------|---------|----------|
| <b>A</b> | 86      | 136      |
| <b>B</b> | 86      | 222      |
| <b>C</b> | 222     | 86       |
| <b>D</b> | 222     | 136      |

**39** A radioactive source emits three types of radiation R, S and T.

The diagram shows an experiment set up to study the penetrating properties of R, S and T.



Which types of radiation are R, S and T?

|          | R                   | S                   | T                   |
|----------|---------------------|---------------------|---------------------|
| <b>A</b> | $\alpha$ -particles | $\beta$ -particles  | $\gamma$ -rays      |
| <b>B</b> | $\alpha$ -particles | $\gamma$ -rays      | $\beta$ -particles  |
| <b>C</b> | $\beta$ -particles  | $\alpha$ -particles | $\gamma$ -rays      |
| <b>D</b> | $\gamma$ -rays      | $\beta$ -particles  | $\alpha$ -particles |

**40** A radioactive source has a half-life of 0.5 hours.

A detector near the source shows a reading of 6000 counts per second.

Background radiation can be ignored.

What is the reading on the detector 1.5 hours later?

- A** 750 counts per second
- B** 1500 counts per second
- C** 2000 counts per second
- D** 3000 counts per second

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**Cambridge International Examinations**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

**0625/11**

Paper 1 Multiple Choice (Core)

**October/November 2017**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)

**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

**DO NOT WRITE IN ANY BARCODES.**

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall =  $10 \text{ m/s}^2$ ).

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **19** printed pages and **1** blank page.



- 1 A student measures the volume of a cork.

He puts some water into a measuring cylinder and then one glass ball. He puts the cork and then a second, identical glass ball into the water as shown.

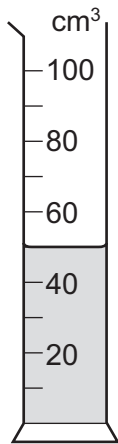


diagram 1

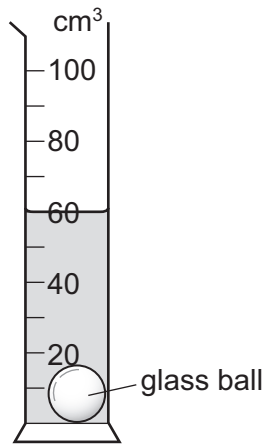


diagram 2

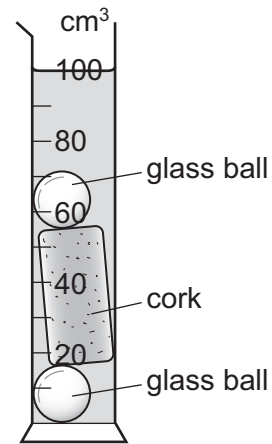


diagram 3

Diagram 1 shows the first water level.

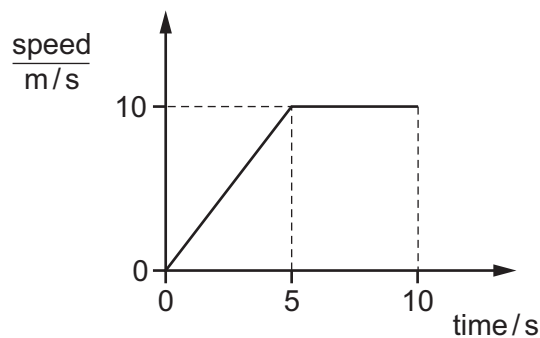
Diagram 2 shows the water level after one glass ball is added.

Diagram 3 shows the water level after the cork and the second glass ball are added.

What is the volume of the cork?

- A**  $30 \text{ cm}^3$       **B**  $40 \text{ cm}^3$       **C**  $50 \text{ cm}^3$       **D**  $100 \text{ cm}^3$

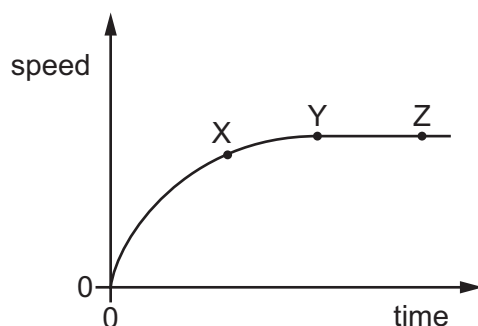
- 2 The graph shows how the speed of a car varies over a period of 10 s.



How far does the car travel during the 10 s?

- A** 10 m      **B** 50 m      **C** 75 m      **D** 100 m

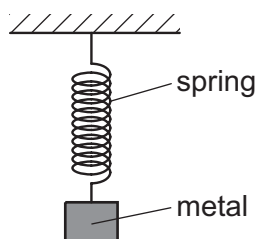
- 3 The diagram shows how the speed of a falling object changes with time.



Which row describes the motion of the object between X and Y, and between Y and Z?

|          | between X and Y | between Y and Z |
|----------|-----------------|-----------------|
| <b>A</b> | accelerating    | at rest         |
| <b>B</b> | accelerating    | constant speed  |
| <b>C</b> | decelerating    | at rest         |
| <b>D</b> | decelerating    | constant speed  |

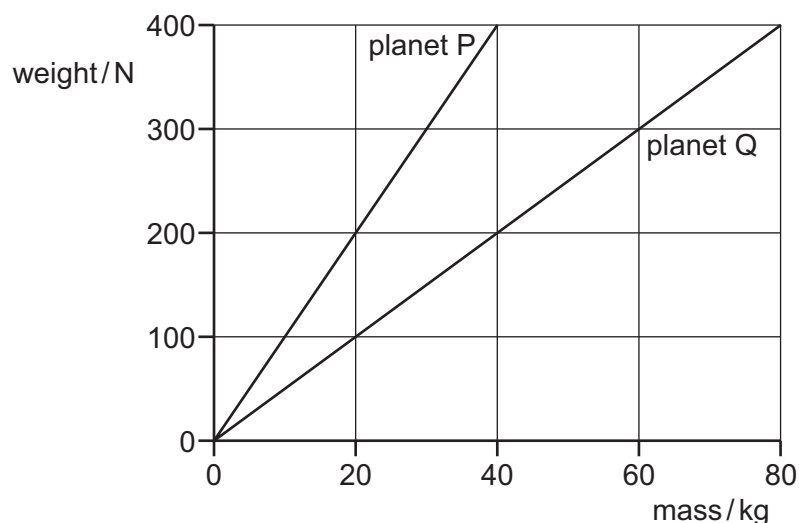
- 4 A spring is stretched by hanging a piece of metal from it.



Which name is given to the force that stretches the spring?

- A** friction
- B** mass
- C** pressure
- D** weight

- 5 The graph shows how weight varies with mass on planet P and on planet Q.



An object weighs 400 N on planet P. The object is taken to planet Q.

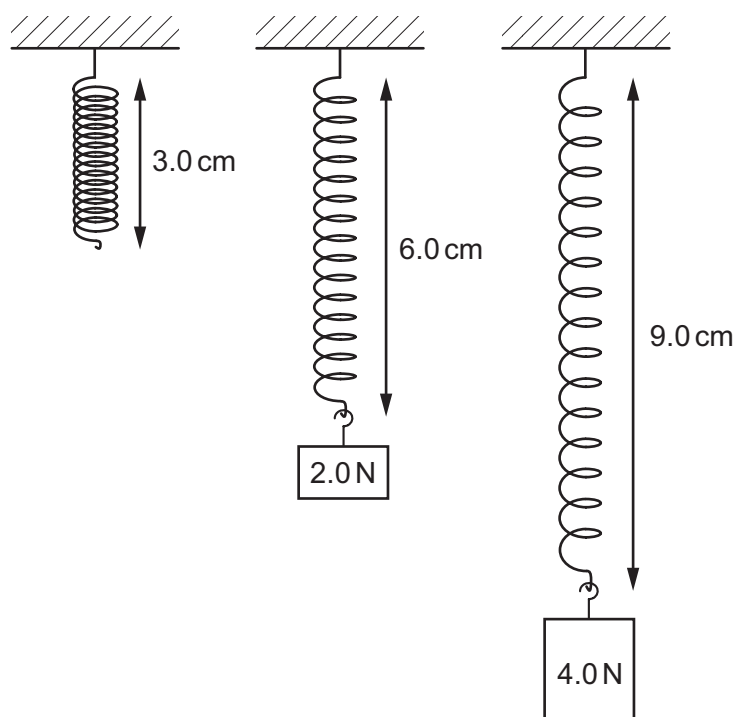
Which row is correct?

|          | mass of object<br>on planet Q / kg | weight of object<br>on planet Q / N |
|----------|------------------------------------|-------------------------------------|
| <b>A</b> | 40                                 | 200                                 |
| <b>B</b> | 40                                 | 400                                 |
| <b>C</b> | 80                                 | 200                                 |
| <b>D</b> | 80                                 | 400                                 |

- 6 What is needed to determine the density of a regularly shaped block?

- A** a balance and a beaker
- B** a balance and a ruler
- C** a measuring cylinder and a beaker
- D** a measuring cylinder and a ruler

- 7 A student conducts an experiment by hanging different loads on a spring. The diagrams show the lengths of the spring with different loads.



Which row gives the correct extensions of the spring?

|          | extension / cm |       |       |
|----------|----------------|-------|-------|
|          | 0 N            | 2.0 N | 4.0 N |
| <b>A</b> | 0              | 3.0   | 3.0   |
| <b>B</b> | 0              | 3.0   | 6.0   |
| <b>C</b> | 3.0            | 3.0   | 3.0   |
| <b>D</b> | 3.0            | 6.0   | 9.0   |

- 8 A car mechanic is trying to loosen a wheel nut. He applies a force to the end of a spanner (wrench) at right-angles to the spanner.

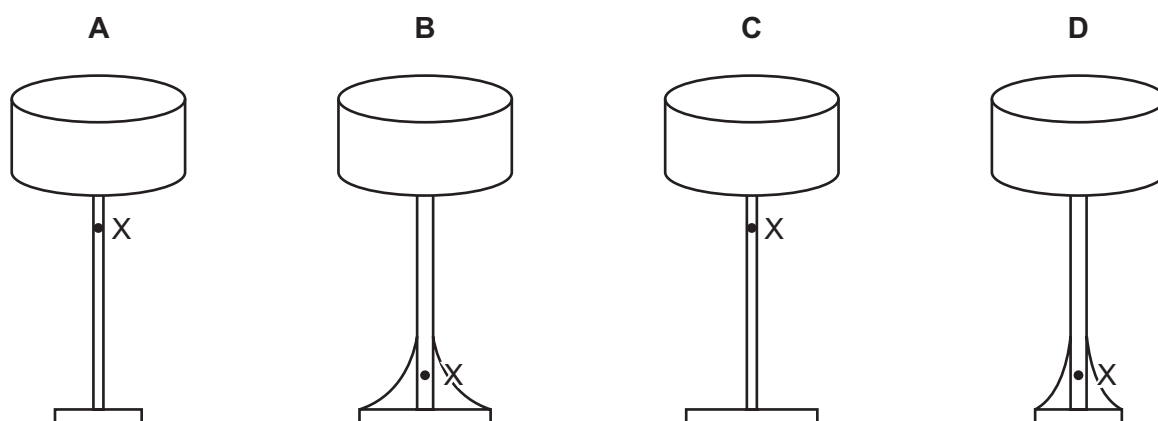
Which method provides the largest turning effect on the nut?

- A** applying double the force to a spanner that is half as long
- B** applying double the force to a spanner that is twice as long
- C** applying double the force to the same spanner
- D** applying the same force to a spanner that is twice as long

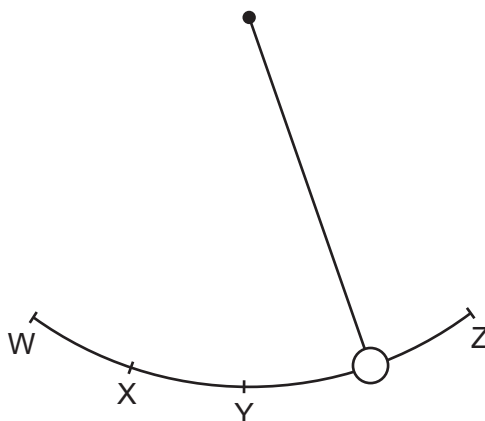


- 9 The diagrams show four table lamps resting on a table. The position of the centre of mass of each lamp is labelled X.

Which lamp is the most stable?



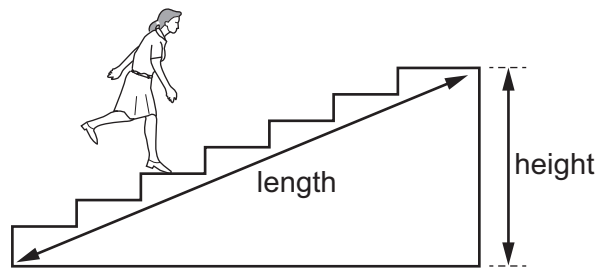
- 10 A pendulum bob swings from W, through points X and Y, to Z and then back to W.



Which statement is correct?

- A The gravitational potential energy of the bob is smallest at W.
- B The gravitational potential energy of the bob is smallest at X.
- C The gravitational potential energy of the bob is smallest at Y.
- D The gravitational potential energy of the bob is the same at all points of the swing.

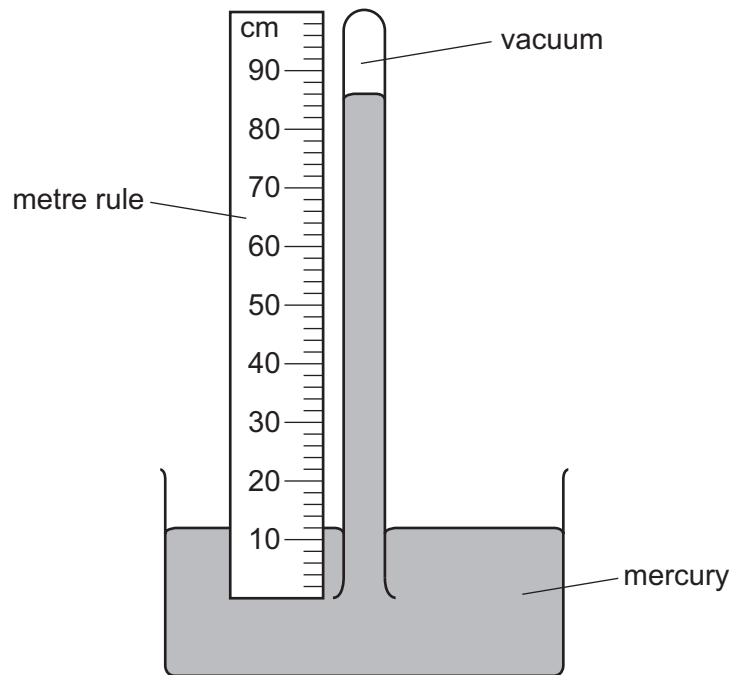
- 11 A student runs up a flight of stairs.



Which information is **not** needed to calculate the rate at which the student is doing work against gravity?

- A the height of the flight of stairs
  - B the length of the flight of stairs
  - C the time taken to run up the stairs
  - D the weight of the student
- 12 What is a manometer used to determine?
- A the difference between two densities
  - B the difference between two forces
  - C the difference between two heights
  - D the difference between two pressures

13 The diagram shows a simple mercury barometer.

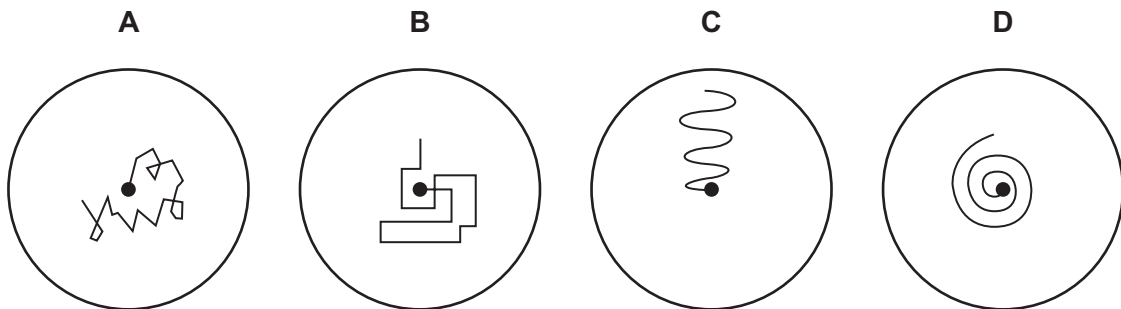


Which length is used to find the value of atmospheric pressure?

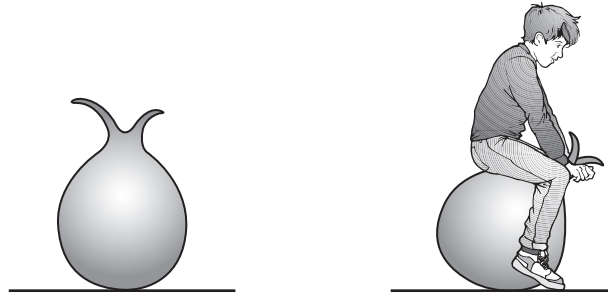
- A** 12 cm      **B** 74 cm      **C** 86 cm      **D** 100 cm

14 A pollen grain in a beaker of still water is viewed through a microscope.

Which diagram shows the most likely movement of the pollen grain?



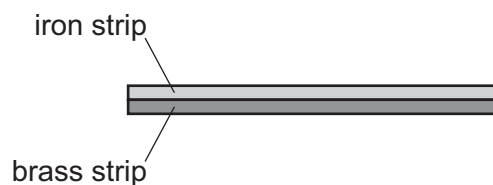
- 15** The diagram shows an air-filled rubber toy. A child sits on the toy and its volume decreases. The temperature of the air in the toy does not change.



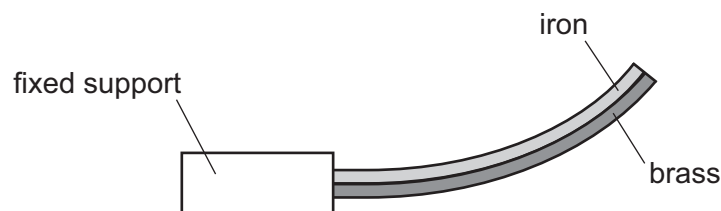
How does the air pressure in the toy change and why?

|          | pressure  | reason                                          |
|----------|-----------|-------------------------------------------------|
| <b>A</b> | decreases | air molecules move more slowly                  |
| <b>B</b> | decreases | air molecules strike the rubber less frequently |
| <b>C</b> | increases | air molecules move more quickly                 |
| <b>D</b> | increases | air molecules strike the rubber more frequently |

- 16** A strip of iron and a strip of brass are firmly attached to each other along their entire length. This combination is a bimetallic strip.



This bimetallic strip is heated and it bends as shown.

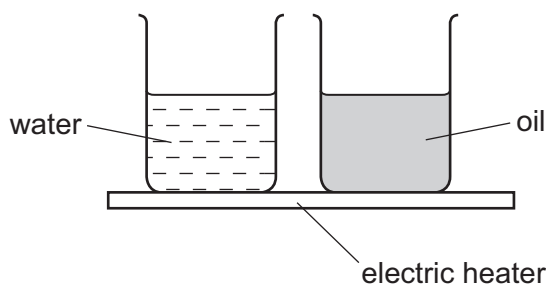


The bimetallic strip is now cooled and becomes straight again.

What causes the bimetallic strip to become straight again?

- A** The brass contracts more than the iron.
- B** The brass expands more than the iron.
- C** The iron contracts more than the brass.
- D** The iron expands more than the brass.

- 17 The diagram shows an electric heater being used to heat a beaker of water and an identical beaker containing oil. Both are heated for one minute.



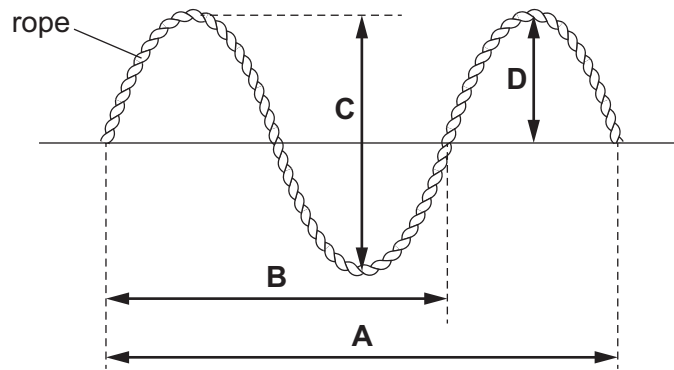
The temperature of the water and the temperature of the oil increase steadily. The increase in temperature of the oil is much greater than that of the water.

Why is this?

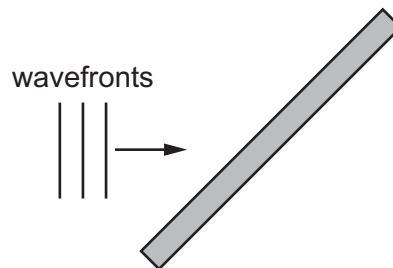
- A Oil has a higher boiling point than water.
  - B Oil has a lower boiling point than water.
  - C The oil has a larger thermal capacity than the water.
  - D The oil has a smaller thermal capacity than the water.
- 18 The air temperature in a cold room is  $15^{\circ}\text{C}$ .
- A heater that contains water at  $50^{\circ}\text{C}$  is used to heat the room.
- By which process is most of the thermal energy transferred throughout the air in the room?
- A conduction
  - B convection
  - C evaporation
  - D radiation
- 19 Which statement about convection currents is correct?
- A Convection currents occur because, when cooled, liquids contract and become more dense.
  - B Convection currents occur because, when warmed, liquids expand and become more dense.
  - C Convection currents only occur in liquids.
  - D Convection currents only occur in solids and liquids.

20 The diagram represents a wave travelling along a rope.

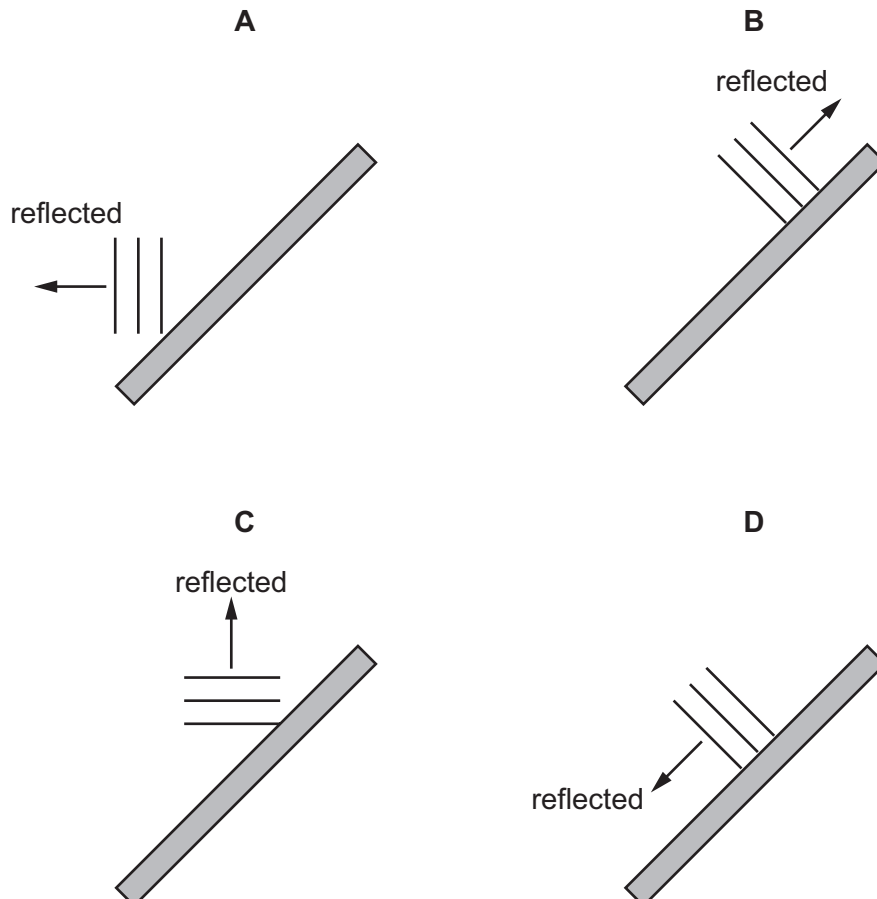
Which labelled arrow indicates the wavelength of the wave?



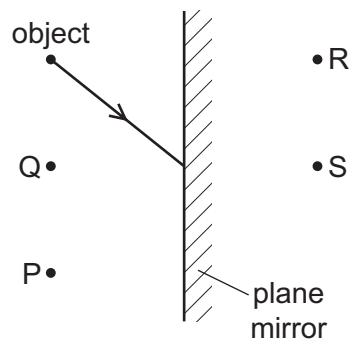
21 The diagram represents plane wavefronts of a water wave about to strike a solid barrier.



Which diagram shows the position of the wavefronts after reflection at the barrier?



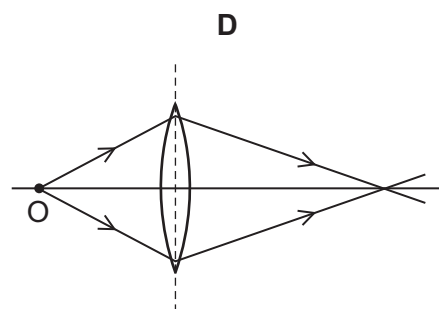
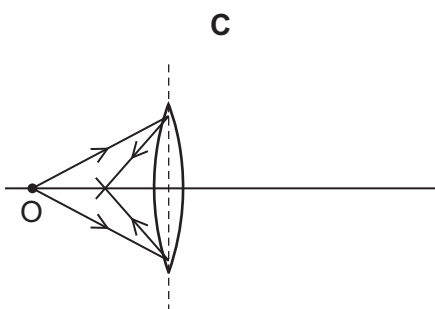
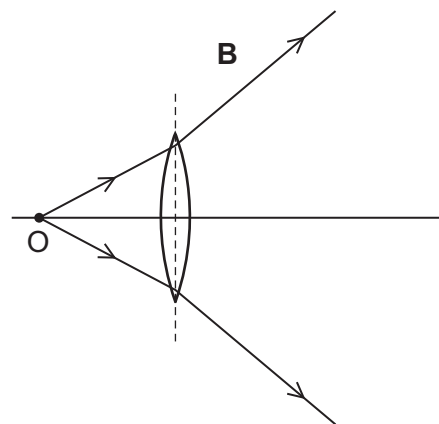
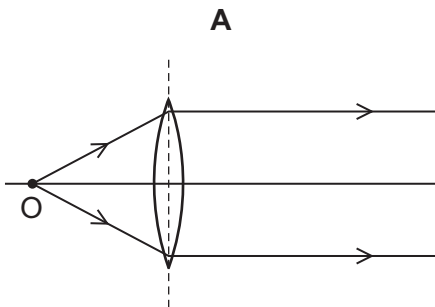
- 22 The diagram shows an object in front of a plane mirror. A ray of light from the object is incident on the mirror.



Through which point does the reflected ray pass, and at which point is the image of the object formed?

|          | point through which reflected ray passes | point at which image is formed |
|----------|------------------------------------------|--------------------------------|
| <b>A</b> | P                                        | R                              |
| <b>B</b> | P                                        | S                              |
| <b>C</b> | Q                                        | R                              |
| <b>D</b> | Q                                        | S                              |

- 23 Which ray diagram shows a converging lens forming a real image of a small object O?



- 24 Visible light, X-rays and microwaves are all components of the electromagnetic spectrum.

Which statement about the waves is correct?

- A In a vacuum, microwaves travel faster than visible light and have a shorter wavelength.
  - B In a vacuum, microwaves travel at the same speed as visible light and have a shorter wavelength.
  - C In a vacuum, X-rays travel faster than visible light and have a shorter wavelength.
  - D In a vacuum, X-rays travel at the same speed as visible light and have a shorter wavelength.
- 25 Space is a vacuum. Waves from stars are used to reveal information about the stars.

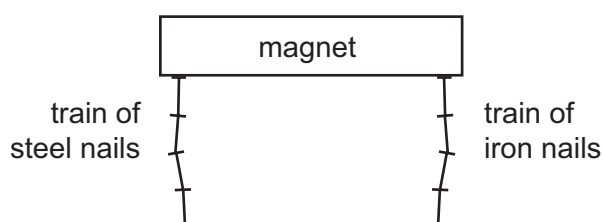
Which type of waves do **not** reveal information about stars?

- A infra-red
  - B radio waves
  - C ultrasound
  - D  $\gamma$ -rays
- 26 A student stands 180 m in front of a vertical, flat cliff and bangs together two pieces of wood to make a short, loud sound.

A timer records the echo of the sound 1.5 seconds after the pieces of wood are banged together.

Based on this result, what is the speed of sound?

- A 120 m/s
  - B 240 m/s
  - C 270 m/s
  - D 540 m/s
- 27 A train of steel nails and a train of iron nails hang from a strong magnet.



The trains are then carefully removed from the magnet.

What happens to the trains?

- A Both trains fall apart.
- B Both trains stay together.
- C Only the train of iron nails falls apart.
- D Only the train of steel nails falls apart.



- 28** The north pole of a bar magnet is placed in turn next to each end of a rod X. One end of the rod moves away from the north pole but the other end moves towards the north pole.

The experiment is repeated with the bar magnet and a different rod Y but this time both ends of the rod move towards the north pole.

From which materials could the rods X and Y be made?

|          | rod X     | rod Y     |
|----------|-----------|-----------|
| <b>A</b> | soft iron | copper    |
| <b>B</b> | soft iron | soft iron |
| <b>C</b> | steel     | copper    |
| <b>D</b> | steel     | soft iron |

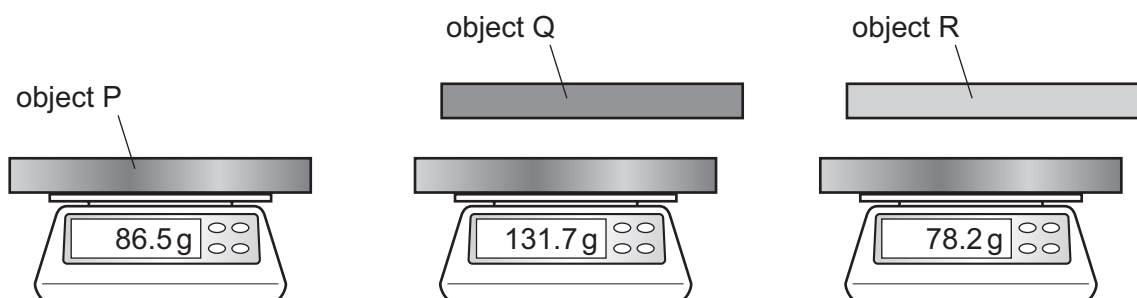
- 29** A student investigates the electrostatic charges on three objects P, Q and R.

The student places object P on the insulated pan of a balance and notes the reading on the balance.

The student then holds object Q a small distance above object P and notes the reading on the balance.

Finally, the student holds object R a small distance above object P and notes the reading on the balance.

The student's observations are shown.



Which row gives possible charges that explain these observations?

|          | charge on P | charge on Q | charge on R |
|----------|-------------|-------------|-------------|
| <b>A</b> | negative    | positive    | negative    |
| <b>B</b> | negative    | negative    | negative    |
| <b>C</b> | positive    | positive    | negative    |
| <b>D</b> | positive    | negative    | positive    |

**30** Which equation is used to calculate resistance?

- A** resistance = p.d. + current
- B** resistance = p.d. – current
- C** resistance = p.d.  $\times$  current
- D** resistance = p.d.  $\div$  current

**31** A student connects a circuit with a power supply, a lamp and an ammeter.

Where must the ammeter be connected?

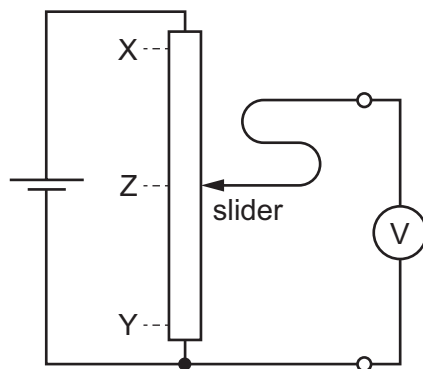
- A** in parallel with the lamp to measure current in the lamp
- B** in parallel with the lamp to measure potential difference across the lamp
- C** in series with the lamp to measure current in the lamp
- D** in series with the lamp to measure potential difference across the lamp

**32** Two  $3.0\,\Omega$  resistors are connected in parallel.

Which statement about their combined resistance is correct?

- A** The combined resistance is less than  $3.0\,\Omega$ .
- B** The combined resistance is  $3.0\,\Omega$ .
- C** The combined resistance is more than  $3.0\,\Omega$  but less than  $6.0\,\Omega$ .
- D** The combined resistance is  $6.0\,\Omega$ .

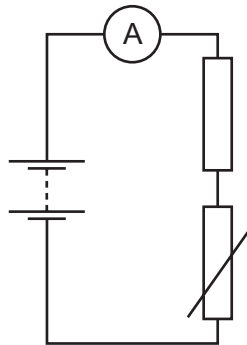
33 The diagram shows a potential divider circuit.



How does the movement of the slider affect the voltmeter reading?

|          | movement<br>of slider | voltmeter<br>reading |
|----------|-----------------------|----------------------|
| <b>A</b> | Z to X                | falls to zero        |
| <b>B</b> | Z to X                | increases            |
| <b>C</b> | Z to Y                | falls to zero        |
| <b>D</b> | Z to Y                | increases            |

- 34** The diagram shows a circuit with a fixed resistor connected in series with a thermistor and an ammeter.

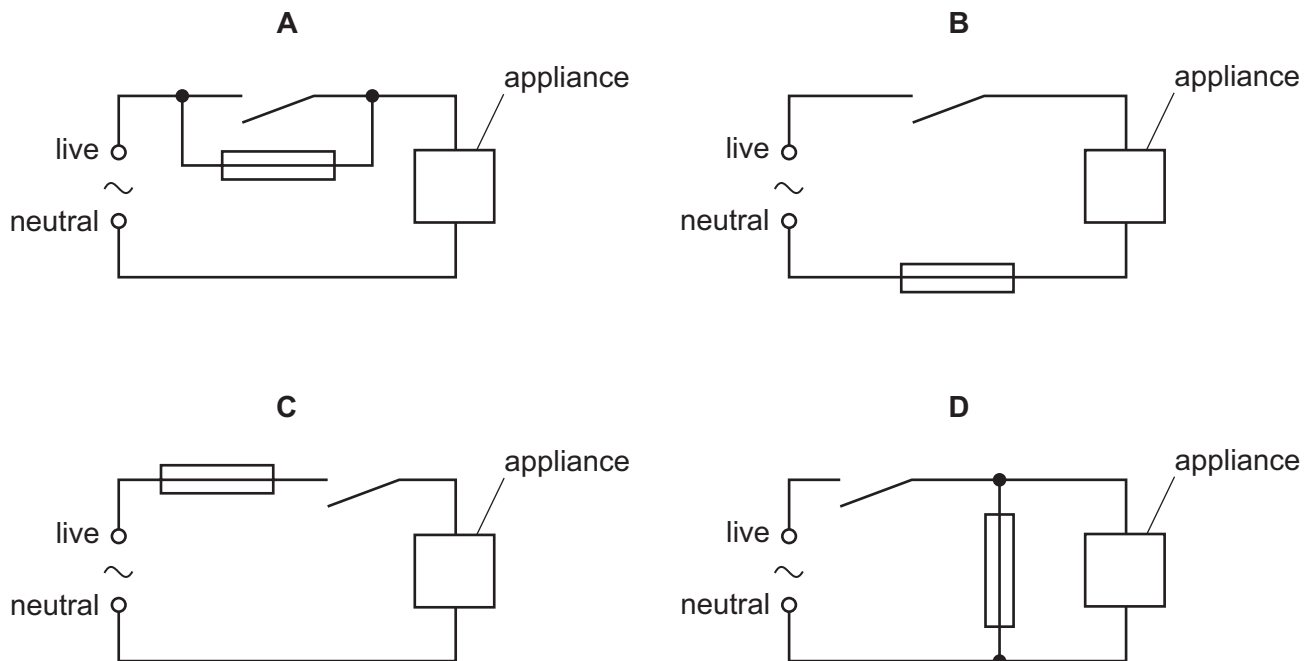


Which row shows how temperature change affects the resistance of the thermistor and the current in the circuit?

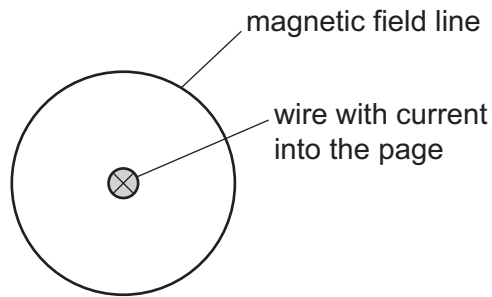
|          | temperature | resistance of thermistor | current in circuit |
|----------|-------------|--------------------------|--------------------|
| <b>A</b> | decreases   | decreases                | increases          |
| <b>B</b> | decreases   | increases                | decreases          |
| <b>C</b> | increases   | decreases                | decreases          |
| <b>D</b> | increases   | increases                | increases          |

- 35** An appliance is connected to a mains supply. Its circuit also contains a switch and a fuse.

Which circuit shows the fuse in the correct position?



- 36** There is an electric current in a straight wire in the direction into the page. This produces a magnetic field around the wire. All the field lines are circles but only one field line is shown.



Which row describes the magnetic field?

|          | direction of the field lines | spacing of the field lines               |
|----------|------------------------------|------------------------------------------|
| <b>A</b> | anti-clockwise               | evenly spaced over the whole field       |
| <b>B</b> | anti-clockwise               | more widely spaced further from the wire |
| <b>C</b> | clockwise                    | evenly spaced over the whole field       |
| <b>D</b> | clockwise                    | more widely spaced further from the wire |

- 37** A wire carrying a current is placed in a magnetic field. The wire experiences a force due to the field.

Which action does **not** change the direction of the force?

- A** increasing the current and reversing the field direction
- B** increasing the current, keeping the field direction constant
- C** reversing the current and increasing the field strength
- D** reversing the current, keeping the field direction constant

38 Different isotopes of chlorine are represented as  $^{35}_{17}\text{Cl}$  and  $^{37}_{17}\text{Cl}$ .

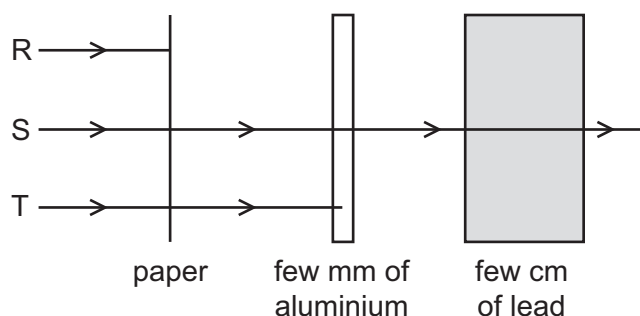
They behave the same way chemically.

Why is this?

- A They have the same nuclide notation.
- B They have the same number of neutrons.
- C They have the same number of nucleons.
- D They have the same number of protons.

39 A radioactive source emits three types of radiation R, S and T.

The diagram shows an experiment set up to study the penetrating properties of R, S and T.



Which types of radiation are R, S and T?

|   | R                   | S                   | T                   |
|---|---------------------|---------------------|---------------------|
| A | $\alpha$ -particles | $\beta$ -particles  | $\gamma$ -rays      |
| B | $\alpha$ -particles | $\gamma$ -rays      | $\beta$ -particles  |
| C | $\beta$ -particles  | $\alpha$ -particles | $\gamma$ -rays      |
| D | $\gamma$ -rays      | $\beta$ -particles  | $\alpha$ -particles |

40 What is meant by the *half-life* of a radioactive isotope?

- A half of the time taken for all of the original nuclei to decay
- B the time taken for half of the original nuclei to decay
- C the time taken for the charges on all the nuclei to halve
- D the time taken for the mass of each nucleus to halve

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**October/November 2017**

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| Question | Answer | Marks |
|----------|--------|-------|
| 1        | A      | 1     |
| 2        | B      | 1     |
| 3        | B      | 1     |
| 4        | D      | 1     |
| 5        | A      | 1     |
| 6        | B      | 1     |
| 7        | B      | 1     |
| 8        | B      | 1     |
| 9        | D      | 1     |
| 10       | C      | 1     |
| 11       | B      | 1     |
| 12       | B      | 1     |
| 13       | B      | 1     |
| 14       | A      | 1     |
| 15       | D      | 1     |
| 16       | A      | 1     |
| 17       | D      | 1     |
| 18       | C      | 1     |
| 19       | D      | 1     |
| 20       | C      | 1     |
| 21       | D      | 1     |
| 22       | C      | 1     |
| 23       | D      | 1     |
| 24       | D      | 1     |
| 25       | C      | 1     |
| 26       | D      | 1     |
| 27       | C      | 1     |
| 28       | B      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | D      | 1     |
| 30       | C      | 1     |
| 31       | A      | 1     |
| 32       | C      | 1     |
| 33       | B      | 1     |
| 34       | C      | 1     |
| 35       | D      | 1     |
| 36       | C      | 1     |
| 37       | A      | 1     |
| 38       | D      | 1     |
| 39       | B      | 1     |
| 40       | D      | 1     |



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**PHYSICS**

**0625/12**

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| 1        | B      | 1     |
| 2        | C      | 1     |
| 3        | D      | 1     |
| 4        | D      | 1     |
| 5        | A      | 1     |
| 6        | B      | 1     |
| 7        | A      | 1     |
| 8        | B      | 1     |
| 9        | C      | 1     |
| 10       | B      | 1     |
| 11       | B      | 1     |
| 12       | D      | 1     |
| 13       | B      | 1     |
| 14       | A      | 1     |
| 15       | D      | 1     |
| 16       | D      | 1     |
| 17       | D      | 1     |
| 18       | B      | 1     |
| 19       | C      | 1     |
| 20       | B      | 1     |
| 21       | A      | 1     |
| 22       | C      | 1     |
| 23       | D      | 1     |
| 24       | D      | 1     |
| 25       | A      | 1     |
| 26       | B      | 1     |
| 27       | C      | 1     |
| 28       | C      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | C      | 1     |
| 30       | C      | 1     |
| 31       | A      | 1     |
| 32       | B      | 1     |
| 33       | B      | 1     |
| 34       | B      | 1     |
| 35       | C      | 1     |
| 36       | C      | 1     |
| 37       | D      | 1     |
| 38       | A      | 1     |
| 39       | B      | 1     |
| 40       | A      | 1     |



**Cambridge Assessment International Education**  
Cambridge International General Certificate of Secondary Education

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**PHYSICS**

**0625/11**

Paper 1 Multiple Choice (Core)

**October/November 2017**

MARK SCHEME

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

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This document consists of **3** printed pages.

| Question | Answer | Marks |
|----------|--------|-------|
| 1        | A      | 1     |
| 2        | C      | 1     |
| 3        | B      | 1     |
| 4        | D      | 1     |
| 5        | A      | 1     |
| 6        | B      | 1     |
| 7        | B      | 1     |
| 8        | B      | 1     |
| 9        | B      | 1     |
| 10       | C      | 1     |
| 11       | B      | 1     |
| 12       | D      | 1     |
| 13       | B      | 1     |
| 14       | A      | 1     |
| 15       | D      | 1     |
| 16       | A      | 1     |
| 17       | D      | 1     |
| 18       | B      | 1     |
| 19       | A      | 1     |
| 20       | B      | 1     |
| 21       | C      | 1     |
| 22       | A      | 1     |
| 23       | D      | 1     |
| 24       | D      | 1     |
| 25       | C      | 1     |
| 26       | B      | 1     |
| 27       | C      | 1     |
| 28       | D      | 1     |

| Question | Answer | Marks |
|----------|--------|-------|
| 29       | C      | 1     |
| 30       | D      | 1     |
| 31       | C      | 1     |
| 32       | A      | 1     |
| 33       | B      | 1     |
| 34       | B      | 1     |
| 35       | C      | 1     |
| 36       | D      | 1     |
| 37       | B      | 1     |
| 38       | D      | 1     |
| 39       | B      | 1     |
| 40       | B      | 1     |





**Cambridge International Examinations**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

Paper 1 Multiple Choice (Core)

**0625/13**

**October/November 2016**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)

**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

**DO NOT WRITE IN ANY BARCODES.**

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall =  $10 \text{ m/s}^2$ ).

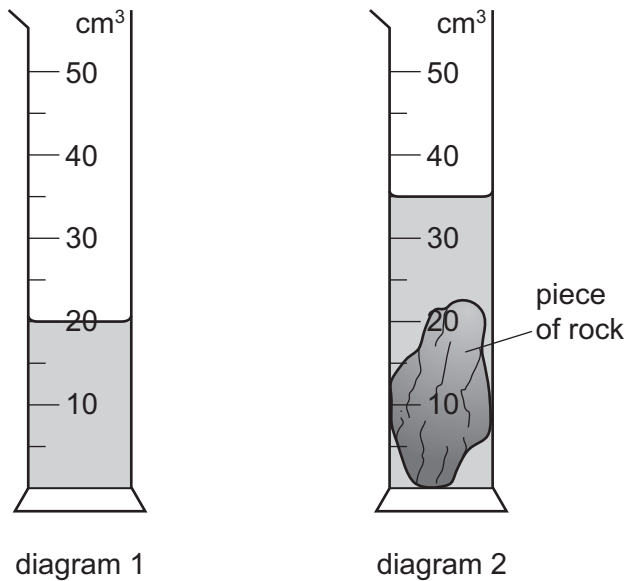
The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **19** printed pages and **1** blank page.



- 1 Diagram 1 shows a measuring cylinder containing some water.

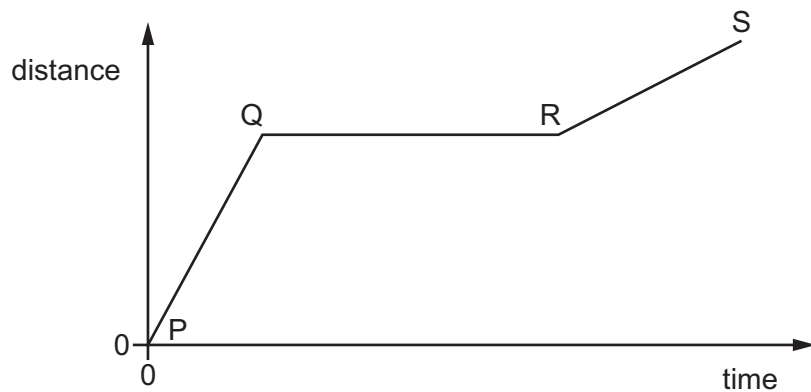
Diagram 2 shows the same measuring cylinder after a piece of rock has been slowly submerged in the water.



What is the volume of the piece of rock?

- A** 15 cm<sup>3</sup>      **B** 20 cm<sup>3</sup>      **C** 23 cm<sup>3</sup>      **D** 35 cm<sup>3</sup>

- 2 The graph shows how the distance travelled by a vehicle changes with time.

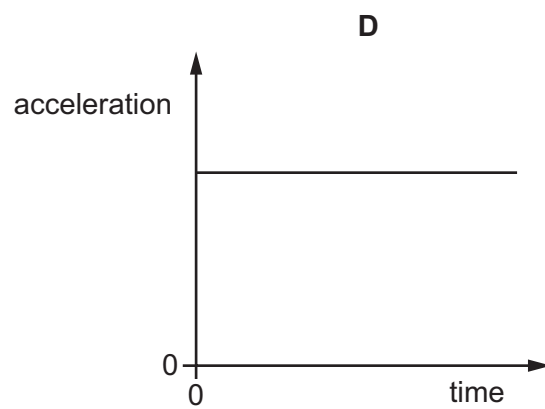
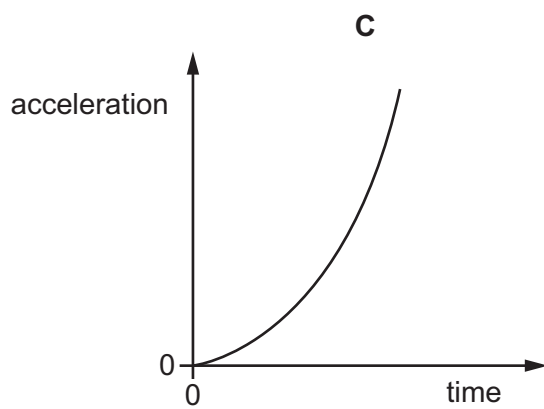
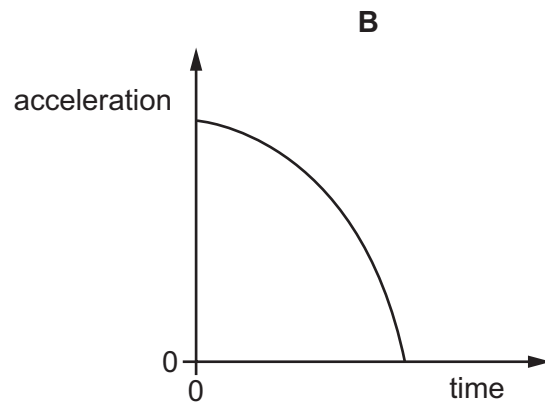
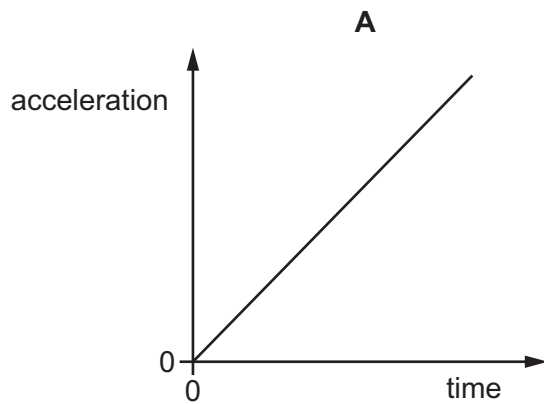


Which row describes the speed of the vehicle in each section of the graph?

|          | P to Q     | Q to R   | R to S     |
|----------|------------|----------|------------|
| <b>A</b> | constant   | zero     | constant   |
| <b>B</b> | constant   | zero     | decreasing |
| <b>C</b> | increasing | constant | decreasing |
| <b>D</b> | increasing | zero     | constant   |

- 3 A stone falls freely from the top of a cliff. Air resistance may be ignored.

Which graph shows how the acceleration of the stone varies with time as it falls?



- 4 Which statement about mass or weight is **not** correct?

- A** Masses can be compared using a balance.
- B** Mass is a force.
- C** Weights can be compared using a balance.
- D** Weight is a force.

- 5 The diagrams show an empty container, and the same container filled with liquid.

The empty container has a mass of 120 g. When filled with the liquid, the total mass of the container and the liquid is 600 g.



empty container  
120 g



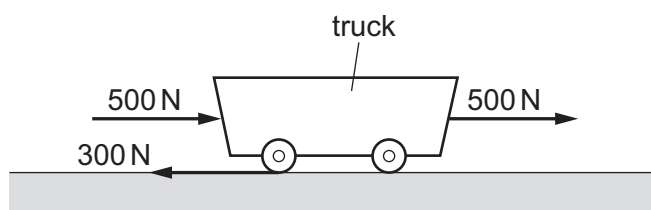
container filled with liquid  
600 g

The volume of liquid in the container is  $600 \text{ cm}^3$ .

What is the density of the liquid?

- A**  $0.020 \text{ g/cm}^3$    **B**  $0.80 \text{ g/cm}^3$    **C**  $1.0 \text{ g/cm}^3$    **D**  $1.2 \text{ g/cm}^3$

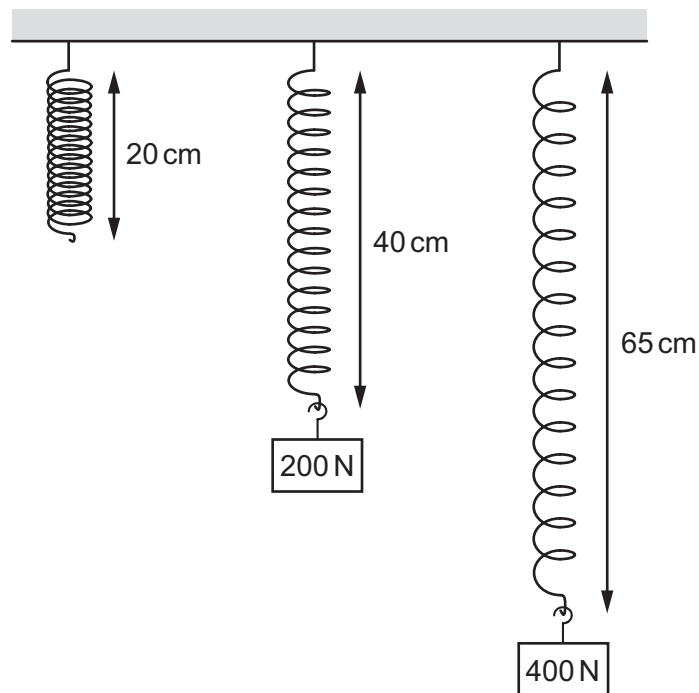
- 6 The diagram shows all the horizontal forces acting on a truck.



What is the size of the resultant horizontal force on the truck?

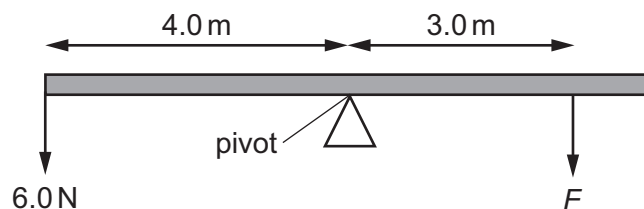
- A** 200 N   **B** 300 N   **C** 700 N   **D** 1000 N

- 7 Different loads are hung on a spring. The diagram shows the length of the spring with and without the loads attached.



What is the extension of the spring when the load is 400 N?

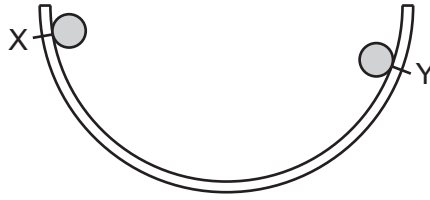
- A** 5 cm      **B** 25 cm      **C** 40 cm      **D** 45 cm
- 8 A uniform bar is pivoted at its centre.



What force  $F$  is needed to balance the bar?

- A** 2.0 N      **B** 4.5 N      **C** 6.0 N      **D** 8.0 N

- 9 A ball is released from point X on a rough semicircular track. It comes to rest for an instant at point Y. The gravitational potential energy of the ball at point X is  $E_X$ . The gravitational potential energy of the ball at point Y is  $E_Y$ .

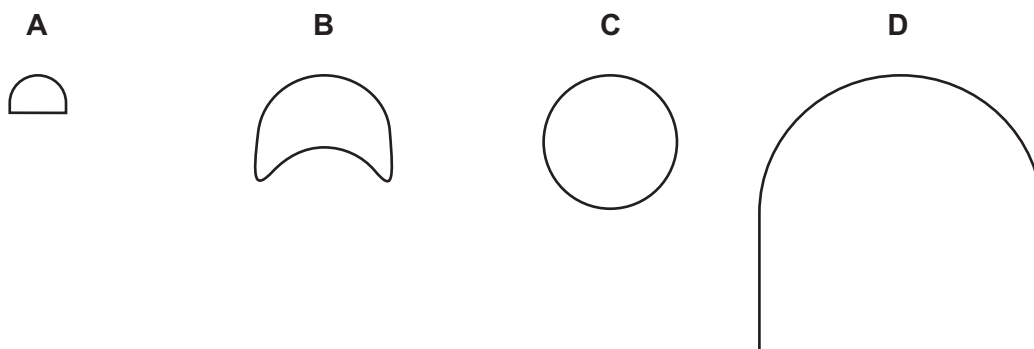


Which equation relates the energy of the ball at point X to its energy at point Y?

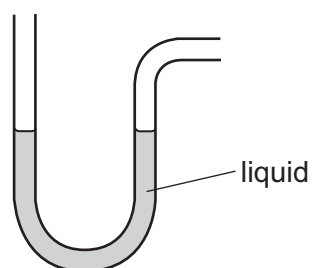
- A  $E_X = E_Y + \text{work done against friction}$
  - B  $E_X = E_Y - \text{work done against friction}$
  - C  $E_X + E_Y = \text{work done against friction}$
  - D  $E_X + E_Y + \text{work done against friction} = 0$
- 10 Two cars P and Q have different engines.
- The input power from the fuel is the same for both engines. The engine in car P is more efficient than the engine in car Q.
- Which statement about the cars or their engines **must** be true?
- A Car P has a higher maximum speed than car Q.
  - B Car P has a lower maximum speed than car Q.
  - C The engine in car P uses less energy per hour than the engine in car Q.
  - D The engine in car P wastes less energy per hour than the engine in car Q.
- 11 A child runs up a set of stairs four times. The time taken for each run is recorded.
- Which time is measured when the child's useful power is greatest?
- A 10 s
  - B 20 s
  - C 30 s
  - D 40 s

- 12 The diagrams show the actual sizes of the heels of four different shoes, as seen from underneath the shoe.

Which heel is **most** likely to cause damage to wooden floors?

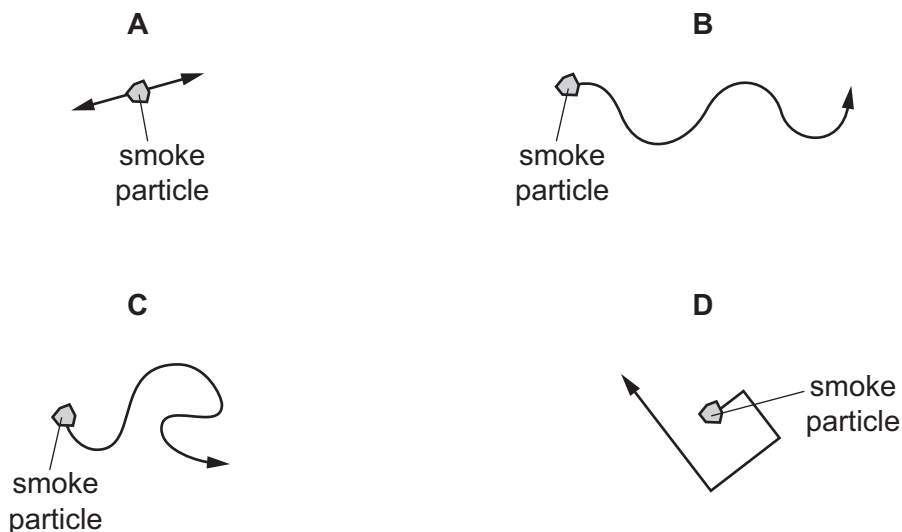


- 13 The diagram shows an instrument used to measure gas pressure.



What is the instrument called?

- A ammeter
  - B barometer
  - C manometer
  - D thermometer
- 14 Which diagram best represents the movement of a smoke particle displaying Brownian motion?



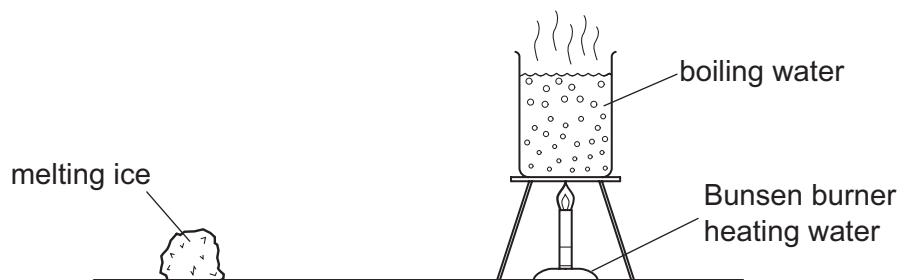
- 15** A cylinder containing trapped air is put into a freezer.



As the air cools, the piston moves to the left as shown.

How has the air changed?

- A** Its density is less.
  - B** Its mass is less.
  - C** The average speed of its molecules is less.
  - D** The average volume of its molecules is less.
- 16** A piece of melting ice at  $0^{\circ}\text{C}$  and a beaker of boiling water are both in a laboratory. The laboratory is at  $20^{\circ}\text{C}$ .



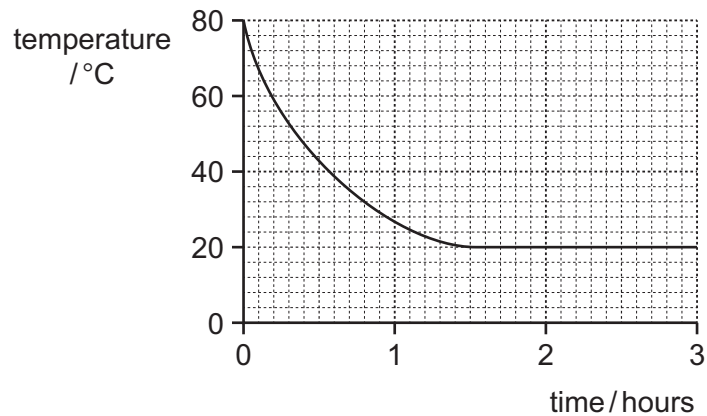
What is happening to the temperature of the melting ice and what is happening to the temperature of the boiling water?

|          | temperature of melting ice | temperature of boiling water |
|----------|----------------------------|------------------------------|
| <b>A</b> | constant                   | constant                     |
| <b>B</b> | constant                   | increasing                   |
| <b>C</b> | increasing                 | constant                     |
| <b>D</b> | increasing                 | increasing                   |



- 17** A liquid-in-glass thermometer works because the liquid shows a change in
- A** colour.
  - B** mass.
  - C** thermal capacity.
  - D** volume.
- 18** Which description applies to infra-red radiation?
- A** longitudinal and electromagnetic
  - B** longitudinal but not electromagnetic
  - C** transverse and electromagnetic
  - D** transverse but not electromagnetic

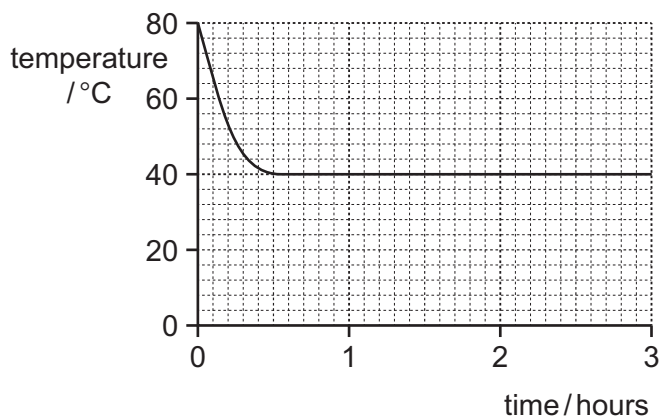
- 19 The diagram shows the temperature-time graph for hot water that is cooling in an uninsulated beaker.



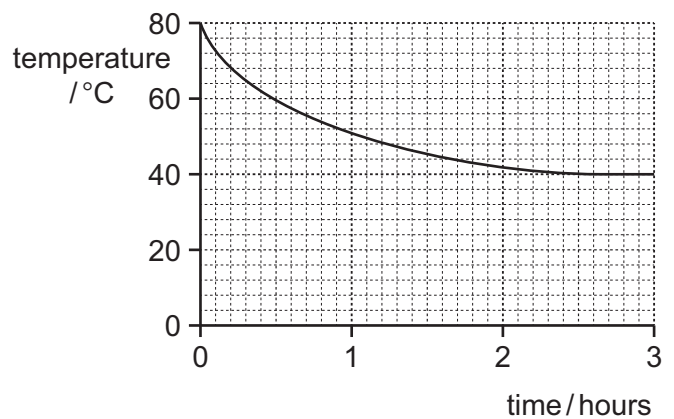
The beaker is now insulated. The same volume of hot water, at the same initial temperature as before, is put into the insulated beaker. The beaker is in the same room as before.

Which is the new temperature-time graph for the water as it cools?

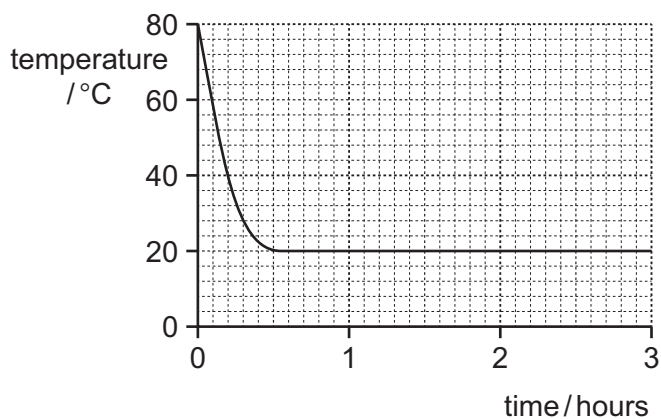
**A**



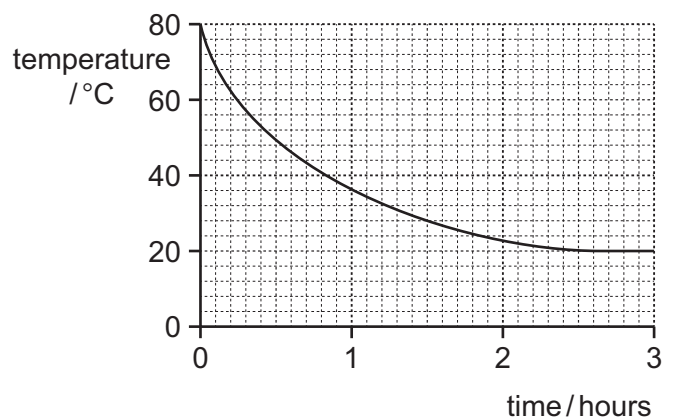
**B**



**C**

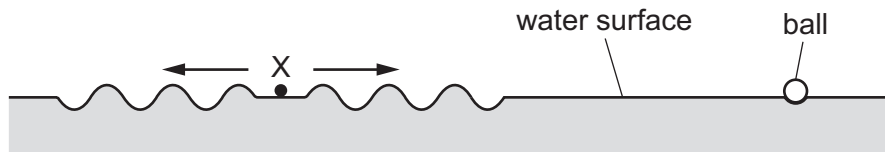


**D**



- 20** A small plastic ball floats on the surface of deep water. A stone is dropped into the water at point X.

A wave spreads out across the water surface, causing the ball to move.

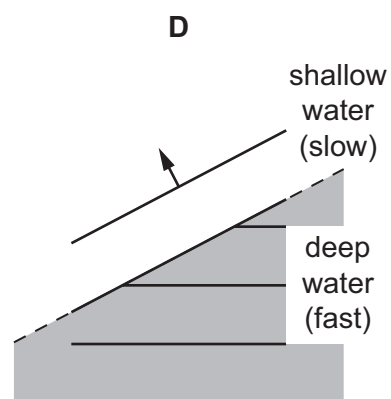
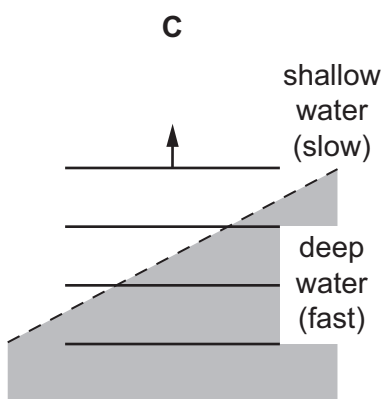
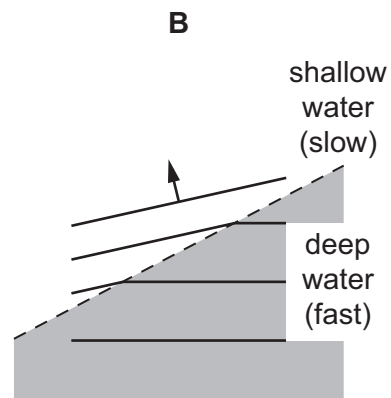
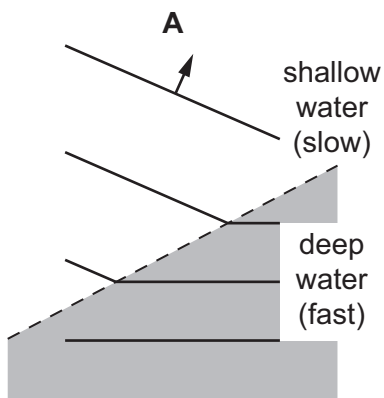


Which type of wave is produced by the stone, and in which direction does the ball move?

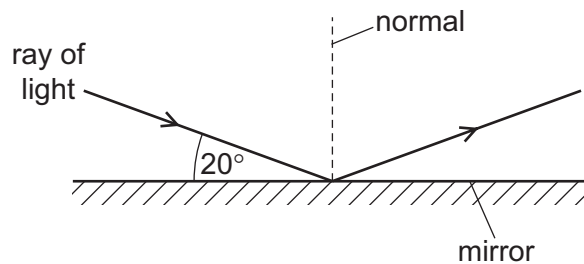
|          | type of wave | direction of movement of ball |
|----------|--------------|-------------------------------|
| <b>A</b> | longitudinal | left and right                |
| <b>B</b> | longitudinal | up and down                   |
| <b>C</b> | transverse   | left and right                |
| <b>D</b> | transverse   | up and down                   |

- 21** Waves travel more slowly on the surface of shallow water than on the surface of deep water.

Which diagram shows a wave travelling from deep water into shallower water?



- 22 Light strikes a plane mirror and is reflected. The angle between the ray of light and the mirror is  $20^\circ$ .



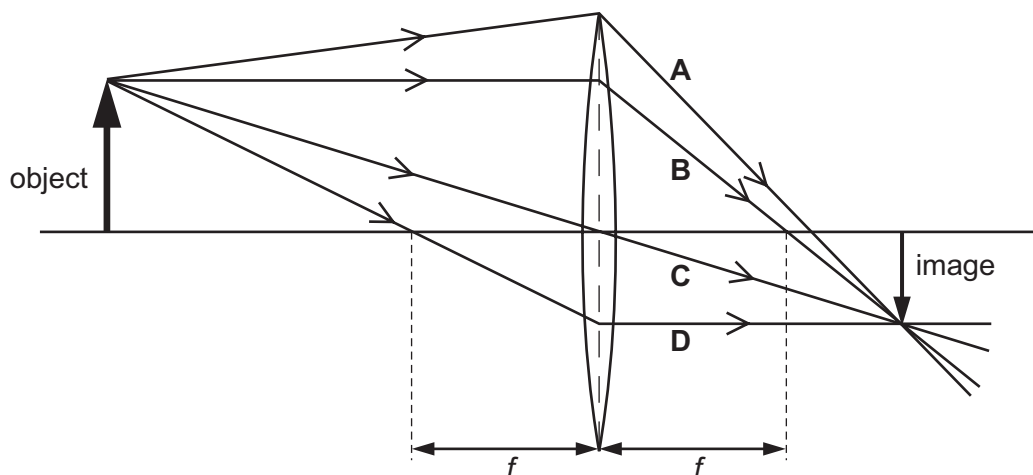
What is the size of the angle of reflection?

- A  $20^\circ$                       B  $70^\circ$                       C  $140^\circ$                       D  $160^\circ$
- 23 A student draws a ray diagram to show how a converging lens, of focal length  $f$ , forms an image of an object.

He draws three rays to locate the image.

He then adds a fourth ray after the position of the image is known.

Which of the labelled rays must have been drawn after the position of the image was known?



- 24 Which group shows four electromagnetic waves in order, from the largest wavelength to the smallest?
- A microwaves, radio waves, X-rays, visible light
- B radio waves, microwaves, visible light, X-rays
- C visible light, X-rays, radio waves, microwaves
- D X-rays, visible light, microwaves, radio waves

- 25** A girl notices that, when she shouts into a cave, she hears an echo.

Which wave property causes the echo?

- A** diffraction
- B** dispersion
- C** reflection
- D** refraction

- 26** A man stands 110m from a high wall. He makes a short, sharp sound and then hears an echo from the wall.

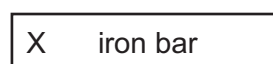
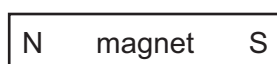
The speed of sound in air is 330 m/s.

How long after making the sound does the man hear the echo?

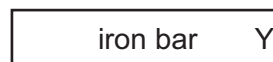
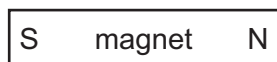
- A** 0.33 s      **B** 0.67 s      **C** 1.5 s      **D** 3.0 s

- 27** In two separate experiments, a magnet is brought near to an unmagnetised iron bar. This causes the bar to become magnetised.

experiment 1



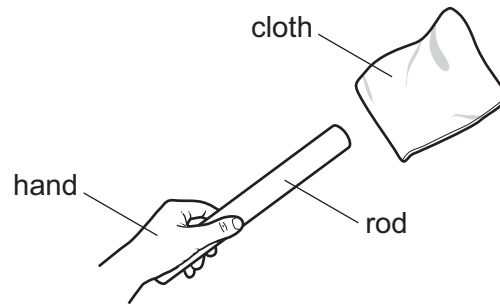
experiment 2



Which magnetic pole is induced at X and at Y?

|          | pole induced at X | pole induced at Y |
|----------|-------------------|-------------------|
| <b>A</b> | N                 | N                 |
| <b>B</b> | N                 | S                 |
| <b>C</b> | S                 | N                 |
| <b>D</b> | S                 | S                 |

28 A student holds a rod in his hand.

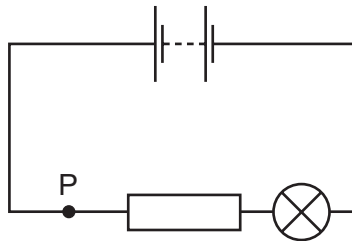


He rubs the rod with a cloth. The rod gains a positive charge.

Of which material could the rod be made, and which transfer of charge has happened?

|          | material of rod | transfer of charge                |
|----------|-----------------|-----------------------------------|
| <b>A</b> | metal           | negative charge from rod to cloth |
| <b>B</b> | metal           | positive charge from cloth to rod |
| <b>C</b> | plastic         | negative charge from rod to cloth |
| <b>D</b> | plastic         | positive charge from cloth to rod |

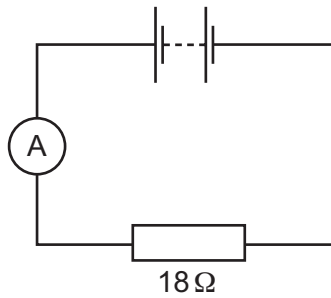
29 The diagram shows a lamp in a circuit.



Which change to the circuit would increase the current in the lamp?

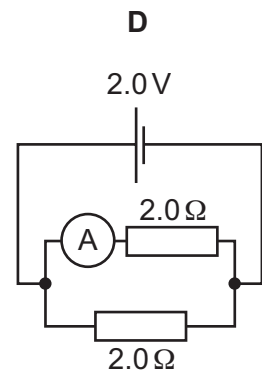
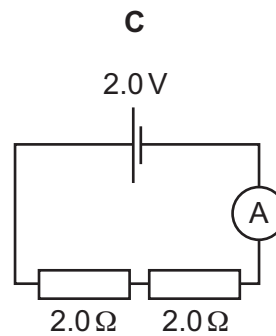
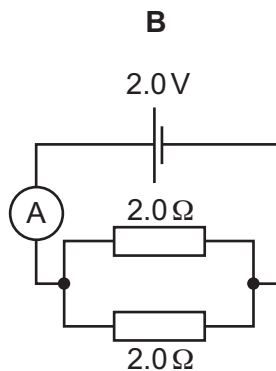
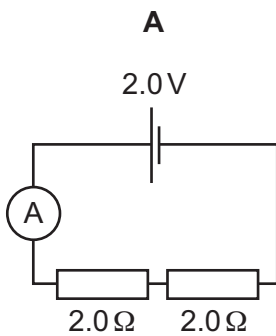
- A** adding another resistor in parallel with the one in the circuit
- B** adding another resistor in series with the one in the circuit
- C** decreasing the electromotive force (e.m.f.) of the battery in the circuit
- D** moving the lamp to point P in the circuit

- 30** An ammeter and an  $18\Omega$  resistor are connected in series with a battery. The reading on the ammeter is  $0.50\text{ A}$ .



What is the electromotive force (e.m.f.) of the battery?

- A**  $9.0\text{ N}$       **B**  $9.0\text{ V}$       **C**  $36\text{ N}$       **D**  $36\text{ V}$
- 31** In which circuit will the ammeter reading be greatest?

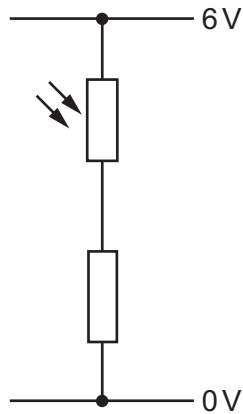


- 32** An electric washing machine is protected by an earth wire.

To which part of the washing machine should the earth wire be connected?

- A** the fuse  
**B** the live wire  
**C** the metal case  
**D** the switch

33 The diagram shows a potential divider.



When brighter light falls on the light-dependent resistor (LDR), its resistance changes.

What happens to the resistance of the LDR and what happens to the current in it?

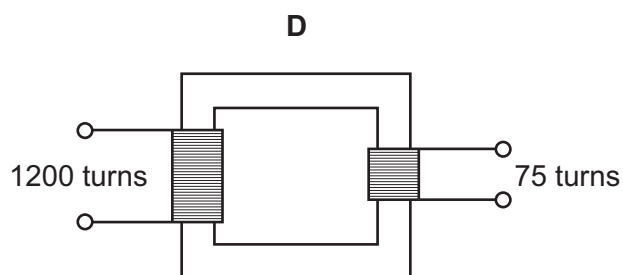
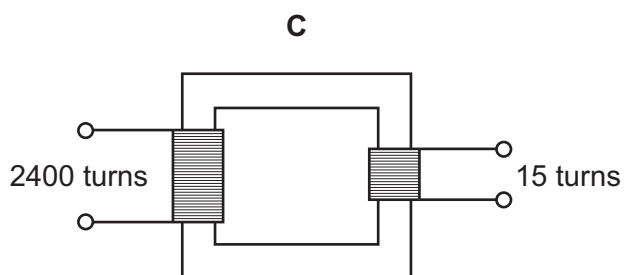
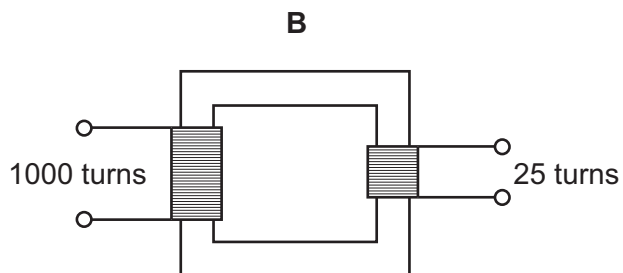
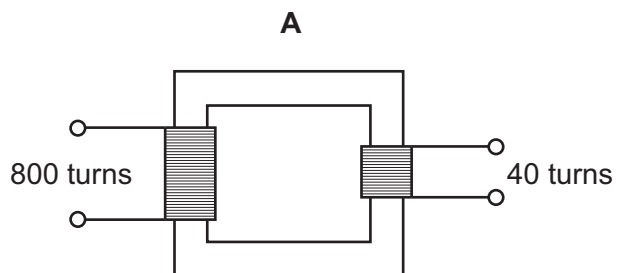
|          | resistance of LDR | current in LDR |
|----------|-------------------|----------------|
| <b>A</b> | decreases         | decreases      |
| <b>B</b> | decreases         | increases      |
| <b>C</b> | increases         | decreases      |
| <b>D</b> | increases         | increases      |

34 Why is a fuse used in an electrical circuit?

- A** so that the current can have only one value
- B** to prevent the current becoming too large
- C** to provide a path to earth if a fault occurs
- D** to save electrical energy



**35** Which transformer can change a 240 V a.c. input into a 15 V a.c. output?



**36** An electric current can produce a heating effect and a magnetic effect.

Which row shows the effect that a relay uses and one application of a relay?

|          | effect used by a relay | one application of a relay                            |
|----------|------------------------|-------------------------------------------------------|
| <b>A</b> | heating effect         | allowing a small current to switch on a large current |
| <b>B</b> | heating effect         | changing the voltage of an a.c. supply                |
| <b>C</b> | magnetic effect        | allowing a small current to switch on a large current |
| <b>D</b> | magnetic effect        | changing the voltage of an a.c. supply                |

**37** A nuclide of sodium can be represented by  ${}_{11}^{23}\text{Na}$ .

Which row gives the numbers of particles in a neutral sodium atom?

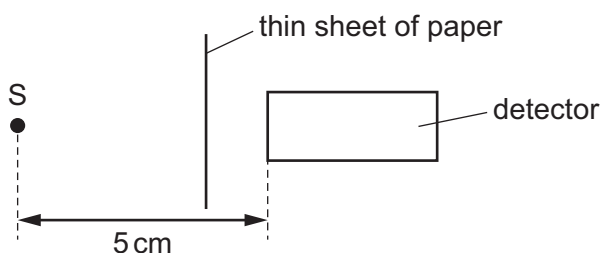
|          | number of electrons | number of protons | number of neutrons |
|----------|---------------------|-------------------|--------------------|
| <b>A</b> | 11                  | 11                | 12                 |
| <b>B</b> | 12                  | 11                | 23                 |
| <b>C</b> | 12                  | 12                | 11                 |
| <b>D</b> | 23                  | 23                | 11                 |

38 Below are four statements about different isotopes of a certain element.

Which statement **must** be true?

- A They have a different number of electrons.
- B They have the same number of neutrons.
- C They have the same number of nucleons.
- D They have the same number of protons.

39 Radioactive source S emits  $\alpha$ -particles,  $\beta$ -particles and  $\gamma$ -rays. A detector is placed 5 cm away from S. A thin sheet of paper is placed as shown in the diagram.



Which emissions from the source can be detected?

- A  $\alpha$ -particles and  $\beta$ -particles only
- B  $\alpha$ -particles and  $\gamma$ -rays only
- C  $\beta$ -particles and  $\gamma$ -rays only
- D  $\alpha$ -particles,  $\beta$ -particles and  $\gamma$ -rays

**40** A radioactive material emits  $\alpha$ -particles.

The table shows the rate of emission of  $\alpha$ -particles from the material at different times.

| time<br>/ hours | rate of emission<br>/ particles per minute |
|-----------------|--------------------------------------------|
| 0.0             | 160                                        |
| 4.0             | 40                                         |
| 8.0             |                                            |

What is the half-life of the material and what is the missing value at 8.0 hours?

|          | half-life<br>/ hours | rate of emission at 8.0 hours<br>/ particles per minute |
|----------|----------------------|---------------------------------------------------------|
| <b>A</b> | 2.0                  | 0                                                       |
| <b>B</b> | 2.0                  | 10                                                      |
| <b>C</b> | 4.0                  | 0                                                       |
| <b>D</b> | 4.0                  | 10                                                      |

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**Cambridge International Examinations**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

**0625/12**

Paper 1 Multiple Choice (Core)

**October/November 2016**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)



**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

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There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

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**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

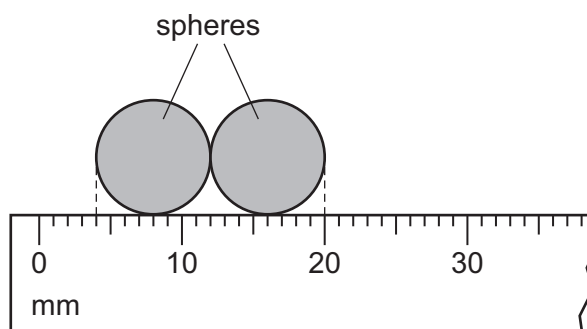
Take the weight of 1.0 kg to be 10 N (acceleration of free fall =  $10 \text{ m/s}^2$ ).

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **17** printed pages and **3** blank pages.

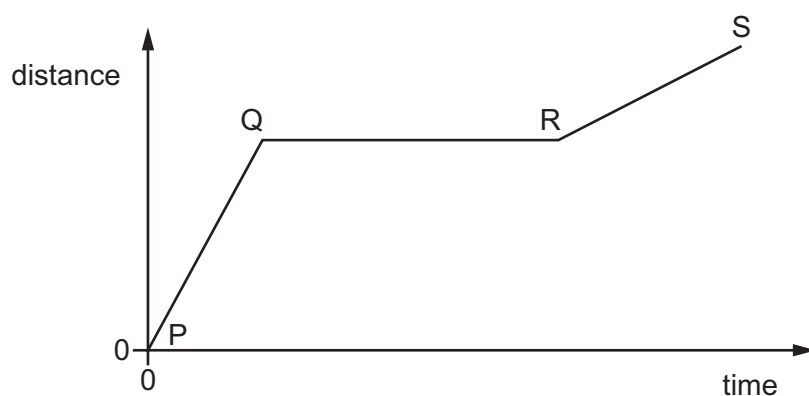


- 1 The diagram shows two identical spheres placed beside a ruler.



What is the radius of one sphere?

- A** 4.0 mm      **B** 5.0 mm      **C** 8.0 mm      **D** 10 mm
- 2 The graph shows how the distance travelled by a vehicle changes with time.

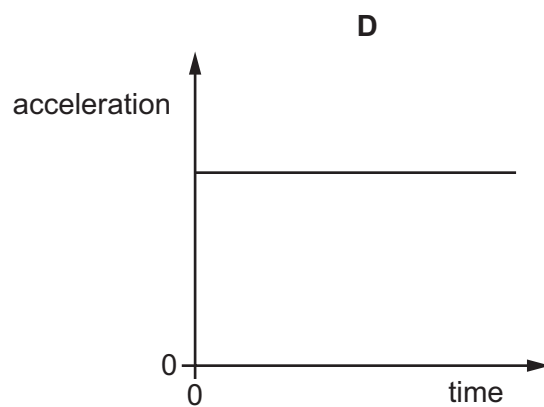
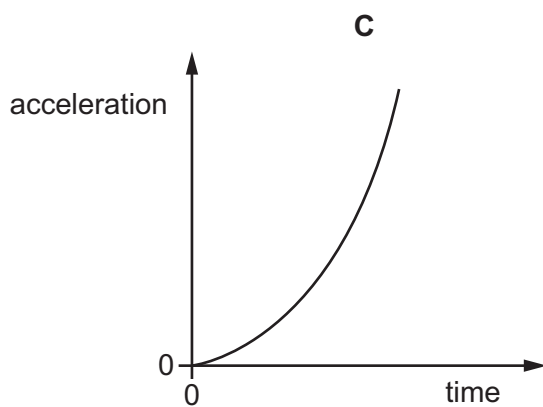
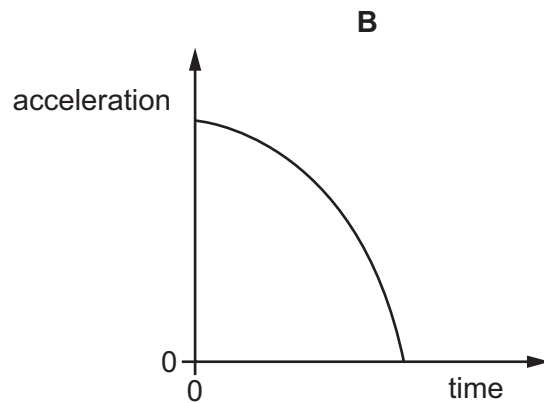
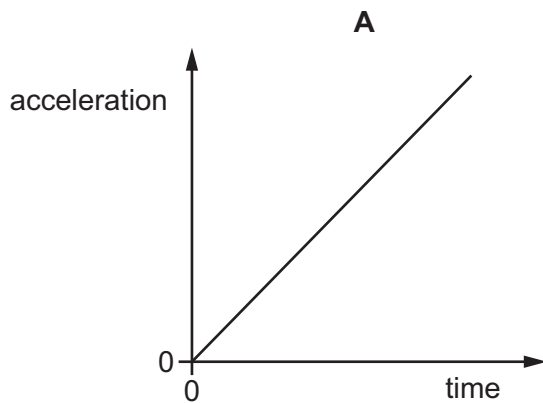


Which row describes the speed of the vehicle in each section of the graph?

|          | P to Q     | Q to R   | R to S     |
|----------|------------|----------|------------|
| <b>A</b> | constant   | zero     | constant   |
| <b>B</b> | constant   | zero     | decreasing |
| <b>C</b> | increasing | constant | decreasing |
| <b>D</b> | increasing | zero     | constant   |

- 3 A stone falls freely from the top of a cliff. Air resistance may be ignored.

Which graph shows how the acceleration of the stone varies with time as it falls?

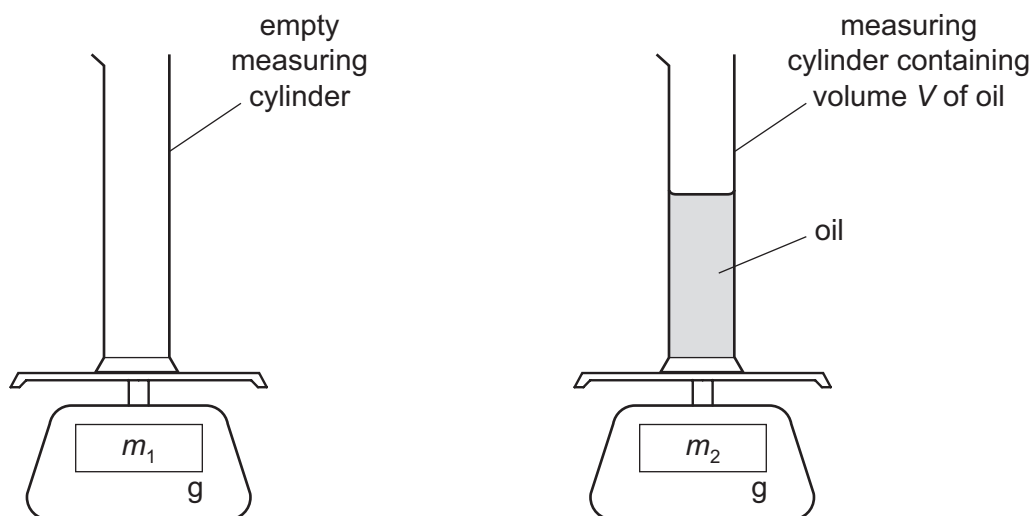


- 4 A student writes about mass and weight.

Which statement is correct?

- A** A ship which is floating has mass but no weight.
- B** Mass is a scientific word that means the same as weight.
- C** Mass is measured in newtons.
- D** The mass of an astronaut is the same on the Moon as on the Earth.

- 5 A student uses a measuring cylinder and a balance to find the density of oil. The diagram shows the arrangement used.

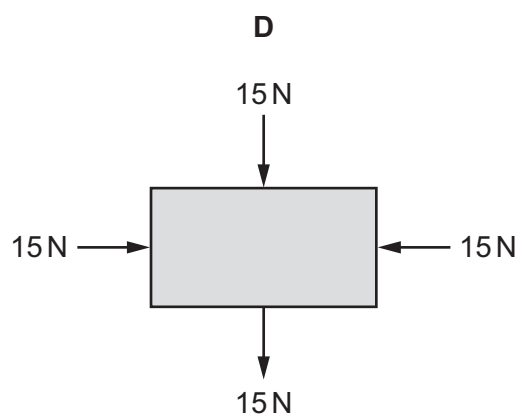
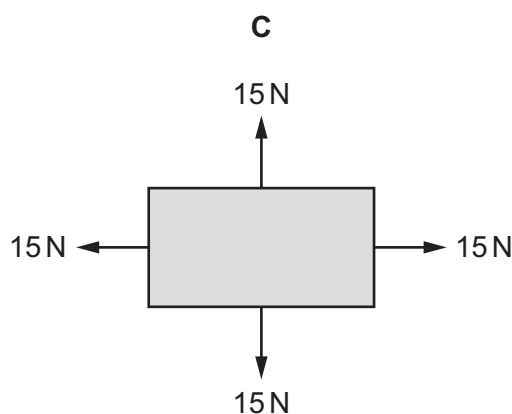
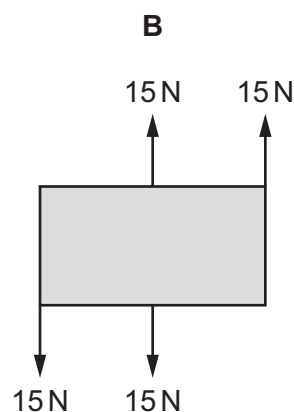
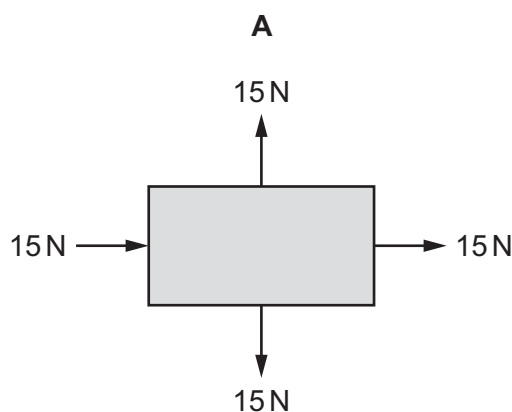


Which calculation gives the density of the oil?

- A  $\frac{V}{m_2}$       B  $\frac{V}{(m_2 - m_1)}$       C  $\frac{m_2}{V}$       D  $\frac{(m_2 - m_1)}{V}$

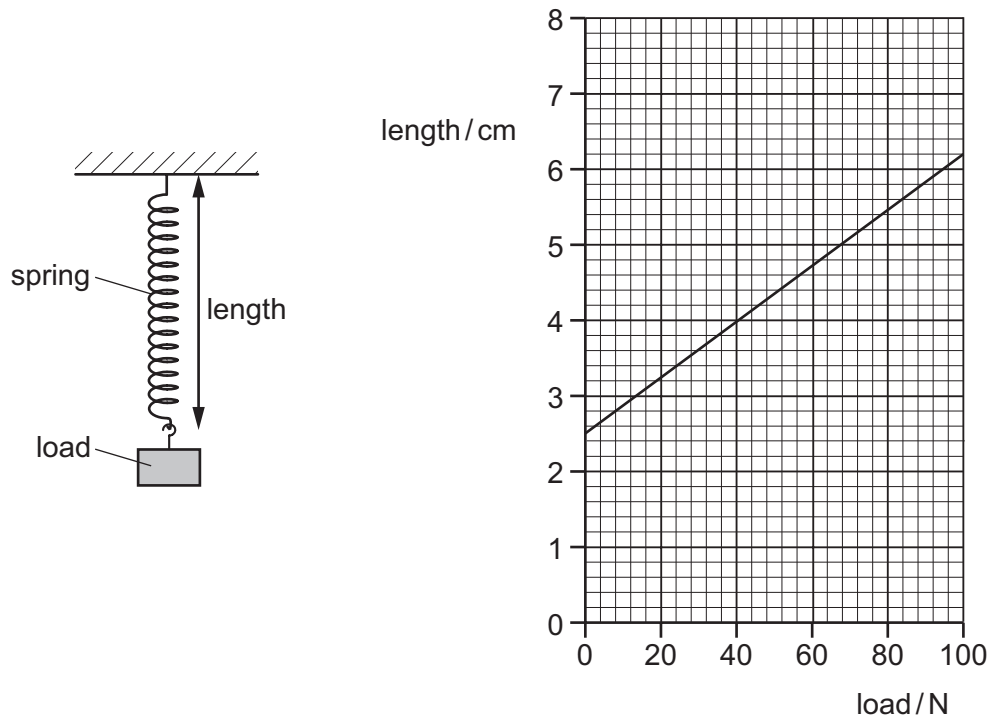
- 6 Each diagram shows all the forces acting on an object.

In which diagram is the object in equilibrium?





- 7 The diagrams show a spring and a graph of the length of the spring against the load applied to it.

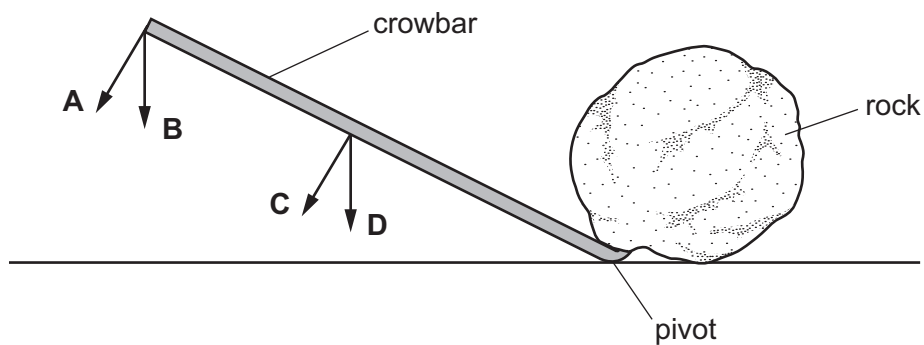


What is the extension of the spring when a load of 40 N is applied to it?

- A** 1.5 cm      **B** 2.5 cm      **C** 4.0 cm      **D** 6.5 cm

- 8 A force is applied to a long crowbar so as to move a large rock.

Which labelled arrow represents the smallest force needed to move the rock?



- 9 What is the main energy change taking place in the battery of a mobile phone (cell phone) as it is being charged but not used?
- A** chemical energy increases  
**B** elastic energy increases  
**C** gravitational potential energy increases  
**D** kinetic energy increases

- 10 Electricity can be generated using different energy resources.

Which energy resource is used to generate electricity without needing any moving parts?

- A geothermal
- B hydroelectric
- C solar
- D water waves

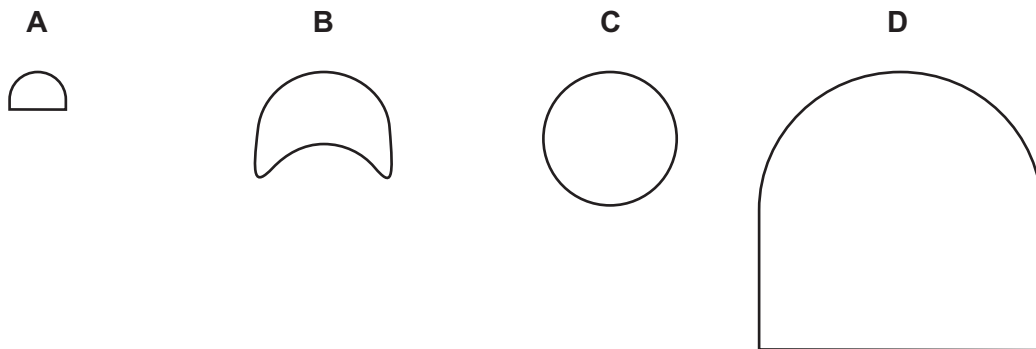
- 11 A child runs up a set of stairs four times. The time taken for each run is recorded.

Which time is measured when the child's useful power is greatest?

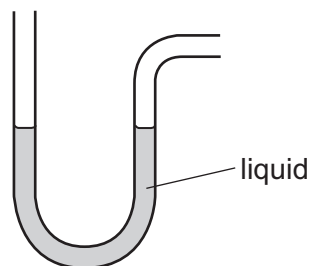
- A 10 s
- B 20 s
- C 30 s
- D 40 s

- 12 The diagrams show the actual sizes of the heels of four different shoes, as seen from underneath the shoe.

Which heel is **most** likely to cause damage to wooden floors?



- 13 The diagram shows an instrument used to measure gas pressure.



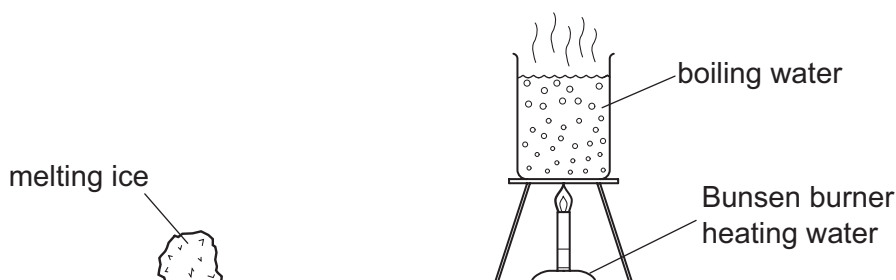
What is the instrument called?

- A ammeter
- B barometer
- C manometer
- D thermometer

- 14 Water can exist in three states: solid (ice), liquid and gas (steam).

In which state, or states, is the motion of the molecules **only** vibrational?

- A ice only
  - B steam only
  - C liquid water only
  - D liquid water and ice only
- 15 Which statement about evaporation is correct?
- A Evaporation causes heating of the liquid.
  - B Evaporation only happens from liquids.
  - C Molecules escape from anywhere in the liquid.
  - D Only the least energetic molecules escape.
- 16 A piece of melting ice at  $0^{\circ}\text{C}$  and a beaker of boiling water are both in a laboratory. The laboratory is at  $20^{\circ}\text{C}$ .



What is happening to the temperature of the melting ice and what is happening to the temperature of the boiling water?

|          | temperature of melting ice | temperature of boiling water |
|----------|----------------------------|------------------------------|
| <b>A</b> | constant                   | constant                     |
| <b>B</b> | constant                   | increasing                   |
| <b>C</b> | increasing                 | constant                     |
| <b>D</b> | increasing                 | increasing                   |

- 17** A liquid-in-glass thermometer is to be calibrated in degrees Celsius.

Where should the bulb of the thermometer be placed to set a fixed point on the temperature scale?

- A** in a mixture of water and ice
- B** in a hot drink
- C** in a Bunsen burner flame
- D** in a refrigerator

- 18** Aluminium foil is attached to the walls of a warm room in a cold country. This reduces the rate of thermal energy loss from inside the warm room.

Which method or methods of thermal energy transfer are reduced by the foil?

- A** conduction and convection only
- B** conduction only
- C** convection and radiation only
- D** radiation only

- 19** On a very cold day, a boy puts one hand on the metal handlebars of his bicycle. He puts the other hand on the rubber hand grip.

The metal feels colder than the rubber hand grip, although they are both at the same temperature.

Why is this?

- A** The metal has a higher melting point than the rubber.
- B** The metal has a lower thermal capacity than the rubber.
- C** The metal is a better thermal conductor than the rubber.
- D** The metal radiates more infra-red radiation than the rubber.

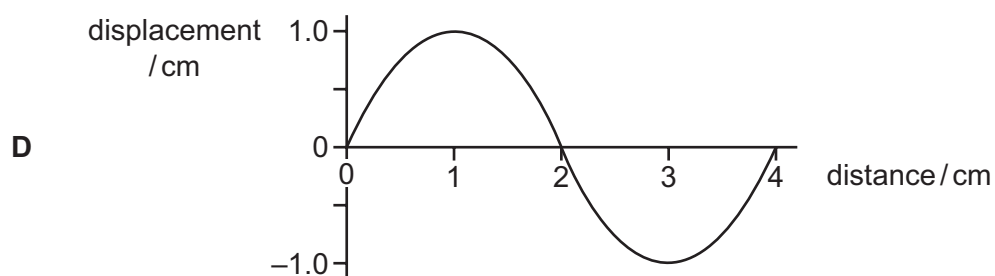
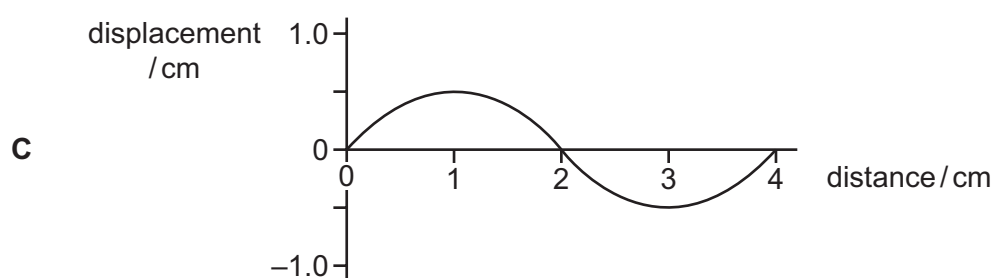
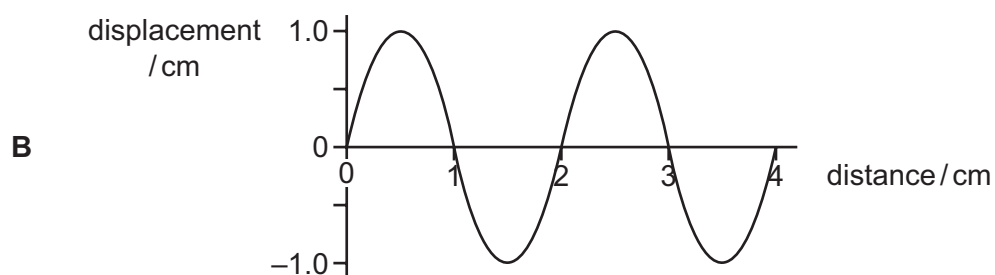
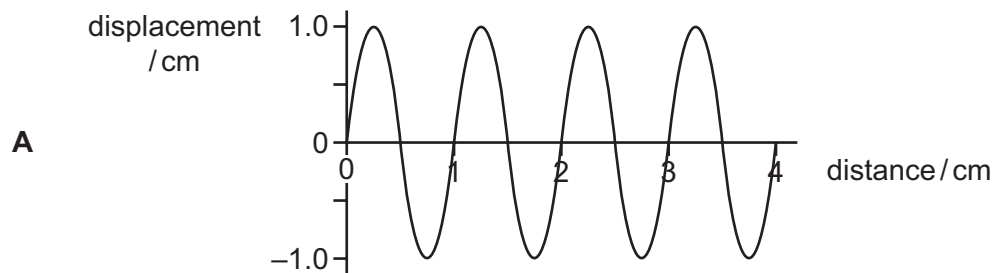
- 20** The box contains a list of four types of wave.

|                |       |             |        |
|----------------|-------|-------------|--------|
| $\gamma$ -rays | sound | ultraviolet | X-rays |
|----------------|-------|-------------|--------|

How many waves in the list are transverse waves?

- A** 1
- B** 2
- C** 3
- D** 4

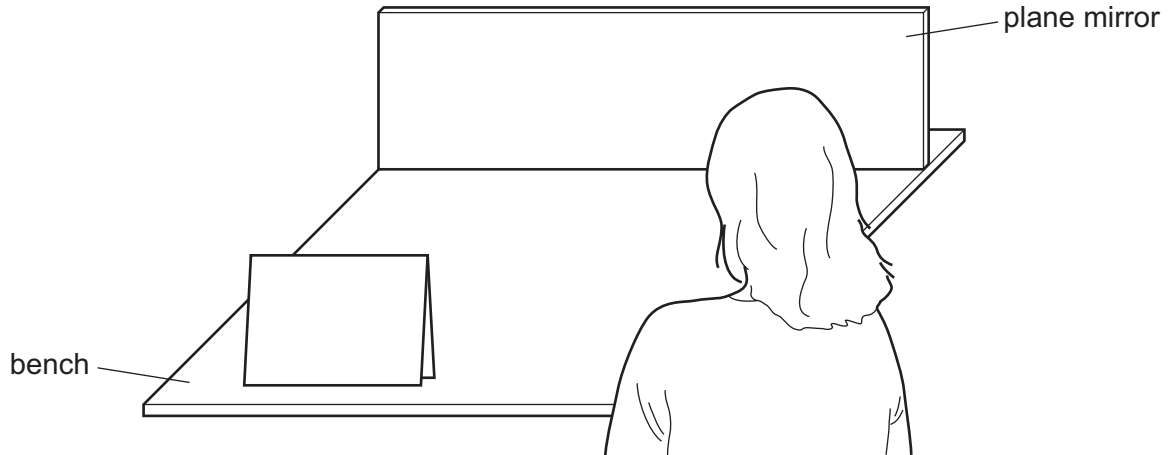
21 Which wave has an amplitude equal to half its wavelength?



22 A card with the letter **R** is made by a student. The letter is drawn on **one side**, as shown.



A plane mirror is mounted vertically on a bench. The student places the card on the bench so that the letter is upright and **facing the mirror**.



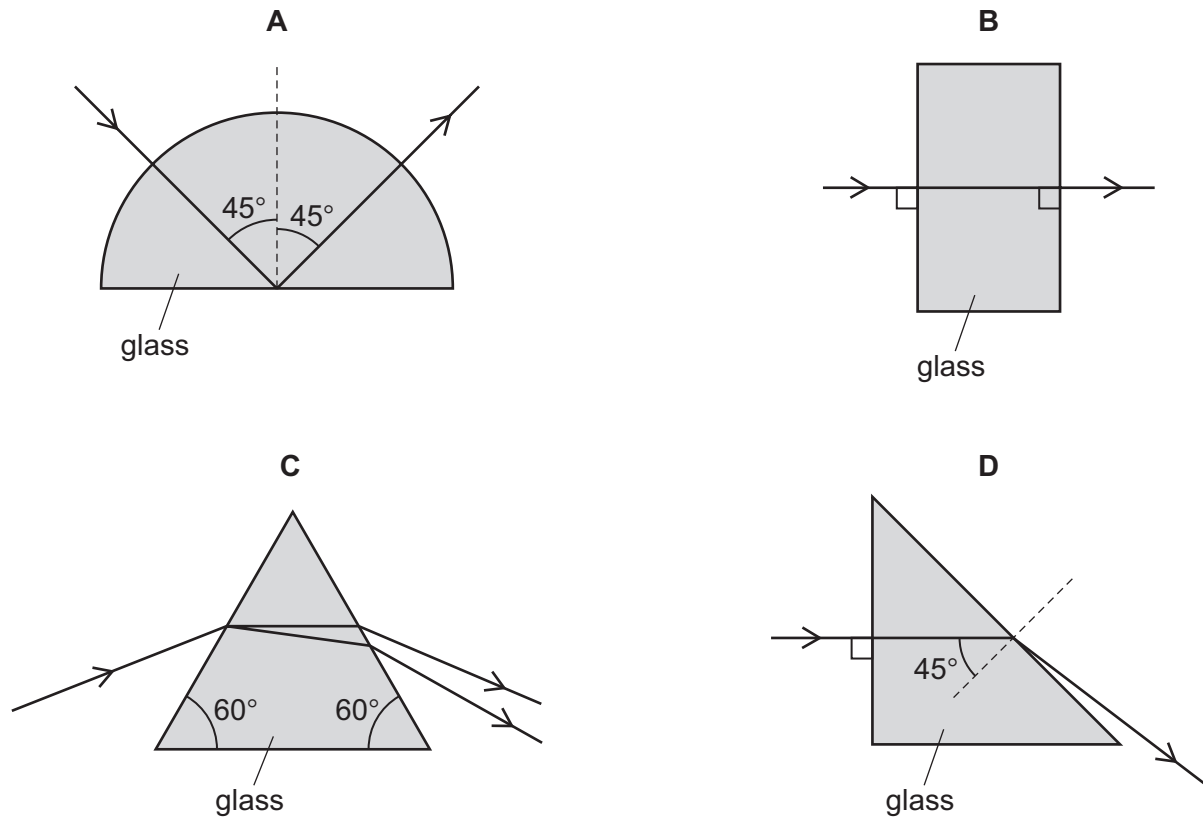
How does the image formed by the mirror appear to the student?



- 23** A student draws four diagrams to represent light passing from air through a glass block and then back into the air.

The critical angle for the glass is  $42^\circ$ .

In which diagram is the path of the light through the glass block **not** correct?



- 24** Which part of the electromagnetic spectrum is used to send television signals from a satellite to Earth?

- A** infra-red
- B** microwaves
- C** ultraviolet
- D** visible light

- 25** A girl notices that, when she shouts into a cave, she hears an echo.

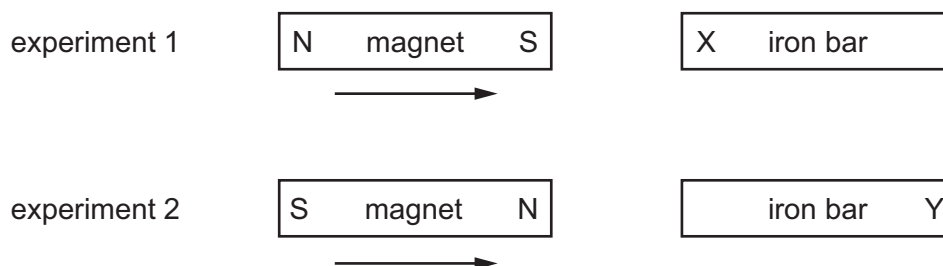
Which wave property causes the echo?

- A** diffraction
- B** dispersion
- C** reflection
- D** refraction

26 Which property of a sound wave affects the loudness of the sound?

- A amplitude
- B frequency
- C speed
- D wavelength

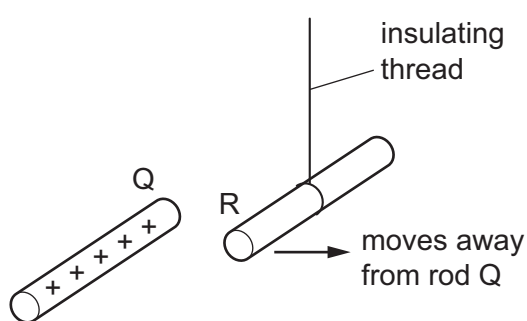
27 In two separate experiments, a magnet is brought near to an unmagnetised iron bar. This causes the bar to become magnetised.



Which magnetic pole is induced at X and at Y?

|          | pole induced at X | pole induced at Y |
|----------|-------------------|-------------------|
| <b>A</b> | N                 | N                 |
| <b>B</b> | N                 | S                 |
| <b>C</b> | S                 | N                 |
| <b>D</b> | S                 | S                 |

28 In the diagram, rod R hangs from an insulating thread.



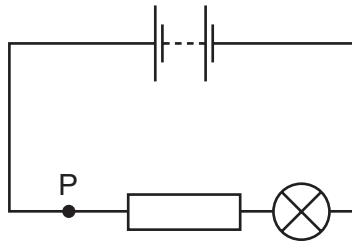
When the positively charged rod Q is brought close to rod R, rod R moves away from rod Q.

Which conclusion can be made from this observation?

- A Rod R is charged, but it is not possible to identify the sign of the charge.
- B Rod R must be positively charged.
- C Rod R must be negatively charged.
- D Rod R is uncharged.

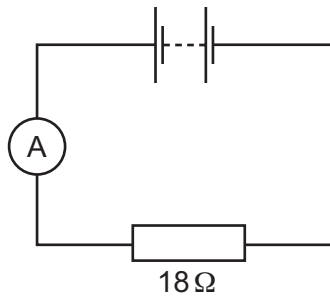


- 29 The diagram shows a lamp in a circuit.



Which change to the circuit would increase the current in the lamp?

- A adding another resistor in parallel with the one in the circuit
  - B adding another resistor in series with the one in the circuit
  - C decreasing the electromotive force (e.m.f.) of the battery in the circuit
  - D moving the lamp to point P in the circuit
- 30 An ammeter and an  $18\Omega$  resistor are connected in series with a battery. The reading on the ammeter is  $0.50\text{ A}$ .

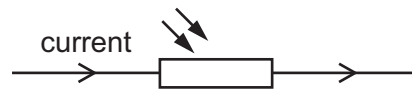


What is the electromotive force (e.m.f.) of the battery?

- A  $9.0\text{ N}$
- B  $9.0\text{ V}$
- C  $36\text{ N}$
- D  $36\text{ V}$

- 31 A source of constant electromotive force (e.m.f.) is connected across a light-dependent resistor (LDR).

There is an electric current in the LDR.

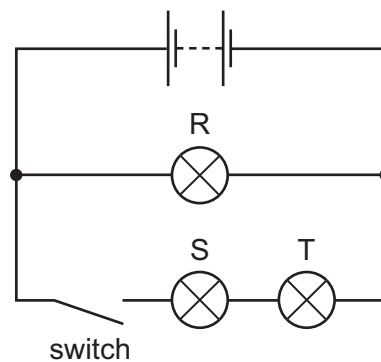


The brightness of the light falling on the LDR is reduced.

Which row shows what happens to the resistance of the LDR and what happens to the current?

|          | resistance | current   |
|----------|------------|-----------|
| <b>A</b> | decreases  | decreases |
| <b>B</b> | decreases  | increases |
| <b>C</b> | increases  | decreases |
| <b>D</b> | increases  | increases |

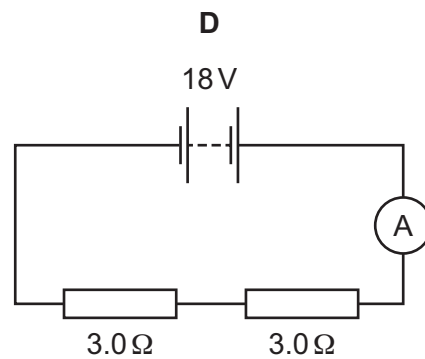
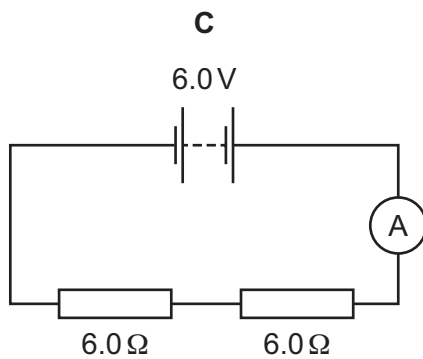
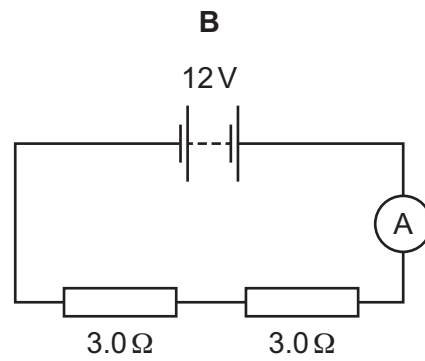
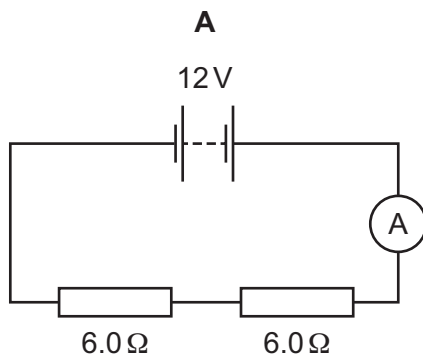
- 32 The diagram shows an electric circuit.



When the switch is open, which lamp or lamps are **not** lit?

- A** R only
- B** S only
- C** R, S and T
- D** S and T only

33 In which circuit does the ammeter read 2.0 A?

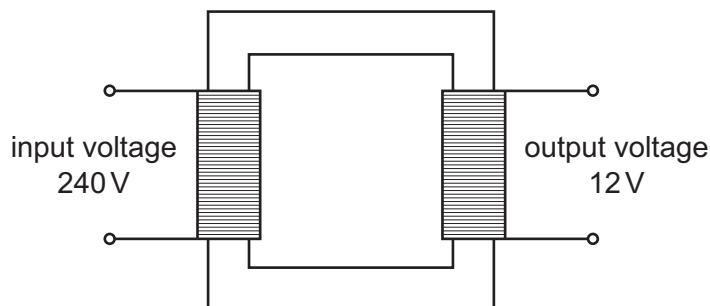


34 An electric kettle has a metal casing. The cable for the kettle contains a wire that is connected to the earth pin of the plug.

Which danger does this guard against?

- A** the cable to the kettle becoming too hot
- B** the casing of the kettle becoming live
- C** the casing of the kettle becoming wet on the outside
- D** the casing of the kettle overheating

- 35** A transformer has an input voltage of 240 V and an output voltage of 12 V.



What is a possible pair of values for the number of turns in the primary coil and the number of turns in the secondary coil?

|          | number of turns in primary coil | number of turns in secondary coil |
|----------|---------------------------------|-----------------------------------|
| <b>A</b> | 12                              | 4800                              |
| <b>B</b> | 240                             | 4800                              |
| <b>C</b> | 4800                            | 12                                |
| <b>D</b> | 4800                            | 240                               |

- 36** An electric current can produce a heating effect and a magnetic effect.

Which row shows the effect that a relay uses and one application of a relay?

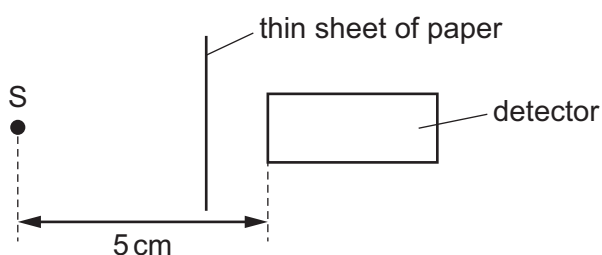
|          | effect used by a relay | one application of a relay                            |
|----------|------------------------|-------------------------------------------------------|
| <b>A</b> | heating effect         | allowing a small current to switch on a large current |
| <b>B</b> | heating effect         | changing the voltage of an a.c. supply                |
| <b>C</b> | magnetic effect        | allowing a small current to switch on a large current |
| <b>D</b> | magnetic effect        | changing the voltage of an a.c. supply                |

- 37** A nuclide of sodium can be represented by  $^{23}_{11}\text{Na}$ .

Which row gives the numbers of particles in a neutral sodium atom?

|          | number of electrons | number of protons | number of neutrons |
|----------|---------------------|-------------------|--------------------|
| <b>A</b> | 11                  | 11                | 12                 |
| <b>B</b> | 12                  | 11                | 23                 |
| <b>C</b> | 12                  | 12                | 11                 |
| <b>D</b> | 23                  | 23                | 11                 |

- 38 Which is the most effective precaution to reduce the risk when handling, storing or using a radioactive source that emits  $\gamma$ -rays?
- A Handle the source for the least possible time.
  - B Have a fire extinguisher nearby when using the source.
  - C Store the source at a low temperature.
  - D Wear plastic safety goggles when handling the source.
- 39 Radioactive source S emits  $\alpha$ -particles,  $\beta$ -particles and  $\gamma$ -rays. A detector is placed 5 cm away from S. A thin sheet of paper is placed as shown in the diagram.



Which emissions from the source can be detected?

- A  $\alpha$ -particles and  $\beta$ -particles only
  - B  $\alpha$ -particles and  $\gamma$ -rays only
  - C  $\beta$ -particles and  $\gamma$ -rays only
  - D  $\alpha$ -particles,  $\beta$ -particles and  $\gamma$ -rays
- 40 A radioactive source has a half-life of 3.0 days. It emits radiation at a rate of 1200 particles per minute.
- At what rate was it emitting radiation 6.0 days **earlier**?
- A 300 particles per minute
  - B 2400 particles per minute
  - C 4800 particles per minute
  - D 7200 particles per minute

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Cambridge International General Certificate of Secondary Education

**PHYSICS**

**0625/11**

Paper 1 Multiple Choice (Core)

**October/November 2016**

**45 minutes**

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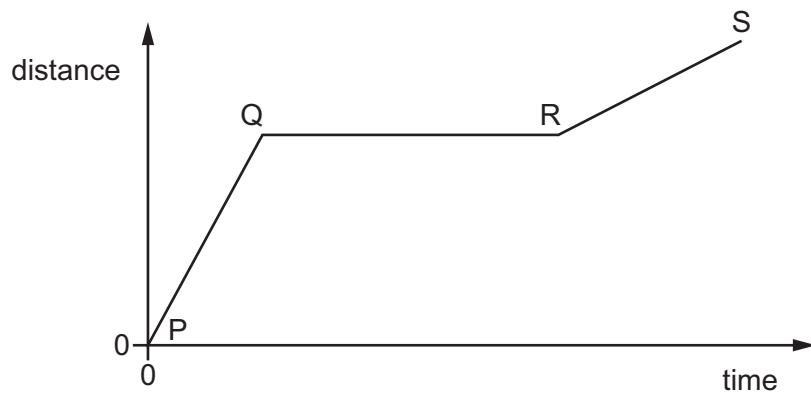


- 1 A student has a can of oil.

Which quantity can be measured using only a measuring cylinder?

- A** density of the oil
- B** mass of the oil
- C** volume of the oil
- D** weight of the oil

- 2 The graph shows how the distance travelled by a vehicle changes with time.

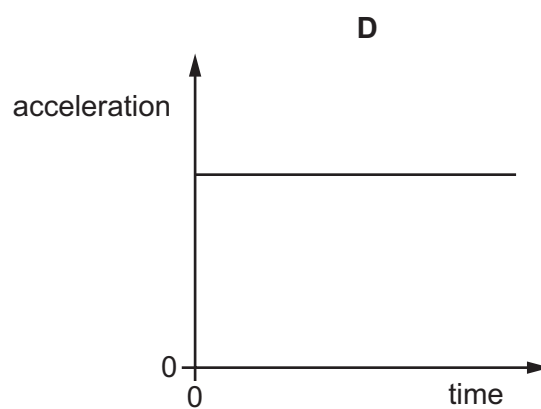
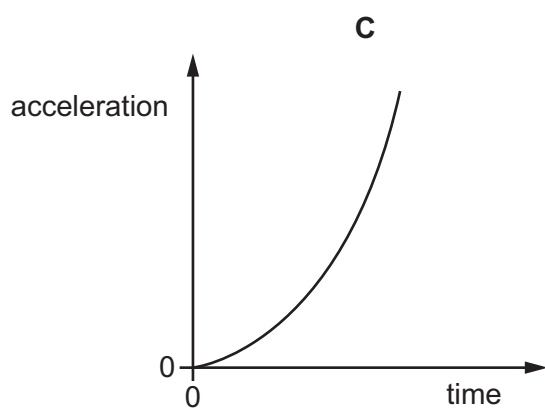
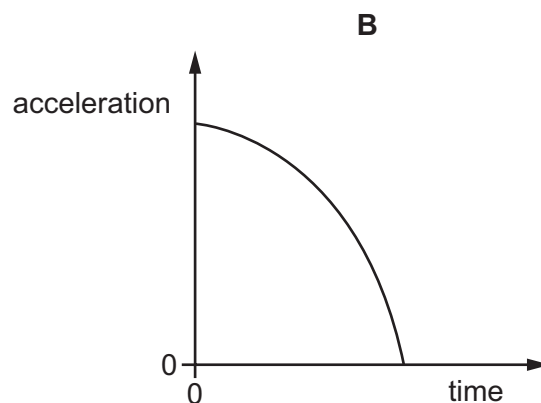
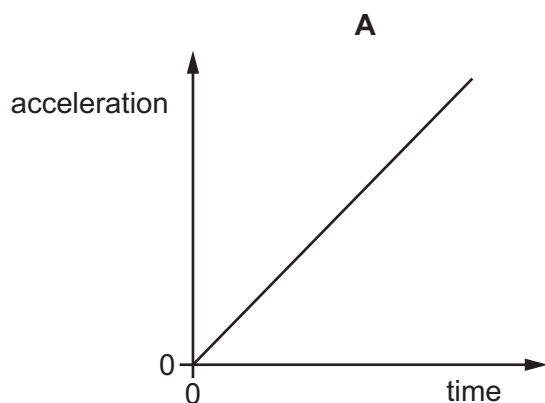


Which row describes the speed of the vehicle in each section of the graph?

|          | P to Q     | Q to R   | R to S     |
|----------|------------|----------|------------|
| <b>A</b> | constant   | zero     | constant   |
| <b>B</b> | constant   | zero     | decreasing |
| <b>C</b> | increasing | constant | decreasing |
| <b>D</b> | increasing | zero     | constant   |

- 3 A stone falls freely from the top of a cliff. Air resistance may be ignored.

Which graph shows how the acceleration of the stone varies with time as it falls?

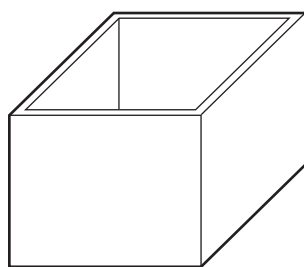


- 4 What name is given to the gravitational force of the Earth on an object?

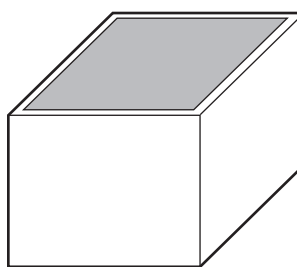
- A** density
- B** mass
- C** volume
- D** weight

- 5 The diagrams show an empty rectangular box, and the same box filled with liquid.

The box has a mass of 60 g when empty. When filled with liquid, the total mass of the box and the liquid is 300 g.



empty box  
60 g



box filled with liquid  
300 g

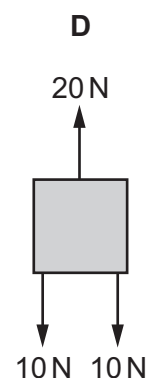
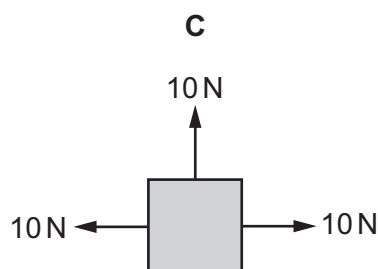
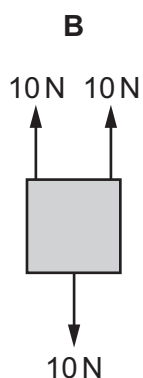
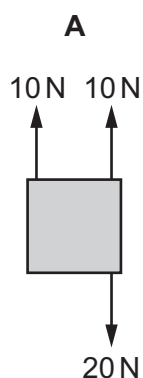
The density of the liquid is  $1.2 \text{ g/cm}^3$ .

What is the volume of the liquid in the box?

- A**  $50 \text{ cm}^3$       **B**  $200 \text{ cm}^3$       **C**  $250 \text{ cm}^3$       **D**  $300 \text{ cm}^3$

- 6 The diagrams show four identical objects. Each object is acted on by only the forces shown.

Which diagram shows an object in equilibrium?



- 7 A student investigates a steel spring. He measures the length of the spring, then he hangs different weights from the spring. He measures the length of the spring for each different weight.

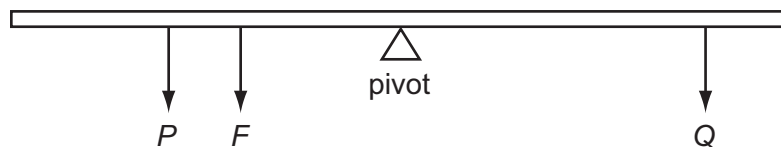
The table shows his results.

| weight / N | length of spring / mm |
|------------|-----------------------|
| 0.0        | 108                   |
| 2.0        | 116                   |
| 4.0        | 124                   |
| 6.0        | 132                   |

A weight of 3.0 N is hung from the spring.

What is the extension of the spring?

- A** 4 mm                      **B** 12 mm                      **C** 40 mm                      **D** 120 mm
- 8 The diagram shows a plank balanced on a pivot. Three forces  $F$ ,  $P$  and  $Q$  act on the plank, as shown.



The force  $F$  is increased, but continues to act at the same distance from the pivot. The plank is no longer balanced.

Which change could make the plank balance again?

- A** decrease  $Q$   
**B** increase  $P$   
**C** move  $P$  further from the pivot  
**D** move  $Q$  further from the pivot
- 9 Which energy transfer takes place when petrol is burned?
- A** chemical to internal energy of surroundings (heat)  
**B** chemical to nuclear  
**C** gravitational potential to internal energy of surroundings (heat)  
**D** kinetic to chemical

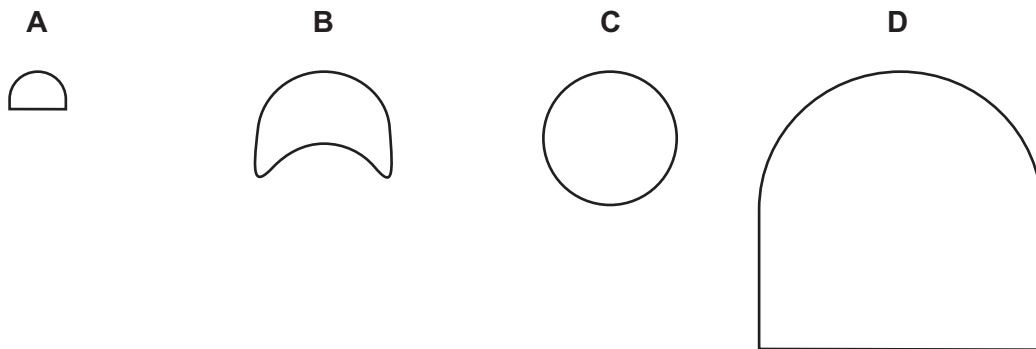
- 10 The box contains the names of eight different energy resources.

|               |            |       |       |
|---------------|------------|-------|-------|
| natural gas   | geothermal | solar | waves |
| hydroelectric | oil        | wind  | coal  |

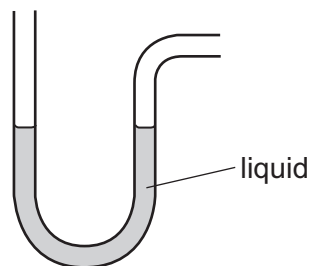
How many of these energy resources are renewable?

- A 3                      B 4                      C 5                      D 6
- 11 A child runs up a set of stairs four times. The time taken for each run is recorded.
- Which time is measured when the child's useful power is greatest?
- A 10 s                      B 20 s                      C 30 s                      D 40 s
- 12 The diagrams show the actual sizes of the heels of four different shoes, as seen from underneath the shoe.

Which heel is **most** likely to cause damage to wooden floors?



- 13 The diagram shows an instrument used to measure gas pressure.



What is the instrument called?

- A ammeter  
B barometer  
C manometer  
D thermometer

- 14** Smoke particles in air are illuminated by a beam of light. The particles are viewed through a microscope. They are seen to move in a random zig-zag way.

What causes this movement?

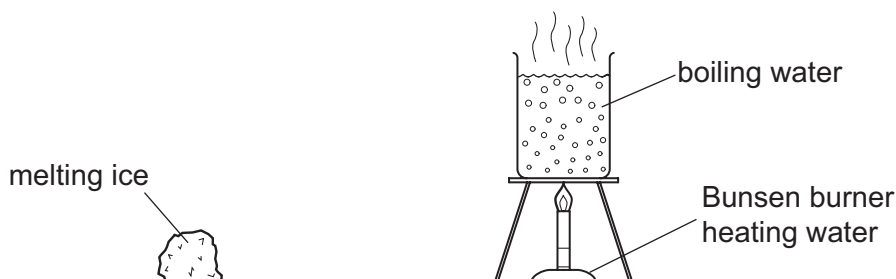
- A** convection currents in the air
- B** impacts of fast-moving air molecules
- C** the energy of the beam of light
- D** vibrations of the atoms in the smoke particles

- 15** When a liquid evaporates, some of its molecules escape from the surface and the temperature of the liquid changes.

Which row describes the escaping molecules and the change in temperature of the liquid?

|          | escaping molecules | temperature of the liquid |
|----------|--------------------|---------------------------|
| <b>A</b> | less energetic     | goes down                 |
| <b>B</b> | less energetic     | goes up                   |
| <b>C</b> | more energetic     | goes down                 |
| <b>D</b> | more energetic     | goes up                   |

- 16** A piece of melting ice at  $0^{\circ}\text{C}$  and a beaker of boiling water are both in a laboratory. The laboratory is at  $20^{\circ}\text{C}$ .



What is happening to the temperature of the melting ice and what is happening to the temperature of the boiling water?

|          | temperature of melting ice | temperature of boiling water |
|----------|----------------------------|------------------------------|
| <b>A</b> | constant                   | constant                     |
| <b>B</b> | constant                   | increasing                   |
| <b>C</b> | increasing                 | constant                     |
| <b>D</b> | increasing                 | increasing                   |

17 Changes in which physical property **cannot** be used for temperature measurement?

- A decay rate of a radioactive source
- B electrical resistance of a solid
- C pressure of a gas
- D volume of a liquid

18 Which statement about infra-red radiation is correct?

- A A black surface is a poor absorber of infra-red radiation.
- B A dull surface is a better emitter of infra-red radiation than a shiny surface.
- C A shiny surface is a better absorber of infra-red radiation than a dull surface.
- D A white surface is a poor reflector of infra-red radiation.

19 A pan is made for heating water on a stove.



From which type of materials should the pan and its handle be made?

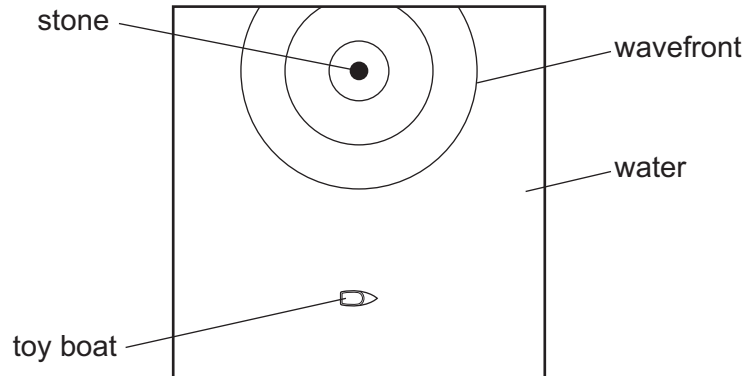
|   | material for pan       | material for handle    |
|---|------------------------|------------------------|
| A | good thermal conductor | good thermal conductor |
| B | good thermal conductor | poor thermal conductor |
| C | poor thermal conductor | good thermal conductor |
| D | poor thermal conductor | poor thermal conductor |



- 20** A toy boat floats on water in a tank. The boat is initially stationary.

A stone is thrown into the tank, which causes a transverse water wave to move across the surface.

The diagram shows the view from above the tank.



How does the boat behave as the wave passes it?

- A** It moves steadily away from where the stone hit the water.
  - B** It moves steadily towards where the stone hit the water.
  - C** It stays the same distance from where the stone hit the water, and vibrates from side to side.
  - D** It stays the same distance from where the stone hit the water, and vibrates up and down.
- 21** Below are four statements about the diffraction of a wave on the surface of water.

Which statement is correct?

- A** Diffraction involves a change in the speed of the wave.
- B** Diffraction involves a change in the wavelength of the wave.
- C** When a wave passes through a gap, its direction changes.
- D** When a wave passes through a gap, its frequency changes.

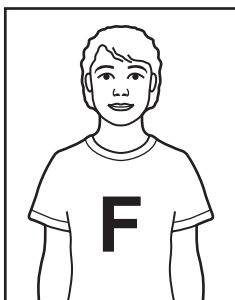
22 A child wears a T-shirt with the letter 'F' written on it.



The child stands in front of a vertical plane mirror.

What does the child see?

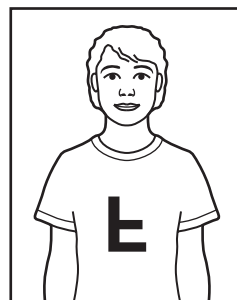
**A**



**B**



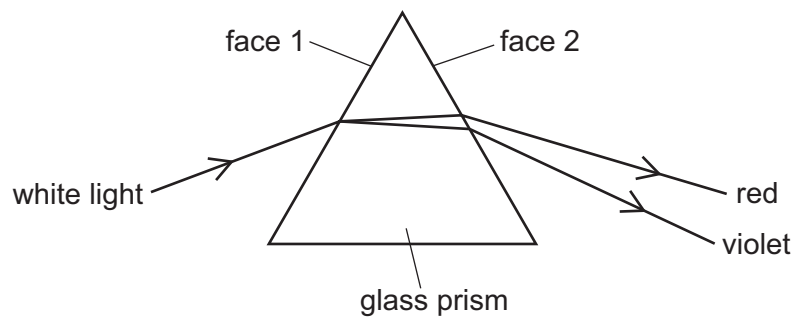
**C**



**D**



23 When white light passes through a glass prism, it disperses as shown in the diagram.



Which row compares the refraction of violet light with the refraction of red light at the faces 1 and 2 of the prism?

|          | face 1               | face 2               |
|----------|----------------------|----------------------|
| <b>A</b> | violet refracts less | violet refracts less |
| <b>B</b> | violet refracts less | violet refracts more |
| <b>C</b> | violet refracts more | violet refracts less |
| <b>D</b> | violet refracts more | violet refracts more |

**24** Radiation from which part of the electromagnetic spectrum is used in the remote controller for a television?

- A** infra-red waves
- B** microwaves
- C** radio waves
- D** ultraviolet waves

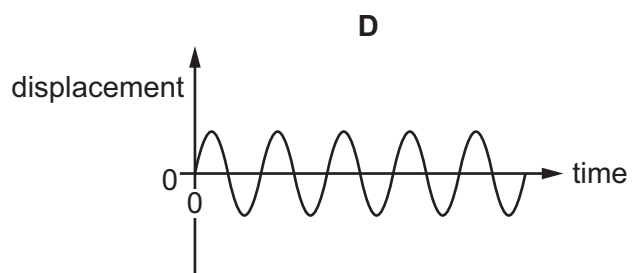
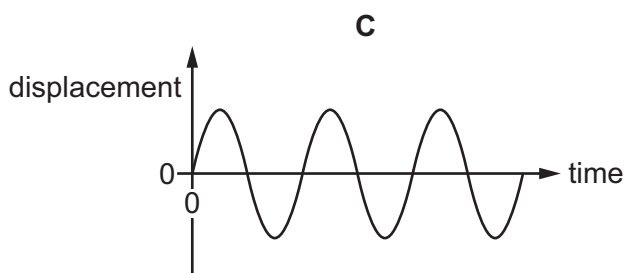
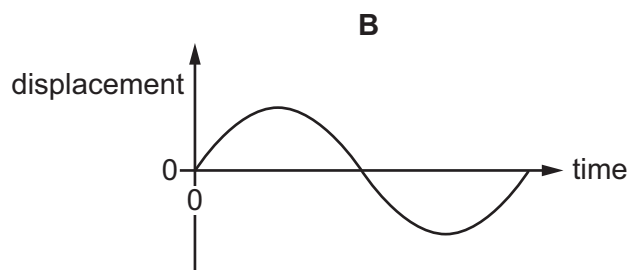
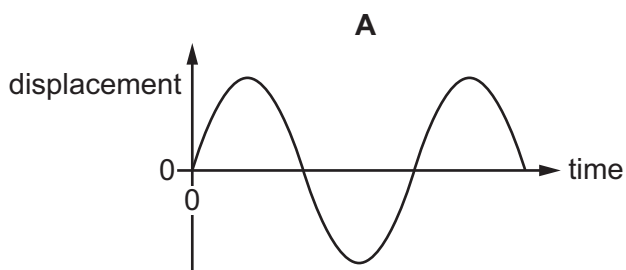
**25** A girl notices that, when she shouts into a cave, she hears an echo.

Which wave property causes the echo?

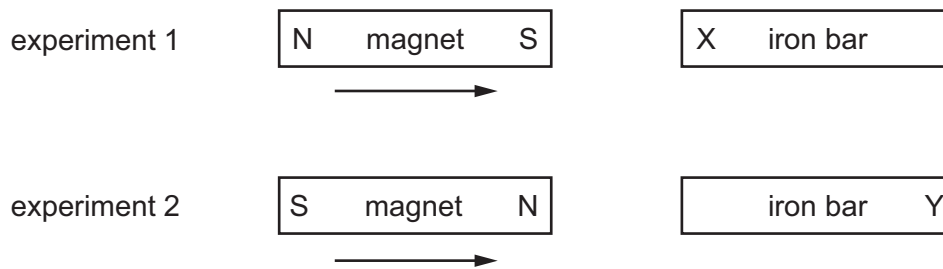
- A** diffraction
- B** dispersion
- C** reflection
- D** refraction

**26** The diagrams represent the displacement in four different sound waves. All the diagrams are drawn to the same scale.

Which diagram represents the sound with the highest pitch?



- 27** In two separate experiments, a magnet is brought near to an unmagnetised iron bar. This causes the bar to become magnetised.



Which magnetic pole is induced at X and at Y?

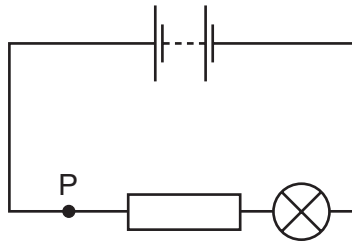
|          | pole induced at X | pole induced at Y |
|----------|-------------------|-------------------|
| <b>A</b> | N                 | N                 |
| <b>B</b> | N                 | S                 |
| <b>C</b> | S                 | N                 |
| <b>D</b> | S                 | S                 |

- 28** A polythene rod is rubbed with a cloth. The rod becomes positively charged because of the movement of charged particles.

Which row gives the name of these charged particles, and the direction in which they move?

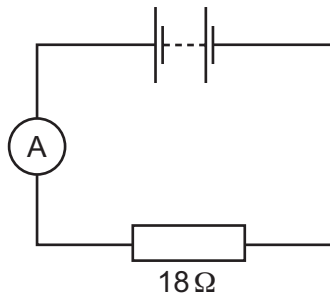
|          | charged particles | direction of movement |
|----------|-------------------|-----------------------|
| <b>A</b> | electrons         | from cloth to rod     |
| <b>B</b> | electrons         | from rod to cloth     |
| <b>C</b> | protons           | from cloth to rod     |
| <b>D</b> | protons           | from rod to cloth     |

- 29 The diagram shows a lamp in a circuit.



Which change to the circuit would increase the current in the lamp?

- A adding another resistor in parallel with the one in the circuit
  - B adding another resistor in series with the one in the circuit
  - C decreasing the electromotive force (e.m.f.) of the battery in the circuit
  - D moving the lamp to point P in the circuit
- 30 An ammeter and an  $18\Omega$  resistor are connected in series with a battery. The reading on the ammeter is  $0.50\text{ A}$ .



What is the electromotive force (e.m.f.) of the battery?

- A  $9.0\text{ N}$
- B  $9.0\text{ V}$
- C  $36\text{ N}$
- D  $36\text{ V}$

- 31 A source of constant electromotive force (e.m.f.) is connected across a thermistor.

There is an electric current in the thermistor.

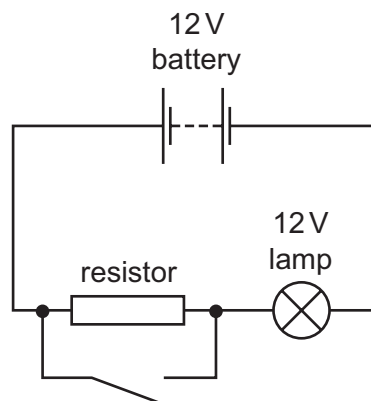


The temperature of the thermistor is reduced.

Which row shows what happens to the resistance of the thermistor and what happens to the current?

|          | resistance     | current   |
|----------|----------------|-----------|
| <b>A</b> | increases      | decreases |
| <b>B</b> | increases      | increases |
| <b>C</b> | stays the same | decreases |
| <b>D</b> | stays the same | increases |

- 32 The diagram shows a circuit containing a battery, a resistor with high resistance, a switch and a lamp.

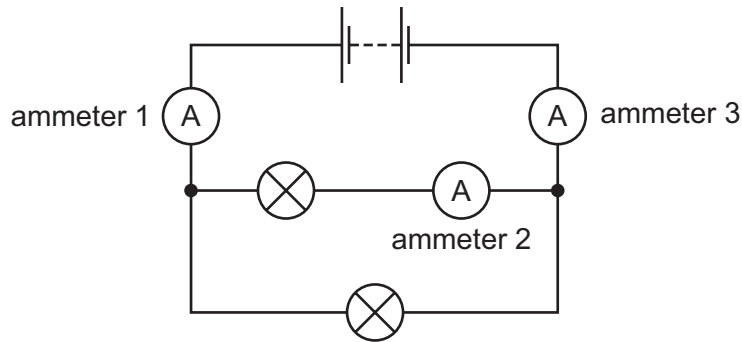


Initially the switch is open.

What happens to the lamp when the switch is closed?

- A** It glows more brightly.
- B** It glows less brightly.
- C** It goes out.
- D** Its brightness does not change.

- 33 The diagram shows a circuit containing two identical lamps and three ammeters.



The current in ammeter 1 is 0.30 A.

Which row gives possible values for the currents in ammeters 2 and 3?

|          | ammeter 2<br>/A | ammeter 3<br>/A |
|----------|-----------------|-----------------|
| <b>A</b> | 0.15            | 0.00            |
| <b>B</b> | 0.15            | 0.30            |
| <b>C</b> | 0.30            | 0.00            |
| <b>D</b> | 0.30            | 0.30            |

- 34 Two electrical appliances are connected to the mains supply.

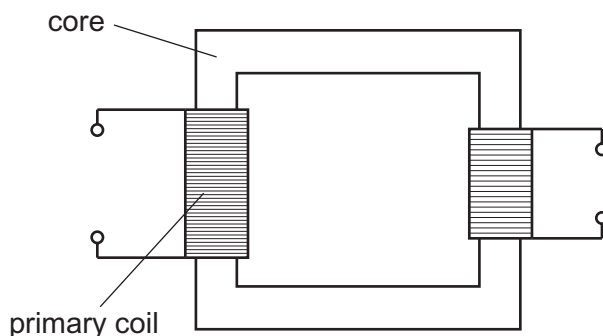
The cable connected to one appliance includes an earth wire.

The cable connected to the second appliance does **not** need an earth wire.

What is a reason for this difference?

- A** One appliance has a metal case, but the other appliance does not.
- B** One appliance is fitted with a fuse, but the other appliance is not.
- C** One appliance is fitted with a switch, but the other appliance is not.
- D** One appliance needs more current than the other appliance.

35 The diagram represents a transformer.



Which row shows materials suitable for making the core and the primary coil?

|          | core  | primary coil |
|----------|-------|--------------|
| <b>A</b> | iron  | copper       |
| <b>B</b> | iron  | plastic      |
| <b>C</b> | steel | copper       |
| <b>D</b> | steel | plastic      |

36 An electric current can produce a heating effect and a magnetic effect.

Which row shows the effect that a relay uses and one application of a relay?

|          | effect used by a relay | one application of a relay                            |
|----------|------------------------|-------------------------------------------------------|
| <b>A</b> | heating effect         | allowing a small current to switch on a large current |
| <b>B</b> | heating effect         | changing the voltage of an a.c. supply                |
| <b>C</b> | magnetic effect        | allowing a small current to switch on a large current |
| <b>D</b> | magnetic effect        | changing the voltage of an a.c. supply                |

37 A nuclide of sodium can be represented by  ${}_{11}^{23}\text{Na}$ .

Which row gives the numbers of particles in a neutral sodium atom?

|          | number of electrons | number of protons | number of neutrons |
|----------|---------------------|-------------------|--------------------|
| <b>A</b> | 11                  | 11                | 12                 |
| <b>B</b> | 12                  | 11                | 23                 |
| <b>C</b> | 12                  | 12                | 11                 |
| <b>D</b> | 23                  | 23                | 11                 |

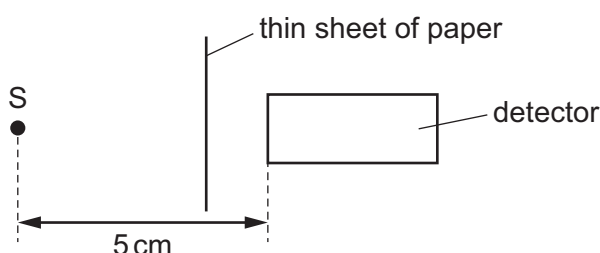


38 Radioactive materials may emit  $\alpha$ -particles or  $\beta$ -particles.

Which statement about the effect of these emissions is correct?

- A Both  $\alpha$ -particles and  $\beta$ -particles cause the nucleus to change into that of a different chemical element.
- B Neither  $\alpha$ -particles nor  $\beta$ -particles cause the nucleus to change into that of a different chemical element.
- C Only  $\alpha$ -particles cause the nucleus to change into that of a different chemical element.
- D Only  $\beta$ -particles cause the nucleus to change into that of a different chemical element.

39 Radioactive source S emits  $\alpha$ -particles,  $\beta$ -particles and  $\gamma$ -rays. A detector is placed 5 cm away from S. A thin sheet of paper is placed as shown in the diagram.



Which emissions from the source can be detected?

- A  $\alpha$ -particles and  $\beta$ -particles only
- B  $\alpha$ -particles and  $\gamma$ -rays only
- C  $\beta$ -particles and  $\gamma$ -rays only
- D  $\alpha$ -particles,  $\beta$ -particles and  $\gamma$ -rays

40 The half-life of a radioactive nuclide is 2.0 hours.

The decay rate of a sample of this nuclide is measured at 1.0 hour intervals. The table shows the measurements, with one value shown as X.

| time /<br>hours | decay rate<br>decays / s |
|-----------------|--------------------------|
| 0               | 240                      |
| 1.0             | 170                      |
| 2.0             | 120                      |
| 3.0             | 85                       |
| 4.0             | X                        |

What is the most likely value of X?

- A 15
- B 42
- C 50
- D 60

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**0625/13**

Paper 1 Multiple Choice

**October/November 2016**

MARK SCHEME

Maximum Mark: 40

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|               | <b>Cambridge IGCSE – October/November 2016</b> | <b>0625</b>     | <b>13</b>    |

| <i>Question Number</i> | <i>Key</i> | <i>Question Number</i> | <i>Key</i> |
|------------------------|------------|------------------------|------------|
| 1                      | A          | 21                     | B          |
| 2                      | A          | 22                     | B          |
| 3                      | D          | 23                     | A          |
| 4                      | B          | 24                     | B          |
| 5                      | B          | 25                     | C          |
| 6                      | C          | 26                     | B          |
| 7                      | D          | 27                     | A          |
| 8                      | D          | 28                     | C          |
| 9                      | A          | 29                     | A          |
| 10                     | D          | 30                     | B          |
| 11                     | A          | 31                     | B          |
| 12                     | A          | 32                     | C          |
| 13                     | C          | 33                     | B          |
| 14                     | D          | 34                     | B          |
| 15                     | C          | 35                     | D          |
| 16                     | A          | 36                     | C          |
| 17                     | D          | 37                     | A          |
| 18                     | C          | 38                     | D          |
| 19                     | D          | 39                     | C          |
| 20                     | D          | 40                     | B          |



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**PHYSICS**

**0625/12**

Paper 1 Multiple Choice

**October/November 2016**

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Maximum Mark: 40

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| <i>Question Number</i> | <i>Key</i> | <i>Question Number</i> | <i>Key</i> |
|------------------------|------------|------------------------|------------|
| 1                      | A          | 21                     | B          |
| 2                      | A          | 22                     | A          |
| 3                      | D          | 23                     | D          |
| 4                      | D          | 24                     | B          |
| 5                      | D          | 25                     | C          |
| 6                      | C          | 26                     | A          |
| 7                      | A          | 27                     | A          |
| 8                      | A          | 28                     | B          |
| 9                      | A          | 29                     | A          |
| 10                     | C          | 30                     | B          |
| 11                     | A          | 31                     | C          |
| 12                     | A          | 32                     | D          |
| 13                     | C          | 33                     | B          |
| 14                     | A          | 34                     | B          |
| 15                     | B          | 35                     | D          |
| 16                     | A          | 36                     | C          |
| 17                     | A          | 37                     | A          |
| 18                     | D          | 38                     | A          |
| 19                     | C          | 39                     | C          |
| 20                     | C          | 40                     | C          |





**Cambridge International Examinations**  
Cambridge International General Certificate of Secondary Education

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**PHYSICS**

**0625/11**

Paper 1 Multiple Choice

**October/November 2016**

MARK SCHEME

Maximum Mark: 40

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **2** printed pages.

|               |                                                |                 |              |
|---------------|------------------------------------------------|-----------------|--------------|
| <b>Page 2</b> | <b>Mark Scheme</b>                             | <b>Syllabus</b> | <b>Paper</b> |
|               | <b>Cambridge IGCSE – October/November 2016</b> | <b>0625</b>     | <b>11</b>    |

| <i>Question Number</i> | <i>Key</i> | <i>Question Number</i> | <i>Key</i> |
|------------------------|------------|------------------------|------------|
| 1                      | C          | 21                     | C          |
| 2                      | A          | 22                     | D          |
| 3                      | D          | 23                     | D          |
| 4                      | D          | 24                     | A          |
| 5                      | B          | 25                     | C          |
| 6                      | D          | 26                     | D          |
| 7                      | B          | 27                     | A          |
| 8                      | D          | 28                     | B          |
| 9                      | A          | 29                     | A          |
| 10                     | C          | 30                     | B          |
| 11                     | A          | 31                     | A          |
| 12                     | A          | 32                     | A          |
| 13                     | C          | 33                     | B          |
| 14                     | B          | 34                     | A          |
| 15                     | C          | 35                     | A          |
| 16                     | A          | 36                     | C          |
| 17                     | A          | 37                     | A          |
| 18                     | B          | 38                     | A          |
| 19                     | B          | 39                     | C          |
| 20                     | D          | 40                     | D          |



**Cambridge International Examinations**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

Paper 1 Multiple Choice

**0625/13**

**October/November 2015**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)



**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

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There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

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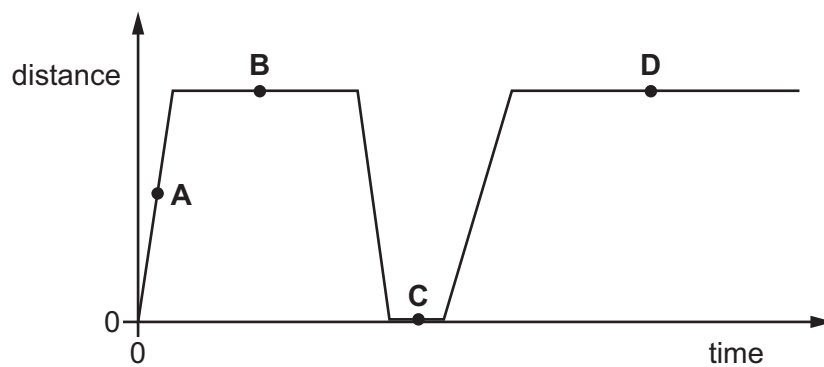
- 1 A student uses a measuring cylinder to measure the volume of a quantity of water.

Which action would make her result **less** accurate?

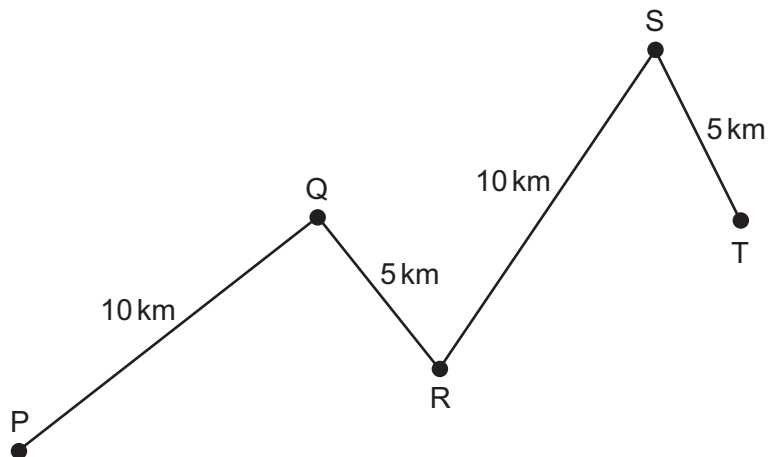
- A making sure her eye is level with the water surface
- B making sure the cylinder is vertical
- C reading the bottom of the meniscus
- D using the largest measuring cylinder possible

- 2 The diagram shows the distance-time graph for a car.

At which labelled point is the car moving with constant speed?



- 3 A car travels along the route PQRST in 30 minutes.



What is the average speed of the car?

- A 10 km/hour
- B 20 km/hour
- C 30 km/hour
- D 60 km/hour

- 4 What is the weight of an object?
- A the force of gravity on the object
  - B the gravitational potential energy of the object
  - C the internal energy of the object
  - D the mass of the object

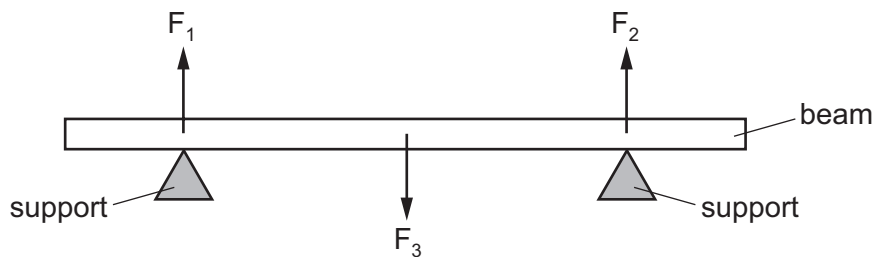
- 5 The mass of a piece of metal is 1200 g.

A measuring cylinder contains  $150 \text{ cm}^3$  of water.

The piece of metal is put into the measuring cylinder. The water level rises to  $250 \text{ cm}^3$  and covers the metal.

What is the density of the metal?

- A  $3.0 \text{ g/cm}^3$       B  $4.8 \text{ g/cm}^3$       C  $8.0 \text{ g/cm}^3$       D  $12.0 \text{ g/cm}^3$
- 6 A heavy beam rests on two supports. The diagram shows the only three forces  $F_1$ ,  $F_2$  and  $F_3$  acting on the beam.



The beam is in equilibrium.

Which statement is correct?

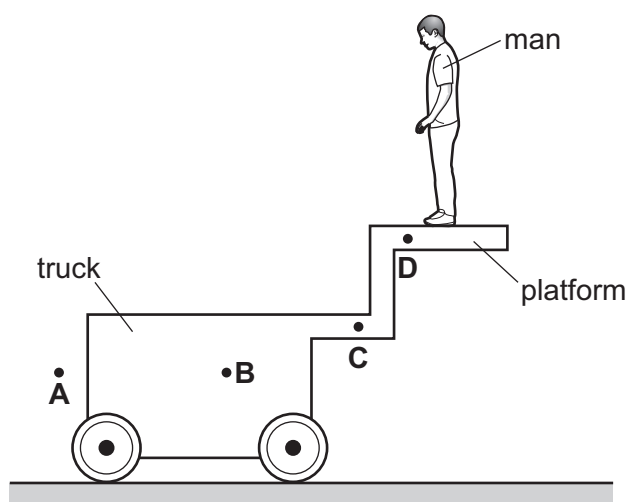
- A All the forces are equal in size.
- B The resultant force on the beam is in the opposite direction to the resultant turning effect.
- C The resultant force on the beam is zero and the resultant turning effect on the beam is zero.
- D The total upward force is twice the total downward force.

- 7 A heavy truck on wheels has a platform attached to it.

A man stands on the platform.

The truck does not fall over.

Which position **A**, **B**, **C** or **D** could be the centre of mass of the whole system (truck, platform and man)?



- 8 Which row gives the energy change in a battery and the energy change in a solar cell?

|          | battery                | solar cell          |
|----------|------------------------|---------------------|
| <b>A</b> | chemical to electrical | electrical to light |
| <b>B</b> | chemical to electrical | light to electrical |
| <b>C</b> | electrical to chemical | electrical to light |
| <b>D</b> | electrical to chemical | light to electrical |

- 9 Electrical energy may be obtained from nuclear fission.

In which order is the energy transferred in this process?

- A** nuclear fuel → generator → reactor and boiler → turbines
- B** nuclear fuel → generator → turbines → reactor and boiler
- C** nuclear fuel → reactor and boiler → generator → turbines
- D** nuclear fuel → reactor and boiler → turbines → generator

- 10 A man is in contact with the floor.

In which of these situations does he produce the least pressure on the floor?

- A kneeling
- B lying flat on his back
- C standing on both feet
- D standing on one foot

- 11 The water in a lake is at  $5^{\circ}\text{C}$ . A diver measures the pressure of the water at two different depths in the lake. He repeats the measurements on a different day when the water is at  $15^{\circ}\text{C}$ .

The density of the water decreases when its temperature increases.

Which combination of depth and temperature produces the greatest water pressure?

|   | depth / m | temperature / $^{\circ}\text{C}$ |
|---|-----------|----------------------------------|
| A | 10        | 5                                |
| B | 10        | 15                               |
| C | 20        | 5                                |
| D | 20        | 15                               |

- 12 A car tyre has a constant volume.

Why does the pressure of the air in the tyre increase when its temperature increases?

- A The air molecules hit each other less often.
- B The air molecules hit the inside of the tyre less often.
- C The average speed of the air molecules in the tyre is greater.
- D There are more air molecules in the tyre.

- 13 A swimmer feels cold after leaving warm water on a warm, windy day.

Why does she feel cold even though the air is warm?

- A The less energetic water molecules on her skin escape quickly.
- B The more energetic water molecules on her skin do not escape quickly.
- C The water on her skin does not evaporate quickly enough to keep her warm.
- D The water on her skin evaporates quickly and cools her skin.

- 14** A circular metal disc is heated.

Which quantity decreases?

- A** its density
  - B** its diameter
  - C** its thickness
  - D** its volume
- 15** The diagram shows a mercury-in-glass thermometer. The scale of the thermometer has not been marked.



The length  $l$  increases uniformly with temperature.

The length  $l$  is measured when the thermometer bulb is placed in water at  $0^{\circ}\text{C}$ , and also when it is in water at  $100^{\circ}\text{C}$ . The table shows the results.

| temperature / $^{\circ}\text{C}$ | length $l$ / cm |
|----------------------------------|-----------------|
| 0                                | 2.0             |
| 100                              | 26.0            |

What is the value of  $l$  when the bulb is placed in water at  $50^{\circ}\text{C}$ ?

- A** 12.0 cm      **B** 13.0 cm      **C** 14.0 cm      **D** 16.0 cm
- 16** A heating engineer fits a heater to the ceiling of an office so that workers in the office are kept warm.

How does thermal energy reach the workers below the heater?

- A** conduction and convection
- B** convection and radiation
- C** convection only
- D** radiation only



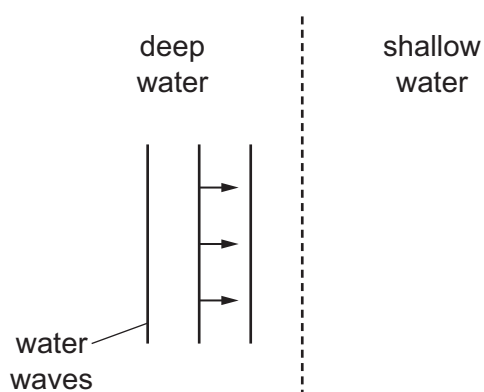
- 17 Which row shows how heating changes the density of air, and the name of the method of energy transfer caused by this density change?

|          | density   | method of energy transfer |
|----------|-----------|---------------------------|
| <b>A</b> | decreases | conduction                |
| <b>B</b> | decreases | convection                |
| <b>C</b> | increases | conduction                |
| <b>D</b> | increases | convection                |

- 18 Which row correctly defines a type of wave and gives a correct example?

|          | wave type    | direction of vibrations                   | example     |
|----------|--------------|-------------------------------------------|-------------|
| <b>A</b> | longitudinal | parallel to direction of wave travel      | radio waves |
| <b>B</b> | longitudinal | perpendicular to direction of wave travel | light waves |
| <b>C</b> | transverse   | parallel to direction of wave travel      | light waves |
| <b>D</b> | transverse   | perpendicular to direction of wave travel | radio waves |

- 19 A water wave moves quickly in deep water.

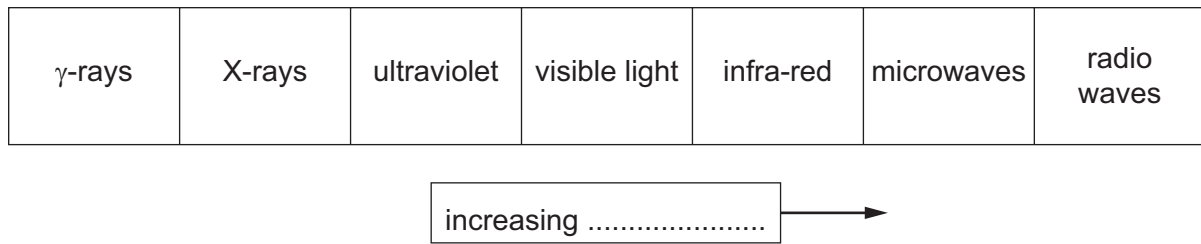


The wave now enters shallow water and its speed decreases.

Which row shows what happens to the frequency of the wave, and what happens to the wavelength of the wave?

|          | frequency       | wavelength      |
|----------|-----------------|-----------------|
| <b>A</b> | decreases       | decreases       |
| <b>B</b> | decreases       | does not change |
| <b>C</b> | does not change | decreases       |
| <b>D</b> | does not change | does not change |

20 The diagram shows the electromagnetic spectrum.

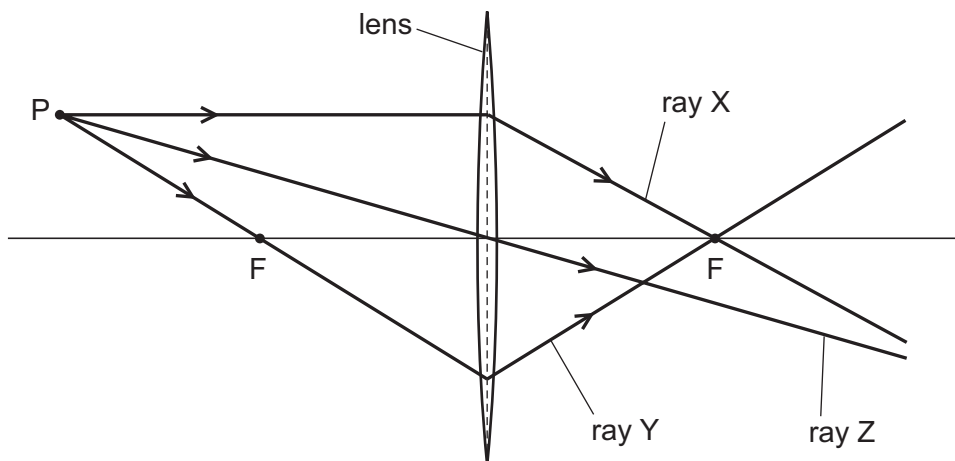


A word is missing from the label below the spectrum.

Which word is missing?

- A** amplitude
- B** frequency
- C** speed
- D** wavelength

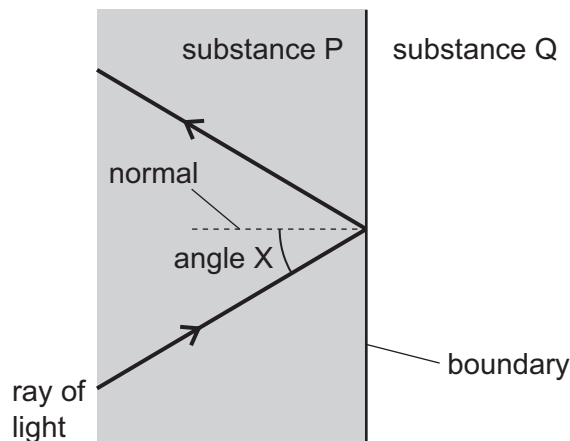
21 A student draws a diagram representing three rays of light from point P passing through a converging lens. Each point labelled F is a principal focus of the lens.



Which of the rays has the student drawn correctly?

- A** ray X and ray Y
- B** ray X and ray Z
- C** ray Y only
- D** ray Z only

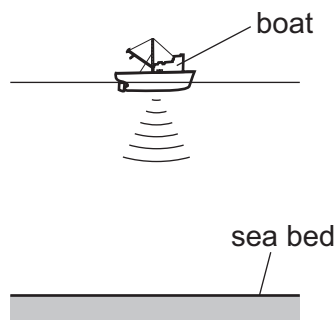
- 22 The diagram shows a ray of light travelling in a substance P. The ray reaches a boundary with a substance Q. Total internal reflection occurs at the boundary.



Which row contains correct statements about angle X and about the optical density of substance Q?

|          | angle X                         | substance Q                 |
|----------|---------------------------------|-----------------------------|
| <b>A</b> | smaller than the critical angle | less dense than substance P |
| <b>B</b> | smaller than the critical angle | more dense than substance P |
| <b>C</b> | greater than the critical angle | less dense than substance P |
| <b>D</b> | greater than the critical angle | more dense than substance P |

- 23 A pulse of sound is produced at the bottom of a boat. The sound travels through the water and is reflected from the sea bed. The sound reaches the boat again after 1.2 s. The speed of sound in the water is 1500 m/s.



How far below the bottom of the boat is the sea bed?

- A** 450 m      **B** 900 m      **C** 1800 m      **D** 3600 m

- 24** Sounds are produced by vibrating objects. A certain object vibrates but a person nearby cannot hear any sound.

Which statement could explain why nothing is heard?

- A** The amplitude of the sound waves is too large.
- B** The frequency of the vibration is too high.
- C** The sound waves are transverse.
- D** The speed of the sound waves is too high.

- 25** Which metal could be used for a permanent magnet and which metal could be used for the core of an electromagnet?

|          | permanent magnet | core of electromagnet |
|----------|------------------|-----------------------|
| <b>A</b> | iron             | copper                |
| <b>B</b> | iron             | steel                 |
| <b>C</b> | steel            | copper                |
| <b>D</b> | steel            | iron                  |

- 26** Which procedure may be used to demagnetise a steel bar?

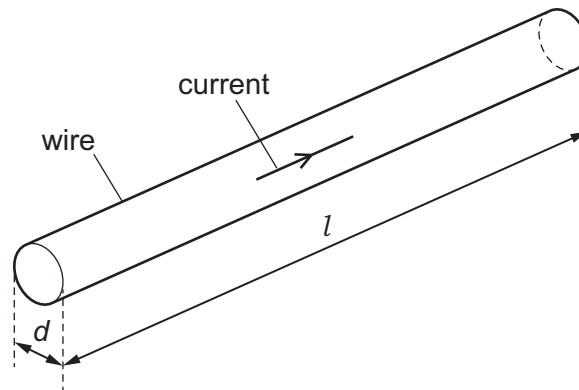
- A** cooling it in a freezer for several hours
- B** earthing it with a copper wire for several seconds
- C** removing it slowly from a coil carrying an alternating current (a.c.)
- D** rubbing it in one direction with a woollen cloth

- 27** A plastic rod is rubbed with a cotton cloth. This process causes the rod and the cloth to become charged. These charges cause a force between the rod and the cloth.

Which row compares the charges on the rod and the cloth, and describes the effect of the force between the rod and the cloth?

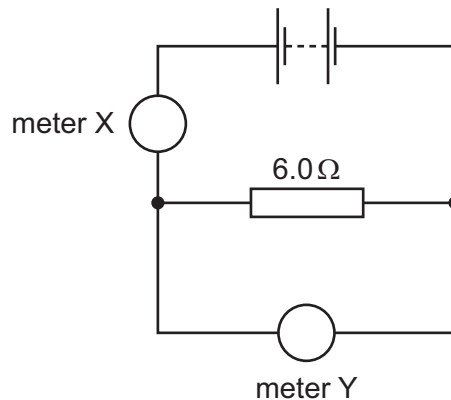
|          | charges on rod and cloth | effect       |
|----------|--------------------------|--------------|
| <b>A</b> | opposite                 | they attract |
| <b>B</b> | opposite                 | they repel   |
| <b>C</b> | the same                 | they attract |
| <b>D</b> | the same                 | they repel   |

- 28 A metal wire of circular cross-section has diameter  $d$  and length  $l$ .



Which pair of changes, if both are carried out, **must** increase the resistance of the wire?

- A decrease  $l$  and decrease  $d$
  - B decrease  $l$  and increase  $d$
  - C increase  $l$  and decrease  $d$
  - D increase  $l$  and increase  $d$
- 29 The circuit shown contains a battery, a  $6.0\ \Omega$  resistor and two meters X and Y.
- One meter records current and one meter records potential difference.



Which row shows possible values for the readings on the meters?

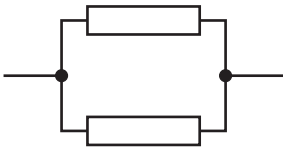
|   | meter X | meter Y |
|---|---------|---------|
| A | 2.0 A   | 12 V    |
| B | 2.0 V   | 12 A    |
| C | 12 A    | 2.0 V   |
| D | 12 V    | 2.0 A   |

- 30** A student designs a circuit to switch on a lamp after a time delay.

Which components are used in a time-delay circuit?

- A** a light-dependent resistor and a relay
- B** a resistor and a capacitor
- C** a resistor and a transformer
- D** a thermistor and a variable resistor

- 31** Identical resistors are connected together to form arrangements X, Y and Z.



arrangement X



arrangement Y

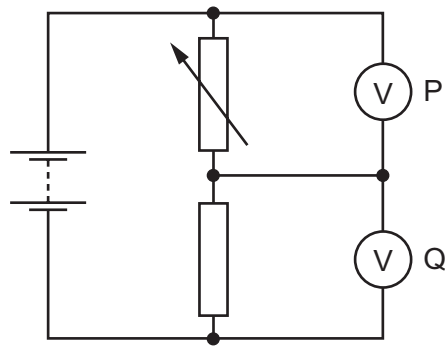


arrangement Z

What is the correct order of the resistances of the arrangements from the largest to the smallest?

- A**  $X \rightarrow Y \rightarrow Z$
- B**  $Y \rightarrow X \rightarrow Z$
- C**  $Z \rightarrow X \rightarrow Y$
- D**  $Z \rightarrow Y \rightarrow X$

32 The diagram shows a potential divider circuit.



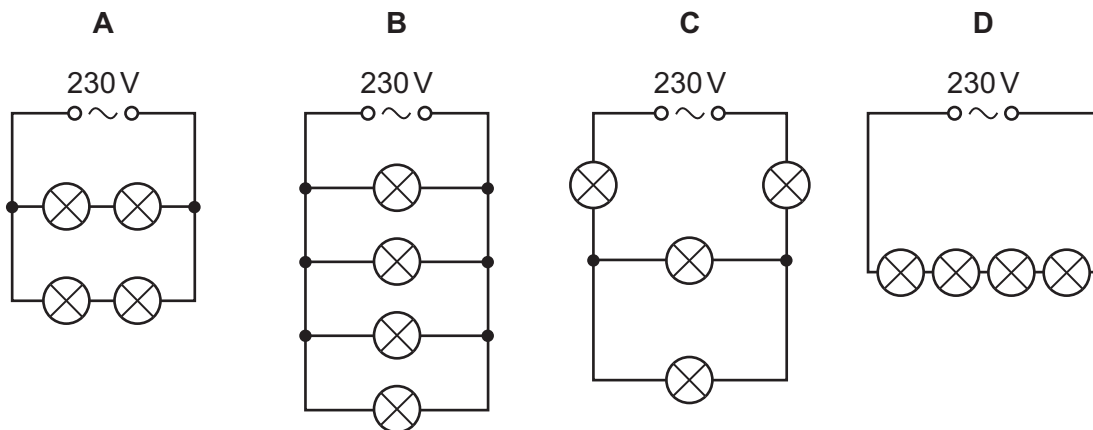
The resistance of the variable resistor is increased.

Which row shows what happens to the readings on voltmeter P and on voltmeter Q?

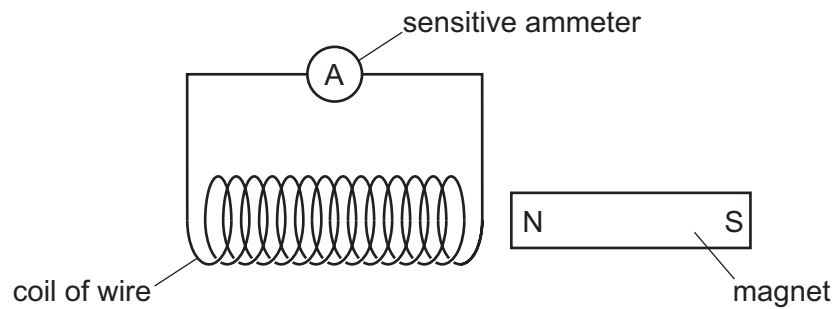
|          | reading on voltmeter P | reading on voltmeter Q |
|----------|------------------------|------------------------|
| <b>A</b> | decreases              | decreases              |
| <b>B</b> | decreases              | increases              |
| <b>C</b> | increases              | decreases              |
| <b>D</b> | increases              | increases              |

33 Four lamps are each labelled '60 W 230 V'.

In which circuit are the lamps connected so that they operate at normal brightness?



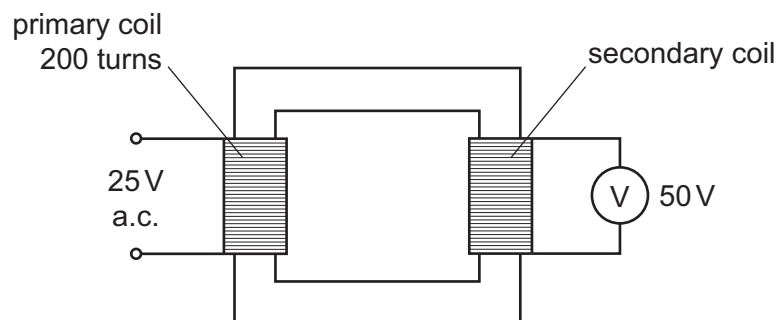
- 34 A student investigates electromagnetic induction. She has a bar magnet and a coil of wire that is connected to a sensitive ammeter.



Which movement does **not** cause a reading on the ammeter?

- A moving the coil to the right
  - B moving both the magnet and the coil to the left at the same speed
  - C moving both the magnet and the coil towards each other at the same speed
  - D moving the magnet to the left
- 35 The primary coil of a transformer has 200 turns. This primary coil is connected to an a.c. power supply of 25 V.

A voltmeter connected across the secondary coil reads 50 V.

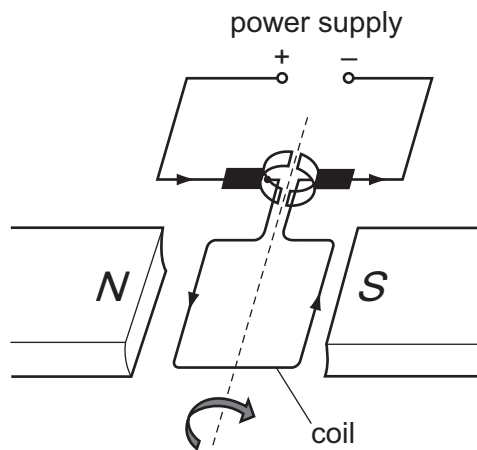


How many turns are on the secondary coil?

- A 25
- B 100
- C 200
- D 400



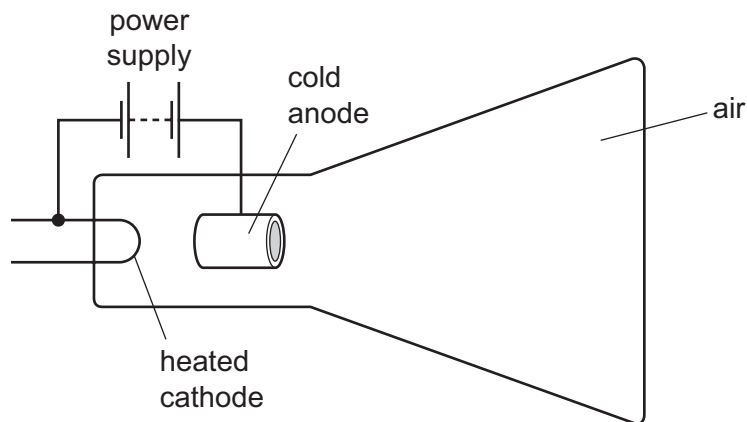
36 A current-carrying coil in a magnetic field experiences a turning effect.



How can the turning effect be increased?

- A Increase the number of turns on the coil.
- B Reduce the size of the current.
- C Reverse the direction of the magnetic field.
- D Use thinner wire for the coil.

37 The diagram shows a cathode-ray tube. The tube is not working.



Which change should be made to make the tube work to produce cathode rays?

- A Change round the connections of the power supply.
- B Heat the anode instead of the cathode.
- C Use an a.c. power supply instead of a d.c. power supply.
- D Use a vacuum in the tube instead of air.

- 38 A radioactive nucleus contains 128 nucleons. It emits a  $\beta$ -particle.

How many nucleons are now in the nucleus?

- A 124                      B 127                      C 128                      D 129

- 39 A student investigates how the radiation from a radioactive source changes with time.

The table shows the results from the detector used by the student.

| time /<br>minutes | count rate /<br>counts per minute |
|-------------------|-----------------------------------|
| 0                 | 340                               |
| 2.0               | 180                               |
| 4.0               | 100                               |
| 6.0               | 60                                |
| 8.0               | 40                                |

The experiment is repeated by many other students, who also measure the count rate every two minutes.

The half-life of the source is known to be exactly 2.0 minutes.

Why is the measured count rate **always greater** than half the previous value?

- A Radioactive emissions occur randomly with time.  
 B The detector used is very close to the source.  
 C There is background radiation present.  
 D The radioactive source is decaying.

- 40 The nuclide notation for radium-226 is  $^{226}_{88}\text{Ra}$ .

How many electrons orbit the nucleus of a neutral atom of radium-226?

- A 0                      B 88                      C 138                      D 226

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**Cambridge International Examinations**  
Cambridge International General Certificate of Secondary Education

**PHYSICS**

Paper 1 Multiple Choice

**0625/12**

**October/November 2015**

**45 minutes**

Additional Materials: Multiple Choice Answer Sheet  
Soft clean eraser  
Soft pencil (type B or HB recommended)



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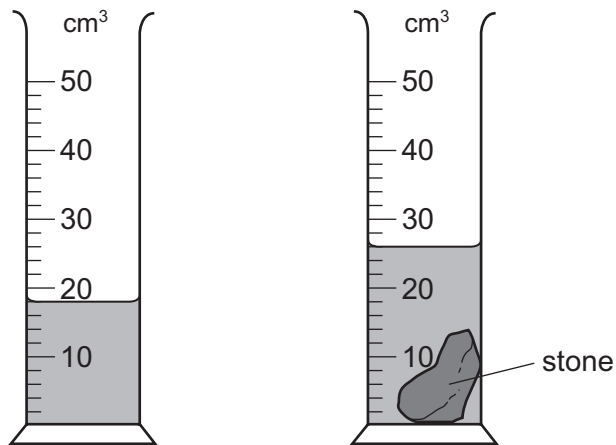
Electronic calculators may be used.

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This document consists of **18** printed pages and **2** blank pages.



- 1 The diagram shows a measuring cylinder used to measure the volume of a small stone.



What is the volume of the stone?

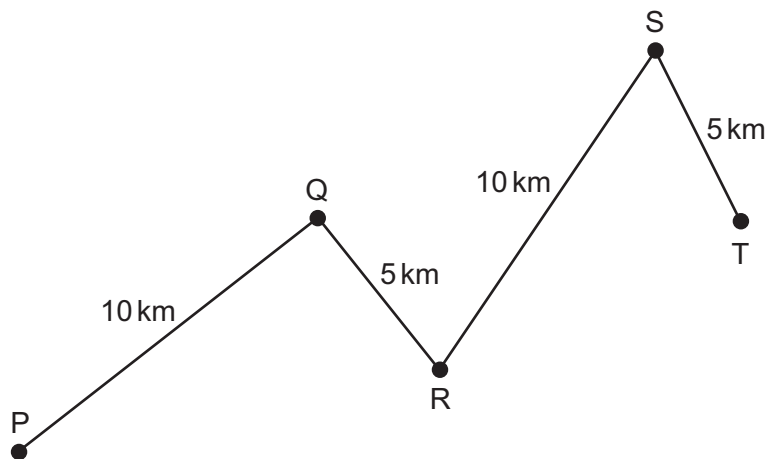
- A** 8 cm<sup>3</sup>      **B** 9 cm<sup>3</sup>      **C** 14 cm<sup>3</sup>      **D** 26 cm<sup>3</sup>
- 2 The table shows the readings on a car speedometer at 5 second intervals.

| time / s | speed<br>km/h |
|----------|---------------|
| 0        | 0             |
| 5        | 30            |
| 10       | 50            |
| 15       | 60            |
| 20       | 65            |

Which row describes the speed and the acceleration of the car?

|          | speed      | acceleration |
|----------|------------|--------------|
| <b>A</b> | decreasing | zero         |
| <b>B</b> | decreasing | not zero     |
| <b>C</b> | increasing | zero         |
| <b>D</b> | increasing | not zero     |

- 3 A car travels along the route PQRST in 30 minutes.



What is the average speed of the car?

- A** 10 km/hour    **B** 20 km/hour    **C** 30 km/hour    **D** 60 km/hour
- 4 Which statement about mass or weight is correct?
- A** Mass is a force.  
**B** Mass is measured in newtons.  
**C** Weight is a force.  
**D** Weight is measured in kilograms.
- 5 Two cylinders are made of the same metal. Both cylinders have the same cross-sectional area but one is longer than the other.



cylinder 1

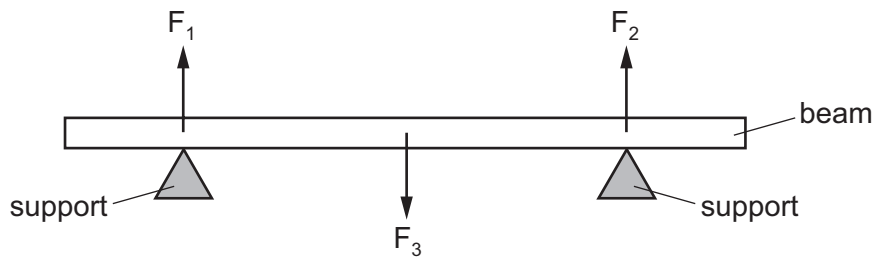


cylinder 2

Which quantity is the same for both cylinders?

- A** density  
**B** mass  
**C** resistance  
**D** volume

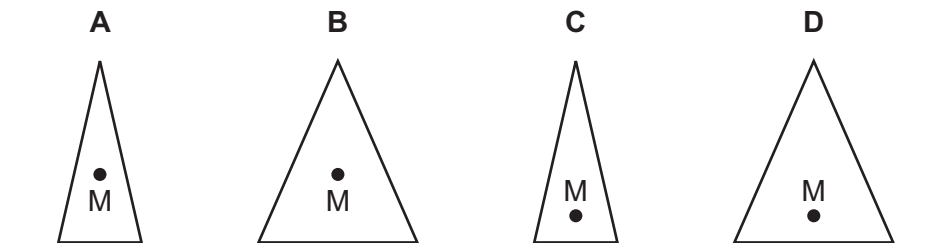
- 6 A heavy beam rests on two supports. The diagram shows the only three forces  $F_1$ ,  $F_2$  and  $F_3$  acting on the beam.



The beam is in equilibrium.

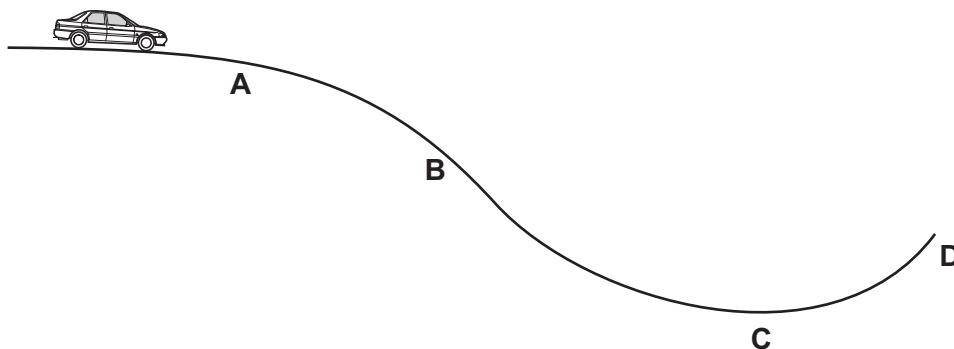
Which statement is correct?

- A** All the forces are equal in size.
  - B** The resultant force on the beam is in the opposite direction to the resultant turning effect.
  - C** The resultant force on the beam is zero and the resultant turning effect on the beam is zero.
  - D** The total upward force is twice the total downward force.
- 7 The diagrams show four solid cones. The centre of mass of each cone is marked by a point labelled M.



- 8 A car is stationary at the top of a hill with the engine switched off. The brakes are released and the car rolls down the hill.

At which labelled point does the car have the greatest kinetic energy? Ignore friction.



- 9 Electrical energy may be obtained from nuclear fission.

In which order is the energy transferred in this process?

- A nuclear fuel → generator → reactor and boiler → turbines
- B nuclear fuel → generator → turbines → reactor and boiler
- C nuclear fuel → reactor and boiler → generator → turbines
- D nuclear fuel → reactor and boiler → turbines → generator

- 10 The table gives four forces and the surface area on which each force acts.

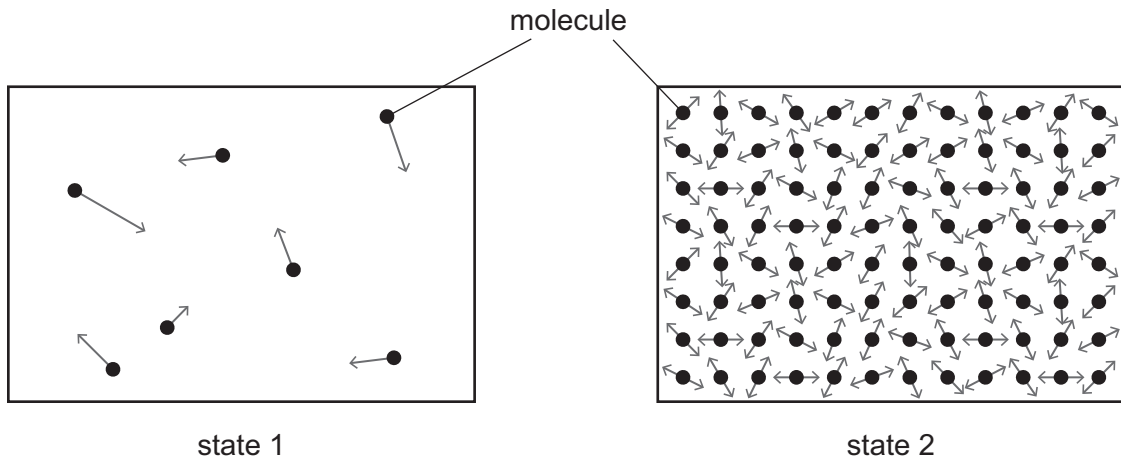
Which row gives the largest pressure on the surface?

|   | force / N | area / m <sup>2</sup> |
|---|-----------|-----------------------|
| A | 20        | 2                     |
| B | 40        | 2                     |
| C | 20        | 4                     |
| D | 40        | 4                     |

- 11 What does a barometer measure?

- A the current in a circuit
- B the density of a liquid
- C the pressure of air
- D the temperature of an object

- 12 The diagrams represent the molecules in two different states of matter. The arrows show the motion of the molecules.



What is state 1, and what is state 2?

|          | state 1 | state 2 |
|----------|---------|---------|
| <b>A</b> | gas     | liquid  |
| <b>B</b> | gas     | solid   |
| <b>C</b> | liquid  | gas     |
| <b>D</b> | liquid  | solid   |

- 13 A swimmer feels cold after leaving warm water on a warm, windy day.

Why does she feel cold even though the air is warm?

- A** The less energetic water molecules on her skin escape quickly.
- B** The more energetic water molecules on her skin do not escape quickly.
- C** The water on her skin does not evaporate quickly enough to keep her warm.
- D** The water on her skin evaporates quickly and cools her skin.
- 14 A circular metal disc is heated.

Which quantity decreases?

- A** its density
- B** its diameter
- C** its thickness
- D** its volume



- 15** A block of copper and a block of lead are heated. The internal energy of each block increases by the same amount.

The block of copper has a lower thermal capacity than the block of lead.

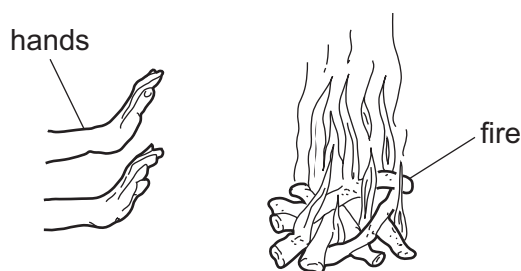
Which conclusion can be made from this information?

- A** The temperature increase of the copper is greater than the temperature increase of the lead.
  - B** The temperature increase of the copper is the same as the temperature increase of the lead.
  - C** The temperature increase of the copper is less than the temperature increase of the lead.
  - D** The melting point of copper is lower than the melting point of lead.
- 16** In a refrigerator, the cooling unit can be fitted either at the top or at the bottom. In an oven, the heater can be fitted either at the top or at the bottom.

Which row shows the best position for the cooling unit and the best position for the heater?

|          | cooling unit | heater |
|----------|--------------|--------|
| <b>A</b> | bottom       | bottom |
| <b>B</b> | bottom       | top    |
| <b>C</b> | top          | bottom |
| <b>D</b> | top          | top    |

- 17** A girl is outdoors. She warms her hands by holding them near a fire, as shown.



How does the heat from the fire reach her hands?

- A** conduction only
- B** convection and conduction
- C** convection and radiation
- D** radiation only